

scenario_A_reproduce_v4

June 8, 2026

- consolidate functions into the [experiments_utils.py](#) file, and call the function here
- change from 100 trials to 1000 trials

```
[3]: from experiments_utils import *
```

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[ ]: CURRENT_BOUNDS = (0.01, 1.5)
    EPSILON = 1e-3

    REF_I_1 = 0.8218
    REF_I_3 = 1.0430
    REF_I = [REF_I_1, REF_I_3]

    # parameter setting
    N_RUNS_A = 1000
    A_BEAMLINE_LEN = 10

    METHODS_A = [
        ("Nelder-Mead", None),
        ("L-BFGS-B", "2-point"),
        ("SLSQP", "2-point"),
        ("COBYLA", None),
        ("trust-constr", "2-point"),
    ]

    A_VARS = {
        1: ["I", "current", lambda x: x],
        3: ["I2", "current", lambda x: x],
    }
    A_OBJ = {
        8: [{"measure": ["x", "alpha"], "goal": 0.0, "weight": 1.0}],
        9: [{"measure": ["y", "alpha"], "goal": 0.0, "weight": 1.0}],
    }
```

```
[ ]: # this is default version
    METHOD_OPTIONS = {
        "L-BFGS-B": {
```

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    "maxcor": 10,
    "ftol": 2.220446049250313e-09,
    "gtol": 1e-5,
    "eps": 1e-8,
    "maxfun": 15000,
    "maxiter": 15000,
    "maxls": 20,
},
"SLSQP": {
    "maxiter": 100,
    "ftol": 1e-6,
    "iprint": 1,
    "disp": False,
    "eps": 1.4901161193847656e-08,
    "finite_diff_rel_step": None,
    "workers": None,
},
"trust-constr": {
    "xtol": 1e-8,
    "gtol": 1e-8,
    "barrier_tol": 1e-8,
    "sparse_jacobian": None,
    "maxiter": 1000,
    "verbose": 0,
    "finite_diff_rel_step": None,
    "initial_constr_penalty": 1.0,
    "initial_tr_radius": 1.0,
    "initial_barrier_parameter": 0.1,
    "initial_barrier_tolerance": 0.1,
    "factorization_method": None,
    "disp": False,
    "workers": None,
},
"Nelder-Mead": {
    "maxiter": None,
    "maxfev": None,
    "disp": False,
    "return_all": False,
    "initial_simplex": None,
    "xatol": 1e-4,
    "fatol": 1e-4,
    "adaptive": False,
},
"COBYLA": {
    "rhobeg": 1.0,
    "tol": 1e-4,
    "maxiter": 1000,

```

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        "disp": 0,
        "catol": None,
        "f_target": float("-inf"),
    },
}

```

```

[4]: results_A, df_A, final_summary = run_scenario_A(CURRENT_BOUNDS, EPSILON,
↳N_RUNS_A, REF_I, A_OBJ, A_VARS, A_BEAMLINE_LEN,
↳METHODS_A,scale="log",METHOD_OPTIONS=METHOD_OPTIONS)
display_results(results_A, REF_I)

```

Scenario A - Nelder-Mead (1000 runs)

Nelder-Mead	run 1/1000	MSE=3.342e-09	nit=37	nfev=73	t=13.04s
Nelder-Mead	run 2/1000	MSE=1.366e-08	nit=36	nfev=69	t=11.84s
Nelder-Mead	run 3/1000	MSE=5.457e-09	nit=46	nfev=88	t=15.55s
Nelder-Mead	run 4/1000	MSE=6.127e-09	nit=40	nfev=75	t=15.10s
Nelder-Mead	run 5/1000	MSE=4.733e-09	nit=38	nfev=74	t=13.45s
Nelder-Mead	run 6/1000	MSE=1.183e-08	nit=30	nfev=58	t=10.59s
Nelder-Mead	run 7/1000	MSE=1.682e-08	nit=42	nfev=82	t=14.73s
Nelder-Mead	run 8/1000	MSE=1.065e-08	nit=53	nfev=103	t=18.36s
Nelder-Mead	run 9/1000	MSE=8.154e-09	nit=44	nfev=86	t=15.44s
Nelder-Mead	run 10/1000	MSE=9.202e-09	nit=52	nfev=99	t=20.99s
Nelder-Mead	run 11/1000	MSE=2.834e-09	nit=42	nfev=78	t=13.84s
Nelder-Mead	run 12/1000	MSE=4.347e-09	nit=49	nfev=94	t=17.92s
Nelder-Mead	run 13/1000	MSE=6.193e-09	nit=60	nfev=116	t=19.02s
Nelder-Mead	run 14/1000	MSE=6.420e-09	nit=34	nfev=67	t=11.34s
Nelder-Mead	run 15/1000	MSE=8.500e-09	nit=32	nfev=61	t=12.08s
Nelder-Mead	run 16/1000	MSE=8.383e-09	nit=38	nfev=74	t=15.58s
Nelder-Mead	run 17/1000	MSE=4.354e-09	nit=46	nfev=85	t=18.95s
Nelder-Mead	run 18/1000	MSE=3.851e-09	nit=46	nfev=86	t=17.17s
Nelder-Mead	run 19/1000	MSE=5.771e-09	nit=31	nfev=60	t=11.81s
Nelder-Mead	run 20/1000	MSE=1.082e-08	nit=52	nfev=94	t=14.94s
Nelder-Mead	run 21/1000	MSE=2.154e-08	nit=37	nfev=73	t=11.65s
Nelder-Mead	run 22/1000	MSE=1.517e-09	nit=38	nfev=72	t=11.26s
Nelder-Mead	run 23/1000	MSE=7.414e-09	nit=45	nfev=85	t=13.36s
Nelder-Mead	run 24/1000	MSE=6.181e-09	nit=50	nfev=97	t=15.23s
Nelder-Mead	run 25/1000	MSE=6.346e-09	nit=44	nfev=87	t=13.82s
Nelder-Mead	run 26/1000	MSE=1.119e-08	nit=43	nfev=82	t=13.61s
Nelder-Mead	run 27/1000	MSE=7.497e-09	nit=60	nfev=114	t=18.41s
Nelder-Mead	run 28/1000	MSE=2.929e-08	nit=38	nfev=71	t=11.13s
Nelder-Mead	run 29/1000	MSE=5.411e-09	nit=40	nfev=77	t=12.12s
Nelder-Mead	run 30/1000	MSE=1.090e-08	nit=44	nfev=82	t=13.80s
Nelder-Mead	run 31/1000	MSE=1.689e-08	nit=33	nfev=63	t=9.84s
Nelder-Mead	run 32/1000	MSE=1.892e-08	nit=42	nfev=79	t=12.38s
Nelder-Mead	run 33/1000	MSE=3.288e-09	nit=34	nfev=63	t=9.91s
Nelder-Mead	run 34/1000	MSE=5.085e-09	nit=48	nfev=87	t=14.13s
Nelder-Mead	run 35/1000	MSE=1.317e-08	nit=43	nfev=83	t=12.94s

Nelder-Mead	run 36/1000	MSE=4.680e-09	nit=35	nfev=69	t=11.22s
Nelder-Mead	run 37/1000	MSE=5.439e-09	nit=42	nfev=83	t=13.68s
Nelder-Mead	run 38/1000	MSE=1.021e-08	nit=52	nfev=97	t=15.41s
Nelder-Mead	run 39/1000	MSE=8.360e-09	nit=39	nfev=73	t=11.22s
Nelder-Mead	run 40/1000	MSE=1.589e+01	nit=28	nfev=52	t=8.22s
Nelder-Mead	run 41/1000	MSE=4.991e-09	nit=36	nfev=71	t=11.44s
Nelder-Mead	run 42/1000	MSE=1.448e-08	nit=36	nfev=68	t=10.84s
Nelder-Mead	run 43/1000	MSE=3.184e-09	nit=43	nfev=82	t=13.05s
Nelder-Mead	run 44/1000	MSE=1.035e-01	nit=13	nfev=25	t=3.75s
Nelder-Mead	run 45/1000	MSE=6.196e-08	nit=30	nfev=60	t=9.27s
Nelder-Mead	run 46/1000	MSE=1.998e-08	nit=31	nfev=60	t=9.55s
Nelder-Mead	run 47/1000	MSE=1.391e-08	nit=38	nfev=75	t=11.78s
Nelder-Mead	run 48/1000	MSE=7.661e-09	nit=37	nfev=69	t=10.82s
Nelder-Mead	run 49/1000	MSE=4.244e-08	nit=34	nfev=64	t=10.15s
Nelder-Mead	run 50/1000	MSE=4.558e-09	nit=68	nfev=128	t=20.09s
Nelder-Mead	run 51/1000	MSE=2.869e-08	nit=35	nfev=67	t=10.60s
Nelder-Mead	run 52/1000	MSE=2.998e-09	nit=36	nfev=69	t=10.80s
Nelder-Mead	run 53/1000	MSE=1.211e-09	nit=47	nfev=88	t=13.99s
Nelder-Mead	run 54/1000	MSE=7.644e-09	nit=44	nfev=84	t=13.26s
Nelder-Mead	run 55/1000	MSE=7.091e-09	nit=53	nfev=101	t=16.03s
Nelder-Mead	run 56/1000	MSE=8.736e-09	nit=31	nfev=57	t=8.87s
Nelder-Mead	run 57/1000	MSE=1.605e-08	nit=38	nfev=71	t=11.22s
Nelder-Mead	run 58/1000	MSE=1.308e-08	nit=46	nfev=88	t=13.89s
Nelder-Mead	run 59/1000	MSE=1.143e-08	nit=44	nfev=87	t=13.42s
Nelder-Mead	run 60/1000	MSE=4.749e-09	nit=49	nfev=92	t=14.54s
Nelder-Mead	run 61/1000	MSE=3.049e-10	nit=56	nfev=104	t=16.29s
Nelder-Mead	run 62/1000	MSE=1.957e-09	nit=36	nfev=70	t=10.90s
Nelder-Mead	run 63/1000	MSE=1.946e-08	nit=34	nfev=67	t=10.60s
Nelder-Mead	run 64/1000	MSE=1.239e-08	nit=43	nfev=84	t=13.04s
Nelder-Mead	run 65/1000	MSE=2.897e-09	nit=53	nfev=99	t=15.81s
Nelder-Mead	run 66/1000	MSE=7.795e-10	nit=36	nfev=67	t=10.63s
Nelder-Mead	run 67/1000	MSE=2.826e-08	nit=34	nfev=67	t=10.51s
Nelder-Mead	run 68/1000	MSE=2.317e-08	nit=33	nfev=64	t=9.92s
Nelder-Mead	run 69/1000	MSE=1.589e+01	nit=26	nfev=50	t=7.81s
Nelder-Mead	run 70/1000	MSE=1.373e-09	nit=45	nfev=86	t=13.61s
Nelder-Mead	run 71/1000	MSE=1.035e-01	nit=16	nfev=31	t=4.83s
Nelder-Mead	run 72/1000	MSE=1.035e-01	nit=13	nfev=25	t=4.01s
Nelder-Mead	run 73/1000	MSE=6.131e-09	nit=39	nfev=76	t=11.83s
Nelder-Mead	run 74/1000	MSE=9.152e-09	nit=38	nfev=74	t=11.93s
Nelder-Mead	run 75/1000	MSE=5.658e-09	nit=35	nfev=66	t=10.38s
Nelder-Mead	run 76/1000	MSE=7.944e-09	nit=37	nfev=70	t=10.75s
Nelder-Mead	run 77/1000	MSE=2.722e-08	nit=33	nfev=63	t=9.94s
Nelder-Mead	run 78/1000	MSE=3.684e-08	nit=38	nfev=72	t=11.58s
Nelder-Mead	run 79/1000	MSE=2.249e-08	nit=40	nfev=78	t=12.18s
Nelder-Mead	run 80/1000	MSE=1.035e-01	nit=13	nfev=25	t=3.97s
Nelder-Mead	run 81/1000	MSE=5.841e-09	nit=39	nfev=75	t=11.70s
Nelder-Mead	run 82/1000	MSE=9.168e-09	nit=34	nfev=65	t=10.33s
Nelder-Mead	run 83/1000	MSE=1.717e-08	nit=29	nfev=57	t=8.98s

Nelder-Mead	run 84/1000	MSE=1.500e-08	nit=41	nfev=78	t=12.20s
Nelder-Mead	run 85/1000	MSE=2.072e-08	nit=41	nfev=76	t=11.79s
Nelder-Mead	run 86/1000	MSE=4.794e-09	nit=39	nfev=77	t=12.04s
Nelder-Mead	run 87/1000	MSE=2.480e-08	nit=30	nfev=60	t=9.36s
Nelder-Mead	run 88/1000	MSE=1.871e-09	nit=33	nfev=62	t=9.76s
Nelder-Mead	run 89/1000	MSE=6.489e-09	nit=40	nfev=75	t=11.86s
Nelder-Mead	run 90/1000	MSE=4.483e-09	nit=31	nfev=60	t=9.83s
Nelder-Mead	run 91/1000	MSE=3.870e-09	nit=58	nfev=113	t=17.86s
Nelder-Mead	run 92/1000	MSE=2.725e-08	nit=28	nfev=55	t=8.59s
Nelder-Mead	run 93/1000	MSE=4.530e-09	nit=36	nfev=70	t=10.90s
Nelder-Mead	run 94/1000	MSE=2.140e-08	nit=40	nfev=77	t=12.41s
Nelder-Mead	run 95/1000	MSE=4.015e-09	nit=31	nfev=59	t=9.52s
Nelder-Mead	run 96/1000	MSE=2.037e-09	nit=43	nfev=81	t=12.64s
Nelder-Mead	run 97/1000	MSE=1.338e-08	nit=33	nfev=65	t=10.37s
Nelder-Mead	run 98/1000	MSE=4.243e-09	nit=43	nfev=77	t=11.92s
Nelder-Mead	run 99/1000	MSE=8.823e-09	nit=35	nfev=69	t=10.94s
Nelder-Mead	run 100/1000	MSE=1.298e-08	nit=38	nfev=71	t=11.30s
Nelder-Mead	run 101/1000	MSE=4.784e-08	nit=37	nfev=69	t=10.86s
Nelder-Mead	run 102/1000	MSE=4.976e-09	nit=48	nfev=89	t=13.88s
Nelder-Mead	run 103/1000	MSE=2.600e-08	nit=35	nfev=68	t=10.51s
Nelder-Mead	run 104/1000	MSE=4.249e-09	nit=47	nfev=88	t=14.05s
Nelder-Mead	run 105/1000	MSE=8.822e-09	nit=34	nfev=66	t=10.30s
Nelder-Mead	run 106/1000	MSE=1.778e-09	nit=48	nfev=93	t=14.80s
Nelder-Mead	run 107/1000	MSE=3.918e-09	nit=51	nfev=95	t=14.93s
Nelder-Mead	run 108/1000	MSE=5.928e-09	nit=41	nfev=78	t=12.33s
Nelder-Mead	run 109/1000	MSE=4.319e-09	nit=40	nfev=77	t=12.07s
Nelder-Mead	run 110/1000	MSE=2.085e-08	nit=33	nfev=64	t=9.90s
Nelder-Mead	run 111/1000	MSE=1.723e-08	nit=41	nfev=77	t=12.03s
Nelder-Mead	run 112/1000	MSE=8.872e-09	nit=37	nfev=72	t=11.63s
Nelder-Mead	run 113/1000	MSE=1.573e-08	nit=33	nfev=63	t=9.89s
Nelder-Mead	run 114/1000	MSE=3.080e-08	nit=34	nfev=66	t=10.44s
Nelder-Mead	run 115/1000	MSE=1.589e+01	nit=20	nfev=36	t=5.61s
Nelder-Mead	run 116/1000	MSE=1.096e-08	nit=41	nfev=78	t=12.33s
Nelder-Mead	run 117/1000	MSE=7.361e-09	nit=47	nfev=91	

t=14.42s						
Nelder-Mead	run 118/1000	MSE=1.015e-08	nit=39	nfev=74		
t=11.75s						
Nelder-Mead	run 119/1000	MSE=6.377e-09	nit=35	nfev=63	t=9.89s	
Nelder-Mead	run 120/1000	MSE=1.553e-08	nit=42	nfev=83		
t=13.16s						
Nelder-Mead	run 121/1000	MSE=4.865e-09	nit=30	nfev=59	t=9.48s	
Nelder-Mead	run 122/1000	MSE=3.361e-09	nit=55	nfev=107		
t=17.13s						
Nelder-Mead	run 123/1000	MSE=4.183e-09	nit=35	nfev=69		
t=10.66s						
Nelder-Mead	run 124/1000	MSE=7.888e-08	nit=31	nfev=59	t=9.28s	
Nelder-Mead	run 125/1000	MSE=1.081e-08	nit=32	nfev=64		
t=10.03s						
Nelder-Mead	run 126/1000	MSE=1.874e-08	nit=40	nfev=75		
t=11.91s						
Nelder-Mead	run 127/1000	MSE=2.441e-08	nit=38	nfev=76		
t=11.82s						
Nelder-Mead	run 128/1000	MSE=7.725e-09	nit=55	nfev=108		
t=16.91s						
Nelder-Mead	run 129/1000	MSE=1.758e-09	nit=26	nfev=49	t=7.88s	
Nelder-Mead	run 130/1000	MSE=2.518e-09	nit=48	nfev=93		
t=14.76s						
Nelder-Mead	run 131/1000	MSE=1.945e-08	nit=37	nfev=69		
t=10.94s						
Nelder-Mead	run 132/1000	MSE=1.035e-01	nit=13	nfev=25	t=4.10s	
Nelder-Mead	run 133/1000	MSE=2.286e-09	nit=43	nfev=84		
t=12.97s						
Nelder-Mead	run 134/1000	MSE=4.295e-09	nit=41	nfev=74		
t=11.91s						
Nelder-Mead	run 135/1000	MSE=4.195e-09	nit=42	nfev=82		
t=13.17s						
Nelder-Mead	run 136/1000	MSE=3.278e-09	nit=37	nfev=70		
t=11.15s						
Nelder-Mead	run 137/1000	MSE=4.127e-09	nit=48	nfev=96		
t=15.14s						
Nelder-Mead	run 138/1000	MSE=1.368e-08	nit=34	nfev=64		
t=10.16s						
Nelder-Mead	run 139/1000	MSE=1.539e-09	nit=32	nfev=63	t=9.83s	
Nelder-Mead	run 140/1000	MSE=9.525e-09	nit=37	nfev=70		
t=10.98s						
Nelder-Mead	run 141/1000	MSE=1.494e-08	nit=49	nfev=94		
t=15.07s						
Nelder-Mead	run 142/1000	MSE=4.235e-09	nit=40	nfev=80		
t=12.66s						
Nelder-Mead	run 143/1000	MSE=1.994e-08	nit=26	nfev=50	t=8.14s	
Nelder-Mead	run 144/1000	MSE=1.209e-08	nit=51	nfev=98		
t=15.45s						

Nelder-Mead t=12.52s	run 145/1000	MSE=9.289e-09	nit=41	nfev=78	
Nelder-Mead t=11.10s	run 146/1000	MSE=2.214e-09	nit=36	nfev=70	
Nelder-Mead t=11.37s	run 147/1000	MSE=8.259e-09	nit=38	nfev=71	
Nelder-Mead	run 148/1000	MSE=1.025e-01	nit=20	nfev=37	t=5.62s
Nelder-Mead	run 149/1000	MSE=3.471e-08	nit=27	nfev=53	t=8.33s
Nelder-Mead	run 150/1000	MSE=1.026e-08	nit=32	nfev=62	t=9.67s
Nelder-Mead t=13.65s	run 151/1000	MSE=6.202e-09	nit=43	nfev=86	
Nelder-Mead	run 152/1000	MSE=2.504e-08	nit=33	nfev=64	t=9.97s
Nelder-Mead t=17.90s	run 153/1000	MSE=5.725e-09	nit=57	nfev=112	
Nelder-Mead t=12.92s	run 154/1000	MSE=1.546e-09	nit=43	nfev=83	
Nelder-Mead t=13.16s	run 155/1000	MSE=1.348e-09	nit=44	nfev=83	
Nelder-Mead t=10.95s	run 156/1000	MSE=5.531e-08	nit=35	nfev=70	
Nelder-Mead	run 157/1000	MSE=2.902e-09	nit=33	nfev=62	t=9.71s
Nelder-Mead t=11.00s	run 158/1000	MSE=2.752e-08	nit=35	nfev=69	
Nelder-Mead t=13.21s	run 159/1000	MSE=9.112e-09	nit=43	nfev=85	
Nelder-Mead t=14.36s	run 160/1000	MSE=4.389e-09	nit=47	nfev=89	
Nelder-Mead t=10.77s	run 161/1000	MSE=2.998e-08	nit=36	nfev=69	
Nelder-Mead t=13.85s	run 162/1000	MSE=1.617e-08	nit=42	nfev=85	
Nelder-Mead t=13.00s	run 163/1000	MSE=9.523e-09	nit=44	nfev=83	
Nelder-Mead t=13.62s	run 164/1000	MSE=1.191e-08	nit=45	nfev=87	
Nelder-Mead t=11.39s	run 165/1000	MSE=7.113e-09	nit=36	nfev=70	
Nelder-Mead	run 166/1000	MSE=9.873e-09	nit=28	nfev=55	t=8.76s
Nelder-Mead t=12.52s	run 167/1000	MSE=4.639e-09	nit=41	nfev=79	
Nelder-Mead t=14.26s	run 168/1000	MSE=1.116e-08	nit=47	nfev=89	
Nelder-Mead t=11.94s	run 169/1000	MSE=1.556e-09	nit=40	nfev=77	
Nelder-Mead t=11.16s	run 170/1000	MSE=1.551e-08	nit=36	nfev=69	
Nelder-Mead t=14.28s	run 171/1000	MSE=4.701e-09	nit=48	nfev=92	

Nelder-Mead t=12.91s	run 172/1000	MSE=2.109e-08	nit=40	nfev=81	
Nelder-Mead t=12.62s	run 173/1000	MSE=6.583e-09	nit=42	nfev=81	
Nelder-Mead t=13.82s	run 174/1000	MSE=1.464e-08	nit=46	nfev=87	
Nelder-Mead	run 175/1000	MSE=9.659e-09	nit=33	nfev=63	t=9.82s
Nelder-Mead t=14.70s	run 176/1000	MSE=7.261e-09	nit=48	nfev=94	
Nelder-Mead t=11.75s	run 177/1000	MSE=1.289e-08	nit=39	nfev=75	
Nelder-Mead t=20.79s	run 178/1000	MSE=1.874e-09	nit=72	nfev=133	
Nelder-Mead t=10.59s	run 179/1000	MSE=1.802e-08	nit=37	nfev=67	
Nelder-Mead	run 180/1000	MSE=2.062e-08	nit=29	nfev=56	t=8.87s
Nelder-Mead t=11.35s	run 181/1000	MSE=9.285e-09	nit=37	nfev=72	
Nelder-Mead t=14.53s	run 182/1000	MSE=1.385e-09	nit=49	nfev=92	
Nelder-Mead t=11.46s	run 183/1000	MSE=2.600e-09	nit=37	nfev=72	
Nelder-Mead t=10.78s	run 184/1000	MSE=7.968e-09	nit=36	nfev=68	
Nelder-Mead	run 185/1000	MSE=6.831e-09	nit=28	nfev=53	t=8.29s
Nelder-Mead t=14.05s	run 186/1000	MSE=2.178e-09	nit=46	nfev=89	
Nelder-Mead t=13.76s	run 187/1000	MSE=1.194e-08	nit=48	nfev=88	
Nelder-Mead t=14.04s	run 188/1000	MSE=3.057e-09	nit=45	nfev=89	
Nelder-Mead t=10.32s	run 189/1000	MSE=4.039e-09	nit=34	nfev=66	
Nelder-Mead	run 190/1000	MSE=1.064e-08	nit=30	nfev=58	t=9.41s
Nelder-Mead t=13.34s	run 191/1000	MSE=2.528e-09	nit=45	nfev=83	
Nelder-Mead t=14.78s	run 192/1000	MSE=6.914e-09	nit=44	nfev=89	
Nelder-Mead t=11.53s	run 193/1000	MSE=3.846e-08	nit=38	nfev=74	
Nelder-Mead t=12.60s	run 194/1000	MSE=3.761e-10	nit=43	nfev=80	
Nelder-Mead t=12.10s	run 195/1000	MSE=9.578e-09	nit=39	nfev=73	
Nelder-Mead t=11.53s	run 196/1000	MSE=6.693e-09	nit=39	nfev=73	
Nelder-Mead	run 197/1000	MSE=1.095e-08	nit=32	nfev=62	t=9.68s
Nelder-Mead	run 198/1000	MSE=5.712e-09	nit=43	nfev=82	

t=12.89s						
Nelder-Mead	run 199/1000	MSE=3.322e-09	nit=32	nfev=59	t=9.48s	
Nelder-Mead	run 200/1000	MSE=6.761e-09	nit=46	nfev=86		
t=13.86s						
Nelder-Mead	run 201/1000	MSE=1.577e-08	nit=37	nfev=72		
t=11.24s						
Nelder-Mead	run 202/1000	MSE=1.406e-08	nit=31	nfev=59	t=9.32s	
Nelder-Mead	run 203/1000	MSE=2.895e-09	nit=58	nfev=107		
t=16.65s						
Nelder-Mead	run 204/1000	MSE=2.006e-09	nit=47	nfev=86		
t=13.75s						
Nelder-Mead	run 205/1000	MSE=9.840e-09	nit=53	nfev=98		
t=15.39s						
Nelder-Mead	run 206/1000	MSE=6.880e-09	nit=42	nfev=81		
t=12.68s						
Nelder-Mead	run 207/1000	MSE=9.754e-08	nit=36	nfev=68		
t=10.58s						
Nelder-Mead	run 208/1000	MSE=1.234e-08	nit=45	nfev=89		
t=14.00s						
Nelder-Mead	run 209/1000	MSE=1.025e-01	nit=22	nfev=41	t=6.62s	
Nelder-Mead	run 210/1000	MSE=5.193e-09	nit=38	nfev=70		
t=10.83s						
Nelder-Mead	run 211/1000	MSE=2.242e-08	nit=49	nfev=92		
t=14.53s						
Nelder-Mead	run 212/1000	MSE=1.352e-08	nit=47	nfev=90		
t=14.07s						
Nelder-Mead	run 213/1000	MSE=1.498e-08	nit=58	nfev=107		
t=17.00s						
Nelder-Mead	run 214/1000	MSE=3.898e-09	nit=44	nfev=85		
t=13.21s						
Nelder-Mead	run 215/1000	MSE=4.141e-09	nit=56	nfev=105		
t=16.52s						
Nelder-Mead	run 216/1000	MSE=2.295e-08	nit=50	nfev=97		
t=15.35s						
Nelder-Mead	run 217/1000	MSE=1.091e-08	nit=54	nfev=103		
t=16.22s						
Nelder-Mead	run 218/1000	MSE=6.448e-09	nit=43	nfev=78		
t=12.15s						
Nelder-Mead	run 219/1000	MSE=8.863e-09	nit=44	nfev=85		
t=13.47s						
Nelder-Mead	run 220/1000	MSE=5.165e-09	nit=43	nfev=84		
t=13.24s						
Nelder-Mead	run 221/1000	MSE=1.025e-01	nit=24	nfev=44	t=6.67s	
Nelder-Mead	run 222/1000	MSE=6.731e-09	nit=46	nfev=88		
t=14.02s						
Nelder-Mead	run 223/1000	MSE=3.652e-09	nit=49	nfev=93		
t=14.55s						
Nelder-Mead	run 224/1000	MSE=1.073e-08	nit=53	nfev=96		

t=15.19s						
Nelder-Mead	run 225/1000	MSE=1.764e-08	nit=42	nfev=78		
t=12.10s						
Nelder-Mead	run 226/1000	MSE=1.646e-08	nit=40	nfev=76		
t=12.05s						
Nelder-Mead	run 227/1000	MSE=4.925e-09	nit=30	nfev=57	t=9.08s	
Nelder-Mead	run 228/1000	MSE=1.714e-08	nit=40	nfev=77		
t=12.03s						
Nelder-Mead	run 229/1000	MSE=5.729e-08	nit=34	nfev=63	t=9.93s	
Nelder-Mead	run 230/1000	MSE=8.857e-09	nit=45	nfev=88		
t=13.89s						
Nelder-Mead	run 231/1000	MSE=1.135e-08	nit=34	nfev=67		
t=10.40s						
Nelder-Mead	run 232/1000	MSE=1.442e-08	nit=53	nfev=100		
t=15.80s						
Nelder-Mead	run 233/1000	MSE=1.318e-08	nit=55	nfev=101		
t=15.90s						
Nelder-Mead	run 234/1000	MSE=4.614e-09	nit=47	nfev=88		
t=13.81s						
Nelder-Mead	run 235/1000	MSE=1.236e-08	nit=42	nfev=82		
t=12.71s						
Nelder-Mead	run 236/1000	MSE=6.013e-09	nit=30	nfev=59	t=9.31s	
Nelder-Mead	run 237/1000	MSE=8.814e-09	nit=41	nfev=78		
t=12.50s						
Nelder-Mead	run 238/1000	MSE=7.442e-08	nit=38	nfev=73		
t=16.24s						
Nelder-Mead	run 239/1000	MSE=6.188e-09	nit=39	nfev=74		
t=11.63s						
Nelder-Mead	run 240/1000	MSE=1.631e-08	nit=32	nfev=62	t=9.97s	
Nelder-Mead	run 241/1000	MSE=3.643e-09	nit=42	nfev=80		
t=12.37s						
Nelder-Mead	run 242/1000	MSE=4.873e-09	nit=47	nfev=88		
t=13.81s						
Nelder-Mead	run 243/1000	MSE=4.330e-08	nit=45	nfev=85		
t=13.14s						
Nelder-Mead	run 244/1000	MSE=6.771e-09	nit=54	nfev=103		
t=15.75s						
Nelder-Mead	run 245/1000	MSE=2.106e-08	nit=41	nfev=78		
t=12.08s						
Nelder-Mead	run 246/1000	MSE=1.027e-08	nit=51	nfev=95		
t=14.44s						
Nelder-Mead	run 247/1000	MSE=6.163e-09	nit=40	nfev=81		
t=12.33s						
Nelder-Mead	run 248/1000	MSE=1.224e-08	nit=49	nfev=91		
t=13.92s						
Nelder-Mead	run 249/1000	MSE=9.492e-10	nit=42	nfev=79		
t=11.96s						
Nelder-Mead	run 250/1000	MSE=1.223e-08	nit=39	nfev=74		

t=11.40s						
Nelder-Mead	run 251/1000	MSE=1.230e-08	nit=33	nfev=65	t=9.95s	
Nelder-Mead	run 252/1000	MSE=1.304e-08	nit=42	nfev=82		
t=12.88s						
Nelder-Mead	run 253/1000	MSE=8.391e-09	nit=35	nfev=69		
t=10.49s						
Nelder-Mead	run 254/1000	MSE=1.972e-08	nit=47	nfev=89		
t=13.47s						
Nelder-Mead	run 255/1000	MSE=3.560e-08	nit=34	nfev=66		
t=10.02s						
Nelder-Mead	run 256/1000	MSE=2.532e-08	nit=48	nfev=92		
t=14.10s						
Nelder-Mead	run 257/1000	MSE=1.501e-09	nit=68	nfev=125		
t=19.08s						
Nelder-Mead	run 258/1000	MSE=1.002e-08	nit=36	nfev=69		
t=10.62s						
Nelder-Mead	run 259/1000	MSE=1.814e-09	nit=48	nfev=92		
t=14.22s						
Nelder-Mead	run 260/1000	MSE=1.183e-08	nit=30	nfev=56	t=8.68s	
Nelder-Mead	run 261/1000	MSE=2.535e-09	nit=37	nfev=71		
t=10.72s						
Nelder-Mead	run 262/1000	MSE=2.790e-09	nit=51	nfev=97		
t=14.90s						
Nelder-Mead	run 263/1000	MSE=1.322e-08	nit=48	nfev=91		
t=13.79s						
Nelder-Mead	run 264/1000	MSE=7.460e-09	nit=46	nfev=89		
t=13.83s						
Nelder-Mead	run 265/1000	MSE=7.030e-09	nit=35	nfev=69		
t=10.51s						
Nelder-Mead	run 266/1000	MSE=8.164e-08	nit=31	nfev=60	t=9.51s	
Nelder-Mead	run 267/1000	MSE=1.177e-08	nit=36	nfev=68		
t=10.58s						
Nelder-Mead	run 268/1000	MSE=1.265e-08	nit=46	nfev=87		
t=13.13s						
Nelder-Mead	run 269/1000	MSE=1.589e+01	nit=24	nfev=41	t=6.33s	
Nelder-Mead	run 270/1000	MSE=3.228e-08	nit=39	nfev=74		
t=11.65s						
Nelder-Mead	run 271/1000	MSE=1.584e-08	nit=38	nfev=74		
t=11.18s						
Nelder-Mead	run 272/1000	MSE=3.651e-09	nit=36	nfev=69		
t=10.54s						
Nelder-Mead	run 273/1000	MSE=6.544e-09	nit=45	nfev=85		
t=12.94s						
Nelder-Mead	run 274/1000	MSE=6.322e-09	nit=44	nfev=82		
t=12.62s						
Nelder-Mead	run 275/1000	MSE=2.336e-08	nit=36	nfev=69		
t=10.48s						
Nelder-Mead	run 276/1000	MSE=1.388e-08	nit=61	nfev=115		

t=17.62s						
Nelder-Mead	run 277/1000	MSE=1.679e-08	nit=46	nfev=91		
t=14.06s						
Nelder-Mead	run 278/1000	MSE=1.027e-09	nit=51	nfev=97		
t=14.67s						
Nelder-Mead	run 279/1000	MSE=6.575e-09	nit=42	nfev=79		
t=11.93s						
Nelder-Mead	run 280/1000	MSE=7.896e-09	nit=43	nfev=82		
t=12.62s						
Nelder-Mead	run 281/1000	MSE=1.206e-08	nit=66	nfev=120		
t=18.69s						
Nelder-Mead	run 282/1000	MSE=4.759e-09	nit=41	nfev=76		
t=11.42s						
Nelder-Mead	run 283/1000	MSE=7.686e-09	nit=37	nfev=69		
t=10.52s						
Nelder-Mead	run 284/1000	MSE=1.026e-09	nit=37	nfev=70		
t=10.82s						
Nelder-Mead	run 285/1000	MSE=2.640e-08	nit=33	nfev=63	t=9.87s	
Nelder-Mead	run 286/1000	MSE=9.964e-09	nit=45	nfev=85		
t=13.17s						
Nelder-Mead	run 287/1000	MSE=2.895e-09	nit=31	nfev=61	t=9.29s	
Nelder-Mead	run 288/1000	MSE=1.101e-08	nit=38	nfev=72		
t=10.97s						
Nelder-Mead	run 289/1000	MSE=1.488e-08	nit=38	nfev=72		
t=11.10s						
Nelder-Mead	run 290/1000	MSE=1.790e-08	nit=53	nfev=103		
t=16.19s						
Nelder-Mead	run 291/1000	MSE=1.219e-08	nit=42	nfev=79		
t=12.35s						
Nelder-Mead	run 292/1000	MSE=2.264e-08	nit=36	nfev=69		
t=10.40s						
Nelder-Mead	run 293/1000	MSE=4.278e-09	nit=54	nfev=102		
t=15.61s						
Nelder-Mead	run 294/1000	MSE=2.175e-08	nit=29	nfev=53	t=7.98s	
Nelder-Mead	run 295/1000	MSE=4.467e-09	nit=40	nfev=75		
t=11.98s						
Nelder-Mead	run 296/1000	MSE=4.357e-09	nit=37	nfev=72		
t=10.96s						
Nelder-Mead	run 297/1000	MSE=5.396e-09	nit=49	nfev=93		
t=15.59s						
Nelder-Mead	run 298/1000	MSE=2.034e-09	nit=46	nfev=92		
t=15.38s						
Nelder-Mead	run 299/1000	MSE=7.748e-08	nit=35	nfev=70		
t=10.59s						
Nelder-Mead	run 300/1000	MSE=3.904e-09	nit=44	nfev=85		
t=12.68s						
Nelder-Mead	run 301/1000	MSE=1.251e-08	nit=45	nfev=87		
t=13.47s						

Nelder-Mead t=11.16s	run 302/1000	MSE=2.044e-08	nit=39	nfev=73	
Nelder-Mead	run 303/1000	MSE=1.025e-01	nit=22	nfev=40	t=6.11s
Nelder-Mead t=11.51s	run 304/1000	MSE=1.190e-08	nit=39	nfev=74	
Nelder-Mead t=14.08s	run 305/1000	MSE=9.455e-09	nit=49	nfev=91	
Nelder-Mead t=10.83s	run 306/1000	MSE=3.986e-08	nit=36	nfev=69	
Nelder-Mead t=18.32s	run 307/1000	MSE=1.598e-08	nit=64	nfev=122	
Nelder-Mead	run 308/1000	MSE=1.740e-08	nit=29	nfev=57	t=8.80s
Nelder-Mead t=10.20s	run 309/1000	MSE=1.950e-09	nit=35	nfev=67	
Nelder-Mead t=14.62s	run 310/1000	MSE=1.093e-08	nit=50	nfev=94	
Nelder-Mead t=12.32s	run 311/1000	MSE=1.395e-09	nit=41	nfev=79	
Nelder-Mead	run 312/1000	MSE=5.597e-08	nit=31	nfev=60	t=9.07s
Nelder-Mead t=11.19s	run 313/1000	MSE=1.303e-08	nit=37	nfev=73	
Nelder-Mead t=13.41s	run 314/1000	MSE=5.725e-09	nit=45	nfev=87	
Nelder-Mead	run 315/1000	MSE=4.035e-09	nit=30	nfev=54	t=8.15s
Nelder-Mead	run 316/1000	MSE=1.025e-01	nit=20	nfev=37	t=5.66s
Nelder-Mead t=12.03s	run 317/1000	MSE=3.607e-08	nit=41	nfev=79	
Nelder-Mead t=10.20s	run 318/1000	MSE=2.620e-09	nit=35	nfev=67	
Nelder-Mead	run 319/1000	MSE=1.856e-08	nit=34	nfev=64	t=9.82s
Nelder-Mead	run 320/1000	MSE=1.025e-01	nit=28	nfev=51	t=7.96s
Nelder-Mead t=15.05s	run 321/1000	MSE=4.035e-09	nit=52	nfev=99	
Nelder-Mead	run 322/1000	MSE=3.977e-09	nit=31	nfev=60	t=9.20s
Nelder-Mead t=10.02s	run 323/1000	MSE=2.042e-08	nit=34	nfev=65	
Nelder-Mead	run 324/1000	MSE=8.000e-08	nit=34	nfev=64	t=9.65s
Nelder-Mead t=11.60s	run 325/1000	MSE=2.919e-09	nit=38	nfev=76	
Nelder-Mead t=11.68s	run 326/1000	MSE=1.092e-08	nit=39	nfev=74	
Nelder-Mead	run 327/1000	MSE=9.144e-09	nit=33	nfev=64	t=9.72s
Nelder-Mead t=10.30s	run 328/1000	MSE=1.464e-08	nit=35	nfev=67	
Nelder-Mead t=14.02s	run 329/1000	MSE=8.883e-09	nit=48	nfev=91	
Nelder-Mead t=14.22s	run 330/1000	MSE=1.059e-08	nit=49	nfev=93	

Nelder-Mead	run 331/1000	MSE=1.056e-07	nit=31	nfev=59	t=8.97s
Nelder-Mead	run 332/1000	MSE=3.027e-09	nit=44	nfev=82	
t=12.35s					
Nelder-Mead	run 333/1000	MSE=1.016e-08	nit=46	nfev=82	
t=12.69s					
Nelder-Mead	run 334/1000	MSE=1.207e-08	nit=30	nfev=58	t=8.82s
Nelder-Mead	run 335/1000	MSE=5.557e-09	nit=33	nfev=64	t=9.65s
Nelder-Mead	run 336/1000	MSE=1.885e-08	nit=38	nfev=73	
t=11.37s					
Nelder-Mead	run 337/1000	MSE=1.511e-08	nit=34	nfev=66	
t=10.50s					
Nelder-Mead	run 338/1000	MSE=1.134e-08	nit=41	nfev=77	
t=11.85s					
Nelder-Mead	run 339/1000	MSE=7.907e-09	nit=33	nfev=64	t=9.76s
Nelder-Mead	run 340/1000	MSE=1.025e-01	nit=22	nfev=39	t=5.71s
Nelder-Mead	run 341/1000	MSE=9.203e-09	nit=37	nfev=70	
t=10.67s					
Nelder-Mead	run 342/1000	MSE=7.268e-09	nit=38	nfev=73	
t=11.19s					
Nelder-Mead	run 343/1000	MSE=1.275e-08	nit=39	nfev=73	
t=11.45s					
Nelder-Mead	run 344/1000	MSE=2.311e-08	nit=32	nfev=62	t=9.88s
Nelder-Mead	run 345/1000	MSE=6.507e-08	nit=42	nfev=81	
t=12.33s					
Nelder-Mead	run 346/1000	MSE=1.067e-08	nit=34	nfev=64	t=9.96s
Nelder-Mead	run 347/1000	MSE=1.565e-08	nit=36	nfev=70	
t=10.70s					
Nelder-Mead	run 348/1000	MSE=2.713e-08	nit=25	nfev=50	t=7.58s
Nelder-Mead	run 349/1000	MSE=2.427e-09	nit=44	nfev=85	
t=12.82s					
Nelder-Mead	run 350/1000	MSE=1.035e-01	nit=12	nfev=23	t=3.63s
Nelder-Mead	run 351/1000	MSE=2.108e-08	nit=49	nfev=91	
t=14.04s					
Nelder-Mead	run 352/1000	MSE=4.626e-09	nit=46	nfev=88	
t=13.83s					
Nelder-Mead	run 353/1000	MSE=3.560e-09	nit=37	nfev=72	
t=11.18s					
Nelder-Mead	run 354/1000	MSE=7.974e-09	nit=33	nfev=65	t=9.87s
Nelder-Mead	run 355/1000	MSE=5.037e-09	nit=32	nfev=62	t=9.52s
Nelder-Mead	run 356/1000	MSE=1.078e-08	nit=36	nfev=69	
t=10.46s					
Nelder-Mead	run 357/1000	MSE=2.767e-09	nit=46	nfev=85	
t=12.90s					
Nelder-Mead	run 358/1000	MSE=4.273e-08	nit=42	nfev=80	
t=12.20s					
Nelder-Mead	run 359/1000	MSE=3.242e-09	nit=56	nfev=109	
t=16.57s					
Nelder-Mead	run 360/1000	MSE=2.335e-08	nit=33	nfev=62	t=9.41s

Nelder-Mead t=13.58s	run 361/1000	MSE=5.044e-09	nit=45	nfev=89	
Nelder-Mead t=11.92s	run 362/1000	MSE=1.305e-09	nit=41	nfev=79	
Nelder-Mead t=15.35s	run 363/1000	MSE=6.565e-09	nit=55	nfev=102	
Nelder-Mead	run 364/1000	MSE=6.395e-09	nit=35	nfev=66	t=9.90s
Nelder-Mead t=10.85s	run 365/1000	MSE=1.229e-08	nit=37	nfev=70	
Nelder-Mead t=10.64s	run 366/1000	MSE=2.623e-09	nit=38	nfev=71	
Nelder-Mead t=11.66s	run 367/1000	MSE=1.438e-08	nit=40	nfev=77	
Nelder-Mead t=15.91s	run 368/1000	MSE=3.205e-09	nit=55	nfev=104	
Nelder-Mead t=15.19s	run 369/1000	MSE=5.759e-09	nit=54	nfev=101	
Nelder-Mead t=10.99s	run 370/1000	MSE=5.637e-08	nit=37	nfev=71	
Nelder-Mead t=14.54s	run 371/1000	MSE=1.164e-08	nit=51	nfev=97	
Nelder-Mead t=14.75s	run 372/1000	MSE=3.571e-09	nit=54	nfev=98	
Nelder-Mead t=13.73s	run 373/1000	MSE=4.062e-09	nit=46	nfev=90	
Nelder-Mead t=10.52s	run 374/1000	MSE=6.002e-09	nit=39	nfev=70	
Nelder-Mead t=13.57s	run 375/1000	MSE=1.385e-08	nit=45	nfev=88	
Nelder-Mead t=12.20s	run 376/1000	MSE=3.273e-09	nit=44	nfev=81	
Nelder-Mead t=10.96s	run 377/1000	MSE=1.607e-08	nit=38	nfev=73	
Nelder-Mead t=14.22s	run 378/1000	MSE=2.741e-09	nit=47	nfev=92	
Nelder-Mead t=12.23s	run 379/1000	MSE=7.454e-09	nit=43	nfev=80	
Nelder-Mead	run 380/1000	MSE=4.583e-08	nit=33	nfev=62	t=9.27s
Nelder-Mead t=17.22s	run 381/1000	MSE=2.790e-09	nit=57	nfev=111	
Nelder-Mead t=12.81s	run 382/1000	MSE=1.238e-08	nit=44	nfev=84	
Nelder-Mead t=13.49s	run 383/1000	MSE=5.754e-08	nit=45	nfev=89	
Nelder-Mead	run 384/1000	MSE=5.262e-09	nit=31	nfev=59	t=9.03s
Nelder-Mead t=12.09s	run 385/1000	MSE=6.309e-09	nit=42	nfev=80	
Nelder-Mead	run 386/1000	MSE=5.517e-09	nit=30	nfev=57	t=8.78s

Nelder-Mead t=12.45s	run 387/1000	MSE=1.404e-09	nit=43	nfev=81	
Nelder-Mead t=10.21s	run 388/1000	MSE=2.084e-07	nit=34	nfev=65	
Nelder-Mead t=11.57s	run 389/1000	MSE=8.788e-09	nit=40	nfev=76	
Nelder-Mead t=12.56s	run 390/1000	MSE=1.713e-08	nit=42	nfev=82	
Nelder-Mead t=10.60s	run 391/1000	MSE=7.196e-10	nit=37	nfev=71	
Nelder-Mead t=11.01s	run 392/1000	MSE=1.683e-08	nit=38	nfev=73	
Nelder-Mead t=13.44s	run 393/1000	MSE=7.031e-09	nit=44	nfev=84	
Nelder-Mead t=10.81s	run 394/1000	MSE=8.733e-09	nit=37	nfev=71	
Nelder-Mead t=10.71s	run 395/1000	MSE=1.232e-08	nit=35	nfev=70	
Nelder-Mead	run 396/1000	MSE=2.587e-08	nit=32	nfev=62	t=9.37s
Nelder-Mead t=11.20s	run 397/1000	MSE=2.253e-08	nit=40	nfev=74	
Nelder-Mead t=12.01s	run 398/1000	MSE=1.335e-08	nit=40	nfev=78	
Nelder-Mead	run 399/1000	MSE=1.486e-08	nit=32	nfev=60	t=9.02s
Nelder-Mead t=10.38s	run 400/1000	MSE=1.089e-08	nit=37	nfev=69	
Nelder-Mead t=11.70s	run 401/1000	MSE=2.026e-08	nit=39	nfev=77	
Nelder-Mead t=10.62s	run 402/1000	MSE=1.750e-08	nit=38	nfev=71	
Nelder-Mead t=13.08s	run 403/1000	MSE=4.218e-09	nit=45	nfev=85	
Nelder-Mead	run 404/1000	MSE=5.341e-09	nit=32	nfev=62	t=9.39s
Nelder-Mead t=12.42s	run 405/1000	MSE=2.426e-08	nit=42	nfev=81	
Nelder-Mead t=15.56s	run 406/1000	MSE=2.148e-09	nit=56	nfev=103	
Nelder-Mead t=10.34s	run 407/1000	MSE=2.628e-09	nit=36	nfev=68	
Nelder-Mead	run 408/1000	MSE=5.752e-09	nit=35	nfev=66	t=9.96s
Nelder-Mead t=12.49s	run 409/1000	MSE=3.424e-09	nit=43	nfev=84	
Nelder-Mead t=13.48s	run 410/1000	MSE=1.094e-08	nit=47	nfev=87	
Nelder-Mead	run 411/1000	MSE=1.837e-08	nit=34	nfev=66	t=9.95s
Nelder-Mead t=11.08s	run 412/1000	MSE=2.030e-08	nit=38	nfev=72	
Nelder-Mead	run 413/1000	MSE=4.165e-09	nit=35	nfev=65	

t=10.09s						
Nelder-Mead	run 414/1000	MSE=1.794e-08	nit=37	nfev=71		
t=10.60s						
Nelder-Mead	run 415/1000	MSE=4.783e-08	nit=38	nfev=71		
t=10.87s						
Nelder-Mead	run 416/1000	MSE=3.834e-09	nit=30	nfev=58	t=8.78s	
Nelder-Mead	run 417/1000	MSE=7.701e-09	nit=27	nfev=52	t=7.78s	
Nelder-Mead	run 418/1000	MSE=2.854e-09	nit=41	nfev=79		
t=11.80s						
Nelder-Mead	run 419/1000	MSE=1.582e-08	nit=36	nfev=70		
t=10.98s						
Nelder-Mead	run 420/1000	MSE=9.538e-09	nit=45	nfev=85		
t=12.94s						
Nelder-Mead	run 421/1000	MSE=3.010e-09	nit=36	nfev=70		
t=10.57s						
Nelder-Mead	run 422/1000	MSE=1.271e-08	nit=42	nfev=80		
t=12.89s						
Nelder-Mead	run 423/1000	MSE=2.902e-10	nit=48	nfev=89		
t=13.38s						
Nelder-Mead	run 424/1000	MSE=8.075e-09	nit=53	nfev=99		
t=14.94s						
Nelder-Mead	run 425/1000	MSE=4.257e-08	nit=31	nfev=59	t=8.96s	
Nelder-Mead	run 426/1000	MSE=1.025e-01	nit=22	nfev=39	t=5.75s	
Nelder-Mead	run 427/1000	MSE=9.176e-09	nit=60	nfev=111		
t=16.93s						
Nelder-Mead	run 428/1000	MSE=1.901e-09	nit=43	nfev=84		
t=12.76s						
Nelder-Mead	run 429/1000	MSE=7.468e-09	nit=66	nfev=125		
t=18.97s						
Nelder-Mead	run 430/1000	MSE=1.025e-01	nit=20	nfev=37	t=5.68s	
Nelder-Mead	run 431/1000	MSE=5.267e-09	nit=33	nfev=64	t=9.63s	
Nelder-Mead	run 432/1000	MSE=1.962e-08	nit=41	nfev=81		
t=12.40s						
Nelder-Mead	run 433/1000	MSE=2.962e-08	nit=33	nfev=66	t=9.79s	
Nelder-Mead	run 434/1000	MSE=8.381e-09	nit=32	nfev=63	t=9.55s	
Nelder-Mead	run 435/1000	MSE=7.760e-09	nit=36	nfev=69		
t=10.70s						
Nelder-Mead	run 436/1000	MSE=7.953e-09	nit=38	nfev=70		
t=10.48s						
Nelder-Mead	run 437/1000	MSE=2.914e-08	nit=29	nfev=56	t=8.65s	
Nelder-Mead	run 438/1000	MSE=5.477e-09	nit=37	nfev=71		
t=10.87s						
Nelder-Mead	run 439/1000	MSE=5.805e-08	nit=45	nfev=83		
t=12.40s						
Nelder-Mead	run 440/1000	MSE=2.139e-08	nit=43	nfev=82		
t=12.39s						
Nelder-Mead	run 441/1000	MSE=1.600e-08	nit=43	nfev=84		
t=12.86s						

Nelder-Mead t=13.37s	run 442/1000	MSE=1.151e-08	nit=46	nfev=87	
Nelder-Mead t=10.50s	run 443/1000	MSE=4.086e-08	nit=36	nfev=70	
Nelder-Mead t=13.01s	run 444/1000	MSE=1.238e-08	nit=43	nfev=85	
Nelder-Mead	run 445/1000	MSE=7.550e-09	nit=32	nfev=63	t=9.63s
Nelder-Mead	run 446/1000	MSE=1.583e-08	nit=33	nfev=64	t=9.85s
Nelder-Mead t=10.18s	run 447/1000	MSE=2.748e-08	nit=34	nfev=67	
Nelder-Mead t=10.52s	run 448/1000	MSE=1.550e-09	nit=36	nfev=69	
Nelder-Mead t=13.51s	run 449/1000	MSE=1.730e-10	nit=48	nfev=91	
Nelder-Mead	run 450/1000	MSE=1.248e-09	nit=35	nfev=64	t=9.84s
Nelder-Mead t=11.00s	run 451/1000	MSE=2.731e-08	nit=37	nfev=72	
Nelder-Mead t=14.06s	run 452/1000	MSE=3.501e-09	nit=50	nfev=92	
Nelder-Mead t=15.25s	run 453/1000	MSE=4.717e-09	nit=52	nfev=100	
Nelder-Mead	run 454/1000	MSE=1.589e+01	nit=30	nfev=58	t=8.59s
Nelder-Mead	run 455/1000	MSE=1.094e-08	nit=32	nfev=62	t=9.35s
Nelder-Mead	run 456/1000	MSE=3.684e-09	nit=28	nfev=54	t=8.13s
Nelder-Mead t=11.75s	run 457/1000	MSE=4.587e-09	nit=38	nfev=75	
Nelder-Mead t=11.80s	run 458/1000	MSE=3.156e-08	nit=43	nfev=78	
Nelder-Mead t=13.63s	run 459/1000	MSE=2.115e-08	nit=47	nfev=89	
Nelder-Mead t=10.04s	run 460/1000	MSE=3.584e-09	nit=35	nfev=67	
Nelder-Mead t=15.16s	run 461/1000	MSE=6.198e-09	nit=52	nfev=98	
Nelder-Mead	run 462/1000	MSE=4.714e-09	nit=33	nfev=63	t=9.41s
Nelder-Mead t=10.09s	run 463/1000	MSE=1.745e-08	nit=34	nfev=67	
Nelder-Mead	run 464/1000	MSE=7.760e-09	nit=32	nfev=63	t=9.60s
Nelder-Mead t=14.61s	run 465/1000	MSE=7.651e-09	nit=52	nfev=98	
Nelder-Mead t=13.09s	run 466/1000	MSE=1.440e-08	nit=44	nfev=85	
Nelder-Mead t=10.55s	run 467/1000	MSE=2.475e-09	nit=37	nfev=70	
Nelder-Mead t=11.10s	run 468/1000	MSE=5.962e-08	nit=38	nfev=72	
Nelder-Mead	run 469/1000	MSE=2.473e-08	nit=28	nfev=53	t=8.21s
Nelder-Mead	run 470/1000	MSE=9.932e-09	nit=38	nfev=73	

t=10.98s						
Nelder-Mead	run 471/1000	MSE=4.487e-09	nit=39	nfev=73		
t=11.12s						
Nelder-Mead	run 472/1000	MSE=2.176e-09	nit=43	nfev=83		
t=12.72s						
Nelder-Mead	run 473/1000	MSE=1.141e-08	nit=40	nfev=76		
t=11.70s						
Nelder-Mead	run 474/1000	MSE=2.053e-08	nit=28	nfev=55	t=8.57s	
Nelder-Mead	run 475/1000	MSE=8.507e-09	nit=39	nfev=73		
t=10.94s						
Nelder-Mead	run 476/1000	MSE=1.964e-08	nit=48	nfev=95		
t=14.45s						
Nelder-Mead	run 477/1000	MSE=7.668e-09	nit=35	nfev=67		
t=10.32s						
Nelder-Mead	run 478/1000	MSE=1.035e-01	nit=13	nfev=25	t=3.88s	
Nelder-Mead	run 479/1000	MSE=1.220e-08	nit=55	nfev=102		
t=15.38s						
Nelder-Mead	run 480/1000	MSE=3.134e-09	nit=37	nfev=72		
t=10.69s						
Nelder-Mead	run 481/1000	MSE=3.269e-09	nit=47	nfev=89		
t=13.51s						
Nelder-Mead	run 482/1000	MSE=5.446e-09	nit=45	nfev=83		
t=12.51s						
Nelder-Mead	run 483/1000	MSE=1.191e-08	nit=34	nfev=65	t=9.96s	
Nelder-Mead	run 484/1000	MSE=7.749e-09	nit=48	nfev=93		
t=14.14s						
Nelder-Mead	run 485/1000	MSE=6.327e-09	nit=44	nfev=84		
t=12.94s						
Nelder-Mead	run 486/1000	MSE=7.267e-09	nit=37	nfev=69		
t=10.29s						
Nelder-Mead	run 487/1000	MSE=1.085e-08	nit=38	nfev=74		
t=11.13s						
Nelder-Mead	run 488/1000	MSE=3.620e-09	nit=35	nfev=65	t=9.71s	
Nelder-Mead	run 489/1000	MSE=2.151e-09	nit=34	nfev=67		
t=10.10s						
Nelder-Mead	run 490/1000	MSE=1.646e-08	nit=46	nfev=90		
t=13.84s						
Nelder-Mead	run 491/1000	MSE=2.526e-08	nit=33	nfev=64	t=9.70s	
Nelder-Mead	run 492/1000	MSE=1.162e-08	nit=30	nfev=55	t=8.32s	
Nelder-Mead	run 493/1000	MSE=6.253e-09	nit=30	nfev=59	t=9.14s	
Nelder-Mead	run 494/1000	MSE=5.853e-09	nit=39	nfev=74		
t=11.80s						
Nelder-Mead	run 495/1000	MSE=4.367e-08	nit=31	nfev=58	t=8.97s	
Nelder-Mead	run 496/1000	MSE=1.175e-09	nit=41	nfev=80		
t=11.86s						
Nelder-Mead	run 497/1000	MSE=2.748e-09	nit=46	nfev=87		
t=13.26s						
Nelder-Mead	run 498/1000	MSE=8.654e-09	nit=47	nfev=86		

t=12.97s						
Nelder-Mead	run 499/1000	MSE=1.023e-08	nit=42	nfev=82		
t=12.74s						
Nelder-Mead	run 500/1000	MSE=3.379e-09	nit=45	nfev=89		
t=13.74s						
Nelder-Mead	run 501/1000	MSE=4.693e-09	nit=42	nfev=78		
t=12.47s						
Nelder-Mead	run 502/1000	MSE=1.829e-09	nit=72	nfev=140		
t=21.00s						
Nelder-Mead	run 503/1000	MSE=1.496e-08	nit=45	nfev=85		
t=12.82s						
Nelder-Mead	run 504/1000	MSE=2.326e-09	nit=43	nfev=82		
t=12.55s						
Nelder-Mead	run 505/1000	MSE=4.070e-09	nit=28	nfev=56	t=8.50s	
Nelder-Mead	run 506/1000	MSE=3.559e-09	nit=33	nfev=66	t=9.89s	
Nelder-Mead	run 507/1000	MSE=9.218e-09	nit=34	nfev=64	t=9.70s	
Nelder-Mead	run 508/1000	MSE=3.897e-08	nit=31	nfev=60	t=9.41s	
Nelder-Mead	run 509/1000	MSE=3.986e-09	nit=37	nfev=72		
t=10.97s						
Nelder-Mead	run 510/1000	MSE=1.072e-08	nit=44	nfev=82		
t=12.45s						
Nelder-Mead	run 511/1000	MSE=1.929e-08	nit=38	nfev=70		
t=10.43s						
Nelder-Mead	run 512/1000	MSE=1.983e-09	nit=54	nfev=100		
t=15.22s						
Nelder-Mead	run 513/1000	MSE=5.486e-09	nit=48	nfev=91		
t=13.89s						
Nelder-Mead	run 514/1000	MSE=1.683e-08	nit=32	nfev=62	t=9.38s	
Nelder-Mead	run 515/1000	MSE=4.458e-09	nit=52	nfev=96		
t=14.84s						
Nelder-Mead	run 516/1000	MSE=4.607e-09	nit=45	nfev=87		
t=12.93s						
Nelder-Mead	run 517/1000	MSE=2.640e-09	nit=47	nfev=92		
t=13.88s						
Nelder-Mead	run 518/1000	MSE=3.789e-08	nit=35	nfev=68		
t=10.23s						
Nelder-Mead	run 519/1000	MSE=1.350e-08	nit=36	nfev=70		
t=10.66s						
Nelder-Mead	run 520/1000	MSE=8.602e-10	nit=43	nfev=86		
t=13.12s						
Nelder-Mead	run 521/1000	MSE=4.616e-09	nit=31	nfev=60	t=9.96s	
Nelder-Mead	run 522/1000	MSE=4.621e-09	nit=50	nfev=95		
t=15.81s						
Nelder-Mead	run 523/1000	MSE=1.280e-08	nit=28	nfev=56	t=8.48s	
Nelder-Mead	run 524/1000	MSE=7.162e-09	nit=30	nfev=60	t=9.12s	
Nelder-Mead	run 525/1000	MSE=1.216e-08	nit=53	nfev=99		
t=15.12s						
Nelder-Mead	run 526/1000	MSE=2.109e-09	nit=49	nfev=93		

t=14.15s						
Nelder-Mead	run 527/1000	MSE=1.347e-08	nit=37	nfev=74		
t=11.15s						
Nelder-Mead	run 528/1000	MSE=2.144e-08	nit=41	nfev=80		
t=12.37s						
Nelder-Mead	run 529/1000	MSE=1.308e-08	nit=45	nfev=85		
t=12.91s						
Nelder-Mead	run 530/1000	MSE=2.593e-09	nit=60	nfev=114		
t=17.44s						
Nelder-Mead	run 531/1000	MSE=1.154e-08	nit=38	nfev=74		
t=11.21s						
Nelder-Mead	run 532/1000	MSE=2.571e-08	nit=42	nfev=80		
t=12.97s						
Nelder-Mead	run 533/1000	MSE=7.395e-09	nit=34	nfev=64	t=9.74s	
Nelder-Mead	run 534/1000	MSE=1.381e-08	nit=32	nfev=63	t=9.52s	
Nelder-Mead	run 535/1000	MSE=8.148e-09	nit=36	nfev=68		
t=10.36s						
Nelder-Mead	run 536/1000	MSE=5.547e-09	nit=49	nfev=93		
t=14.22s						
Nelder-Mead	run 537/1000	MSE=8.014e-09	nit=41	nfev=80		
t=12.49s						
Nelder-Mead	run 538/1000	MSE=1.600e-08	nit=37	nfev=69		
t=10.33s						
Nelder-Mead	run 539/1000	MSE=2.176e-08	nit=35	nfev=65	t=9.80s	
Nelder-Mead	run 540/1000	MSE=2.358e-08	nit=44	nfev=83		
t=12.76s						
Nelder-Mead	run 541/1000	MSE=1.515e-08	nit=47	nfev=94		
t=14.22s						
Nelder-Mead	run 542/1000	MSE=2.234e-08	nit=29	nfev=57	t=8.62s	
Nelder-Mead	run 543/1000	MSE=6.701e-09	nit=57	nfev=106		
t=16.92s						
Nelder-Mead	run 544/1000	MSE=1.262e-08	nit=43	nfev=80		
t=13.19s						
Nelder-Mead	run 545/1000	MSE=1.508e-08	nit=39	nfev=76		
t=12.02s						
Nelder-Mead	run 546/1000	MSE=3.938e-09	nit=36	nfev=67		
t=10.27s						
Nelder-Mead	run 547/1000	MSE=2.622e-09	nit=44	nfev=87		
t=13.25s						
Nelder-Mead	run 548/1000	MSE=5.722e-09	nit=37	nfev=72		
t=11.43s						
Nelder-Mead	run 549/1000	MSE=9.936e-09	nit=50	nfev=97		
t=14.66s						
Nelder-Mead	run 550/1000	MSE=5.914e-09	nit=30	nfev=55	t=8.73s	
Nelder-Mead	run 551/1000	MSE=8.709e-08	nit=38	nfev=75		
t=11.31s						
Nelder-Mead	run 552/1000	MSE=8.408e-09	nit=39	nfev=74		
t=11.47s						

Nelder-Mead t=13.48s	run 553/1000	MSE=5.485e-09	nit=45	nfev=88	
Nelder-Mead t=12.14s	run 554/1000	MSE=5.585e-09	nit=44	nfev=81	
Nelder-Mead t=12.09s	run 555/1000	MSE=8.216e-09	nit=41	nfev=79	
Nelder-Mead t=12.55s	run 556/1000	MSE=3.026e-08	nit=42	nfev=82	
Nelder-Mead t=11.94s	run 557/1000	MSE=4.132e-09	nit=42	nfev=78	
Nelder-Mead t=11.37s	run 558/1000	MSE=5.903e-09	nit=38	nfev=74	
Nelder-Mead t=11.91s	run 559/1000	MSE=3.368e-08	nit=40	nfev=78	
Nelder-Mead t=13.27s	run 560/1000	MSE=4.135e-09	nit=48	nfev=87	
Nelder-Mead t=10.44s	run 561/1000	MSE=9.603e-09	nit=37	nfev=69	
Nelder-Mead t=13.22s	run 562/1000	MSE=2.761e-08	nit=43	nfev=87	
Nelder-Mead t=14.47s	run 563/1000	MSE=4.496e-09	nit=49	nfev=95	
Nelder-Mead t=12.49s	run 564/1000	MSE=1.871e-08	nit=42	nfev=82	
Nelder-Mead t=13.47s	run 565/1000	MSE=2.631e-09	nit=46	nfev=88	
Nelder-Mead t=10.68s	run 566/1000	MSE=1.060e-08	nit=37	nfev=71	
Nelder-Mead	run 567/1000	MSE=4.843e-09	nit=35	nfev=65	t=9.86s
Nelder-Mead t=13.84s	run 568/1000	MSE=1.443e-08	nit=42	nfev=80	
Nelder-Mead t=19.48s	run 569/1000	MSE=5.715e-09	nit=52	nfev=98	
Nelder-Mead t=13.49s	run 570/1000	MSE=6.736e-09	nit=46	nfev=89	
Nelder-Mead t=10.60s	run 571/1000	MSE=1.442e-08	nit=36	nfev=69	
Nelder-Mead	run 572/1000	MSE=1.041e-08	nit=34	nfev=65	t=9.70s
Nelder-Mead t=12.07s	run 573/1000	MSE=1.026e-08	nit=43	nfev=81	
Nelder-Mead	run 574/1000	MSE=1.454e-08	nit=34	nfev=63	t=9.59s
Nelder-Mead	run 575/1000	MSE=1.644e-07	nit=33	nfev=65	t=9.95s
Nelder-Mead t=12.25s	run 576/1000	MSE=3.780e-09	nit=43	nfev=83	
Nelder-Mead t=12.92s	run 577/1000	MSE=5.988e-09	nit=46	nfev=86	
Nelder-Mead t=13.45s	run 578/1000	MSE=1.317e-09	nit=47	nfev=89	

Nelder-Mead t=11.04s	run 579/1000	MSE=1.058e-08	nit=38	nfev=74	
Nelder-Mead	run 580/1000	MSE=2.658e-09	nit=30	nfev=57	t=8.64s
Nelder-Mead t=12.62s	run 581/1000	MSE=1.822e-09	nit=43	nfev=84	
Nelder-Mead t=12.11s	run 582/1000	MSE=1.051e-08	nit=43	nfev=82	
Nelder-Mead	run 583/1000	MSE=8.864e-09	nit=27	nfev=54	t=8.22s
Nelder-Mead	run 584/1000	MSE=7.365e-08	nit=30	nfev=59	t=8.96s
Nelder-Mead t=13.13s	run 585/1000	MSE=5.342e-09	nit=45	nfev=89	
Nelder-Mead t=11.70s	run 586/1000	MSE=2.976e-08	nit=40	nfev=76	
Nelder-Mead	run 587/1000	MSE=1.025e-01	nit=24	nfev=42	t=6.22s
Nelder-Mead t=11.38s	run 588/1000	MSE=6.430e-09	nit=41	nfev=77	
Nelder-Mead	run 589/1000	MSE=2.624e-09	nit=33	nfev=63	t=9.41s
Nelder-Mead t=11.97s	run 590/1000	MSE=8.004e-09	nit=40	nfev=79	
Nelder-Mead	run 591/1000	MSE=1.331e-08	nit=34	nfev=65	t=9.65s
Nelder-Mead	run 592/1000	MSE=3.566e-09	nit=35	nfev=66	t=9.68s
Nelder-Mead t=15.31s	run 593/1000	MSE=1.263e-08	nit=56	nfev=105	
Nelder-Mead t=10.34s	run 594/1000	MSE=7.785e-09	nit=35	nfev=65	
Nelder-Mead	run 595/1000	MSE=1.295e-08	nit=36	nfev=68	t=9.93s
Nelder-Mead	run 596/1000	MSE=1.025e-01	nit=22	nfev=40	t=5.90s
Nelder-Mead t=12.76s	run 597/1000	MSE=3.748e-09	nit=43	nfev=85	
Nelder-Mead t=12.43s	run 598/1000	MSE=1.353e-09	nit=45	nfev=83	
Nelder-Mead t=10.13s	run 599/1000	MSE=1.533e-08	nit=36	nfev=68	
Nelder-Mead t=11.38s	run 600/1000	MSE=1.922e-08	nit=38	nfev=75	
Nelder-Mead t=11.46s	run 601/1000	MSE=5.881e-09	nit=39	nfev=76	
Nelder-Mead t=13.28s	run 602/1000	MSE=7.850e-09	nit=46	nfev=88	
Nelder-Mead	run 603/1000	MSE=1.589e+01	nit=26	nfev=50	t=7.40s
Nelder-Mead t=14.28s	run 604/1000	MSE=1.122e-09	nit=49	nfev=94	
Nelder-Mead t=11.90s	run 605/1000	MSE=6.908e-09	nit=40	nfev=76	
Nelder-Mead	run 606/1000	MSE=6.656e-09	nit=32	nfev=63	t=9.34s
Nelder-Mead t=15.97s	run 607/1000	MSE=1.353e-09	nit=57	nfev=107	
Nelder-Mead	run 608/1000	MSE=5.279e-09	nit=40	nfev=80	

t=11.81s						
Nelder-Mead	run 609/1000	MSE=8.010e-09	nit=40	nfev=79		
t=11.71s						
Nelder-Mead	run 610/1000	MSE=2.682e-08	nit=26	nfev=48	t=7.34s	
Nelder-Mead	run 611/1000	MSE=7.430e-08	nit=34	nfev=66	t=9.71s	
Nelder-Mead	run 612/1000	MSE=1.388e-09	nit=31	nfev=56	t=8.60s	
Nelder-Mead	run 613/1000	MSE=2.971e-09	nit=33	nfev=65	t=9.74s	
Nelder-Mead	run 614/1000	MSE=5.094e-09	nit=37	nfev=70		
t=10.38s						
Nelder-Mead	run 615/1000	MSE=9.210e-09	nit=35	nfev=67		
t=11.91s						
Nelder-Mead	run 616/1000	MSE=1.025e-01	nit=21	nfev=38	t=6.31s	
Nelder-Mead	run 617/1000	MSE=5.913e-08	nit=38	nfev=72		
t=11.25s						
Nelder-Mead	run 618/1000	MSE=4.405e-09	nit=28	nfev=54	t=8.13s	
Nelder-Mead	run 619/1000	MSE=1.757e-08	nit=36	nfev=70		
t=10.41s						
Nelder-Mead	run 620/1000	MSE=1.136e-08	nit=26	nfev=50	t=7.56s	
Nelder-Mead	run 621/1000	MSE=2.106e-08	nit=31	nfev=58	t=8.54s	
Nelder-Mead	run 622/1000	MSE=1.775e-09	nit=39	nfev=76		
t=11.25s						
Nelder-Mead	run 623/1000	MSE=4.990e-09	nit=39	nfev=76		
t=11.39s						
Nelder-Mead	run 624/1000	MSE=2.756e-08	nit=46	nfev=89		
t=13.36s						
Nelder-Mead	run 625/1000	MSE=2.586e-08	nit=45	nfev=85		
t=12.82s						
Nelder-Mead	run 626/1000	MSE=2.461e-09	nit=39	nfev=76		
t=11.28s						
Nelder-Mead	run 627/1000	MSE=1.025e-01	nit=21	nfev=38	t=5.68s	
Nelder-Mead	run 628/1000	MSE=1.855e-09	nit=34	nfev=70		
t=11.18s						
Nelder-Mead	run 629/1000	MSE=6.338e-09	nit=43	nfev=81		
t=12.15s						
Nelder-Mead	run 630/1000	MSE=4.235e-09	nit=50	nfev=95		
t=14.34s						
Nelder-Mead	run 631/1000	MSE=1.588e-08	nit=36	nfev=68		
t=10.00s						
Nelder-Mead	run 632/1000	MSE=6.128e-09	nit=40	nfev=75		
t=11.11s						
Nelder-Mead	run 633/1000	MSE=3.587e-09	nit=47	nfev=89		
t=13.07s						
Nelder-Mead	run 634/1000	MSE=5.954e-09	nit=43	nfev=81		
t=12.27s						
Nelder-Mead	run 635/1000	MSE=3.221e-09	nit=35	nfev=69		
t=10.02s						
Nelder-Mead	run 636/1000	MSE=6.521e-09	nit=39	nfev=77		
t=11.33s						

Nelder-Mead t=14.64s	run 637/1000	MSE=6.257e-09	nit=51	nfev=96	
Nelder-Mead	run 638/1000	MSE=1.025e-01	nit=22	nfev=39	t=6.05s
Nelder-Mead	run 639/1000	MSE=4.585e-09	nit=35	nfev=67	t=9.93s
Nelder-Mead t=10.92s	run 640/1000	MSE=1.906e-08	nit=39	nfev=72	
Nelder-Mead	run 641/1000	MSE=6.493e-09	nit=36	nfev=68	t=9.94s
Nelder-Mead t=10.42s	run 642/1000	MSE=5.011e-09	nit=37	nfev=71	
Nelder-Mead	run 643/1000	MSE=8.232e-09	nit=31	nfev=60	t=8.84s
Nelder-Mead t=11.39s	run 644/1000	MSE=4.193e-08	nit=40	nfev=77	
Nelder-Mead t=10.80s	run 645/1000	MSE=7.338e-09	nit=38	nfev=74	
Nelder-Mead t=17.05s	run 646/1000	MSE=2.603e-08	nit=55	nfev=110	
Nelder-Mead t=10.22s	run 647/1000	MSE=1.456e-08	nit=35	nfev=68	
Nelder-Mead t=12.07s	run 648/1000	MSE=4.160e-09	nit=45	nfev=81	
Nelder-Mead t=12.40s	run 649/1000	MSE=2.848e-09	nit=42	nfev=80	
Nelder-Mead t=11.23s	run 650/1000	MSE=8.689e-09	nit=41	nfev=74	
Nelder-Mead	run 651/1000	MSE=2.042e-08	nit=32	nfev=62	t=9.05s
Nelder-Mead t=11.83s	run 652/1000	MSE=5.147e-09	nit=41	nfev=79	
Nelder-Mead t=11.54s	run 653/1000	MSE=2.133e-09	nit=41	nfev=77	
Nelder-Mead t=14.71s	run 654/1000	MSE=1.573e-08	nit=52	nfev=97	
Nelder-Mead	run 655/1000	MSE=1.035e-01	nit=13	nfev=25	t=3.74s
Nelder-Mead t=12.16s	run 656/1000	MSE=1.914e-09	nit=41	nfev=82	
Nelder-Mead t=14.95s	run 657/1000	MSE=6.472e-09	nit=52	nfev=100	
Nelder-Mead t=12.44s	run 658/1000	MSE=4.208e-09	nit=43	nfev=81	
Nelder-Mead	run 659/1000	MSE=1.025e-01	nit=23	nfev=41	t=5.97s
Nelder-Mead t=18.86s	run 660/1000	MSE=5.793e-09	nit=68	nfev=127	
Nelder-Mead t=10.76s	run 661/1000	MSE=5.367e-09	nit=39	nfev=72	
Nelder-Mead	run 662/1000	MSE=2.683e-09	nit=32	nfev=62	t=9.07s
Nelder-Mead t=10.73s	run 663/1000	MSE=4.439e-09	nit=37	nfev=71	
Nelder-Mead t=17.19s	run 664/1000	MSE=2.704e-08	nit=58	nfev=114	

Nelder-Mead t=11.46s	run 665/1000	MSE=3.804e-09	nit=40	nfev=77	
Nelder-Mead t=14.84s	run 666/1000	MSE=2.269e-09	nit=52	nfev=100	
Nelder-Mead	run 667/1000	MSE=8.371e-09	nit=34	nfev=67	t=9.89s
Nelder-Mead t=12.05s	run 668/1000	MSE=4.387e-09	nit=41	nfev=80	
Nelder-Mead t=10.09s	run 669/1000	MSE=2.805e-08	nit=38	nfev=69	
Nelder-Mead	run 670/1000	MSE=2.284e-08	nit=31	nfev=60	t=8.86s
Nelder-Mead	run 671/1000	MSE=1.716e-08	nit=34	nfev=63	t=9.28s
Nelder-Mead t=12.80s	run 672/1000	MSE=5.254e-08	nit=47	nfev=86	
Nelder-Mead t=12.01s	run 673/1000	MSE=7.468e-08	nit=43	nfev=83	
Nelder-Mead	run 674/1000	MSE=4.728e-09	nit=32	nfev=61	t=9.26s
Nelder-Mead	run 675/1000	MSE=7.886e-08	nit=31	nfev=61	t=8.96s
Nelder-Mead t=11.83s	run 676/1000	MSE=8.156e-09	nit=41	nfev=78	
Nelder-Mead t=13.81s	run 677/1000	MSE=5.983e-09	nit=49	nfev=93	
Nelder-Mead t=11.82s	run 678/1000	MSE=5.284e-09	nit=43	nfev=81	
Nelder-Mead t=10.85s	run 679/1000	MSE=5.968e-09	nit=39	nfev=72	
Nelder-Mead t=11.26s	run 680/1000	MSE=1.653e-08	nit=40	nfev=77	
Nelder-Mead t=13.15s	run 681/1000	MSE=1.479e-09	nit=47	nfev=89	
Nelder-Mead t=10.01s	run 682/1000	MSE=1.087e-08	nit=36	nfev=68	
Nelder-Mead t=15.95s	run 683/1000	MSE=2.800e-09	nit=57	nfev=108	
Nelder-Mead t=10.10s	run 684/1000	MSE=4.311e-08	nit=34	nfev=66	
Nelder-Mead t=10.00s	run 685/1000	MSE=2.897e-08	nit=36	nfev=68	
Nelder-Mead t=11.75s	run 686/1000	MSE=8.844e-09	nit=42	nfev=80	
Nelder-Mead t=12.91s	run 687/1000	MSE=3.630e-09	nit=47	nfev=87	
Nelder-Mead t=10.27s	run 688/1000	MSE=4.748e-08	nit=36	nfev=69	
Nelder-Mead t=14.55s	run 689/1000	MSE=1.386e-07	nit=50	nfev=97	
Nelder-Mead t=13.23s	run 690/1000	MSE=9.414e-09	nit=45	nfev=89	
Nelder-Mead	run 691/1000	MSE=5.607e-09	nit=33	nfev=64	t=9.55s

Nelder-Mead	run 692/1000	MSE=4.375e-09	nit=34	nfev=66	t=9.82s
Nelder-Mead	run 693/1000	MSE=4.748e-09	nit=32	nfev=62	t=9.02s
Nelder-Mead	run 694/1000	MSE=1.061e-08	nit=52	nfev=100	
t=14.48s					
Nelder-Mead	run 695/1000	MSE=1.522e-09	nit=37	nfev=71	
t=10.69s					
Nelder-Mead	run 696/1000	MSE=2.466e-09	nit=36	nfev=72	
t=10.64s					
Nelder-Mead	run 697/1000	MSE=1.062e-08	nit=49	nfev=93	
t=13.92s					
Nelder-Mead	run 698/1000	MSE=1.431e-08	nit=53	nfev=96	
t=14.45s					
Nelder-Mead	run 699/1000	MSE=7.731e-09	nit=35	nfev=68	
t=10.00s					
Nelder-Mead	run 700/1000	MSE=5.113e-09	nit=29	nfev=59	t=8.84s
Nelder-Mead	run 701/1000	MSE=6.610e-09	nit=36	nfev=70	
t=10.65s					
Nelder-Mead	run 702/1000	MSE=1.032e-08	nit=48	nfev=93	
t=13.98s					
Nelder-Mead	run 703/1000	MSE=1.035e-01	nit=13	nfev=25	t=3.72s
Nelder-Mead	run 704/1000	MSE=3.157e-08	nit=52	nfev=96	
t=13.92s					
Nelder-Mead	run 705/1000	MSE=1.010e-08	nit=44	nfev=88	
t=13.15s					
Nelder-Mead	run 706/1000	MSE=4.405e-09	nit=44	nfev=79	
t=12.11s					
Nelder-Mead	run 707/1000	MSE=8.472e-09	nit=36	nfev=69	
t=10.03s					
Nelder-Mead	run 708/1000	MSE=1.118e-08	nit=31	nfev=59	t=8.88s
Nelder-Mead	run 709/1000	MSE=8.139e-09	nit=46	nfev=88	
t=13.25s					
Nelder-Mead	run 710/1000	MSE=4.028e-09	nit=39	nfev=72	
t=10.66s					
Nelder-Mead	run 711/1000	MSE=1.703e-08	nit=43	nfev=80	
t=12.52s					
Nelder-Mead	run 712/1000	MSE=1.744e-08	nit=38	nfev=73	
t=10.83s					
Nelder-Mead	run 713/1000	MSE=1.035e-01	nit=13	nfev=25	t=3.68s
Nelder-Mead	run 714/1000	MSE=2.623e-09	nit=40	nfev=79	
t=11.78s					
Nelder-Mead	run 715/1000	MSE=7.098e-09	nit=38	nfev=69	
t=10.11s					
Nelder-Mead	run 716/1000	MSE=6.066e-08	nit=50	nfev=96	
t=14.36s					
Nelder-Mead	run 717/1000	MSE=3.952e-09	nit=40	nfev=74	
t=10.78s					
Nelder-Mead	run 718/1000	MSE=1.397e-08	nit=40	nfev=76	
t=11.43s					

Nelder-Mead	run 719/1000	MSE=1.415e-08	nit=24	nfev=48	t=7.34s
Nelder-Mead	run 720/1000	MSE=2.883e-09	nit=37	nfev=68	t=9.84s
Nelder-Mead	run 721/1000	MSE=4.919e-09	nit=69	nfev=130	t=19.45s
Nelder-Mead	run 722/1000	MSE=5.194e-09	nit=33	nfev=63	t=9.28s
Nelder-Mead	run 723/1000	MSE=3.863e-08	nit=35	nfev=65	t=9.59s
Nelder-Mead	run 724/1000	MSE=6.085e-09	nit=43	nfev=82	t=12.26s
Nelder-Mead	run 725/1000	MSE=1.223e-08	nit=36	nfev=70	t=10.23s
Nelder-Mead	run 726/1000	MSE=3.614e-08	nit=57	nfev=114	t=16.78s
Nelder-Mead	run 727/1000	MSE=8.486e-09	nit=41	nfev=80	t=11.74s
Nelder-Mead	run 728/1000	MSE=9.198e-09	nit=49	nfev=95	t=14.29s
Nelder-Mead	run 729/1000	MSE=4.666e-08	nit=37	nfev=69	t=10.60s
Nelder-Mead	run 730/1000	MSE=5.263e-09	nit=46	nfev=89	t=13.01s
Nelder-Mead	run 731/1000	MSE=2.440e-09	nit=41	nfev=80	t=11.97s
Nelder-Mead	run 732/1000	MSE=5.011e-09	nit=39	nfev=72	t=10.70s
Nelder-Mead	run 733/1000	MSE=7.786e-09	nit=43	nfev=82	t=12.23s
Nelder-Mead	run 734/1000	MSE=6.193e-09	nit=43	nfev=80	t=11.93s
Nelder-Mead	run 735/1000	MSE=5.817e-08	nit=37	nfev=70	t=10.43s
Nelder-Mead	run 736/1000	MSE=1.637e-08	nit=26	nfev=53	t=8.07s
Nelder-Mead	run 737/1000	MSE=1.035e-01	nit=13	nfev=25	t=3.67s
Nelder-Mead	run 738/1000	MSE=1.452e-08	nit=38	nfev=72	t=10.65s
Nelder-Mead	run 739/1000	MSE=8.686e-09	nit=36	nfev=69	t=10.31s
Nelder-Mead	run 740/1000	MSE=3.997e-09	nit=44	nfev=85	t=12.59s
Nelder-Mead	run 741/1000	MSE=2.778e-09	nit=31	nfev=62	t=9.17s
Nelder-Mead	run 742/1000	MSE=1.268e-08	nit=41	nfev=79	t=11.76s
Nelder-Mead	run 743/1000	MSE=3.471e-08	nit=35	nfev=66	t=9.80s
Nelder-Mead	run 744/1000	MSE=1.269e-08	nit=34	nfev=63	t=9.24s
Nelder-Mead	run 745/1000	MSE=5.702e-08	nit=29	nfev=57	t=8.48s
Nelder-Mead	run 746/1000	MSE=2.173e-09	nit=41	nfev=75	t=11.57s
Nelder-Mead	run 747/1000	MSE=5.458e-09	nit=45	nfev=84	t=12.31s

Nelder-Mead	run 748/1000	MSE=1.589e+01	nit=27	nfev=50	t=7.32s
Nelder-Mead	run 749/1000	MSE=2.286e-08	nit=41	nfev=80	
t=12.16s					
Nelder-Mead	run 750/1000	MSE=1.035e-01	nit=13	nfev=25	t=3.53s
Nelder-Mead	run 751/1000	MSE=1.240e-08	nit=68	nfev=128	
t=19.04s					
Nelder-Mead	run 752/1000	MSE=1.035e-01	nit=13	nfev=25	t=3.65s
Nelder-Mead	run 753/1000	MSE=6.696e-09	nit=43	nfev=85	
t=12.55s					
Nelder-Mead	run 754/1000	MSE=8.190e-09	nit=36	nfev=71	
t=10.41s					
Nelder-Mead	run 755/1000	MSE=3.789e-09	nit=46	nfev=87	
t=13.50s					
Nelder-Mead	run 756/1000	MSE=2.647e-09	nit=44	nfev=83	
t=12.59s					
Nelder-Mead	run 757/1000	MSE=9.302e-09	nit=32	nfev=63	t=9.50s
Nelder-Mead	run 758/1000	MSE=3.867e-09	nit=42	nfev=78	
t=11.54s					
Nelder-Mead	run 759/1000	MSE=4.738e-09	nit=48	nfev=87	
t=12.92s					
Nelder-Mead	run 760/1000	MSE=1.647e-08	nit=34	nfev=64	t=9.69s
Nelder-Mead	run 761/1000	MSE=3.102e-08	nit=40	nfev=78	
t=12.03s					
Nelder-Mead	run 762/1000	MSE=1.896e-09	nit=32	nfev=60	t=8.87s
Nelder-Mead	run 763/1000	MSE=1.844e-08	nit=37	nfev=70	
t=10.36s					
Nelder-Mead	run 764/1000	MSE=6.701e-08	nit=39	nfev=77	
t=11.51s					
Nelder-Mead	run 765/1000	MSE=1.278e-08	nit=37	nfev=71	
t=10.41s					
Nelder-Mead	run 766/1000	MSE=8.666e-09	nit=37	nfev=71	
t=10.63s					
Nelder-Mead	run 767/1000	MSE=4.208e-08	nit=43	nfev=82	
t=12.22s					
Nelder-Mead	run 768/1000	MSE=4.619e-09	nit=48	nfev=94	
t=14.07s					
Nelder-Mead	run 769/1000	MSE=1.802e-08	nit=37	nfev=71	
t=10.42s					
Nelder-Mead	run 770/1000	MSE=2.818e-08	nit=31	nfev=60	t=8.88s
Nelder-Mead	run 771/1000	MSE=1.349e-08	nit=34	nfev=68	
t=10.54s					
Nelder-Mead	run 772/1000	MSE=3.783e-09	nit=47	nfev=91	
t=13.32s					
Nelder-Mead	run 773/1000	MSE=3.957e-08	nit=35	nfev=68	
t=10.10s					
Nelder-Mead	run 774/1000	MSE=2.319e-09	nit=35	nfev=68	
t=10.31s					
Nelder-Mead	run 775/1000	MSE=1.227e-08	nit=32	nfev=60	t=8.73s

Nelder-Mead	run 776/1000	MSE=9.300e-09	nit=35	nfev=66	t=9.63s
Nelder-Mead	run 777/1000	MSE=5.189e-09	nit=54	nfev=105	
t=15.60s					
Nelder-Mead	run 778/1000	MSE=9.941e-09	nit=39	nfev=74	
t=11.08s					
Nelder-Mead	run 779/1000	MSE=1.671e-08	nit=43	nfev=83	
t=12.31s					
Nelder-Mead	run 780/1000	MSE=7.378e-09	nit=40	nfev=76	
t=11.05s					
Nelder-Mead	run 781/1000	MSE=2.615e-08	nit=45	nfev=86	
t=12.85s					
Nelder-Mead	run 782/1000	MSE=2.592e-08	nit=33	nfev=64	t=9.46s
Nelder-Mead	run 783/1000	MSE=3.593e-09	nit=44	nfev=84	
t=12.57s					
Nelder-Mead	run 784/1000	MSE=2.316e-08	nit=48	nfev=91	
t=13.80s					
Nelder-Mead	run 785/1000	MSE=6.571e-09	nit=39	nfev=77	
t=11.43s					
Nelder-Mead	run 786/1000	MSE=3.660e-09	nit=49	nfev=92	
t=14.09s					
Nelder-Mead	run 787/1000	MSE=3.651e-09	nit=45	nfev=85	
t=12.48s					
Nelder-Mead	run 788/1000	MSE=2.937e-09	nit=42	nfev=81	
t=12.18s					
Nelder-Mead	run 789/1000	MSE=6.498e-09	nit=60	nfev=112	
t=16.71s					
Nelder-Mead	run 790/1000	MSE=6.332e-09	nit=31	nfev=59	t=8.59s
Nelder-Mead	run 791/1000	MSE=7.805e-09	nit=42	nfev=78	
t=11.77s					
Nelder-Mead	run 792/1000	MSE=6.865e-09	nit=35	nfev=66	t=9.76s
Nelder-Mead	run 793/1000	MSE=1.118e-09	nit=50	nfev=97	
t=14.46s					
Nelder-Mead	run 794/1000	MSE=9.744e-09	nit=34	nfev=66	t=9.72s
Nelder-Mead	run 795/1000	MSE=9.342e-08	nit=58	nfev=110	
t=16.29s					
Nelder-Mead	run 796/1000	MSE=2.415e-09	nit=39	nfev=77	
t=11.40s					
Nelder-Mead	run 797/1000	MSE=2.343e-08	nit=40	nfev=75	
t=11.37s					
Nelder-Mead	run 798/1000	MSE=4.392e-09	nit=53	nfev=98	
t=14.65s					
Nelder-Mead	run 799/1000	MSE=2.094e-08	nit=41	nfev=79	
t=11.54s					
Nelder-Mead	run 800/1000	MSE=3.866e-09	nit=55	nfev=106	
t=15.71s					
Nelder-Mead	run 801/1000	MSE=7.380e-09	nit=31	nfev=61	t=8.96s
Nelder-Mead	run 802/1000	MSE=3.054e-08	nit=43	nfev=82	
t=12.29s					

Nelder-Mead t=15.61s	run 803/1000	MSE=9.898e-09	nit=55	nfev=105	
Nelder-Mead t=10.36s	run 804/1000	MSE=4.064e-09	nit=38	nfev=71	
Nelder-Mead t=11.16s	run 805/1000	MSE=6.318e-09	nit=39	nfev=74	
Nelder-Mead t=10.98s	run 806/1000	MSE=5.178e-09	nit=36	nfev=71	
Nelder-Mead t=14.75s	run 807/1000	MSE=9.562e-09	nit=51	nfev=99	
Nelder-Mead t=10.16s	run 808/1000	MSE=3.909e-09	nit=35	nfev=68	
Nelder-Mead t=12.96s	run 809/1000	MSE=3.515e-09	nit=47	nfev=87	
Nelder-Mead t=14.91s	run 810/1000	MSE=2.974e-09	nit=54	nfev=101	
Nelder-Mead t=16.01s	run 811/1000	MSE=4.969e-09	nit=57	nfev=106	
Nelder-Mead t=12.47s	run 812/1000	MSE=1.081e-08	nit=46	nfev=83	
Nelder-Mead t=15.14s	run 813/1000	MSE=4.012e-09	nit=56	nfev=104	
Nelder-Mead t=12.84s	run 814/1000	MSE=2.916e-09	nit=44	nfev=83	
Nelder-Mead	run 815/1000	MSE=2.396e-08	nit=22	nfev=44	t=6.64s
Nelder-Mead t=14.88s	run 816/1000	MSE=6.488e-10	nit=56	nfev=101	
Nelder-Mead t=12.05s	run 817/1000	MSE=3.453e-09	nit=42	nfev=82	
Nelder-Mead t=13.27s	run 818/1000	MSE=4.045e-09	nit=47	nfev=89	
Nelder-Mead t=12.63s	run 819/1000	MSE=5.264e-09	nit=44	nfev=85	
Nelder-Mead t=11.06s	run 820/1000	MSE=1.348e-07	nit=38	nfev=75	
Nelder-Mead t=14.79s	run 821/1000	MSE=3.465e-09	nit=53	nfev=100	
Nelder-Mead	run 822/1000	MSE=1.216e-08	nit=31	nfev=62	t=9.33s
Nelder-Mead	run 823/1000	MSE=4.271e-09	nit=28	nfev=55	t=8.02s
Nelder-Mead	run 824/1000	MSE=1.035e-01	nit=13	nfev=25	t=3.81s
Nelder-Mead t=13.97s	run 825/1000	MSE=7.185e-10	nit=50	nfev=96	
Nelder-Mead	run 826/1000	MSE=5.530e-09	nit=32	nfev=63	t=9.45s
Nelder-Mead t=16.49s	run 827/1000	MSE=2.887e-09	nit=57	nfev=110	
Nelder-Mead t=10.83s	run 828/1000	MSE=6.451e-09	nit=39	nfev=74	
Nelder-Mead	run 829/1000	MSE=3.468e-09	nit=83	nfev=161	

t=23.77s						
Nelder-Mead	run 830/1000	MSE=6.592e-09	nit=28	nfev=54	t=8.04s	
Nelder-Mead	run 831/1000	MSE=3.452e-09	nit=48	nfev=92		
t=13.62s						
Nelder-Mead	run 832/1000	MSE=1.228e-08	nit=47	nfev=90		
t=13.39s						
Nelder-Mead	run 833/1000	MSE=3.927e-08	nit=31	nfev=59	t=8.72s	
Nelder-Mead	run 834/1000	MSE=1.033e-08	nit=37	nfev=71		
t=10.32s						
Nelder-Mead	run 835/1000	MSE=3.863e-09	nit=32	nfev=61	t=8.97s	
Nelder-Mead	run 836/1000	MSE=5.221e-09	nit=51	nfev=98		
t=14.50s						
Nelder-Mead	run 837/1000	MSE=1.053e-09	nit=57	nfev=109		
t=16.30s						
Nelder-Mead	run 838/1000	MSE=4.458e-08	nit=40	nfev=75		
t=10.98s						
Nelder-Mead	run 839/1000	MSE=2.182e-08	nit=33	nfev=63	t=9.23s	
Nelder-Mead	run 840/1000	MSE=6.679e-09	nit=42	nfev=81		
t=12.07s						
Nelder-Mead	run 841/1000	MSE=9.728e-09	nit=50	nfev=97		
t=14.47s						
Nelder-Mead	run 842/1000	MSE=2.967e-09	nit=48	nfev=90		
t=13.94s						
Nelder-Mead	run 843/1000	MSE=3.394e-09	nit=53	nfev=99		
t=14.38s						
Nelder-Mead	run 844/1000	MSE=3.384e-09	nit=39	nfev=79		
t=12.04s						
Nelder-Mead	run 845/1000	MSE=7.587e-09	nit=46	nfev=84		
t=12.45s						
Nelder-Mead	run 846/1000	MSE=1.777e-09	nit=30	nfev=59	t=8.73s	
Nelder-Mead	run 847/1000	MSE=1.579e-08	nit=38	nfev=71		
t=10.55s						
Nelder-Mead	run 848/1000	MSE=4.629e-10	nit=47	nfev=91		
t=13.50s						
Nelder-Mead	run 849/1000	MSE=6.793e-09	nit=48	nfev=89		
t=13.20s						
Nelder-Mead	run 850/1000	MSE=1.704e-08	nit=52	nfev=94		
t=13.69s						
Nelder-Mead	run 851/1000	MSE=1.903e-09	nit=42	nfev=80		
t=12.05s						
Nelder-Mead	run 852/1000	MSE=3.803e-09	nit=46	nfev=87		
t=12.70s						
Nelder-Mead	run 853/1000	MSE=1.035e-01	nit=13	nfev=25	t=3.80s	
Nelder-Mead	run 854/1000	MSE=4.680e-09	nit=33	nfev=65	t=9.53s	
Nelder-Mead	run 855/1000	MSE=5.700e-09	nit=34	nfev=64	t=9.57s	
Nelder-Mead	run 856/1000	MSE=1.289e-08	nit=39	nfev=75		
t=11.56s						
Nelder-Mead	run 857/1000	MSE=7.862e-09	nit=37	nfev=72		

t=10.77s						
Nelder-Mead	run 858/1000	MSE=1.482e-09	nit=33	nfev=63	t=9.69s	
Nelder-Mead	run 859/1000	MSE=5.645e-09	nit=53	nfev=95		
t=14.16s						
Nelder-Mead	run 860/1000	MSE=7.414e-09	nit=52	nfev=97		
t=14.89s						
Nelder-Mead	run 861/1000	MSE=5.259e-09	nit=39	nfev=76		
t=11.25s						
Nelder-Mead	run 862/1000	MSE=5.124e-09	nit=34	nfev=67		
t=10.02s						
Nelder-Mead	run 863/1000	MSE=2.287e-08	nit=31	nfev=60	t=8.97s	
Nelder-Mead	run 864/1000	MSE=1.141e-08	nit=43	nfev=82		
t=12.43s						
Nelder-Mead	run 865/1000	MSE=5.003e-09	nit=48	nfev=92		
t=13.65s						
Nelder-Mead	run 866/1000	MSE=9.388e-08	nit=36	nfev=71		
t=10.54s						
Nelder-Mead	run 867/1000	MSE=1.035e-01	nit=13	nfev=25	t=3.77s	
Nelder-Mead	run 868/1000	MSE=1.487e-08	nit=45	nfev=87		
t=13.08s						
Nelder-Mead	run 869/1000	MSE=1.035e-01	nit=13	nfev=25	t=3.59s	
Nelder-Mead	run 870/1000	MSE=3.231e-09	nit=38	nfev=73		
t=10.56s						
Nelder-Mead	run 871/1000	MSE=2.242e-08	nit=29	nfev=56	t=8.20s	
Nelder-Mead	run 872/1000	MSE=1.035e-01	nit=13	nfev=25	t=3.78s	
Nelder-Mead	run 873/1000	MSE=1.399e-08	nit=35	nfev=65	t=9.63s	
Nelder-Mead	run 874/1000	MSE=6.105e-09	nit=40	nfev=77		
t=11.41s						
Nelder-Mead	run 875/1000	MSE=2.424e-08	nit=41	nfev=79		
t=11.63s						
Nelder-Mead	run 876/1000	MSE=4.581e-09	nit=58	nfev=108		
t=15.99s						
Nelder-Mead	run 877/1000	MSE=1.154e-08	nit=31	nfev=60	t=8.80s	
Nelder-Mead	run 878/1000	MSE=7.207e-09	nit=46	nfev=87		
t=13.02s						
Nelder-Mead	run 879/1000	MSE=1.903e-09	nit=48	nfev=92		
t=13.71s						
Nelder-Mead	run 880/1000	MSE=4.293e-08	nit=41	nfev=73		
t=10.72s						
Nelder-Mead	run 881/1000	MSE=5.368e-09	nit=27	nfev=53	t=7.81s	
Nelder-Mead	run 882/1000	MSE=2.431e-09	nit=53	nfev=98		
t=14.52s						
Nelder-Mead	run 883/1000	MSE=3.003e-09	nit=46	nfev=91		
t=13.56s						
Nelder-Mead	run 884/1000	MSE=1.978e-08	nit=34	nfev=63	t=9.32s	
Nelder-Mead	run 885/1000	MSE=2.896e-08	nit=33	nfev=62	t=9.30s	
Nelder-Mead	run 886/1000	MSE=1.597e-08	nit=49	nfev=89		
t=13.32s						

Nelder-Mead t=11.86s	run 887/1000	MSE=1.792e-08	nit=41	nfev=81	
Nelder-Mead t=11.66s	run 888/1000	MSE=9.971e-09	nit=40	nfev=78	
Nelder-Mead	run 889/1000	MSE=9.985e-09	nit=34	nfev=65	t=9.50s
Nelder-Mead t=13.19s	run 890/1000	MSE=1.565e-08	nit=46	nfev=88	
Nelder-Mead t=10.01s	run 891/1000	MSE=2.497e-08	nit=35	nfev=67	
Nelder-Mead t=15.22s	run 892/1000	MSE=8.693e-09	nit=53	nfev=102	
Nelder-Mead t=21.89s	run 893/1000	MSE=3.901e-09	nit=75	nfev=142	
Nelder-Mead	run 894/1000	MSE=5.642e-09	nit=35	nfev=68	t=9.97s
Nelder-Mead t=12.74s	run 895/1000	MSE=2.963e-08	nit=46	nfev=87	
Nelder-Mead t=12.60s	run 896/1000	MSE=9.039e-08	nit=44	nfev=86	
Nelder-Mead t=12.77s	run 897/1000	MSE=1.434e-09	nit=48	nfev=85	
Nelder-Mead t=10.02s	run 898/1000	MSE=7.793e-09	nit=38	nfev=69	
Nelder-Mead t=10.15s	run 899/1000	MSE=2.562e-09	nit=36	nfev=69	
Nelder-Mead t=11.59s	run 900/1000	MSE=2.112e-08	nit=39	nfev=76	
Nelder-Mead	run 901/1000	MSE=2.703e-08	nit=27	nfev=51	t=7.49s
Nelder-Mead t=14.21s	run 902/1000	MSE=4.684e-08	nit=51	nfev=95	
Nelder-Mead	run 903/1000	MSE=8.256e-09	nit=30	nfev=56	t=8.27s
Nelder-Mead	run 904/1000	MSE=2.727e-09	nit=32	nfev=61	t=8.88s
Nelder-Mead t=10.38s	run 905/1000	MSE=3.684e-08	nit=37	nfev=70	
Nelder-Mead	run 906/1000	MSE=9.810e-09	nit=35	nfev=67	t=9.93s
Nelder-Mead t=14.49s	run 907/1000	MSE=6.888e-09	nit=53	nfev=98	
Nelder-Mead t=14.51s	run 908/1000	MSE=5.029e-09	nit=49	nfev=97	
Nelder-Mead	run 909/1000	MSE=1.385e-08	nit=31	nfev=60	t=9.18s
Nelder-Mead	run 910/1000	MSE=2.326e-09	nit=30	nfev=55	t=8.98s
Nelder-Mead t=12.27s	run 911/1000	MSE=1.688e-08	nit=30	nfev=56	
Nelder-Mead t=10.67s	run 912/1000	MSE=1.020e-08	nit=37	nfev=71	
Nelder-Mead	run 913/1000	MSE=1.035e-01	nit=13	nfev=25	t=3.61s
Nelder-Mead	run 914/1000	MSE=1.665e-07	nit=31	nfev=58	t=8.55s
Nelder-Mead t=10.22s	run 915/1000	MSE=5.410e-09	nit=34	nfev=65	

Nelder-Mead	run 916/1000	MSE=1.360e-08	nit=26	nfev=50	t=7.49s
Nelder-Mead	run 917/1000	MSE=2.024e-08	nit=40	nfev=80	
t=11.95s					
Nelder-Mead	run 918/1000	MSE=1.035e-01	nit=13	nfev=25	t=3.80s
Nelder-Mead	run 919/1000	MSE=1.906e-08	nit=33	nfev=64	t=9.56s
Nelder-Mead	run 920/1000	MSE=6.446e-09	nit=42	nfev=78	
t=11.69s					
Nelder-Mead	run 921/1000	MSE=6.438e-09	nit=33	nfev=63	t=9.36s
Nelder-Mead	run 922/1000	MSE=5.980e-08	nit=40	nfev=77	
t=11.54s					
Nelder-Mead	run 923/1000	MSE=1.458e-09	nit=58	nfev=107	
t=16.30s					
Nelder-Mead	run 924/1000	MSE=2.133e-08	nit=41	nfev=75	
t=10.94s					
Nelder-Mead	run 925/1000	MSE=1.091e-08	nit=42	nfev=83	
t=12.27s					
Nelder-Mead	run 926/1000	MSE=8.344e-09	nit=30	nfev=59	t=9.17s
Nelder-Mead	run 927/1000	MSE=9.167e-09	nit=42	nfev=82	
t=11.94s					
Nelder-Mead	run 928/1000	MSE=1.054e-07	nit=35	nfev=69	
t=10.35s					
Nelder-Mead	run 929/1000	MSE=1.547e-08	nit=54	nfev=104	
t=15.55s					
Nelder-Mead	run 930/1000	MSE=6.562e-09	nit=52	nfev=95	
t=13.90s					
Nelder-Mead	run 931/1000	MSE=8.040e-09	nit=64	nfev=114	
t=16.98s					
Nelder-Mead	run 932/1000	MSE=5.892e-09	nit=38	nfev=76	
t=11.26s					
Nelder-Mead	run 933/1000	MSE=8.058e-09	nit=28	nfev=56	t=8.67s
Nelder-Mead	run 934/1000	MSE=2.258e-08	nit=47	nfev=89	
t=13.14s					
Nelder-Mead	run 935/1000	MSE=3.538e-09	nit=46	nfev=89	
t=12.94s					
Nelder-Mead	run 936/1000	MSE=7.905e-09	nit=55	nfev=102	
t=15.19s					
Nelder-Mead	run 937/1000	MSE=4.501e-08	nit=27	nfev=54	t=8.19s
Nelder-Mead	run 938/1000	MSE=3.387e-09	nit=46	nfev=86	
t=12.61s					
Nelder-Mead	run 939/1000	MSE=3.660e-08	nit=38	nfev=68	t=9.93s
Nelder-Mead	run 940/1000	MSE=2.616e-08	nit=33	nfev=65	t=9.81s
Nelder-Mead	run 941/1000	MSE=5.529e-08	nit=39	nfev=72	
t=10.72s					
Nelder-Mead	run 942/1000	MSE=1.814e-08	nit=44	nfev=81	
t=12.09s					
Nelder-Mead	run 943/1000	MSE=4.525e-09	nit=42	nfev=82	
t=11.96s					
Nelder-Mead	run 944/1000	MSE=4.508e-08	nit=49	nfev=90	

t=13.40s						
Nelder-Mead	run 945/1000	MSE=2.552e-09	nit=35	nfev=68		
t=10.86s						
Nelder-Mead	run 946/1000	MSE=9.953e-09	nit=35	nfev=69		
t=10.15s						
Nelder-Mead	run 947/1000	MSE=4.783e-09	nit=35	nfev=66	t=9.98s	
Nelder-Mead	run 948/1000	MSE=4.853e-09	nit=53	nfev=98		
t=14.43s						
Nelder-Mead	run 949/1000	MSE=1.888e-08	nit=37	nfev=75		
t=11.37s						
Nelder-Mead	run 950/1000	MSE=1.629e-08	nit=40	nfev=77		
t=11.29s						
Nelder-Mead	run 951/1000	MSE=4.781e-09	nit=57	nfev=107		
t=15.78s						
Nelder-Mead	run 952/1000	MSE=2.771e-09	nit=40	nfev=77		
t=11.51s						
Nelder-Mead	run 953/1000	MSE=2.964e-09	nit=33	nfev=63	t=9.42s	
Nelder-Mead	run 954/1000	MSE=3.096e-09	nit=38	nfev=72		
t=10.75s						
Nelder-Mead	run 955/1000	MSE=1.035e-01	nit=13	nfev=25	t=3.77s	
Nelder-Mead	run 956/1000	MSE=5.756e-09	nit=44	nfev=81		
t=11.86s						
Nelder-Mead	run 957/1000	MSE=4.577e-09	nit=51	nfev=95		
t=14.44s						
Nelder-Mead	run 958/1000	MSE=1.118e-08	nit=36	nfev=71		
t=10.51s						
Nelder-Mead	run 959/1000	MSE=6.437e-09	nit=39	nfev=74		
t=10.94s						
Nelder-Mead	run 960/1000	MSE=7.641e-09	nit=39	nfev=75		
t=10.89s						
Nelder-Mead	run 961/1000	MSE=6.503e-10	nit=35	nfev=70		
t=10.24s						
Nelder-Mead	run 962/1000	MSE=1.357e-08	nit=46	nfev=88		
t=13.33s						
Nelder-Mead	run 963/1000	MSE=3.811e-09	nit=30	nfev=58	t=8.59s	
Nelder-Mead	run 964/1000	MSE=6.177e-08	nit=27	nfev=52	t=7.73s	
Nelder-Mead	run 965/1000	MSE=1.334e-08	nit=37	nfev=72		
t=11.63s						
Nelder-Mead	run 966/1000	MSE=8.432e-08	nit=31	nfev=60	t=9.04s	
Nelder-Mead	run 967/1000	MSE=1.185e-08	nit=27	nfev=53	t=7.78s	
Nelder-Mead	run 968/1000	MSE=7.567e-09	nit=33	nfev=65	t=9.51s	
Nelder-Mead	run 969/1000	MSE=3.290e-08	nit=45	nfev=90		
t=13.59s						
Nelder-Mead	run 970/1000	MSE=3.884e-08	nit=38	nfev=75		
t=11.08s						
Nelder-Mead	run 971/1000	MSE=1.936e-09	nit=40	nfev=73		
t=11.01s						
Nelder-Mead	run 972/1000	MSE=1.425e-08	nit=32	nfev=63	t=9.26s	

Nelder-Mead t=13.60s	run 973/1000	MSE=2.422e-09	nit=46	nfev=91	
Nelder-Mead t=12.92s	run 974/1000	MSE=8.026e-09	nit=44	nfev=84	
Nelder-Mead t=12.86s	run 975/1000	MSE=4.929e-09	nit=45	nfev=85	
Nelder-Mead	run 976/1000	MSE=1.141e-08	nit=32	nfev=61	t=8.98s
Nelder-Mead t=12.50s	run 977/1000	MSE=2.753e-09	nit=45	nfev=85	
Nelder-Mead	run 978/1000	MSE=1.206e-07	nit=28	nfev=55	t=8.08s
Nelder-Mead	run 979/1000	MSE=2.696e-09	nit=33	nfev=66	t=9.77s
Nelder-Mead	run 980/1000	MSE=6.671e-08	nit=33	nfev=65	t=9.47s
Nelder-Mead t=11.56s	run 981/1000	MSE=1.249e-08	nit=37	nfev=71	
Nelder-Mead	run 982/1000	MSE=5.270e-09	nit=29	nfev=55	t=9.04s
Nelder-Mead t=13.70s	run 983/1000	MSE=2.143e-09	nit=47	nfev=93	
Nelder-Mead	run 984/1000	MSE=9.868e-09	nit=33	nfev=63	t=9.42s
Nelder-Mead	run 985/1000	MSE=1.035e-01	nit=13	nfev=25	t=3.53s
Nelder-Mead t=15.19s	run 986/1000	MSE=1.313e-08	nit=55	nfev=103	
Nelder-Mead	run 987/1000	MSE=1.025e-01	nit=22	nfev=39	t=5.83s
Nelder-Mead	run 988/1000	MSE=2.016e-09	nit=29	nfev=56	t=8.25s
Nelder-Mead	run 989/1000	MSE=1.426e-08	nit=31	nfev=62	t=9.09s
Nelder-Mead t=10.91s	run 990/1000	MSE=4.372e-09	nit=40	nfev=75	
Nelder-Mead t=11.52s	run 991/1000	MSE=1.771e-08	nit=41	nfev=76	
Nelder-Mead	run 992/1000	MSE=2.986e-09	nit=25	nfev=50	t=7.34s
Nelder-Mead t=12.36s	run 993/1000	MSE=8.170e-09	nit=46	nfev=85	
Nelder-Mead	run 994/1000	MSE=3.300e-09	nit=34	nfev=65	t=9.73s
Nelder-Mead	run 995/1000	MSE=7.514e-09	nit=33	nfev=62	t=9.11s
Nelder-Mead t=12.46s	run 996/1000	MSE=8.686e-09	nit=45	nfev=85	
Nelder-Mead t=14.07s	run 997/1000	MSE=5.361e-09	nit=50	nfev=95	
Nelder-Mead t=11.11s	run 998/1000	MSE=8.843e-09	nit=37	nfev=71	
Nelder-Mead t=11.11s	run 999/1000	MSE=9.513e-09	nit=38	nfev=73	
Nelder-Mead t=11.65s	run 1000/1000	MSE=3.318e-09	nit=41	nfev=80	

Scenario A - L-BFGS-B+jac (1000 runs)

L-BFGS-B	run 1/1000	MSE=1.224e-13	nit=10	nfev=51	t=7.57s
L-BFGS-B	run 2/1000	MSE=5.367e-13	nit=13	nfev=57	t=8.42s
L-BFGS-B	run 3/1000	MSE=5.157e-13	nit=13	nfev=63	t=9.33s

L-BFGS-B	run 4/1000	MSE=6.484e-14	nit=10	nfev=42	t=6.10s
L-BFGS-B	run 5/1000	MSE=1.090e-12	nit=16	nfev=78	t=11.73s
L-BFGS-B	run 6/1000	MSE=1.158e-13	nit=9	nfev=39	t=5.76s
L-BFGS-B	run 7/1000	MSE=4.887e-13	nit=15	nfev=69	t=10.01s
L-BFGS-B	run 8/1000	MSE=1.399e-13	nit=9	nfev=45	t=6.74s
L-BFGS-B	run 9/1000	MSE=4.898e-13	nit=12	nfev=42	t=6.00s
L-BFGS-B	run 10/1000	MSE=1.263e-13	nit=10	nfev=51	t=7.76s
L-BFGS-B	run 11/1000	MSE=6.628e-13	nit=7	nfev=36	t=5.08s
L-BFGS-B	run 12/1000	MSE=3.524e-13	nit=13	nfev=45	t=6.71s
L-BFGS-B	run 13/1000	MSE=1.803e-13	nit=9	nfev=51	t=7.66s
L-BFGS-B	run 14/1000	MSE=3.233e-11	nit=14	nfev=48	t=7.01s
L-BFGS-B	run 15/1000	MSE=1.555e-12	nit=10	nfev=45	t=6.59s
L-BFGS-B	run 16/1000	MSE=1.580e-12	nit=14	nfev=48	t=7.06s
L-BFGS-B	run 17/1000	MSE=1.125e-13	nit=14	nfev=69	t=10.14s
L-BFGS-B	run 18/1000	MSE=4.838e-13	nit=13	nfev=60	t=8.71s
L-BFGS-B	run 19/1000	MSE=2.505e-12	nit=16	nfev=60	t=8.71s
L-BFGS-B	run 20/1000	MSE=6.123e-13	nit=11	nfev=69	t=10.13s
L-BFGS-B	run 21/1000	MSE=2.144e-12	nit=12	nfev=51	t=7.51s
L-BFGS-B	run 22/1000	MSE=5.510e-13	nit=12	nfev=51	t=7.67s
L-BFGS-B	run 23/1000	MSE=1.471e-14	nit=9	nfev=48	t=7.10s
L-BFGS-B	run 24/1000	MSE=3.892e-13	nit=7	nfev=39	t=5.81s
L-BFGS-B	run 25/1000	MSE=4.752e-13	nit=14	nfev=48	t=7.09s
L-BFGS-B	run 26/1000	MSE=3.120e-14	nit=14	nfev=54	t=8.40s
L-BFGS-B	run 27/1000	MSE=1.191e-13	nit=10	nfev=48	t=7.07s
L-BFGS-B	run 28/1000	MSE=1.186e-13	nit=7	nfev=48	t=7.07s
L-BFGS-B	run 29/1000	MSE=3.875e-13	nit=12	nfev=66	t=9.64s
L-BFGS-B	run 30/1000	MSE=4.410e-13	nit=16	nfev=90	t=13.12s
L-BFGS-B	run 31/1000	MSE=1.681e-12	nit=11	nfev=42	t=6.28s
L-BFGS-B	run 32/1000	MSE=1.335e-13	nit=12	nfev=45	t=6.66s
L-BFGS-B	run 33/1000	MSE=3.642e-15	nit=13	nfev=48	t=7.59s
L-BFGS-B	run 34/1000	MSE=1.107e-13	nit=9	nfev=51	t=7.73s
L-BFGS-B	run 35/1000	MSE=4.850e-13	nit=14	nfev=66	t=9.63s
L-BFGS-B	run 36/1000	MSE=4.511e-13	nit=14	nfev=48	t=7.04s
L-BFGS-B	run 37/1000	MSE=5.749e-13	nit=11	nfev=39	t=5.89s
L-BFGS-B	run 38/1000	MSE=4.671e-12	nit=15	nfev=57	t=8.30s
L-BFGS-B	run 39/1000	MSE=5.684e-13	nit=14	nfev=54	t=7.98s
L-BFGS-B	run 40/1000	MSE=5.936e-13	nit=15	nfev=51	t=7.64s
L-BFGS-B	run 41/1000	MSE=4.413e-12	nit=12	nfev=48	t=7.05s
L-BFGS-B	run 42/1000	MSE=2.372e-13	nit=9	nfev=72	t=10.52s
L-BFGS-B	run 43/1000	MSE=2.433e-12	nit=10	nfev=39	t=5.78s
L-BFGS-B	run 44/1000	MSE=4.720e-13	nit=14	nfev=48	t=7.12s
L-BFGS-B	run 45/1000	MSE=4.684e-13	nit=13	nfev=51	t=7.64s
L-BFGS-B	run 46/1000	MSE=1.168e-13	nit=11	nfev=48	t=7.03s
L-BFGS-B	run 47/1000	MSE=1.933e-13	nit=9	nfev=48	t=6.96s
L-BFGS-B	run 48/1000	MSE=4.557e-13	nit=8	nfev=54	t=8.04s
L-BFGS-B	run 49/1000	MSE=7.064e-13	nit=10	nfev=57	t=8.39s
L-BFGS-B	run 50/1000	MSE=3.311e-13	nit=10	nfev=54	t=8.23s
L-BFGS-B	run 51/1000	MSE=1.028e-11	nit=11	nfev=48	t=7.00s

L-BFGS-B	run 52/1000	MSE=8.389e-13	nit=12	nfev=54	t=7.94s
L-BFGS-B	run 53/1000	MSE=1.419e-12	nit=11	nfev=54	t=7.99s
L-BFGS-B	run 54/1000	MSE=4.889e-13	nit=14	nfev=72	t=10.64s
L-BFGS-B	run 55/1000	MSE=3.112e-12	nit=9	nfev=69	t=10.24s
L-BFGS-B	run 56/1000	MSE=1.432e-13	nit=8	nfev=36	t=5.41s
L-BFGS-B	run 57/1000	MSE=2.349e-13	nit=16	nfev=54	t=7.97s
L-BFGS-B	run 58/1000	MSE=1.164e-13	nit=9	nfev=48	t=7.42s
L-BFGS-B	run 59/1000	MSE=1.345e-13	nit=9	nfev=45	t=6.89s
L-BFGS-B	run 60/1000	MSE=5.959e-13	nit=18	nfev=72	t=10.87s
L-BFGS-B	run 61/1000	MSE=4.546e-13	nit=12	nfev=45	t=6.61s
L-BFGS-B	run 62/1000	MSE=1.031e-12	nit=11	nfev=45	t=6.63s
L-BFGS-B	run 63/1000	MSE=5.385e-14	nit=13	nfev=51	t=7.51s
L-BFGS-B	run 64/1000	MSE=8.358e-13	nit=12	nfev=42	t=6.01s
L-BFGS-B	run 65/1000	MSE=4.461e-13	nit=12	nfev=60	t=8.80s
L-BFGS-B	run 66/1000	MSE=2.172e-13	nit=14	nfev=51	t=7.64s
L-BFGS-B	run 67/1000	MSE=1.151e-13	nit=11	nfev=63	t=9.70s
L-BFGS-B	run 68/1000	MSE=1.750e-13	nit=10	nfev=48	t=6.92s
L-BFGS-B	run 69/1000	MSE=5.136e-13	nit=11	nfev=54	t=7.88s
L-BFGS-B	run 70/1000	MSE=4.665e-13	nit=11	nfev=48	t=7.07s
L-BFGS-B	run 71/1000	MSE=5.870e-13	nit=14	nfev=54	t=7.89s
L-BFGS-B	run 72/1000	MSE=1.224e-11	nit=11	nfev=48	t=7.00s
L-BFGS-B	run 73/1000	MSE=3.736e-13	nit=12	nfev=42	t=6.37s
L-BFGS-B	run 74/1000	MSE=9.595e-14	nit=7	nfev=48	t=7.17s
L-BFGS-B	run 75/1000	MSE=7.633e-12	nit=15	nfev=54	t=7.95s
L-BFGS-B	run 76/1000	MSE=1.231e-13	nit=10	nfev=45	t=6.64s
L-BFGS-B	run 77/1000	MSE=4.186e-13	nit=9	nfev=39	t=5.69s
L-BFGS-B	run 78/1000	MSE=4.448e-13	nit=18	nfev=87	t=12.67s
L-BFGS-B	run 79/1000	MSE=1.148e-11	nit=10	nfev=48	t=7.28s
L-BFGS-B	run 80/1000	MSE=4.397e-13	nit=13	nfev=57	t=8.26s
L-BFGS-B	run 81/1000	MSE=4.493e-13	nit=14	nfev=54	t=8.32s
L-BFGS-B	run 82/1000	MSE=1.468e-13	nit=9	nfev=57	t=8.35s
L-BFGS-B	run 83/1000	MSE=2.123e-12	nit=11	nfev=42	t=6.15s
L-BFGS-B	run 84/1000	MSE=5.698e-13	nit=7	nfev=48	t=7.03s
L-BFGS-B	run 85/1000	MSE=4.582e-13	nit=13	nfev=75	t=10.93s
L-BFGS-B	run 86/1000	MSE=1.179e-13	nit=14	nfev=57	t=8.28s
L-BFGS-B	run 87/1000	MSE=4.226e-13	nit=9	nfev=39	t=5.79s
L-BFGS-B	run 88/1000	MSE=7.934e-13	nit=8	nfev=36	t=5.33s
L-BFGS-B	run 89/1000	MSE=1.543e-11	nit=8	nfev=63	t=9.30s
L-BFGS-B	run 90/1000	MSE=4.123e-11	nit=11	nfev=42	t=6.15s
L-BFGS-B	run 91/1000	MSE=5.664e-13	nit=12	nfev=42	t=6.32s
L-BFGS-B	run 92/1000	MSE=1.227e-13	nit=10	nfev=45	t=6.56s
L-BFGS-B	run 93/1000	MSE=4.755e-15	nit=13	nfev=48	t=6.99s
L-BFGS-B	run 94/1000	MSE=8.434e-13	nit=13	nfev=75	t=11.00s
L-BFGS-B	run 95/1000	MSE=2.077e-13	nit=17	nfev=60	t=8.73s
L-BFGS-B	run 96/1000	MSE=2.257e-13	nit=10	nfev=48	t=7.18s
L-BFGS-B	run 97/1000	MSE=4.501e-13	nit=14	nfev=54	t=7.95s
L-BFGS-B	run 98/1000	MSE=1.179e-13	nit=10	nfev=42	t=6.17s
L-BFGS-B	run 99/1000	MSE=6.654e-11	nit=13	nfev=51	t=7.49s

L-BFGS-B	run 100/1000	MSE=4.759e-13	nit=15	nfev=87	
t=12.81s					
L-BFGS-B	run 101/1000	MSE=4.705e-13	nit=15	nfev=69	
t=10.12s					
L-BFGS-B	run 102/1000	MSE=1.185e-13	nit=10	nfev=75	
t=11.26s					
L-BFGS-B	run 103/1000	MSE=1.261e-13	nit=7	nfev=42	t=6.25s
L-BFGS-B	run 104/1000	MSE=5.539e-12	nit=14	nfev=72	
t=10.49s					
L-BFGS-B	run 105/1000	MSE=9.032e-13	nit=12	nfev=48	t=7.15s
L-BFGS-B	run 106/1000	MSE=4.790e-13	nit=10	nfev=48	t=6.99s
L-BFGS-B	run 107/1000	MSE=8.989e-13	nit=12	nfev=42	t=6.16s
L-BFGS-B	run 108/1000	MSE=2.838e-12	nit=11	nfev=66	t=9.62s
L-BFGS-B	run 109/1000	MSE=3.754e-12	nit=12	nfev=54	t=8.25s
L-BFGS-B	run 110/1000	MSE=1.589e-13	nit=14	nfev=54	t=8.01s
L-BFGS-B	run 111/1000	MSE=4.777e-12	nit=13	nfev=66	t=9.74s
L-BFGS-B	run 112/1000	MSE=1.762e-13	nit=10	nfev=57	t=8.41s
L-BFGS-B	run 113/1000	MSE=5.304e-13	nit=9	nfev=42	t=6.12s
L-BFGS-B	run 114/1000	MSE=4.652e-12	nit=8	nfev=42	t=6.16s
L-BFGS-B	run 115/1000	MSE=6.750e-13	nit=15	nfev=51	t=7.50s
L-BFGS-B	run 116/1000	MSE=1.033e-13	nit=10	nfev=51	t=7.70s
L-BFGS-B	run 117/1000	MSE=1.773e-12	nit=19	nfev=96	
t=13.92s					
L-BFGS-B	run 118/1000	MSE=6.000e-14	nit=9	nfev=48	t=7.16s
L-BFGS-B	run 119/1000	MSE=3.945e-13	nit=11	nfev=54	t=7.91s
L-BFGS-B	run 120/1000	MSE=4.530e-13	nit=16	nfev=78	
t=11.54s					
L-BFGS-B	run 121/1000	MSE=4.521e-13	nit=18	nfev=66	t=9.61s
L-BFGS-B	run 122/1000	MSE=1.067e-13	nit=9	nfev=39	t=5.77s
L-BFGS-B	run 123/1000	MSE=5.992e-13	nit=17	nfev=60	t=8.85s
L-BFGS-B	run 124/1000	MSE=9.626e-14	nit=9	nfev=42	t=6.13s
L-BFGS-B	run 125/1000	MSE=5.444e-13	nit=11	nfev=51	t=7.48s
L-BFGS-B	run 126/1000	MSE=1.530e-13	nit=10	nfev=60	t=9.00s
L-BFGS-B	run 127/1000	MSE=1.353e-13	nit=10	nfev=39	t=5.78s
L-BFGS-B	run 128/1000	MSE=8.086e-10	nit=16	nfev=78	
t=11.36s					
L-BFGS-B	run 129/1000	MSE=1.231e-12	nit=7	nfev=33	t=4.86s
L-BFGS-B	run 130/1000	MSE=1.599e-13	nit=13	nfev=48	t=7.09s
L-BFGS-B	run 131/1000	MSE=2.103e-13	nit=9	nfev=51	t=7.56s
L-BFGS-B	run 132/1000	MSE=4.919e-13	nit=14	nfev=57	t=8.30s
L-BFGS-B	run 133/1000	MSE=5.089e-13	nit=11	nfev=54	t=7.98s
L-BFGS-B	run 134/1000	MSE=2.687e-12	nit=11	nfev=51	t=7.38s
L-BFGS-B	run 135/1000	MSE=7.284e-13	nit=12	nfev=48	t=7.48s
L-BFGS-B	run 136/1000	MSE=4.416e-13	nit=15	nfev=57	t=8.46s
L-BFGS-B	run 137/1000	MSE=4.849e-14	nit=14	nfev=75	
t=11.17s					
L-BFGS-B	run 138/1000	MSE=1.015e-13	nit=8	nfev=36	t=5.13s
L-BFGS-B	run 139/1000	MSE=5.224e-13	nit=11	nfev=42	t=6.20s

L-BFGS-B	run 140/1000	MSE=4.434e-13	nit=13	nfev=51	t=7.47s
L-BFGS-B	run 141/1000	MSE=4.981e-13	nit=16	nfev=84	
t=12.54s					
L-BFGS-B	run 142/1000	MSE=9.534e-12	nit=11	nfev=39	t=5.56s
L-BFGS-B	run 143/1000	MSE=1.955e-12	nit=11	nfev=42	t=6.12s
L-BFGS-B	run 144/1000	MSE=7.729e-14	nit=19	nfev=78	
t=11.64s					
L-BFGS-B	run 145/1000	MSE=1.715e-12	nit=16	nfev=90	
t=13.63s					
L-BFGS-B	run 146/1000	MSE=1.093e-12	nit=18	nfev=66	t=9.95s
L-BFGS-B	run 147/1000	MSE=2.740e-11	nit=13	nfev=45	t=6.86s
L-BFGS-B	run 148/1000	MSE=3.214e-13	nit=15	nfev=66	t=9.89s
L-BFGS-B	run 149/1000	MSE=4.753e-13	nit=17	nfev=81	
t=11.79s					
L-BFGS-B	run 150/1000	MSE=1.607e-12	nit=10	nfev=42	t=6.20s
L-BFGS-B	run 151/1000	MSE=5.262e-13	nit=12	nfev=54	t=7.99s
L-BFGS-B	run 152/1000	MSE=1.152e-13	nit=10	nfev=39	t=5.76s
L-BFGS-B	run 153/1000	MSE=4.185e-13	nit=14	nfev=54	t=7.87s
L-BFGS-B	run 154/1000	MSE=1.019e-13	nit=11	nfev=45	t=6.54s
L-BFGS-B	run 155/1000	MSE=3.090e-13	nit=10	nfev=51	t=7.53s
L-BFGS-B	run 156/1000	MSE=8.822e-14	nit=9	nfev=51	t=7.73s
L-BFGS-B	run 157/1000	MSE=5.248e-13	nit=10	nfev=42	t=6.23s
L-BFGS-B	run 158/1000	MSE=4.603e-13	nit=10	nfev=63	t=9.24s
L-BFGS-B	run 159/1000	MSE=4.504e-13	nit=11	nfev=54	t=7.92s
L-BFGS-B	run 160/1000	MSE=4.632e-13	nit=13	nfev=45	t=6.65s
L-BFGS-B	run 161/1000	MSE=1.884e-12	nit=13	nfev=78	
t=11.51s					
L-BFGS-B	run 162/1000	MSE=1.207e-12	nit=10	nfev=48	t=7.02s
L-BFGS-B	run 163/1000	MSE=9.970e-13	nit=12	nfev=54	t=7.89s
L-BFGS-B	run 164/1000	MSE=2.819e-12	nit=11	nfev=81	
t=11.86s					
L-BFGS-B	run 165/1000	MSE=5.665e-13	nit=14	nfev=48	t=7.14s
L-BFGS-B	run 166/1000	MSE=5.749e-13	nit=12	nfev=45	t=6.66s
L-BFGS-B	run 167/1000	MSE=4.541e-13	nit=12	nfev=45	t=6.55s
L-BFGS-B	run 168/1000	MSE=4.356e-13	nit=12	nfev=45	t=6.57s
L-BFGS-B	run 169/1000	MSE=1.107e-13	nit=9	nfev=48	t=7.20s
L-BFGS-B	run 170/1000	MSE=1.417e-12	nit=14	nfev=69	
t=10.13s					
L-BFGS-B	run 171/1000	MSE=1.719e-12	nit=14	nfev=69	
t=10.10s					
L-BFGS-B	run 172/1000	MSE=6.263e-13	nit=9	nfev=45	t=6.55s
L-BFGS-B	run 173/1000	MSE=1.784e-12	nit=14	nfev=48	t=7.09s
L-BFGS-B	run 174/1000	MSE=1.175e-13	nit=11	nfev=51	t=7.51s
L-BFGS-B	run 175/1000	MSE=4.942e-13	nit=11	nfev=42	t=6.36s
L-BFGS-B	run 176/1000	MSE=4.330e-13	nit=14	nfev=54	t=7.95s
L-BFGS-B	run 177/1000	MSE=4.581e-13	nit=9	nfev=45	t=6.74s
L-BFGS-B	run 178/1000	MSE=4.825e-13	nit=14	nfev=57	t=8.48s
L-BFGS-B	run 179/1000	MSE=4.556e-13	nit=17	nfev=66	t=9.88s

L-BFGS-B	run 180/1000	MSE=4.736e-13	nit=11	nfev=45	t=6.44s
L-BFGS-B	run 181/1000	MSE=4.285e-12	nit=14	nfev=48	t=7.08s
L-BFGS-B	run 182/1000	MSE=1.187e-13	nit=10	nfev=51	t=7.66s
L-BFGS-B	run 183/1000	MSE=5.094e-13	nit=15	nfev=69	
t=10.52s					
L-BFGS-B	run 184/1000	MSE=1.106e-12	nit=11	nfev=45	t=6.45s
L-BFGS-B	run 185/1000	MSE=3.620e-12	nit=9	nfev=39	t=5.88s
L-BFGS-B	run 186/1000	MSE=5.091e-13	nit=16	nfev=72	
t=10.56s					
L-BFGS-B	run 187/1000	MSE=8.787e-14	nit=9	nfev=72	
t=10.61s					
L-BFGS-B	run 188/1000	MSE=1.543e-12	nit=15	nfev=51	t=7.48s
L-BFGS-B	run 189/1000	MSE=2.679e-12	nit=14	nfev=48	t=7.24s
L-BFGS-B	run 190/1000	MSE=5.412e-13	nit=11	nfev=45	t=6.60s
L-BFGS-B	run 191/1000	MSE=1.857e-11	nit=14	nfev=48	t=6.90s
L-BFGS-B	run 192/1000	MSE=5.745e-13	nit=12	nfev=54	t=8.26s
L-BFGS-B	run 193/1000	MSE=2.866e-14	nit=9	nfev=45	t=6.71s
L-BFGS-B	run 194/1000	MSE=1.895e-13	nit=15	nfev=90	
t=13.22s					
L-BFGS-B	run 195/1000	MSE=3.784e-13	nit=11	nfev=51	t=7.45s
L-BFGS-B	run 196/1000	MSE=3.333e-13	nit=18	nfev=63	t=9.27s
L-BFGS-B	run 197/1000	MSE=4.930e-13	nit=9	nfev=39	t=5.84s
L-BFGS-B	run 198/1000	MSE=1.216e-13	nit=13	nfev=66	t=9.67s
L-BFGS-B	run 199/1000	MSE=8.708e-15	nit=9	nfev=36	t=5.31s
L-BFGS-B	run 200/1000	MSE=3.121e-13	nit=11	nfev=54	t=7.99s
L-BFGS-B	run 201/1000	MSE=4.186e-13	nit=17	nfev=69	
t=10.14s					
L-BFGS-B	run 202/1000	MSE=3.574e-13	nit=14	nfev=48	t=7.15s
L-BFGS-B	run 203/1000	MSE=1.173e-13	nit=10	nfev=57	t=8.31s
L-BFGS-B	run 204/1000	MSE=4.710e-13	nit=15	nfev=75	
t=10.89s					
L-BFGS-B	run 205/1000	MSE=6.599e-14	nit=8	nfev=63	t=9.17s
L-BFGS-B	run 206/1000	MSE=4.404e-13	nit=12	nfev=57	t=8.59s
L-BFGS-B	run 207/1000	MSE=2.666e-13	nit=8	nfev=39	t=5.64s
L-BFGS-B	run 208/1000	MSE=8.965e-13	nit=12	nfev=54	t=7.92s
L-BFGS-B	run 209/1000	MSE=7.151e-15	nit=14	nfev=57	t=8.36s
L-BFGS-B	run 210/1000	MSE=1.111e-12	nit=10	nfev=57	t=8.32s
L-BFGS-B	run 211/1000	MSE=4.554e-13	nit=14	nfev=63	t=9.47s
L-BFGS-B	run 212/1000	MSE=2.122e-12	nit=10	nfev=39	t=5.55s
L-BFGS-B	run 213/1000	MSE=2.840e-14	nit=11	nfev=45	t=6.71s
L-BFGS-B	run 214/1000	MSE=1.378e-13	nit=15	nfev=57	t=8.42s
L-BFGS-B	run 215/1000	MSE=3.520e-13	nit=18	nfev=90	
t=13.24s					
L-BFGS-B	run 216/1000	MSE=1.250e-13	nit=9	nfev=48	t=7.09s
L-BFGS-B	run 217/1000	MSE=4.080e-13	nit=12	nfev=63	t=9.29s
L-BFGS-B	run 218/1000	MSE=1.529e-13	nit=13	nfev=57	t=8.33s
L-BFGS-B	run 219/1000	MSE=1.783e-12	nit=15	nfev=75	
t=10.93s					

L-BFGS-B	run 220/1000	MSE=2.333e-14	nit=9	nfev=45	t=6.45s
L-BFGS-B	run 221/1000	MSE=4.221e-13	nit=10	nfev=51	t=7.56s
L-BFGS-B	run 222/1000	MSE=7.338e-13	nit=15	nfev=72	
t=10.62s					
L-BFGS-B	run 223/1000	MSE=1.179e-13	nit=10	nfev=51	t=7.45s
L-BFGS-B	run 224/1000	MSE=2.215e-11	nit=8	nfev=42	t=6.18s
L-BFGS-B	run 225/1000	MSE=4.553e-13	nit=17	nfev=75	
t=11.27s					
L-BFGS-B	run 226/1000	MSE=4.660e-13	nit=9	nfev=45	t=6.41s
L-BFGS-B	run 227/1000	MSE=8.386e-13	nit=13	nfev=48	t=7.11s
L-BFGS-B	run 228/1000	MSE=1.417e-12	nit=14	nfev=60	t=8.78s
L-BFGS-B	run 229/1000	MSE=1.133e-11	nit=14	nfev=69	
t=10.18s					
L-BFGS-B	run 230/1000	MSE=5.410e-13	nit=12	nfev=57	t=8.41s
L-BFGS-B	run 231/1000	MSE=4.426e-13	nit=12	nfev=45	t=6.71s
L-BFGS-B	run 232/1000	MSE=6.175e-14	nit=13	nfev=66	t=9.84s
L-BFGS-B	run 233/1000	MSE=3.095e-12	nit=8	nfev=48	t=7.03s
L-BFGS-B	run 234/1000	MSE=1.780e-13	nit=15	nfev=54	t=7.93s
L-BFGS-B	run 235/1000	MSE=4.783e-13	nit=15	nfev=75	
t=11.10s					
L-BFGS-B	run 236/1000	MSE=5.416e-13	nit=13	nfev=45	t=6.74s
L-BFGS-B	run 237/1000	MSE=5.595e-13	nit=11	nfev=39	t=5.81s
L-BFGS-B	run 238/1000	MSE=5.233e-13	nit=14	nfev=63	t=9.17s
L-BFGS-B	run 239/1000	MSE=4.918e-13	nit=11	nfev=39	t=5.80s
L-BFGS-B	run 240/1000	MSE=1.407e-12	nit=15	nfev=54	t=7.93s
L-BFGS-B	run 241/1000	MSE=1.954e-14	nit=16	nfev=63	t=9.15s
L-BFGS-B	run 242/1000	MSE=5.918e-12	nit=11	nfev=42	t=6.06s
L-BFGS-B	run 243/1000	MSE=3.388e-13	nit=16	nfev=78	
t=11.84s					
L-BFGS-B	run 244/1000	MSE=8.710e-14	nit=9	nfev=42	t=6.18s
L-BFGS-B	run 245/1000	MSE=1.646e-13	nit=12	nfev=48	t=7.01s
L-BFGS-B	run 246/1000	MSE=7.019e-14	nit=8	nfev=42	t=6.00s
L-BFGS-B	run 247/1000	MSE=1.129e-13	nit=10	nfev=39	t=6.00s
L-BFGS-B	run 248/1000	MSE=4.844e-13	nit=14	nfev=75	
t=11.46s					
L-BFGS-B	run 249/1000	MSE=1.520e-12	nit=16	nfev=57	t=8.56s
L-BFGS-B	run 250/1000	MSE=3.999e-13	nit=15	nfev=51	t=7.54s
L-BFGS-B	run 251/1000	MSE=2.413e-12	nit=12	nfev=48	t=6.94s
L-BFGS-B	run 252/1000	MSE=2.348e-12	nit=9	nfev=48	t=7.03s
L-BFGS-B	run 253/1000	MSE=2.427e-13	nit=13	nfev=48	t=7.13s
L-BFGS-B	run 254/1000	MSE=5.834e-15	nit=9	nfev=51	t=7.54s
L-BFGS-B	run 255/1000	MSE=1.453e-13	nit=8	nfev=39	t=6.28s
L-BFGS-B	run 256/1000	MSE=1.388e-13	nit=11	nfev=51	t=7.99s
L-BFGS-B	run 257/1000	MSE=5.930e-13	nit=10	nfev=36	t=5.37s
L-BFGS-B	run 258/1000	MSE=4.820e-13	nit=14	nfev=81	
t=11.87s					
L-BFGS-B	run 259/1000	MSE=1.585e-11	nit=12	nfev=60	t=8.91s
L-BFGS-B	run 260/1000	MSE=4.938e-13	nit=12	nfev=45	t=6.60s

L-BFGS-B	run 261/1000	MSE=9.025e-12	nit=13	nfev=45	t=6.61s
L-BFGS-B	run 262/1000	MSE=9.927e-13	nit=7	nfev=51	t=7.45s
L-BFGS-B	run 263/1000	MSE=2.217e-13	nit=9	nfev=51	t=7.75s
L-BFGS-B	run 264/1000	MSE=8.208e-13	nit=11	nfev=54	t=9.13s
L-BFGS-B	run 265/1000	MSE=5.586e-13	nit=7	nfev=51	t=7.53s
L-BFGS-B	run 266/1000	MSE=6.262e-13	nit=15	nfev=72	
t=10.62s					
L-BFGS-B	run 267/1000	MSE=4.864e-13	nit=10	nfev=72	
t=10.38s					
L-BFGS-B	run 268/1000	MSE=4.839e-13	nit=14	nfev=72	
t=10.73s					
L-BFGS-B	run 269/1000	MSE=1.209e-11	nit=15	nfev=54	t=7.97s
L-BFGS-B	run 270/1000	MSE=1.234e-13	nit=11	nfev=39	t=5.67s
L-BFGS-B	run 271/1000	MSE=1.158e-13	nit=8	nfev=69	
t=10.30s					
L-BFGS-B	run 272/1000	MSE=2.444e-13	nit=13	nfev=81	
t=12.05s					
L-BFGS-B	run 273/1000	MSE=3.424e-13	nit=14	nfev=48	t=7.04s
L-BFGS-B	run 274/1000	MSE=1.099e-13	nit=11	nfev=75	
t=11.16s					
L-BFGS-B	run 275/1000	MSE=1.010e-13	nit=11	nfev=48	t=6.89s
L-BFGS-B	run 276/1000	MSE=9.229e-13	nit=13	nfev=45	t=6.73s
L-BFGS-B	run 277/1000	MSE=1.726e-12	nit=13	nfev=60	t=8.76s
L-BFGS-B	run 278/1000	MSE=4.423e-13	nit=11	nfev=42	t=6.25s
L-BFGS-B	run 279/1000	MSE=2.473e-13	nit=10	nfev=42	t=6.22s
L-BFGS-B	run 280/1000	MSE=3.714e-13	nit=11	nfev=42	t=6.32s
L-BFGS-B	run 281/1000	MSE=1.848e-12	nit=9	nfev=66	t=9.83s
L-BFGS-B	run 282/1000	MSE=1.464e-13	nit=8	nfev=48	t=7.17s
L-BFGS-B	run 283/1000	MSE=8.835e-12	nit=11	nfev=45	t=6.69s
L-BFGS-B	run 284/1000	MSE=1.685e-12	nit=12	nfev=45	t=6.72s
L-BFGS-B	run 285/1000	MSE=1.040e-13	nit=11	nfev=51	t=7.64s
L-BFGS-B	run 286/1000	MSE=2.713e-13	nit=14	nfev=81	
t=11.98s					
L-BFGS-B	run 287/1000	MSE=4.857e-13	nit=15	nfev=51	t=7.85s
L-BFGS-B	run 288/1000	MSE=6.290e-13	nit=13	nfev=51	t=7.70s
L-BFGS-B	run 289/1000	MSE=5.318e-13	nit=13	nfev=54	t=8.10s
L-BFGS-B	run 290/1000	MSE=2.813e-12	nit=12	nfev=48	t=7.12s
L-BFGS-B	run 291/1000	MSE=9.072e-13	nit=11	nfev=39	t=5.77s
L-BFGS-B	run 292/1000	MSE=4.324e-13	nit=15	nfev=72	
t=10.64s					
L-BFGS-B	run 293/1000	MSE=4.785e-13	nit=13	nfev=45	t=6.64s
L-BFGS-B	run 294/1000	MSE=2.084e-12	nit=11	nfev=42	t=6.32s
L-BFGS-B	run 295/1000	MSE=2.680e-13	nit=14	nfev=48	t=7.72s
L-BFGS-B	run 296/1000	MSE=5.185e-13	nit=14	nfev=51	t=7.99s
L-BFGS-B	run 297/1000	MSE=1.179e-13	nit=13	nfev=69	
t=10.35s					
L-BFGS-B	run 298/1000	MSE=4.371e-13	nit=14	nfev=63	t=9.36s
L-BFGS-B	run 299/1000	MSE=3.503e-10	nit=10	nfev=51	t=8.14s

L-BFGS-B	run 300/1000	MSE=3.071e-12	nit=13	nfev=45	t=6.63s
L-BFGS-B	run 301/1000	MSE=8.080e-13	nit=12	nfev=54	t=7.92s
L-BFGS-B	run 302/1000	MSE=1.353e-12	nit=13	nfev=60	t=9.11s
L-BFGS-B	run 303/1000	MSE=4.176e-13	nit=14	nfev=66	t=9.77s
L-BFGS-B	run 304/1000	MSE=1.186e-11	nit=11	nfev=45	t=6.70s
L-BFGS-B	run 305/1000	MSE=1.915e-13	nit=10	nfev=51	t=7.57s
L-BFGS-B	run 306/1000	MSE=4.574e-13	nit=17	nfev=78	
t=11.59s					
L-BFGS-B	run 307/1000	MSE=1.581e-12	nit=12	nfev=63	t=9.48s
L-BFGS-B	run 308/1000	MSE=4.683e-13	nit=11	nfev=42	t=6.08s
L-BFGS-B	run 309/1000	MSE=3.276e-12	nit=12	nfev=45	t=6.72s
L-BFGS-B	run 310/1000	MSE=4.363e-13	nit=16	nfev=78	
t=11.84s					
L-BFGS-B	run 311/1000	MSE=3.421e-13	nit=10	nfev=48	t=7.01s
L-BFGS-B	run 312/1000	MSE=4.687e-13	nit=8	nfev=48	t=6.97s
L-BFGS-B	run 313/1000	MSE=4.752e-13	nit=13	nfev=60	t=8.80s
L-BFGS-B	run 314/1000	MSE=1.341e-12	nit=16	nfev=66	t=9.80s
L-BFGS-B	run 315/1000	MSE=6.559e-13	nit=13	nfev=48	t=7.10s
L-BFGS-B	run 316/1000	MSE=1.464e-11	nit=18	nfev=72	
t=10.64s					
L-BFGS-B	run 317/1000	MSE=1.660e-13	nit=12	nfev=54	t=8.00s
L-BFGS-B	run 318/1000	MSE=1.902e-11	nit=8	nfev=51	t=7.61s
L-BFGS-B	run 319/1000	MSE=9.607e-14	nit=9	nfev=42	t=6.21s
L-BFGS-B	run 320/1000	MSE=4.205e-12	nit=11	nfev=57	t=8.53s
L-BFGS-B	run 321/1000	MSE=1.423e-13	nit=14	nfev=69	
t=10.51s					
L-BFGS-B	run 322/1000	MSE=2.058e-13	nit=14	nfev=48	t=7.07s
L-BFGS-B	run 323/1000	MSE=1.789e-12	nit=13	nfev=63	t=9.29s
L-BFGS-B	run 324/1000	MSE=4.840e-12	nit=13	nfev=60	t=9.04s
L-BFGS-B	run 325/1000	MSE=5.643e-13	nit=8	nfev=33	t=4.90s
L-BFGS-B	run 326/1000	MSE=4.540e-13	nit=11	nfev=42	t=6.19s
L-BFGS-B	run 327/1000	MSE=1.128e-12	nit=12	nfev=54	t=7.96s
L-BFGS-B	run 328/1000	MSE=6.246e-12	nit=10	nfev=48	t=7.29s
L-BFGS-B	run 329/1000	MSE=1.132e-13	nit=8	nfev=48	t=6.86s
L-BFGS-B	run 330/1000	MSE=1.589e-12	nit=18	nfev=69	
t=10.44s					
L-BFGS-B	run 331/1000	MSE=4.751e-13	nit=8	nfev=48	t=7.11s
L-BFGS-B	run 332/1000	MSE=4.241e-12	nit=19	nfev=72	
t=10.79s					
L-BFGS-B	run 333/1000	MSE=1.149e-13	nit=9	nfev=69	
t=10.16s					
L-BFGS-B	run 334/1000	MSE=6.008e-14	nit=13	nfev=48	t=7.18s
L-BFGS-B	run 335/1000	MSE=3.423e-13	nit=19	nfev=78	
t=11.51s					
L-BFGS-B	run 336/1000	MSE=9.958e-13	nit=12	nfev=42	t=6.26s
L-BFGS-B	run 337/1000	MSE=4.467e-13	nit=11	nfev=42	t=6.54s
L-BFGS-B	run 338/1000	MSE=2.855e-14	nit=9	nfev=69	
t=10.26s					

L-BFGS-B	run 339/1000	MSE=2.772e-12	nit=17	nfev=69	
t=10.38s					
L-BFGS-B	run 340/1000	MSE=4.421e-13	nit=13	nfev=63	t=9.23s
L-BFGS-B	run 341/1000	MSE=9.303e-12	nit=9	nfev=36	t=5.36s
L-BFGS-B	run 342/1000	MSE=2.228e-13	nit=15	nfev=51	t=7.62s
L-BFGS-B	run 343/1000	MSE=1.175e-12	nit=11	nfev=81	
t=12.01s					
L-BFGS-B	run 344/1000	MSE=1.410e-12	nit=11	nfev=42	t=6.27s
L-BFGS-B	run 345/1000	MSE=1.187e-13	nit=13	nfev=48	t=7.17s
L-BFGS-B	run 346/1000	MSE=4.901e-13	nit=8	nfev=48	t=6.99s
L-BFGS-B	run 347/1000	MSE=8.096e-14	nit=10	nfev=45	t=6.60s
L-BFGS-B	run 348/1000	MSE=1.195e-12	nit=9	nfev=36	t=5.44s
L-BFGS-B	run 349/1000	MSE=1.658e-13	nit=9	nfev=45	t=6.65s
L-BFGS-B	run 350/1000	MSE=4.637e-13	nit=13	nfev=45	t=6.86s
L-BFGS-B	run 351/1000	MSE=4.270e-13	nit=16	nfev=78	
t=11.53s					
L-BFGS-B	run 352/1000	MSE=1.078e-13	nit=9	nfev=51	t=7.69s
L-BFGS-B	run 353/1000	MSE=1.604e-13	nit=13	nfev=51	t=7.92s
L-BFGS-B	run 354/1000	MSE=4.689e-13	nit=17	nfev=63	t=9.16s
L-BFGS-B	run 355/1000	MSE=4.132e-13	nit=15	nfev=60	t=8.89s
L-BFGS-B	run 356/1000	MSE=2.301e-12	nit=11	nfev=42	t=6.17s
L-BFGS-B	run 357/1000	MSE=1.388e-13	nit=8	nfev=66	t=9.84s
L-BFGS-B	run 358/1000	MSE=1.157e-13	nit=9	nfev=75	
t=11.10s					
L-BFGS-B	run 359/1000	MSE=1.012e-13	nit=9	nfev=45	t=6.68s
L-BFGS-B	run 360/1000	MSE=1.205e-13	nit=8	nfev=39	t=9.25s
L-BFGS-B	run 361/1000	MSE=4.078e-13	nit=12	nfev=48	t=8.19s
L-BFGS-B	run 362/1000	MSE=1.056e-13	nit=13	nfev=45	t=6.87s
L-BFGS-B	run 363/1000	MSE=1.232e-13	nit=13	nfev=51	t=7.97s
L-BFGS-B	run 364/1000	MSE=2.970e-13	nit=11	nfev=42	t=6.46s
L-BFGS-B	run 365/1000	MSE=1.918e-12	nit=13	nfev=60	t=8.82s
L-BFGS-B	run 366/1000	MSE=9.903e-13	nit=15	nfev=66	t=9.95s
L-BFGS-B	run 367/1000	MSE=6.908e-13	nit=15	nfev=57	t=8.42s
L-BFGS-B	run 368/1000	MSE=4.683e-13	nit=16	nfev=81	
t=12.33s					
L-BFGS-B	run 369/1000	MSE=9.953e-14	nit=10	nfev=54	t=8.06s
L-BFGS-B	run 370/1000	MSE=4.228e-13	nit=14	nfev=63	t=9.35s
L-BFGS-B	run 371/1000	MSE=4.545e-13	nit=16	nfev=78	
t=11.53s					
L-BFGS-B	run 372/1000	MSE=3.280e-12	nit=12	nfev=51	t=7.64s
L-BFGS-B	run 373/1000	MSE=1.768e-16	nit=9	nfev=51	t=7.58s
L-BFGS-B	run 374/1000	MSE=3.526e-13	nit=10	nfev=39	t=5.93s
L-BFGS-B	run 375/1000	MSE=3.173e-13	nit=14	nfev=48	t=7.06s
L-BFGS-B	run 376/1000	MSE=4.521e-13	nit=16	nfev=81	
t=11.86s					
L-BFGS-B	run 377/1000	MSE=4.568e-13	nit=9	nfev=48	t=7.16s
L-BFGS-B	run 378/1000	MSE=4.939e-13	nit=14	nfev=54	t=8.12s
L-BFGS-B	run 379/1000	MSE=2.336e-13	nit=10	nfev=51	t=7.46s

L-BFGS-B	run 380/1000	MSE=7.733e-12	nit=13	nfev=75	
t=11.01s					
L-BFGS-B	run 381/1000	MSE=1.735e-13	nit=17	nfev=84	
t=12.64s					
L-BFGS-B	run 382/1000	MSE=1.106e-13	nit=9	nfev=48	t=7.15s
L-BFGS-B	run 383/1000	MSE=1.828e-11	nit=8	nfev=57	t=8.53s
L-BFGS-B	run 384/1000	MSE=1.627e-12	nit=9	nfev=42	t=6.40s
L-BFGS-B	run 385/1000	MSE=1.531e-12	nit=14	nfev=81	
t=12.03s					
L-BFGS-B	run 386/1000	MSE=4.715e-13	nit=11	nfev=45	t=6.93s
L-BFGS-B	run 387/1000	MSE=4.265e-13	nit=7	nfev=48	t=7.19s
L-BFGS-B	run 388/1000	MSE=4.982e-13	nit=14	nfev=57	t=8.73s
L-BFGS-B	run 389/1000	MSE=2.825e-13	nit=14	nfev=54	t=7.96s
L-BFGS-B	run 390/1000	MSE=9.624e-14	nit=9	nfev=48	t=7.15s
L-BFGS-B	run 391/1000	MSE=6.072e-14	nit=15	nfev=72	
t=11.09s					
L-BFGS-B	run 392/1000	MSE=1.631e-12	nit=13	nfev=57	t=8.78s
L-BFGS-B	run 393/1000	MSE=4.930e-13	nit=13	nfev=45	t=6.76s
L-BFGS-B	run 394/1000	MSE=9.177e-13	nit=9	nfev=39	t=5.91s
L-BFGS-B	run 395/1000	MSE=4.958e-13	nit=14	nfev=72	
t=10.74s					
L-BFGS-B	run 396/1000	MSE=7.605e-14	nit=9	nfev=39	t=5.86s
L-BFGS-B	run 397/1000	MSE=6.225e-12	nit=13	nfev=60	t=9.00s
L-BFGS-B	run 398/1000	MSE=1.613e-13	nit=13	nfev=48	t=7.12s
L-BFGS-B	run 399/1000	MSE=1.064e-12	nit=18	nfev=63	t=9.70s
L-BFGS-B	run 400/1000	MSE=4.704e-13	nit=14	nfev=66	t=9.81s
L-BFGS-B	run 401/1000	MSE=2.777e-13	nit=11	nfev=45	t=6.79s
L-BFGS-B	run 402/1000	MSE=4.955e-13	nit=10	nfev=42	t=6.42s
L-BFGS-B	run 403/1000	MSE=5.990e-13	nit=20	nfev=93	
t=14.14s					
L-BFGS-B	run 404/1000	MSE=7.548e-13	nit=11	nfev=48	t=7.02s
L-BFGS-B	run 405/1000	MSE=1.459e-13	nit=9	nfev=54	t=7.93s
L-BFGS-B	run 406/1000	MSE=3.540e-02	nit=7	nfev=69	
t=10.44s					
L-BFGS-B	run 407/1000	MSE=3.882e-13	nit=18	nfev=66	t=9.82s
L-BFGS-B	run 408/1000	MSE=4.421e-13	nit=7	nfev=48	t=7.05s
L-BFGS-B	run 409/1000	MSE=4.928e-13	nit=10	nfev=48	t=7.56s
L-BFGS-B	run 410/1000	MSE=1.951e-13	nit=10	nfev=51	t=7.62s
L-BFGS-B	run 411/1000	MSE=1.073e-13	nit=9	nfev=39	t=5.63s
L-BFGS-B	run 412/1000	MSE=4.560e-13	nit=12	nfev=78	
t=11.79s					
L-BFGS-B	run 413/1000	MSE=4.241e-13	nit=11	nfev=48	t=7.26s
L-BFGS-B	run 414/1000	MSE=1.684e-12	nit=14	nfev=63	t=9.25s
L-BFGS-B	run 415/1000	MSE=7.464e-13	nit=11	nfev=45	t=6.72s
L-BFGS-B	run 416/1000	MSE=4.715e-13	nit=11	nfev=48	t=7.39s
L-BFGS-B	run 417/1000	MSE=4.558e-13	nit=11	nfev=42	t=6.50s
L-BFGS-B	run 418/1000	MSE=2.948e-12	nit=11	nfev=45	t=6.49s
L-BFGS-B	run 419/1000	MSE=2.466e-11	nit=11	nfev=48	t=7.11s

L-BFGS-B	run 420/1000	MSE=6.886e-13	nit=10	nfev=48	t=7.10s
L-BFGS-B	run 421/1000	MSE=1.629e-12	nit=10	nfev=39	t=5.87s
L-BFGS-B	run 422/1000	MSE=1.333e-13	nit=15	nfev=54	t=8.05s
L-BFGS-B	run 423/1000	MSE=1.011e-13	nit=11	nfev=51	t=7.66s
L-BFGS-B	run 424/1000	MSE=7.611e-14	nit=15	nfev=60	t=8.86s
L-BFGS-B	run 425/1000	MSE=1.183e-13	nit=9	nfev=36	t=5.44s
L-BFGS-B	run 426/1000	MSE=3.568e-13	nit=14	nfev=63	t=9.38s
L-BFGS-B	run 427/1000	MSE=5.410e-13	nit=13	nfev=66	t=9.67s
L-BFGS-B	run 428/1000	MSE=6.742e-13	nit=14	nfev=75	
t=11.30s					
L-BFGS-B	run 429/1000	MSE=4.986e-13	nit=13	nfev=45	t=6.53s
L-BFGS-B	run 430/1000	MSE=1.385e-12	nit=15	nfev=72	
t=10.74s					
L-BFGS-B	run 431/1000	MSE=3.003e-14	nit=12	nfev=69	
t=10.18s					
L-BFGS-B	run 432/1000	MSE=4.213e-13	nit=14	nfev=63	t=9.36s
L-BFGS-B	run 433/1000	MSE=2.572e-12	nit=17	nfev=63	t=9.22s
L-BFGS-B	run 434/1000	MSE=4.554e-13	nit=10	nfev=45	t=6.65s
L-BFGS-B	run 435/1000	MSE=4.843e-13	nit=12	nfev=48	t=7.25s
L-BFGS-B	run 436/1000	MSE=1.341e-11	nit=13	nfev=51	t=7.60s
L-BFGS-B	run 437/1000	MSE=6.433e-13	nit=11	nfev=42	t=6.32s
L-BFGS-B	run 438/1000	MSE=5.179e-12	nit=13	nfev=45	t=6.69s
L-BFGS-B	run 439/1000	MSE=4.539e-13	nit=9	nfev=57	t=8.61s
L-BFGS-B	run 440/1000	MSE=2.294e-12	nit=16	nfev=66	t=9.72s
L-BFGS-B	run 441/1000	MSE=4.790e-13	nit=19	nfev=96	
t=14.11s					
L-BFGS-B	run 442/1000	MSE=1.118e-13	nit=10	nfev=48	t=7.16s
L-BFGS-B	run 443/1000	MSE=1.288e-13	nit=8	nfev=48	t=7.29s
L-BFGS-B	run 444/1000	MSE=3.587e-13	nit=12	nfev=45	t=6.52s
L-BFGS-B	run 445/1000	MSE=2.691e-13	nit=22	nfev=84	
t=12.58s					
L-BFGS-B	run 446/1000	MSE=1.094e-13	nit=9	nfev=45	t=6.71s
L-BFGS-B	run 447/1000	MSE=4.658e-13	nit=15	nfev=72	
t=10.62s					
L-BFGS-B	run 448/1000	MSE=1.090e-12	nit=13	nfev=51	t=7.45s
L-BFGS-B	run 449/1000	MSE=1.547e-12	nit=12	nfev=42	t=6.40s
L-BFGS-B	run 450/1000	MSE=3.874e-14	nit=15	nfev=60	t=8.92s
L-BFGS-B	run 451/1000	MSE=1.603e-12	nit=13	nfev=66	t=9.69s
L-BFGS-B	run 452/1000	MSE=5.529e-13	nit=14	nfev=60	t=9.03s
L-BFGS-B	run 453/1000	MSE=4.663e-13	nit=13	nfev=51	t=7.60s
L-BFGS-B	run 454/1000	MSE=2.432e-12	nit=13	nfev=57	t=8.36s
L-BFGS-B	run 455/1000	MSE=1.128e-13	nit=15	nfev=63	t=9.37s
L-BFGS-B	run 456/1000	MSE=9.578e-13	nit=9	nfev=39	t=5.93s
L-BFGS-B	run 457/1000	MSE=6.047e-13	nit=15	nfev=75	
t=11.27s					
L-BFGS-B	run 458/1000	MSE=1.152e-13	nit=10	nfev=39	t=5.64s
L-BFGS-B	run 459/1000	MSE=6.922e-13	nit=20	nfev=93	
t=13.87s					

L-BFGS-B	run 460/1000	MSE=1.104e-13	nit=13	nfev=45	t=6.67s
L-BFGS-B	run 461/1000	MSE=2.578e-13	nit=11	nfev=57	t=8.44s
L-BFGS-B	run 462/1000	MSE=3.481e-12	nit=10	nfev=42	t=6.18s
L-BFGS-B	run 463/1000	MSE=2.219e-13	nit=11	nfev=45	t=7.04s
L-BFGS-B	run 464/1000	MSE=4.629e-13	nit=10	nfev=42	t=6.19s
L-BFGS-B	run 465/1000	MSE=9.332e-13	nit=12	nfev=48	t=6.83s
L-BFGS-B	run 466/1000	MSE=6.433e-12	nit=13	nfev=57	t=8.37s
L-BFGS-B	run 467/1000	MSE=5.324e-13	nit=10	nfev=66	
t=10.15s					
L-BFGS-B	run 468/1000	MSE=5.198e-13	nit=11	nfev=54	t=7.94s
L-BFGS-B	run 469/1000	MSE=2.107e-13	nit=9	nfev=36	t=5.10s
L-BFGS-B	run 470/1000	MSE=1.094e-13	nit=9	nfev=39	t=5.75s
L-BFGS-B	run 471/1000	MSE=5.590e-13	nit=11	nfev=60	t=9.11s
L-BFGS-B	run 472/1000	MSE=9.314e-13	nit=15	nfev=75	
t=11.36s					
L-BFGS-B	run 473/1000	MSE=1.002e-12	nit=17	nfev=81	
t=11.85s					
L-BFGS-B	run 474/1000	MSE=1.774e-13	nit=10	nfev=39	t=5.85s
L-BFGS-B	run 475/1000	MSE=8.686e-13	nit=13	nfev=51	t=7.53s
L-BFGS-B	run 476/1000	MSE=1.184e-13	nit=10	nfev=48	t=7.02s
L-BFGS-B	run 477/1000	MSE=5.334e-13	nit=11	nfev=45	t=6.53s
L-BFGS-B	run 478/1000	MSE=1.135e-13	nit=10	nfev=51	t=7.58s
L-BFGS-B	run 479/1000	MSE=4.540e-13	nit=17	nfev=66	
t=10.06s					
L-BFGS-B	run 480/1000	MSE=4.845e-13	nit=13	nfev=60	t=8.81s
L-BFGS-B	run 481/1000	MSE=7.068e-12	nit=10	nfev=51	t=7.44s
L-BFGS-B	run 482/1000	MSE=3.828e-13	nit=13	nfev=54	t=8.22s
L-BFGS-B	run 483/1000	MSE=1.756e-12	nit=13	nfev=45	t=6.56s
L-BFGS-B	run 484/1000	MSE=1.674e-12	nit=12	nfev=54	t=7.89s
L-BFGS-B	run 485/1000	MSE=1.193e-12	nit=12	nfev=51	t=7.49s
L-BFGS-B	run 486/1000	MSE=1.359e-13	nit=9	nfev=60	t=8.90s
L-BFGS-B	run 487/1000	MSE=1.484e-12	nit=13	nfev=45	t=6.62s
L-BFGS-B	run 488/1000	MSE=5.935e-13	nit=11	nfev=45	t=6.74s
L-BFGS-B	run 489/1000	MSE=3.683e-12	nit=10	nfev=39	t=5.60s
L-BFGS-B	run 490/1000	MSE=1.462e-13	nit=11	nfev=39	t=5.81s
L-BFGS-B	run 491/1000	MSE=1.206e-13	nit=10	nfev=42	t=6.33s
L-BFGS-B	run 492/1000	MSE=6.596e-13	nit=10	nfev=39	t=5.81s
L-BFGS-B	run 493/1000	MSE=4.010e-13	nit=12	nfev=45	t=6.62s
L-BFGS-B	run 494/1000	MSE=7.762e-13	nit=14	nfev=72	
t=10.74s					
L-BFGS-B	run 495/1000	MSE=1.194e-12	nit=13	nfev=51	t=7.66s
L-BFGS-B	run 496/1000	MSE=2.057e-13	nit=10	nfev=51	t=7.44s
L-BFGS-B	run 497/1000	MSE=1.144e-13	nit=11	nfev=69	
t=10.17s					
L-BFGS-B	run 498/1000	MSE=2.731e-13	nit=15	nfev=51	t=7.67s
L-BFGS-B	run 499/1000	MSE=2.451e-13	nit=13	nfev=57	t=8.48s
L-BFGS-B	run 500/1000	MSE=1.155e-13	nit=15	nfev=63	t=9.20s
L-BFGS-B	run 501/1000	MSE=3.533e-13	nit=12	nfev=93	

t=13.87s						
L-BFGS-B	run 502/1000	MSE=1.225e-13	nit=10	nfev=51	t=7.49s	
L-BFGS-B	run 503/1000	MSE=4.704e-13	nit=13	nfev=54	t=8.04s	
L-BFGS-B	run 504/1000	MSE=6.309e-13	nit=14	nfev=75		
t=11.19s						
L-BFGS-B	run 505/1000	MSE=3.143e-13	nit=14	nfev=51	t=7.59s	
L-BFGS-B	run 506/1000	MSE=2.076e-13	nit=12	nfev=45	t=6.65s	
L-BFGS-B	run 507/1000	MSE=1.054e-12	nit=13	nfev=48	t=7.12s	
L-BFGS-B	run 508/1000	MSE=5.865e-12	nit=15	nfev=72		
t=10.70s						
L-BFGS-B	run 509/1000	MSE=4.417e-13	nit=14	nfev=54	t=8.00s	
L-BFGS-B	run 510/1000	MSE=1.142e-12	nit=12	nfev=54	t=7.86s	
L-BFGS-B	run 511/1000	MSE=1.425e-12	nit=10	nfev=39	t=5.81s	
L-BFGS-B	run 512/1000	MSE=3.760e-13	nit=13	nfev=57	t=8.59s	
L-BFGS-B	run 513/1000	MSE=1.182e-13	nit=11	nfev=51	t=7.50s	
L-BFGS-B	run 514/1000	MSE=4.568e-13	nit=14	nfev=75		
t=10.91s						
L-BFGS-B	run 515/1000	MSE=7.062e-13	nit=11	nfev=54	t=8.06s	
L-BFGS-B	run 516/1000	MSE=4.549e-13	nit=13	nfev=45	t=6.72s	
L-BFGS-B	run 517/1000	MSE=1.432e-13	nit=8	nfev=42	t=6.41s	
L-BFGS-B	run 518/1000	MSE=1.199e-13	nit=12	nfev=60	t=8.80s	
L-BFGS-B	run 519/1000	MSE=3.303e-14	nit=9	nfev=42	t=6.30s	
L-BFGS-B	run 520/1000	MSE=2.988e-14	nit=10	nfev=54	t=8.28s	
L-BFGS-B	run 521/1000	MSE=4.444e-13	nit=11	nfev=42	t=6.26s	
L-BFGS-B	run 522/1000	MSE=4.138e-13	nit=11	nfev=54	t=7.72s	
L-BFGS-B	run 523/1000	MSE=2.907e-11	nit=14	nfev=57	t=8.48s	
L-BFGS-B	run 524/1000	MSE=1.036e-12	nit=10	nfev=42	t=6.22s	
L-BFGS-B	run 525/1000	MSE=7.240e-13	nit=16	nfev=84		
t=12.57s						
L-BFGS-B	run 526/1000	MSE=2.053e-14	nit=8	nfev=45	t=6.87s	
L-BFGS-B	run 527/1000	MSE=4.540e-13	nit=9	nfev=36	t=5.42s	
L-BFGS-B	run 528/1000	MSE=1.319e-12	nit=10	nfev=48	t=7.15s	
L-BFGS-B	run 529/1000	MSE=5.871e-13	nit=12	nfev=51	t=7.51s	
L-BFGS-B	run 530/1000	MSE=8.037e-14	nit=9	nfev=45	t=6.72s	
L-BFGS-B	run 531/1000	MSE=9.044e-14	nit=12	nfev=51	t=7.58s	
L-BFGS-B	run 532/1000	MSE=1.451e-12	nit=11	nfev=42	t=6.30s	
L-BFGS-B	run 533/1000	MSE=4.494e-13	nit=13	nfev=51	t=7.46s	
L-BFGS-B	run 534/1000	MSE=2.960e-11	nit=14	nfev=51	t=7.57s	
L-BFGS-B	run 535/1000	MSE=4.739e-13	nit=15	nfev=69		
t=10.38s						
L-BFGS-B	run 536/1000	MSE=2.314e-13	nit=12	nfev=42	t=6.28s	
L-BFGS-B	run 537/1000	MSE=6.550e-13	nit=12	nfev=42	t=6.19s	
L-BFGS-B	run 538/1000	MSE=1.621e-12	nit=10	nfev=57	t=8.43s	
L-BFGS-B	run 539/1000	MSE=4.720e-12	nit=12	nfev=45	t=7.02s	
L-BFGS-B	run 540/1000	MSE=6.383e-13	nit=10	nfev=72		
t=10.94s						
L-BFGS-B	run 541/1000	MSE=4.807e-13	nit=13	nfev=72		
t=10.59s						

L-BFGS-B	run 542/1000	MSE=3.764e-13	nit=11	nfev=42	t=6.30s
L-BFGS-B	run 543/1000	MSE=4.399e-13	nit=13	nfev=57	t=8.49s
L-BFGS-B	run 544/1000	MSE=1.137e-12	nit=9	nfev=72	
t=10.64s					
L-BFGS-B	run 545/1000	MSE=1.852e-13	nit=11	nfev=42	t=6.25s
L-BFGS-B	run 546/1000	MSE=3.640e-12	nit=14	nfev=48	t=7.42s
L-BFGS-B	run 547/1000	MSE=6.911e-13	nit=11	nfev=42	t=6.34s
L-BFGS-B	run 548/1000	MSE=3.969e-13	nit=11	nfev=54	t=7.96s
L-BFGS-B	run 549/1000	MSE=4.844e-13	nit=12	nfev=51	t=7.66s
L-BFGS-B	run 550/1000	MSE=4.676e-13	nit=13	nfev=57	t=8.67s
L-BFGS-B	run 551/1000	MSE=2.381e-12	nit=14	nfev=66	t=9.70s
L-BFGS-B	run 552/1000	MSE=5.420e-13	nit=15	nfev=51	t=7.48s
L-BFGS-B	run 553/1000	MSE=8.307e-14	nit=9	nfev=45	t=6.73s
L-BFGS-B	run 554/1000	MSE=6.821e-14	nit=13	nfev=45	t=6.83s
L-BFGS-B	run 555/1000	MSE=6.950e-13	nit=12	nfev=45	t=6.50s
L-BFGS-B	run 556/1000	MSE=1.669e-12	nit=11	nfev=39	t=5.83s
L-BFGS-B	run 557/1000	MSE=3.895e-14	nit=9	nfev=42	t=6.24s
L-BFGS-B	run 558/1000	MSE=3.443e-13	nit=16	nfev=54	t=8.20s
L-BFGS-B	run 559/1000	MSE=5.277e-14	nit=8	nfev=48	t=7.28s
L-BFGS-B	run 560/1000	MSE=4.893e-13	nit=10	nfev=66	t=9.70s
L-BFGS-B	run 561/1000	MSE=5.068e-13	nit=7	nfev=48	t=7.02s
L-BFGS-B	run 562/1000	MSE=1.029e-12	nit=9	nfev=45	t=6.87s
L-BFGS-B	run 563/1000	MSE=1.371e-13	nit=8	nfev=51	t=7.59s
L-BFGS-B	run 564/1000	MSE=5.773e-13	nit=10	nfev=51	t=7.53s
L-BFGS-B	run 565/1000	MSE=2.930e-13	nit=10	nfev=48	t=7.38s
L-BFGS-B	run 566/1000	MSE=1.252e-12	nit=17	nfev=78	
t=11.47s					
L-BFGS-B	run 567/1000	MSE=1.615e-12	nit=11	nfev=45	t=6.60s
L-BFGS-B	run 568/1000	MSE=4.686e-13	nit=16	nfev=75	
t=10.91s					
L-BFGS-B	run 569/1000	MSE=4.877e-13	nit=11	nfev=42	t=6.27s
L-BFGS-B	run 570/1000	MSE=6.398e-14	nit=14	nfev=57	t=8.53s
L-BFGS-B	run 571/1000	MSE=4.691e-13	nit=15	nfev=69	
t=10.10s					
L-BFGS-B	run 572/1000	MSE=1.051e-11	nit=14	nfev=54	t=8.01s
L-BFGS-B	run 573/1000	MSE=1.037e-11	nit=10	nfev=45	t=6.55s
L-BFGS-B	run 574/1000	MSE=1.725e-13	nit=12	nfev=45	t=6.65s
L-BFGS-B	run 575/1000	MSE=1.503e-12	nit=10	nfev=45	t=6.59s
L-BFGS-B	run 576/1000	MSE=2.138e-13	nit=14	nfev=51	t=7.64s
L-BFGS-B	run 577/1000	MSE=1.191e-12	nit=14	nfev=75	
t=11.17s					
L-BFGS-B	run 578/1000	MSE=5.245e-13	nit=14	nfev=72	
t=10.65s					
L-BFGS-B	run 579/1000	MSE=2.277e-13	nit=11	nfev=42	t=6.16s
L-BFGS-B	run 580/1000	MSE=1.640e-12	nit=11	nfev=39	t=5.68s
L-BFGS-B	run 581/1000	MSE=1.872e-15	nit=9	nfev=51	t=7.83s
L-BFGS-B	run 582/1000	MSE=1.053e-13	nit=9	nfev=51	t=7.52s
L-BFGS-B	run 583/1000	MSE=3.811e-13	nit=11	nfev=45	t=6.72s

L-BFGS-B	run 584/1000	MSE=5.923e-13	nit=15	nfev=87	
t=13.34s					
L-BFGS-B	run 585/1000	MSE=7.711e-13	nit=10	nfev=39	t=5.59s
L-BFGS-B	run 586/1000	MSE=6.787e-13	nit=14	nfev=66	
t=10.01s					
L-BFGS-B	run 587/1000	MSE=1.685e-12	nit=16	nfev=60	t=8.96s
L-BFGS-B	run 588/1000	MSE=7.388e-13	nit=12	nfev=45	t=6.76s
L-BFGS-B	run 589/1000	MSE=5.341e-13	nit=10	nfev=39	t=5.61s
L-BFGS-B	run 590/1000	MSE=1.584e-13	nit=9	nfev=60	t=8.92s
L-BFGS-B	run 591/1000	MSE=4.451e-13	nit=15	nfev=54	t=7.85s
L-BFGS-B	run 592/1000	MSE=9.313e-13	nit=10	nfev=39	t=6.00s
L-BFGS-B	run 593/1000	MSE=2.758e-12	nit=15	nfev=60	t=8.88s
L-BFGS-B	run 594/1000	MSE=4.931e-13	nit=14	nfev=54	t=7.95s
L-BFGS-B	run 595/1000	MSE=4.401e-13	nit=13	nfev=45	t=6.72s
L-BFGS-B	run 596/1000	MSE=5.853e-13	nit=10	nfev=54	t=8.20s
L-BFGS-B	run 597/1000	MSE=4.915e-13	nit=11	nfev=51	t=7.41s
L-BFGS-B	run 598/1000	MSE=5.402e-13	nit=15	nfev=75	
t=10.94s					
L-BFGS-B	run 599/1000	MSE=2.987e-13	nit=14	nfev=48	t=7.21s
L-BFGS-B	run 600/1000	MSE=5.618e-12	nit=11	nfev=51	t=7.51s
L-BFGS-B	run 601/1000	MSE=4.018e-12	nit=11	nfev=48	t=7.19s
L-BFGS-B	run 602/1000	MSE=1.422e-12	nit=10	nfev=57	t=8.50s
L-BFGS-B	run 603/1000	MSE=1.879e-13	nit=13	nfev=45	t=6.89s
L-BFGS-B	run 604/1000	MSE=1.260e-12	nit=11	nfev=42	t=6.15s
L-BFGS-B	run 605/1000	MSE=4.458e-13	nit=15	nfev=63	t=9.19s
L-BFGS-B	run 606/1000	MSE=1.147e-12	nit=11	nfev=42	t=6.11s
L-BFGS-B	run 607/1000	MSE=3.163e-12	nit=13	nfev=45	t=6.86s
L-BFGS-B	run 608/1000	MSE=1.319e-13	nit=9	nfev=42	t=6.11s
L-BFGS-B	run 609/1000	MSE=4.709e-13	nit=12	nfev=51	t=7.50s
L-BFGS-B	run 610/1000	MSE=1.141e-13	nit=6	nfev=30	t=4.49s
L-BFGS-B	run 611/1000	MSE=1.999e-13	nit=9	nfev=36	t=5.24s
L-BFGS-B	run 612/1000	MSE=4.750e-13	nit=11	nfev=45	t=6.63s
L-BFGS-B	run 613/1000	MSE=3.530e-12	nit=10	nfev=39	t=5.81s
L-BFGS-B	run 614/1000	MSE=5.744e-13	nit=10	nfev=57	t=8.45s
L-BFGS-B	run 615/1000	MSE=4.317e-13	nit=19	nfev=69	
t=10.36s					
L-BFGS-B	run 616/1000	MSE=1.033e-13	nit=11	nfev=54	t=8.22s
L-BFGS-B	run 617/1000	MSE=1.362e-12	nit=14	nfev=60	t=8.80s
L-BFGS-B	run 618/1000	MSE=4.212e-13	nit=14	nfev=48	t=7.15s
L-BFGS-B	run 619/1000	MSE=4.965e-13	nit=8	nfev=57	t=8.53s
L-BFGS-B	run 620/1000	MSE=4.572e-13	nit=16	nfev=63	t=9.38s
L-BFGS-B	run 621/1000	MSE=1.204e-13	nit=9	nfev=39	t=5.75s
L-BFGS-B	run 622/1000	MSE=2.855e-12	nit=13	nfev=48	t=7.10s
L-BFGS-B	run 623/1000	MSE=1.308e-13	nit=17	nfev=63	t=9.83s
L-BFGS-B	run 624/1000	MSE=6.849e-12	nit=11	nfev=48	t=7.07s
L-BFGS-B	run 625/1000	MSE=4.844e-12	nit=13	nfev=60	t=8.79s
L-BFGS-B	run 626/1000	MSE=4.728e-13	nit=12	nfev=75	
t=11.31s					

L-BFGS-B	run 627/1000	MSE=1.256e-13	nit=11	nfev=63	t=9.78s
L-BFGS-B	run 628/1000	MSE=6.074e-12	nit=12	nfev=48	t=7.10s
L-BFGS-B	run 629/1000	MSE=4.517e-13	nit=11	nfev=57	t=8.39s
L-BFGS-B	run 630/1000	MSE=3.849e-13	nit=9	nfev=48	t=7.34s
L-BFGS-B	run 631/1000	MSE=4.626e-13	nit=12	nfev=75	
t=11.14s					
L-BFGS-B	run 632/1000	MSE=1.139e-13	nit=18	nfev=99	
t=14.85s					
L-BFGS-B	run 633/1000	MSE=7.618e-14	nit=10	nfev=48	t=7.23s
L-BFGS-B	run 634/1000	MSE=5.742e-13	nit=10	nfev=54	t=8.48s
L-BFGS-B	run 635/1000	MSE=4.565e-13	nit=8	nfev=51	t=7.65s
L-BFGS-B	run 636/1000	MSE=3.963e-13	nit=13	nfev=48	t=7.27s
L-BFGS-B	run 637/1000	MSE=4.389e-13	nit=12	nfev=48	t=7.31s
L-BFGS-B	run 638/1000	MSE=4.446e-13	nit=11	nfev=42	t=6.08s
L-BFGS-B	run 639/1000	MSE=4.227e-13	nit=13	nfev=48	t=7.15s
L-BFGS-B	run 640/1000	MSE=1.138e-12	nit=12	nfev=45	t=6.74s
L-BFGS-B	run 641/1000	MSE=6.817e-13	nit=9	nfev=51	t=7.76s
L-BFGS-B	run 642/1000	MSE=5.159e-13	nit=13	nfev=54	t=7.98s
L-BFGS-B	run 643/1000	MSE=1.373e-12	nit=13	nfev=45	t=6.65s
L-BFGS-B	run 644/1000	MSE=4.834e-13	nit=14	nfev=66	t=9.83s
L-BFGS-B	run 645/1000	MSE=2.200e-12	nit=7	nfev=39	t=5.86s
L-BFGS-B	run 646/1000	MSE=1.444e-13	nit=14	nfev=60	t=8.80s
L-BFGS-B	run 647/1000	MSE=8.854e-12	nit=12	nfev=57	t=8.97s
L-BFGS-B	run 648/1000	MSE=8.572e-14	nit=10	nfev=57	t=8.77s
L-BFGS-B	run 649/1000	MSE=6.700e-13	nit=12	nfev=54	t=8.11s
L-BFGS-B	run 650/1000	MSE=3.572e-12	nit=11	nfev=69	
t=10.08s					
L-BFGS-B	run 651/1000	MSE=7.627e-13	nit=12	nfev=45	t=6.83s
L-BFGS-B	run 652/1000	MSE=3.666e-14	nit=10	nfev=54	t=8.14s
L-BFGS-B	run 653/1000	MSE=1.063e-13	nit=13	nfev=63	t=9.27s
L-BFGS-B	run 654/1000	MSE=4.447e-13	nit=10	nfev=45	t=6.71s
L-BFGS-B	run 655/1000	MSE=6.691e-13	nit=15	nfev=57	t=8.59s
L-BFGS-B	run 656/1000	MSE=4.648e-13	nit=11	nfev=42	t=6.41s
L-BFGS-B	run 657/1000	MSE=9.166e-14	nit=18	nfev=90	
t=13.32s					
L-BFGS-B	run 658/1000	MSE=1.181e-13	nit=9	nfev=60	t=8.95s
L-BFGS-B	run 659/1000	MSE=6.510e-14	nit=11	nfev=54	t=8.16s
L-BFGS-B	run 660/1000	MSE=1.539e-14	nit=9	nfev=84	
t=12.33s					
L-BFGS-B	run 661/1000	MSE=7.115e-14	nit=15	nfev=57	t=8.57s
L-BFGS-B	run 662/1000	MSE=5.620e-12	nit=13	nfev=48	t=7.40s
L-BFGS-B	run 663/1000	MSE=7.231e-12	nit=11	nfev=45	t=6.61s
L-BFGS-B	run 664/1000	MSE=2.203e-12	nit=10	nfev=48	t=7.01s
L-BFGS-B	run 665/1000	MSE=5.601e-13	nit=9	nfev=51	t=7.55s
L-BFGS-B	run 666/1000	MSE=1.226e-13	nit=12	nfev=51	t=7.75s
L-BFGS-B	run 667/1000	MSE=1.066e-12	nit=9	nfev=39	t=6.43s
L-BFGS-B	run 668/1000	MSE=1.400e-13	nit=9	nfev=42	t=6.40s
L-BFGS-B	run 669/1000	MSE=4.814e-13	nit=14	nfev=63	t=9.57s

L-BFGS-B	run 670/1000	MSE=2.593e-13	nit=10	nfev=39	t=5.87s
L-BFGS-B	run 671/1000	MSE=5.932e-13	nit=13	nfev=54	t=7.94s
L-BFGS-B	run 672/1000	MSE=1.213e-13	nit=10	nfev=39	t=5.81s
L-BFGS-B	run 673/1000	MSE=7.759e-13	nit=13	nfev=57	t=8.47s
L-BFGS-B	run 674/1000	MSE=1.406e-12	nit=10	nfev=45	t=6.86s
L-BFGS-B	run 675/1000	MSE=7.249e-13	nit=14	nfev=69	
t=10.10s					
L-BFGS-B	run 676/1000	MSE=2.187e-13	nit=15	nfev=51	t=7.48s
L-BFGS-B	run 677/1000	MSE=4.523e-13	nit=11	nfev=63	t=9.52s
L-BFGS-B	run 678/1000	MSE=6.392e-13	nit=9	nfev=69	
t=10.61s					
L-BFGS-B	run 679/1000	MSE=1.844e-13	nit=15	nfev=51	t=7.48s
L-BFGS-B	run 680/1000	MSE=2.030e-12	nit=10	nfev=42	t=6.04s
L-BFGS-B	run 681/1000	MSE=1.080e-02	nit=8	nfev=72	
t=11.08s					
L-BFGS-B	run 682/1000	MSE=1.178e-13	nit=9	nfev=48	t=7.07s
L-BFGS-B	run 683/1000	MSE=3.519e-13	nit=10	nfev=54	t=7.97s
L-BFGS-B	run 684/1000	MSE=1.216e-12	nit=14	nfev=66	t=9.89s
L-BFGS-B	run 685/1000	MSE=4.988e-13	nit=14	nfev=66	t=9.71s
L-BFGS-B	run 686/1000	MSE=4.576e-13	nit=20	nfev=81	
t=11.84s					
L-BFGS-B	run 687/1000	MSE=4.679e-13	nit=11	nfev=42	t=6.59s
L-BFGS-B	run 688/1000	MSE=6.627e-13	nit=12	nfev=57	t=8.62s
L-BFGS-B	run 689/1000	MSE=1.137e-13	nit=10	nfev=51	t=7.52s
L-BFGS-B	run 690/1000	MSE=2.938e-12	nit=14	nfev=48	t=7.07s
L-BFGS-B	run 691/1000	MSE=1.768e-11	nit=10	nfev=42	t=6.30s
L-BFGS-B	run 692/1000	MSE=9.200e-13	nit=11	nfev=54	t=8.10s
L-BFGS-B	run 693/1000	MSE=1.241e-12	nit=17	nfev=63	t=9.34s
L-BFGS-B	run 694/1000	MSE=3.661e-13	nit=15	nfev=51	t=7.59s
L-BFGS-B	run 695/1000	MSE=1.726e-15	nit=8	nfev=42	t=6.47s
L-BFGS-B	run 696/1000	MSE=2.522e-12	nit=17	nfev=60	t=8.85s
L-BFGS-B	run 697/1000	MSE=2.730e-12	nit=10	nfev=69	
t=10.20s					
L-BFGS-B	run 698/1000	MSE=1.056e-12	nit=10	nfev=45	t=6.78s
L-BFGS-B	run 699/1000	MSE=1.262e-13	nit=7	nfev=48	t=7.33s
L-BFGS-B	run 700/1000	MSE=2.777e-12	nit=15	nfev=54	t=8.01s
L-BFGS-B	run 701/1000	MSE=4.577e-13	nit=8	nfev=51	t=7.46s
L-BFGS-B	run 702/1000	MSE=4.445e-13	nit=16	nfev=78	
t=11.73s					
L-BFGS-B	run 703/1000	MSE=4.509e-13	nit=10	nfev=66	
t=10.04s					
L-BFGS-B	run 704/1000	MSE=6.709e-11	nit=13	nfev=60	t=8.86s
L-BFGS-B	run 705/1000	MSE=1.240e-13	nit=10	nfev=45	t=6.75s
L-BFGS-B	run 706/1000	MSE=4.642e-13	nit=12	nfev=45	t=6.70s
L-BFGS-B	run 707/1000	MSE=4.623e-13	nit=13	nfev=48	t=7.07s
L-BFGS-B	run 708/1000	MSE=4.405e-13	nit=13	nfev=57	t=8.35s
L-BFGS-B	run 709/1000	MSE=4.538e-13	nit=16	nfev=78	
t=11.93s					

L-BFGS-B	run 710/1000	MSE=9.008e-12	nit=14	nfev=48	t=7.11s
L-BFGS-B	run 711/1000	MSE=2.833e-13	nit=9	nfev=51	t=7.58s
L-BFGS-B	run 712/1000	MSE=7.925e-13	nit=14	nfev=66	
t=10.15s					
L-BFGS-B	run 713/1000	MSE=1.005e-13	nit=12	nfev=57	t=8.60s
L-BFGS-B	run 714/1000	MSE=9.573e-14	nit=15	nfev=51	t=7.50s
L-BFGS-B	run 715/1000	MSE=9.114e-14	nit=9	nfev=36	t=5.32s
L-BFGS-B	run 716/1000	MSE=4.170e-13	nit=14	nfev=63	t=9.35s
L-BFGS-B	run 717/1000	MSE=6.481e-13	nit=14	nfev=60	t=8.92s
L-BFGS-B	run 718/1000	MSE=8.143e-14	nit=8	nfev=39	t=5.75s
L-BFGS-B	run 719/1000	MSE=5.614e-13	nit=12	nfev=45	t=6.50s
L-BFGS-B	run 720/1000	MSE=1.175e-13	nit=12	nfev=57	t=8.35s
L-BFGS-B	run 721/1000	MSE=4.687e-13	nit=15	nfev=63	t=9.81s
L-BFGS-B	run 722/1000	MSE=7.315e-13	nit=15	nfev=54	t=8.03s
L-BFGS-B	run 723/1000	MSE=1.319e-13	nit=9	nfev=39	t=5.57s
L-BFGS-B	run 724/1000	MSE=2.066e-14	nit=10	nfev=45	t=6.72s
L-BFGS-B	run 725/1000	MSE=3.166e-13	nit=13	nfev=45	t=6.81s
L-BFGS-B	run 726/1000	MSE=7.563e-14	nit=9	nfev=51	t=7.54s
L-BFGS-B	run 727/1000	MSE=2.827e-12	nit=10	nfev=45	t=6.71s
L-BFGS-B	run 728/1000	MSE=3.246e-13	nit=13	nfev=60	t=9.00s
L-BFGS-B	run 729/1000	MSE=7.152e-14	nit=13	nfev=60	t=9.04s
L-BFGS-B	run 730/1000	MSE=5.918e-14	nit=10	nfev=51	t=7.51s
L-BFGS-B	run 731/1000	MSE=5.674e-14	nit=17	nfev=75	
t=11.09s					
L-BFGS-B	run 732/1000	MSE=3.742e-13	nit=12	nfev=42	t=6.38s
L-BFGS-B	run 733/1000	MSE=1.157e-12	nit=13	nfev=57	t=8.55s
L-BFGS-B	run 734/1000	MSE=6.979e-03	nit=8	nfev=72	
t=10.76s					
L-BFGS-B	run 735/1000	MSE=4.159e-12	nit=14	nfev=60	t=9.07s
L-BFGS-B	run 736/1000	MSE=3.329e-12	nit=13	nfev=45	t=6.75s
L-BFGS-B	run 737/1000	MSE=2.587e-13	nit=9	nfev=60	t=8.85s
L-BFGS-B	run 738/1000	MSE=1.095e-13	nit=8	nfev=51	t=7.62s
L-BFGS-B	run 739/1000	MSE=9.898e-12	nit=10	nfev=42	t=6.29s
L-BFGS-B	run 740/1000	MSE=5.422e-13	nit=13	nfev=66	t=9.94s
L-BFGS-B	run 741/1000	MSE=2.225e-13	nit=16	nfev=75	
t=11.18s					
L-BFGS-B	run 742/1000	MSE=1.446e-13	nit=15	nfev=57	t=9.12s
L-BFGS-B	run 743/1000	MSE=4.297e-13	nit=13	nfev=60	t=9.28s
L-BFGS-B	run 744/1000	MSE=7.240e-12	nit=12	nfev=45	t=7.33s
L-BFGS-B	run 745/1000	MSE=3.095e-13	nit=10	nfev=42	t=6.38s
L-BFGS-B	run 746/1000	MSE=2.867e-12	nit=15	nfev=51	t=7.71s
L-BFGS-B	run 747/1000	MSE=3.900e-14	nit=10	nfev=39	t=5.79s
L-BFGS-B	run 748/1000	MSE=4.298e-13	nit=12	nfev=57	t=8.37s
L-BFGS-B	run 749/1000	MSE=8.677e-14	nit=13	nfev=72	
t=10.91s					
L-BFGS-B	run 750/1000	MSE=4.907e-13	nit=15	nfev=78	
t=11.75s					
L-BFGS-B	run 751/1000	MSE=2.129e-12	nit=12	nfev=42	t=6.26s

L-BFGS-B	run 752/1000	MSE=9.187e-14	nit=10	nfev=48	t=7.21s
L-BFGS-B	run 753/1000	MSE=1.052e-13	nit=11	nfev=69	
t=10.33s					
L-BFGS-B	run 754/1000	MSE=5.302e-13	nit=10	nfev=39	t=5.88s
L-BFGS-B	run 755/1000	MSE=1.191e-13	nit=9	nfev=48	t=7.08s
L-BFGS-B	run 756/1000	MSE=3.897e-13	nit=12	nfev=60	t=8.92s
L-BFGS-B	run 757/1000	MSE=3.914e-13	nit=8	nfev=51	t=7.90s
L-BFGS-B	run 758/1000	MSE=3.913e-13	nit=15	nfev=54	t=7.85s
L-BFGS-B	run 759/1000	MSE=1.553e-12	nit=12	nfev=93	
t=14.18s					
L-BFGS-B	run 760/1000	MSE=4.610e-13	nit=8	nfev=42	t=6.29s
L-BFGS-B	run 761/1000	MSE=1.686e-13	nit=10	nfev=42	t=6.02s
L-BFGS-B	run 762/1000	MSE=5.059e-13	nit=10	nfev=42	t=6.21s
L-BFGS-B	run 763/1000	MSE=4.641e-13	nit=15	nfev=96	
t=14.51s					
L-BFGS-B	run 764/1000	MSE=1.005e-13	nit=9	nfev=45	t=6.95s
L-BFGS-B	run 765/1000	MSE=4.404e-13	nit=13	nfev=45	t=6.60s
L-BFGS-B	run 766/1000	MSE=1.122e-11	nit=13	nfev=45	t=6.68s
L-BFGS-B	run 767/1000	MSE=4.634e-13	nit=15	nfev=75	
t=11.48s					
L-BFGS-B	run 768/1000	MSE=1.221e-13	nit=14	nfev=51	t=7.60s
L-BFGS-B	run 769/1000	MSE=2.854e-15	nit=13	nfev=51	t=7.58s
L-BFGS-B	run 770/1000	MSE=3.183e-12	nit=15	nfev=54	t=8.18s
L-BFGS-B	run 771/1000	MSE=2.019e-11	nit=16	nfev=54	t=8.42s
L-BFGS-B	run 772/1000	MSE=9.740e-12	nit=9	nfev=63	t=9.28s
L-BFGS-B	run 773/1000	MSE=6.383e-13	nit=14	nfev=69	
t=10.18s					
L-BFGS-B	run 774/1000	MSE=4.603e-13	nit=8	nfev=51	t=7.65s
L-BFGS-B	run 775/1000	MSE=2.811e-13	nit=22	nfev=84	
t=12.69s					
L-BFGS-B	run 776/1000	MSE=1.472e-12	nit=12	nfev=72	
t=11.43s					
L-BFGS-B	run 777/1000	MSE=1.193e-13	nit=12	nfev=57	t=8.81s
L-BFGS-B	run 778/1000	MSE=7.049e-13	nit=15	nfev=51	t=7.52s
L-BFGS-B	run 779/1000	MSE=1.641e-13	nit=9	nfev=45	t=6.63s
L-BFGS-B	run 780/1000	MSE=4.717e-13	nit=13	nfev=51	t=7.65s
L-BFGS-B	run 781/1000	MSE=5.688e-13	nit=12	nfev=54	t=8.32s
L-BFGS-B	run 782/1000	MSE=4.754e-13	nit=8	nfev=51	t=7.50s
L-BFGS-B	run 783/1000	MSE=1.177e-13	nit=8	nfev=48	t=7.11s
L-BFGS-B	run 784/1000	MSE=6.084e-13	nit=15	nfev=72	
t=10.83s					
L-BFGS-B	run 785/1000	MSE=8.346e-13	nit=14	nfev=48	t=7.28s
L-BFGS-B	run 786/1000	MSE=4.807e-13	nit=14	nfev=48	t=7.08s
L-BFGS-B	run 787/1000	MSE=5.434e-13	nit=15	nfev=75	
t=11.13s					
L-BFGS-B	run 788/1000	MSE=6.197e-13	nit=12	nfev=42	t=6.49s
L-BFGS-B	run 789/1000	MSE=2.816e-12	nit=12	nfev=60	t=8.89s
L-BFGS-B	run 790/1000	MSE=5.083e-13	nit=12	nfev=54	t=8.12s

L-BFGS-B	run 791/1000	MSE=9.711e-14	nit=10	nfev=48	t=7.66s
L-BFGS-B	run 792/1000	MSE=1.837e-13	nit=9	nfev=48	t=7.29s
L-BFGS-B	run 793/1000	MSE=5.926e-13	nit=12	nfev=54	t=7.96s
L-BFGS-B	run 794/1000	MSE=1.014e-12	nit=16	nfev=57	t=8.37s
L-BFGS-B	run 795/1000	MSE=5.402e-12	nit=10	nfev=45	t=6.99s
L-BFGS-B	run 796/1000	MSE=2.070e-11	nit=12	nfev=42	t=6.42s
L-BFGS-B	run 797/1000	MSE=4.413e-13	nit=16	nfev=69	
t=10.13s					
L-BFGS-B	run 798/1000	MSE=6.876e-13	nit=11	nfev=54	t=8.09s
L-BFGS-B	run 799/1000	MSE=5.164e-13	nit=13	nfev=57	t=8.52s
L-BFGS-B	run 800/1000	MSE=7.970e-13	nit=13	nfev=45	t=6.71s
L-BFGS-B	run 801/1000	MSE=6.231e-12	nit=19	nfev=72	
t=10.61s					
L-BFGS-B	run 802/1000	MSE=8.299e-14	nit=11	nfev=39	t=6.07s
L-BFGS-B	run 803/1000	MSE=3.783e-13	nit=15	nfev=66	
t=10.00s					
L-BFGS-B	run 804/1000	MSE=7.773e-13	nit=12	nfev=45	t=6.65s
L-BFGS-B	run 805/1000	MSE=1.167e-12	nit=16	nfev=57	t=8.41s
L-BFGS-B	run 806/1000	MSE=9.987e-14	nit=16	nfev=63	t=9.65s
L-BFGS-B	run 807/1000	MSE=4.852e-14	nit=7	nfev=42	t=6.42s
L-BFGS-B	run 808/1000	MSE=8.122e-13	nit=15	nfev=69	
t=10.19s					
L-BFGS-B	run 809/1000	MSE=2.860e-13	nit=11	nfev=45	t=6.81s
L-BFGS-B	run 810/1000	MSE=4.578e-13	nit=13	nfev=51	t=7.67s
L-BFGS-B	run 811/1000	MSE=4.662e-13	nit=15	nfev=99	
t=14.97s					
L-BFGS-B	run 812/1000	MSE=3.208e-13	nit=8	nfev=45	t=6.90s
L-BFGS-B	run 813/1000	MSE=1.604e-13	nit=13	nfev=96	
t=15.00s					
L-BFGS-B	run 814/1000	MSE=1.203e-13	nit=11	nfev=39	t=5.59s
L-BFGS-B	run 815/1000	MSE=4.473e-13	nit=7	nfev=36	t=5.45s
L-BFGS-B	run 816/1000	MSE=2.716e-13	nit=21	nfev=90	
t=13.79s					
L-BFGS-B	run 817/1000	MSE=5.485e-13	nit=15	nfev=60	t=8.85s
L-BFGS-B	run 818/1000	MSE=3.339e-13	nit=9	nfev=63	t=9.38s
L-BFGS-B	run 819/1000	MSE=1.522e-12	nit=13	nfev=63	t=9.63s
L-BFGS-B	run 820/1000	MSE=4.652e-13	nit=15	nfev=69	
t=10.17s					
L-BFGS-B	run 821/1000	MSE=2.458e-13	nit=14	nfev=60	t=8.78s
L-BFGS-B	run 822/1000	MSE=5.672e-13	nit=14	nfev=60	t=9.41s
L-BFGS-B	run 823/1000	MSE=4.692e-13	nit=11	nfev=45	t=6.57s
L-BFGS-B	run 824/1000	MSE=1.161e-13	nit=13	nfev=72	
t=10.75s					
L-BFGS-B	run 825/1000	MSE=1.107e-12	nit=11	nfev=54	t=8.12s
L-BFGS-B	run 826/1000	MSE=4.483e-13	nit=15	nfev=66	
t=10.11s					
L-BFGS-B	run 827/1000	MSE=1.433e-12	nit=9	nfev=51	t=7.47s
L-BFGS-B	run 828/1000	MSE=2.070e-12	nit=14	nfev=75	

t=11.47s						
L-BFGS-B	run 829/1000	MSE=1.436e-12	nit=12	nfev=42	t=6.38s	
L-BFGS-B	run 830/1000	MSE=2.290e-11	nit=8	nfev=33	t=4.96s	
L-BFGS-B	run 831/1000	MSE=1.208e-12	nit=10	nfev=48	t=6.82s	
L-BFGS-B	run 832/1000	MSE=7.682e-12	nit=15	nfev=69		
t=13.28s						
L-BFGS-B	run 833/1000	MSE=5.591e-13	nit=14	nfev=66		
t=10.94s						
L-BFGS-B	run 834/1000	MSE=1.657e-12	nit=13	nfev=66	t=9.86s	
L-BFGS-B	run 835/1000	MSE=1.193e-13	nit=10	nfev=63	t=9.85s	
L-BFGS-B	run 836/1000	MSE=4.226e-13	nit=13	nfev=60		
t=11.06s						
L-BFGS-B	run 837/1000	MSE=1.374e-13	nit=10	nfev=45	t=6.51s	
L-BFGS-B	run 838/1000	MSE=5.171e-13	nit=13	nfev=57	t=8.69s	
L-BFGS-B	run 839/1000	MSE=3.683e-13	nit=7	nfev=48	t=7.53s	
L-BFGS-B	run 840/1000	MSE=1.544e-13	nit=10	nfev=48	t=7.17s	
L-BFGS-B	run 841/1000	MSE=5.413e-13	nit=13	nfev=45	t=6.66s	
L-BFGS-B	run 842/1000	MSE=2.113e-12	nit=16	nfev=75		
t=11.31s						
L-BFGS-B	run 843/1000	MSE=7.612e-13	nit=13	nfev=45	t=6.81s	
L-BFGS-B	run 844/1000	MSE=3.019e-13	nit=15	nfev=78		
t=11.64s						
L-BFGS-B	run 845/1000	MSE=1.389e-13	nit=11	nfev=48	t=7.20s	
L-BFGS-B	run 846/1000	MSE=3.153e-12	nit=11	nfev=42	t=6.22s	
L-BFGS-B	run 847/1000	MSE=2.809e-13	nit=10	nfev=66		
t=10.16s						
L-BFGS-B	run 848/1000	MSE=2.440e-13	nit=14	nfev=48	t=7.97s	
L-BFGS-B	run 849/1000	MSE=4.040e-13	nit=16	nfev=75		
t=11.50s						
L-BFGS-B	run 850/1000	MSE=4.498e-13	nit=15	nfev=57	t=8.54s	
L-BFGS-B	run 851/1000	MSE=5.504e-13	nit=12	nfev=51	t=7.68s	
L-BFGS-B	run 852/1000	MSE=2.195e-12	nit=10	nfev=51	t=7.88s	
L-BFGS-B	run 853/1000	MSE=1.371e-13	nit=10	nfev=51	t=7.87s	
L-BFGS-B	run 854/1000	MSE=1.739e-10	nit=13	nfev=48	t=7.14s	
L-BFGS-B	run 855/1000	MSE=1.463e-13	nit=15	nfev=54	t=7.97s	
L-BFGS-B	run 856/1000	MSE=4.531e-13	nit=17	nfev=57	t=8.45s	
L-BFGS-B	run 857/1000	MSE=5.227e-12	nit=13	nfev=45	t=6.98s	
L-BFGS-B	run 858/1000	MSE=1.948e-13	nit=16	nfev=57	t=8.41s	
L-BFGS-B	run 859/1000	MSE=2.876e-13	nit=13	nfev=54	t=7.99s	
L-BFGS-B	run 860/1000	MSE=6.607e-13	nit=9	nfev=48	t=7.15s	
L-BFGS-B	run 861/1000	MSE=1.683e-12	nit=7	nfev=51	t=7.80s	
L-BFGS-B	run 862/1000	MSE=2.949e-13	nit=11	nfev=42	t=6.27s	
L-BFGS-B	run 863/1000	MSE=7.825e-14	nit=9	nfev=42	t=6.06s	
L-BFGS-B	run 864/1000	MSE=1.133e-13	nit=9	nfev=84		
t=12.58s						
L-BFGS-B	run 865/1000	MSE=3.452e-13	nit=10	nfev=45	t=7.80s	
L-BFGS-B	run 866/1000	MSE=8.554e-13	nit=13	nfev=60	t=9.13s	
L-BFGS-B	run 867/1000	MSE=3.273e-13	nit=11	nfev=42	t=6.30s	

L-BFGS-B	run 868/1000	MSE=4.415e-13	nit=11	nfev=39	t=5.62s
L-BFGS-B	run 869/1000	MSE=4.465e-13	nit=10	nfev=69	
t=10.75s					
L-BFGS-B	run 870/1000	MSE=1.746e-12	nit=16	nfev=57	t=8.52s
L-BFGS-B	run 871/1000	MSE=4.906e-12	nit=14	nfev=48	t=6.85s
L-BFGS-B	run 872/1000	MSE=8.115e-12	nit=14	nfev=84	
t=12.80s					
L-BFGS-B	run 873/1000	MSE=6.980e-13	nit=10	nfev=45	t=6.66s
L-BFGS-B	run 874/1000	MSE=6.382e-15	nit=10	nfev=51	t=7.82s
L-BFGS-B	run 875/1000	MSE=1.139e-13	nit=10	nfev=51	t=7.75s
L-BFGS-B	run 876/1000	MSE=4.945e-13	nit=13	nfev=51	t=7.80s
L-BFGS-B	run 877/1000	MSE=5.620e-13	nit=12	nfev=45	t=6.48s
L-BFGS-B	run 878/1000	MSE=4.453e-13	nit=14	nfev=51	t=7.53s
L-BFGS-B	run 879/1000	MSE=6.240e-13	nit=10	nfev=66	
t=10.25s					
L-BFGS-B	run 880/1000	MSE=6.801e-13	nit=14	nfev=72	
t=10.48s					
L-BFGS-B	run 881/1000	MSE=4.649e-13	nit=18	nfev=75	
t=11.17s					
L-BFGS-B	run 882/1000	MSE=1.409e-13	nit=11	nfev=45	t=6.84s
L-BFGS-B	run 883/1000	MSE=7.170e-12	nit=11	nfev=57	t=8.52s
L-BFGS-B	run 884/1000	MSE=1.639e-13	nit=12	nfev=57	t=8.39s
L-BFGS-B	run 885/1000	MSE=1.283e-13	nit=10	nfev=54	t=8.18s
L-BFGS-B	run 886/1000	MSE=3.960e-11	nit=14	nfev=48	t=7.39s
L-BFGS-B	run 887/1000	MSE=1.495e-12	nit=13	nfev=45	t=7.08s
L-BFGS-B	run 888/1000	MSE=9.081e-14	nit=9	nfev=60	t=9.30s
L-BFGS-B	run 889/1000	MSE=9.856e-13	nit=7	nfev=51	t=7.67s
L-BFGS-B	run 890/1000	MSE=3.706e-13	nit=12	nfev=42	t=6.56s
L-BFGS-B	run 891/1000	MSE=4.801e-13	nit=11	nfev=42	t=6.27s
L-BFGS-B	run 892/1000	MSE=7.932e-13	nit=13	nfev=45	t=6.74s
L-BFGS-B	run 893/1000	MSE=4.668e-13	nit=16	nfev=81	
t=12.23s					
L-BFGS-B	run 894/1000	MSE=2.609e-13	nit=12	nfev=54	t=8.20s
L-BFGS-B	run 895/1000	MSE=3.938e-13	nit=11	nfev=51	t=7.63s
L-BFGS-B	run 896/1000	MSE=4.374e-13	nit=14	nfev=78	
t=11.59s					
L-BFGS-B	run 897/1000	MSE=1.699e-13	nit=10	nfev=51	t=7.87s
L-BFGS-B	run 898/1000	MSE=4.084e-13	nit=16	nfev=57	t=8.32s
L-BFGS-B	run 899/1000	MSE=5.028e-13	nit=13	nfev=51	t=7.77s
L-BFGS-B	run 900/1000	MSE=4.562e-13	nit=13	nfev=48	t=7.40s
L-BFGS-B	run 901/1000	MSE=1.124e-13	nit=7	nfev=39	t=6.12s
L-BFGS-B	run 902/1000	MSE=7.226e-13	nit=16	nfev=78	
t=11.57s					
L-BFGS-B	run 903/1000	MSE=4.279e-13	nit=10	nfev=42	t=6.26s
L-BFGS-B	run 904/1000	MSE=9.543e-13	nit=11	nfev=42	t=6.33s
L-BFGS-B	run 905/1000	MSE=6.456e-13	nit=15	nfev=72	
t=10.70s					
L-BFGS-B	run 906/1000	MSE=4.580e-13	nit=8	nfev=45	t=6.78s

L-BFGS-B	run 907/1000	MSE=1.192e-01	nit=12	nfev=75	
t=11.57s					
L-BFGS-B	run 908/1000	MSE=4.104e-13	nit=12	nfev=51	t=7.84s
L-BFGS-B	run 909/1000	MSE=2.586e-13	nit=19	nfev=66	t=9.73s
L-BFGS-B	run 910/1000	MSE=1.664e-12	nit=12	nfev=45	t=6.67s
L-BFGS-B	run 911/1000	MSE=1.335e-13	nit=10	nfev=39	t=5.71s
L-BFGS-B	run 912/1000	MSE=2.136e-12	nit=10	nfev=42	t=6.52s
L-BFGS-B	run 913/1000	MSE=5.154e-13	nit=11	nfev=75	
t=10.98s					
L-BFGS-B	run 914/1000	MSE=1.375e-13	nit=12	nfev=51	t=7.53s
L-BFGS-B	run 915/1000	MSE=4.501e-13	nit=12	nfev=51	t=7.66s
L-BFGS-B	run 916/1000	MSE=5.491e-13	nit=10	nfev=39	t=5.99s
L-BFGS-B	run 917/1000	MSE=8.237e-13	nit=12	nfev=54	t=8.41s
L-BFGS-B	run 918/1000	MSE=2.478e-13	nit=9	nfev=51	t=7.62s
L-BFGS-B	run 919/1000	MSE=4.647e-13	nit=14	nfev=66	
t=10.12s					
L-BFGS-B	run 920/1000	MSE=1.417e-13	nit=10	nfev=45	t=6.63s
L-BFGS-B	run 921/1000	MSE=1.129e-12	nit=14	nfev=57	t=8.39s
L-BFGS-B	run 922/1000	MSE=9.361e-14	nit=9	nfev=51	t=7.58s
L-BFGS-B	run 923/1000	MSE=2.672e-12	nit=9	nfev=48	t=7.54s
L-BFGS-B	run 924/1000	MSE=7.487e-14	nit=15	nfev=54	t=7.92s
L-BFGS-B	run 925/1000	MSE=1.209e-12	nit=13	nfev=69	
t=10.21s					
L-BFGS-B	run 926/1000	MSE=5.114e-13	nit=11	nfev=45	t=6.76s
L-BFGS-B	run 927/1000	MSE=1.243e-12	nit=15	nfev=51	t=7.62s
L-BFGS-B	run 928/1000	MSE=4.760e-13	nit=13	nfev=60	t=8.97s
L-BFGS-B	run 929/1000	MSE=8.321e-13	nit=7	nfev=66	t=9.65s
L-BFGS-B	run 930/1000	MSE=4.770e-13	nit=11	nfev=54	t=8.33s
L-BFGS-B	run 931/1000	MSE=4.994e-13	nit=12	nfev=42	t=6.19s
L-BFGS-B	run 932/1000	MSE=2.184e-13	nit=7	nfev=48	t=7.43s
L-BFGS-B	run 933/1000	MSE=9.862e-13	nit=14	nfev=57	t=8.47s
L-BFGS-B	run 934/1000	MSE=2.356e-13	nit=10	nfev=54	t=8.30s
L-BFGS-B	run 935/1000	MSE=4.464e-13	nit=13	nfev=51	t=7.51s
L-BFGS-B	run 936/1000	MSE=5.072e-13	nit=13	nfev=45	t=6.73s
L-BFGS-B	run 937/1000	MSE=6.238e-12	nit=8	nfev=39	t=5.60s
L-BFGS-B	run 938/1000	MSE=1.061e-12	nit=9	nfev=48	t=7.26s
L-BFGS-B	run 939/1000	MSE=6.989e-13	nit=12	nfev=48	t=7.05s
L-BFGS-B	run 940/1000	MSE=1.176e-13	nit=11	nfev=57	t=8.65s
L-BFGS-B	run 941/1000	MSE=1.620e-13	nit=11	nfev=63	t=9.44s
L-BFGS-B	run 942/1000	MSE=6.981e-13	nit=11	nfev=39	t=5.81s
L-BFGS-B	run 943/1000	MSE=1.185e-13	nit=11	nfev=48	t=7.11s
L-BFGS-B	run 944/1000	MSE=5.214e-13	nit=11	nfev=69	
t=10.24s					
L-BFGS-B	run 945/1000	MSE=3.569e-13	nit=14	nfev=51	t=7.63s
L-BFGS-B	run 946/1000	MSE=4.780e-13	nit=14	nfev=48	t=7.42s
L-BFGS-B	run 947/1000	MSE=8.464e-14	nit=9	nfev=54	t=8.05s
L-BFGS-B	run 948/1000	MSE=4.474e-13	nit=13	nfev=48	t=8.18s
L-BFGS-B	run 949/1000	MSE=1.124e-12	nit=14	nfev=51	t=7.91s

L-BFGS-B	run 950/1000	MSE=6.002e-13	nit=10	nfev=42	t=6.37s
L-BFGS-B	run 951/1000	MSE=4.212e-13	nit=13	nfev=48	t=7.11s
L-BFGS-B	run 952/1000	MSE=1.222e-12	nit=13	nfev=48	t=7.10s
L-BFGS-B	run 953/1000	MSE=3.836e-13	nit=10	nfev=39	t=5.92s
L-BFGS-B	run 954/1000	MSE=7.279e-14	nit=9	nfev=42	t=6.47s
L-BFGS-B	run 955/1000	MSE=3.257e-12	nit=13	nfev=45	t=6.65s
L-BFGS-B	run 956/1000	MSE=4.132e-13	nit=11	nfev=42	t=6.34s
L-BFGS-B	run 957/1000	MSE=1.429e-12	nit=11	nfev=54	t=8.00s
L-BFGS-B	run 958/1000	MSE=1.226e-12	nit=11	nfev=54	t=8.34s
L-BFGS-B	run 959/1000	MSE=8.574e-12	nit=12	nfev=51	t=7.61s
L-BFGS-B	run 960/1000	MSE=4.558e-13	nit=15	nfev=51	t=7.53s
L-BFGS-B	run 961/1000	MSE=1.742e-12	nit=10	nfev=39	t=6.03s
L-BFGS-B	run 962/1000	MSE=1.265e-13	nit=14	nfev=51	t=7.89s
L-BFGS-B	run 963/1000	MSE=2.104e-12	nit=12	nfev=45	t=6.45s
L-BFGS-B	run 964/1000	MSE=6.627e-14	nit=10	nfev=42	t=6.50s
L-BFGS-B	run 965/1000	MSE=4.556e-13	nit=14	nfev=48	t=7.09s
L-BFGS-B	run 966/1000	MSE=1.184e-13	nit=9	nfev=42	t=6.49s
L-BFGS-B	run 967/1000	MSE=4.726e-13	nit=12	nfev=45	t=6.81s
L-BFGS-B	run 968/1000	MSE=4.826e-13	nit=11	nfev=51	t=7.68s
L-BFGS-B	run 969/1000	MSE=1.322e-13	nit=8	nfev=48	t=6.93s
L-BFGS-B	run 970/1000	MSE=1.358e-13	nit=9	nfev=39	t=5.81s
L-BFGS-B	run 971/1000	MSE=9.778e-14	nit=16	nfev=54	t=8.32s
L-BFGS-B	run 972/1000	MSE=4.328e-15	nit=16	nfev=54	t=7.94s
L-BFGS-B	run 973/1000	MSE=4.544e-12	nit=18	nfev=84	
t=12.59s					
L-BFGS-B	run 974/1000	MSE=6.115e-13	nit=14	nfev=69	
t=10.56s					
L-BFGS-B	run 975/1000	MSE=7.396e-13	nit=13	nfev=45	t=6.83s
L-BFGS-B	run 976/1000	MSE=1.136e-13	nit=8	nfev=39	t=6.57s
L-BFGS-B	run 977/1000	MSE=1.426e-13	nit=12	nfev=51	t=7.74s
L-BFGS-B	run 978/1000	MSE=5.451e-13	nit=14	nfev=66	t=9.82s
L-BFGS-B	run 979/1000	MSE=4.674e-13	nit=17	nfev=57	t=8.61s
L-BFGS-B	run 980/1000	MSE=6.491e-13	nit=13	nfev=60	t=8.99s
L-BFGS-B	run 981/1000	MSE=5.835e-13	nit=16	nfev=54	t=8.27s
L-BFGS-B	run 982/1000	MSE=4.248e-13	nit=10	nfev=45	t=6.73s
L-BFGS-B	run 983/1000	MSE=2.782e-13	nit=9	nfev=51	t=7.64s
L-BFGS-B	run 984/1000	MSE=2.790e-12	nit=17	nfev=60	t=9.10s
L-BFGS-B	run 985/1000	MSE=2.381e-12	nit=9	nfev=63	t=9.41s
L-BFGS-B	run 986/1000	MSE=2.590e-13	nit=13	nfev=78	
t=11.68s					
L-BFGS-B	run 987/1000	MSE=4.486e-13	nit=12	nfev=51	t=7.39s
L-BFGS-B	run 988/1000	MSE=5.762e-13	nit=11	nfev=42	t=6.51s
L-BFGS-B	run 989/1000	MSE=9.030e-13	nit=14	nfev=54	t=8.05s
L-BFGS-B	run 990/1000	MSE=4.538e-13	nit=11	nfev=42	t=6.27s
L-BFGS-B	run 991/1000	MSE=2.346e-13	nit=10	nfev=57	t=8.46s
L-BFGS-B	run 992/1000	MSE=4.653e-13	nit=11	nfev=42	t=6.44s
L-BFGS-B	run 993/1000	MSE=6.169e-13	nit=15	nfev=51	t=7.56s
L-BFGS-B	run 994/1000	MSE=4.865e-13	nit=13	nfev=45	t=6.65s

L-BFGS-B	run 995/1000	MSE=4.093e-13	nit=15	nfev=54	t=7.97s
L-BFGS-B	run 996/1000	MSE=3.185e-13	nit=9	nfev=39	t=6.01s
L-BFGS-B	run 997/1000	MSE=1.172e-13	nit=9	nfev=48	t=7.23s
L-BFGS-B	run 998/1000	MSE=3.822e-13	nit=13	nfev=57	t=8.44s
L-BFGS-B	run 999/1000	MSE=8.445e-12	nit=12	nfev=57	t=8.24s
L-BFGS-B	run 1000/1000	MSE=5.124e-13	nit=12	nfev=54	

t=8.34s

Scenario A - SLSQP+jac (1000 runs)

SLSQP	run 1/1000	MSE=2.241e-07	nit=9	nfev=29	t=4.30s
SLSQP	run 2/1000	MSE=1.960e-07	nit=10	nfev=33	t=4.73s
SLSQP	run 3/1000	MSE=9.980e-08	nit=12	nfev=38	t=5.77s
SLSQP	run 4/1000	MSE=1.441e-08	nit=15	nfev=51	t=7.75s
SLSQP	run 5/1000	MSE=8.728e-08	nit=10	nfev=33	t=5.25s
SLSQP	run 6/1000	MSE=1.649e-08	nit=9	nfev=30	t=4.35s
SLSQP	run 7/1000	MSE=2.544e-09	nit=11	nfev=34	t=5.15s
SLSQP	run 8/1000	MSE=1.242e-07	nit=13	nfev=41	t=6.08s
SLSQP	run 9/1000	MSE=5.523e-07	nit=6	nfev=20	t=2.91s
SLSQP	run 10/1000	MSE=8.211e-08	nit=13	nfev=41	t=6.14s
SLSQP	run 11/1000	MSE=4.671e-08	nit=5	nfev=18	t=2.65s
SLSQP	run 12/1000	MSE=2.149e-07	nit=9	nfev=28	t=4.30s
SLSQP	run 13/1000	MSE=1.332e-09	nit=20	nfev=68	t=10.18s
SLSQP	run 14/1000	MSE=3.664e-08	nit=12	nfev=40	t=6.11s
SLSQP	run 15/1000	MSE=1.043e-07	nit=8	nfev=27	t=3.87s
SLSQP	run 16/1000	MSE=1.813e-08	nit=12	nfev=39	t=5.94s
SLSQP	run 17/1000	MSE=3.354e-08	nit=10	nfev=33	t=5.11s
SLSQP	run 18/1000	MSE=2.733e-07	nit=13	nfev=42	t=6.02s
SLSQP	run 19/1000	MSE=2.790e-09	nit=14	nfev=46	t=6.92s
SLSQP	run 20/1000	MSE=1.996e-07	nit=14	nfev=46	t=6.91s
SLSQP	run 21/1000	MSE=1.276e-08	nit=10	nfev=32	t=5.01s
SLSQP	run 22/1000	MSE=6.277e-08	nit=11	nfev=38	t=5.60s
SLSQP	run 23/1000	MSE=2.126e-09	nit=12	nfev=41	t=6.00s
SLSQP	run 24/1000	MSE=3.326e-07	nit=12	nfev=38	t=5.71s
SLSQP	run 25/1000	MSE=7.184e-08	nit=13	nfev=44	t=6.61s
SLSQP	run 26/1000	MSE=1.760e-07	nit=11	nfev=36	t=5.34s
SLSQP	run 27/1000	MSE=3.048e-07	nit=15	nfev=48	t=7.30s
SLSQP	run 28/1000	MSE=4.339e-09	nit=12	nfev=40	t=6.06s
SLSQP	run 29/1000	MSE=4.210e-10	nit=9	nfev=30	t=4.57s
SLSQP	run 30/1000	MSE=6.335e-09	nit=13	nfev=42	t=6.28s
SLSQP	run 31/1000	MSE=2.676e-07	nit=7	nfev=25	t=4.00s
SLSQP	run 32/1000	MSE=5.611e-07	nit=11	nfev=35	t=5.67s
SLSQP	run 33/1000	MSE=5.534e-08	nit=14	nfev=47	t=7.38s
SLSQP	run 34/1000	MSE=2.009e-07	nit=18	nfev=62	t=9.82s
SLSQP	run 35/1000	MSE=5.174e-08	nit=12	nfev=40	t=5.78s
SLSQP	run 36/1000	MSE=7.714e-08	nit=16	nfev=55	t=8.42s
SLSQP	run 37/1000	MSE=1.006e-07	nit=12	nfev=41	t=6.05s
SLSQP	run 38/1000	MSE=2.232e-08	nit=13	nfev=43	t=6.38s
SLSQP	run 39/1000	MSE=1.327e-09	nit=8	nfev=28	t=4.32s

SLSQP	run 40/1000	MSE=2.173e-07	nit=13	nfev=43	t=6.29s
SLSQP	run 41/1000	MSE=3.254e-07	nit=10	nfev=33	t=4.92s
SLSQP	run 42/1000	MSE=1.488e-08	nit=8	nfev=26	t=4.01s
SLSQP	run 43/1000	MSE=3.681e-08	nit=6	nfev=20	t=2.83s
SLSQP	run 44/1000	MSE=1.010e-07	nit=14	nfev=47	t=7.08s
SLSQP	run 45/1000	MSE=9.349e-08	nit=9	nfev=30	t=4.40s
SLSQP	run 46/1000	MSE=1.260e-07	nit=12	nfev=42	t=6.68s
SLSQP	run 47/1000	MSE=1.872e-08	nit=11	nfev=36	t=5.53s
SLSQP	run 48/1000	MSE=1.365e-08	nit=11	nfev=37	t=5.50s
SLSQP	run 49/1000	MSE=8.027e-08	nit=8	nfev=26	t=3.81s
SLSQP	run 50/1000	MSE=1.939e-07	nit=13	nfev=44	t=6.59s
SLSQP	run 51/1000	MSE=6.166e-08	nit=11	nfev=36	t=5.41s
SLSQP	run 52/1000	MSE=2.865e-09	nit=15	nfev=50	t=7.77s
SLSQP	run 53/1000	MSE=1.040e-07	nit=14	nfev=45	t=6.79s
SLSQP	run 54/1000	MSE=2.283e-07	nit=11	nfev=34	t=4.90s
SLSQP	run 55/1000	MSE=9.042e-10	nit=9	nfev=29	t=4.46s
SLSQP	run 56/1000	MSE=2.849e-09	nit=8	nfev=29	t=4.30s
SLSQP	run 57/1000	MSE=4.836e-07	nit=13	nfev=42	t=6.37s
SLSQP	run 58/1000	MSE=2.990e-07	nit=13	nfev=42	t=6.26s
SLSQP	run 59/1000	MSE=3.408e-07	nit=16	nfev=54	t=7.98s
SLSQP	run 60/1000	MSE=3.606e-07	nit=11	nfev=37	t=5.59s
SLSQP	run 61/1000	MSE=4.766e-09	nit=7	nfev=23	t=3.32s
SLSQP	run 62/1000	MSE=3.503e-07	nit=8	nfev=26	t=3.86s
SLSQP	run 63/1000	MSE=4.776e-07	nit=12	nfev=39	t=5.98s
SLSQP	run 64/1000	MSE=3.262e-08	nit=17	nfev=57	t=8.44s
SLSQP	run 65/1000	MSE=1.071e-07	nit=13	nfev=43	t=6.51s
SLSQP	run 66/1000	MSE=4.248e-07	nit=10	nfev=32	t=4.75s
SLSQP	run 67/1000	MSE=1.361e-11	nit=7	nfev=25	t=3.93s
SLSQP	run 68/1000	MSE=2.864e-07	nit=7	nfev=24	t=3.92s
SLSQP	run 69/1000	MSE=6.281e-09	nit=11	nfev=36	t=5.22s
SLSQP	run 70/1000	MSE=6.933e-08	nit=20	nfev=66	t=9.95s
SLSQP	run 71/1000	MSE=1.962e-07	nit=15	nfev=49	t=7.39s
SLSQP	run 72/1000	MSE=6.071e-08	nit=12	nfev=38	t=5.73s
SLSQP	run 73/1000	MSE=1.456e-08	nit=7	nfev=24	t=3.61s
SLSQP	run 74/1000	MSE=1.415e-08	nit=12	nfev=38	t=5.63s
SLSQP	run 75/1000	MSE=1.086e-08	nit=20	nfev=65	t=9.66s
SLSQP	run 76/1000	MSE=3.077e-09	nit=11	nfev=38	t=5.69s
SLSQP	run 77/1000	MSE=1.666e-07	nit=8	nfev=27	t=4.18s
SLSQP	run 78/1000	MSE=1.515e-07	nit=9	nfev=31	t=4.53s
SLSQP	run 79/1000	MSE=2.877e-09	nit=16	nfev=53	t=7.79s
SLSQP	run 80/1000	MSE=6.514e-08	nit=12	nfev=40	t=6.27s
SLSQP	run 81/1000	MSE=1.582e-08	nit=11	nfev=36	t=5.36s
SLSQP	run 82/1000	MSE=1.429e-09	nit=8	nfev=30	t=4.43s
SLSQP	run 83/1000	MSE=5.702e-09	nit=8	nfev=29	t=4.59s
SLSQP	run 84/1000	MSE=1.975e-07	nit=8	nfev=27	t=3.90s
SLSQP	run 85/1000	MSE=5.765e-08	nit=10	nfev=33	t=4.87s
SLSQP	run 86/1000	MSE=3.198e-07	nit=10	nfev=33	t=5.00s
SLSQP	run 87/1000	MSE=1.839e-08	nit=11	nfev=37	t=5.42s

SLSQP	run 88/1000	MSE=3.934e-07	nit=8	nfev=27	t=4.13s
SLSQP	run 89/1000	MSE=4.820e-09	nit=11	nfev=35	t=5.19s
SLSQP	run 90/1000	MSE=7.475e-09	nit=11	nfev=37	t=5.59s
SLSQP	run 91/1000	MSE=2.499e-08	nit=7	nfev=24	t=3.73s
SLSQP	run 92/1000	MSE=1.726e-08	nit=7	nfev=25	t=3.64s
SLSQP	run 93/1000	MSE=1.222e-08	nit=11	nfev=35	t=5.54s
SLSQP	run 94/1000	MSE=3.180e-08	nit=13	nfev=42	t=6.31s
SLSQP	run 95/1000	MSE=3.392e-08	nit=15	nfev=53	t=8.22s
SLSQP	run 96/1000	MSE=2.203e-09	nit=10	nfev=34	t=4.79s
SLSQP	run 97/1000	MSE=6.892e-09	nit=10	nfev=33	t=5.13s
SLSQP	run 98/1000	MSE=1.926e-07	nit=12	nfev=42	t=6.33s
SLSQP	run 99/1000	MSE=3.959e-07	nit=11	nfev=35	t=5.08s
SLSQP	run 100/1000	MSE=2.365e-08	nit=11	nfev=38	t=6.01s
SLSQP	run 101/1000	MSE=1.621e-10	nit=13	nfev=44	t=6.57s
SLSQP	run 102/1000	MSE=9.153e-08	nit=9	nfev=29	t=4.12s
SLSQP	run 103/1000	MSE=3.186e-10	nit=7	nfev=26	t=3.90s
SLSQP	run 104/1000	MSE=7.473e-08	nit=11	nfev=34	t=4.89s
SLSQP	run 105/1000	MSE=3.732e-07	nit=10	nfev=34	t=5.29s
SLSQP	run 106/1000	MSE=1.036e-06	nit=8	nfev=27	t=4.36s
SLSQP	run 107/1000	MSE=8.246e-09	nit=12	nfev=43	t=6.52s
SLSQP	run 108/1000	MSE=2.566e-08	nit=10	nfev=32	t=4.89s
SLSQP	run 109/1000	MSE=3.989e-07	nit=11	nfev=37	t=5.71s
SLSQP	run 110/1000	MSE=1.928e-08	nit=9	nfev=30	t=4.35s
SLSQP	run 111/1000	MSE=4.793e-07	nit=10	nfev=31	t=4.79s
SLSQP	run 112/1000	MSE=1.531e-07	nit=14	nfev=47	t=6.99s
SLSQP	run 113/1000	MSE=5.925e-08	nit=8	nfev=27	t=3.88s
SLSQP	run 114/1000	MSE=4.146e-07	nit=8	nfev=31	t=4.71s
SLSQP	run 115/1000	MSE=1.334e-08	nit=13	nfev=41	t=6.15s
SLSQP	run 116/1000	MSE=1.849e-08	nit=20	nfev=67	
t=10.25s					
SLSQP	run 117/1000	MSE=5.481e-08	nit=9	nfev=30	t=4.34s
SLSQP	run 118/1000	MSE=4.200e-07	nit=10	nfev=32	t=4.75s
SLSQP	run 119/1000	MSE=9.680e-08	nit=14	nfev=46	t=6.84s
SLSQP	run 120/1000	MSE=5.133e-07	nit=11	nfev=36	t=5.20s
SLSQP	run 121/1000	MSE=8.373e-09	nit=20	nfev=68	
t=10.35s					
SLSQP	run 122/1000	MSE=1.296e-07	nit=13	nfev=45	t=6.90s
SLSQP	run 123/1000	MSE=3.629e-07	nit=16	nfev=52	t=7.81s
SLSQP	run 124/1000	MSE=4.034e-08	nit=8	nfev=27	t=3.97s
SLSQP	run 125/1000	MSE=1.682e-08	nit=8	nfev=28	t=4.34s
SLSQP	run 126/1000	MSE=4.624e-07	nit=9	nfev=30	t=4.47s
SLSQP	run 127/1000	MSE=2.621e-08	nit=14	nfev=47	t=7.00s
SLSQP	run 128/1000	MSE=8.078e-10	nit=11	nfev=36	t=5.42s
SLSQP	run 129/1000	MSE=6.065e-10	nit=3	nfev=14	t=2.19s
SLSQP	run 130/1000	MSE=2.368e-09	nit=14	nfev=49	t=7.25s
SLSQP	run 131/1000	MSE=5.154e-08	nit=15	nfev=49	t=7.55s
SLSQP	run 132/1000	MSE=1.008e-07	nit=16	nfev=53	t=7.80s
SLSQP	run 133/1000	MSE=2.964e-07	nit=13	nfev=43	t=6.36s

SLSQP	run 134/1000	MSE=2.902e-07	nit=13	nfev=46	t=6.80s
SLSQP	run 135/1000	MSE=1.006e-09	nit=9	nfev=31	t=4.59s
SLSQP	run 136/1000	MSE=7.755e-08	nit=12	nfev=40	t=6.03s
SLSQP	run 137/1000	MSE=9.365e-08	nit=12	nfev=41	t=6.18s
SLSQP	run 138/1000	MSE=3.309e-08	nit=8	nfev=29	t=4.24s
SLSQP	run 139/1000	MSE=3.026e-07	nit=14	nfev=47	t=7.09s
SLSQP	run 140/1000	MSE=9.325e-09	nit=12	nfev=41	t=6.21s
SLSQP	run 141/1000	MSE=1.729e-11	nit=14	nfev=47	t=7.19s
SLSQP	run 142/1000	MSE=1.580e-08	nit=8	nfev=25	t=3.56s
SLSQP	run 143/1000	MSE=2.810e-09	nit=7	nfev=25	t=3.86s
SLSQP	run 144/1000	MSE=2.838e-08	nit=17	nfev=56	t=8.33s
SLSQP	run 145/1000	MSE=1.132e-07	nit=10	nfev=33	t=4.73s
SLSQP	run 146/1000	MSE=1.173e-07	nit=10	nfev=32	t=4.97s
SLSQP	run 147/1000	MSE=2.633e-07	nit=15	nfev=53	t=7.86s
SLSQP	run 148/1000	MSE=1.967e-07	nit=17	nfev=58	t=8.66s
SLSQP	run 149/1000	MSE=2.069e-07	nit=10	nfev=33	t=5.02s
SLSQP	run 150/1000	MSE=1.351e-07	nit=13	nfev=44	t=6.86s
SLSQP	run 151/1000	MSE=2.032e-07	nit=12	nfev=40	t=5.80s
SLSQP	run 152/1000	MSE=4.290e-08	nit=10	nfev=32	t=4.89s
SLSQP	run 153/1000	MSE=3.946e-10	nit=6	nfev=22	t=3.28s
SLSQP	run 154/1000	MSE=1.002e-08	nit=15	nfev=50	t=7.54s
SLSQP	run 155/1000	MSE=3.802e-09	nit=12	nfev=39	t=5.89s
SLSQP	run 156/1000	MSE=4.605e-09	nit=14	nfev=47	t=7.15s
SLSQP	run 157/1000	MSE=4.643e-07	nit=8	nfev=27	t=4.02s
SLSQP	run 158/1000	MSE=6.369e-10	nit=11	nfev=38	t=5.48s
SLSQP	run 159/1000	MSE=4.151e-07	nit=11	nfev=35	t=5.29s
SLSQP	run 160/1000	MSE=1.809e-07	nit=13	nfev=42	t=6.45s
SLSQP	run 161/1000	MSE=4.844e-08	nit=13	nfev=43	t=6.87s
SLSQP	run 162/1000	MSE=1.913e-07	nit=16	nfev=54	t=8.09s
SLSQP	run 163/1000	MSE=3.808e-07	nit=11	nfev=37	t=5.29s
SLSQP	run 164/1000	MSE=5.189e-08	nit=11	nfev=36	t=5.35s
SLSQP	run 165/1000	MSE=2.353e-08	nit=12	nfev=39	t=5.81s
SLSQP	run 166/1000	MSE=2.546e-07	nit=13	nfev=41	t=5.98s
SLSQP	run 167/1000	MSE=2.397e-09	nit=8	nfev=26	t=4.01s
SLSQP	run 168/1000	MSE=6.862e-08	nit=7	nfev=23	t=3.35s
SLSQP	run 169/1000	MSE=2.318e-09	nit=11	nfev=36	t=5.51s
SLSQP	run 170/1000	MSE=9.004e-08	nit=10	nfev=33	t=4.97s
SLSQP	run 171/1000	MSE=1.571e-08	nit=17	nfev=55	t=8.07s
SLSQP	run 172/1000	MSE=1.064e-08	nit=12	nfev=38	t=5.71s
SLSQP	run 173/1000	MSE=6.347e-09	nit=10	nfev=34	t=5.04s
SLSQP	run 174/1000	MSE=2.626e-08	nit=13	nfev=42	t=6.71s
SLSQP	run 175/1000	MSE=2.670e-08	nit=8	nfev=27	t=4.05s
SLSQP	run 176/1000	MSE=4.906e-09	nit=7	nfev=24	t=3.57s
SLSQP	run 177/1000	MSE=6.322e-10	nit=9	nfev=33	t=4.94s
SLSQP	run 178/1000	MSE=1.010e-07	nit=13	nfev=43	t=6.47s
SLSQP	run 179/1000	MSE=3.024e-07	nit=13	nfev=45	t=6.63s
SLSQP	run 180/1000	MSE=4.725e-07	nit=8	nfev=26	t=3.95s
SLSQP	run 181/1000	MSE=3.759e-08	nit=12	nfev=39	t=5.87s

SLSQP	run 182/1000	MSE=1.607e-08	nit=14	nfev=50	t=7.64s
SLSQP	run 183/1000	MSE=1.470e-07	nit=10	nfev=33	t=4.93s
SLSQP	run 184/1000	MSE=3.390e-07	nit=11	nfev=38	t=5.58s
SLSQP	run 185/1000	MSE=1.525e-11	nit=6	nfev=21	t=3.36s
SLSQP	run 186/1000	MSE=2.650e-07	nit=11	nfev=36	t=5.37s
SLSQP	run 187/1000	MSE=4.705e-07	nit=10	nfev=32	t=4.80s
SLSQP	run 188/1000	MSE=1.837e-07	nit=15	nfev=52	t=7.83s
SLSQP	run 189/1000	MSE=1.371e-07	nit=13	nfev=42	t=6.34s
SLSQP	run 190/1000	MSE=1.091e-07	nit=9	nfev=30	t=4.41s
SLSQP	run 191/1000	MSE=2.734e-08	nit=15	nfev=51	t=7.68s
SLSQP	run 192/1000	MSE=3.229e-10	nit=10	nfev=34	t=5.11s
SLSQP	run 193/1000	MSE=9.476e-09	nit=12	nfev=38	t=5.93s
SLSQP	run 194/1000	MSE=5.477e-08	nit=11	nfev=36	t=5.17s
SLSQP	run 195/1000	MSE=4.858e-08	nit=13	nfev=45	t=6.68s
SLSQP	run 196/1000	MSE=2.709e-08	nit=24	nfev=81	
t=12.18s					
SLSQP	run 197/1000	MSE=6.095e-09	nit=12	nfev=42	t=6.06s
SLSQP	run 198/1000	MSE=4.084e-07	nit=8	nfev=27	t=4.07s
SLSQP	run 199/1000	MSE=1.083e-07	nit=9	nfev=30	t=4.52s
SLSQP	run 200/1000	MSE=1.527e-07	nit=15	nfev=52	t=7.68s
SLSQP	run 201/1000	MSE=4.113e-09	nit=11	nfev=37	t=5.70s
SLSQP	run 202/1000	MSE=2.971e-09	nit=10	nfev=31	t=4.72s
SLSQP	run 203/1000	MSE=3.236e-08	nit=18	nfev=60	t=8.88s
SLSQP	run 204/1000	MSE=1.185e-08	nit=12	nfev=40	t=6.03s
SLSQP	run 205/1000	MSE=5.193e-08	nit=9	nfev=29	t=4.25s
SLSQP	run 206/1000	MSE=1.198e-08	nit=15	nfev=50	t=7.59s
SLSQP	run 207/1000	MSE=1.952e-07	nit=14	nfev=48	t=8.78s
SLSQP	run 208/1000	MSE=2.072e-08	nit=13	nfev=42	t=6.85s
SLSQP	run 209/1000	MSE=4.167e-07	nit=17	nfev=58	t=8.76s
SLSQP	run 210/1000	MSE=1.385e-07	nit=11	nfev=37	t=5.49s
SLSQP	run 211/1000	MSE=8.762e-08	nit=13	nfev=42	t=6.24s
SLSQP	run 212/1000	MSE=8.884e-10	nit=10	nfev=33	t=4.93s
SLSQP	run 213/1000	MSE=2.468e-09	nit=17	nfev=59	t=8.85s
SLSQP	run 214/1000	MSE=5.133e-10	nit=12	nfev=39	t=5.79s
SLSQP	run 215/1000	MSE=1.515e-08	nit=16	nfev=54	t=8.03s
SLSQP	run 216/1000	MSE=3.197e-08	nit=14	nfev=46	t=6.62s
SLSQP	run 217/1000	MSE=6.410e-08	nit=13	nfev=43	t=6.38s
SLSQP	run 218/1000	MSE=2.731e-08	nit=19	nfev=63	t=9.39s
SLSQP	run 219/1000	MSE=1.578e-06	nit=5	nfev=17	t=2.65s
SLSQP	run 220/1000	MSE=3.967e-07	nit=15	nfev=51	t=7.48s
SLSQP	run 221/1000	MSE=3.939e-08	nit=12	nfev=38	t=5.60s
SLSQP	run 222/1000	MSE=5.821e-09	nit=18	nfev=58	t=8.51s
SLSQP	run 223/1000	MSE=2.335e-07	nit=19	nfev=64	t=9.48s
SLSQP	run 224/1000	MSE=3.239e-09	nit=15	nfev=51	t=7.50s
SLSQP	run 225/1000	MSE=1.361e-08	nit=9	nfev=31	t=4.93s
SLSQP	run 226/1000	MSE=4.486e-07	nit=6	nfev=22	t=3.19s
SLSQP	run 227/1000	MSE=5.840e-08	nit=13	nfev=45	t=6.86s
SLSQP	run 228/1000	MSE=1.264e-07	nit=10	nfev=34	t=5.19s

SLSQP	run 229/1000	MSE=3.968e-08	nit=10	nfev=33	t=4.90s
SLSQP	run 230/1000	MSE=1.059e-07	nit=13	nfev=41	t=6.21s
SLSQP	run 231/1000	MSE=8.091e-09	nit=9	nfev=33	t=5.02s
SLSQP	run 232/1000	MSE=2.619e-09	nit=9	nfev=30	t=4.72s
SLSQP	run 233/1000	MSE=1.974e-08	nit=11	nfev=36	t=5.41s
SLSQP	run 234/1000	MSE=2.986e-10	nit=13	nfev=40	t=6.08s
SLSQP	run 235/1000	MSE=2.247e-08	nit=10	nfev=31	t=4.62s
SLSQP	run 236/1000	MSE=4.383e-08	nit=14	nfev=46	t=6.76s
SLSQP	run 237/1000	MSE=4.228e-08	nit=8	nfev=27	t=4.12s
SLSQP	run 238/1000	MSE=3.669e-09	nit=13	nfev=40	t=5.91s
SLSQP	run 239/1000	MSE=1.388e-08	nit=7	nfev=23	t=3.56s
SLSQP	run 240/1000	MSE=1.088e-07	nit=12	nfev=38	t=5.62s
SLSQP	run 241/1000	MSE=1.518e-07	nit=9	nfev=31	t=4.63s
SLSQP	run 242/1000	MSE=1.855e-09	nit=15	nfev=52	t=7.83s
SLSQP	run 243/1000	MSE=5.173e-07	nit=12	nfev=39	t=5.82s
SLSQP	run 244/1000	MSE=7.974e-10	nit=10	nfev=34	t=4.87s
SLSQP	run 245/1000	MSE=2.607e-07	nit=19	nfev=64	
t=10.09s					
SLSQP	run 246/1000	MSE=2.148e-09	nit=12	nfev=41	t=5.87s
SLSQP	run 247/1000	MSE=4.999e-07	nit=14	nfev=47	t=7.33s
SLSQP	run 248/1000	MSE=3.116e-07	nit=11	nfev=37	t=5.73s
SLSQP	run 249/1000	MSE=5.403e-07	nit=10	nfev=33	t=4.94s
SLSQP	run 250/1000	MSE=1.342e-08	nit=12	nfev=38	t=5.62s
SLSQP	run 251/1000	MSE=7.679e-09	nit=7	nfev=24	t=3.74s
SLSQP	run 252/1000	MSE=7.185e-08	nit=14	nfev=47	t=6.93s
SLSQP	run 253/1000	MSE=5.191e-09	nit=10	nfev=34	t=5.07s
SLSQP	run 254/1000	MSE=4.864e-07	nit=14	nfev=47	t=6.99s
SLSQP	run 255/1000	MSE=9.467e-10	nit=8	nfev=28	t=4.37s
SLSQP	run 256/1000	MSE=9.239e-08	nit=15	nfev=49	t=7.26s
SLSQP	run 257/1000	MSE=1.985e-06	nit=9	nfev=33	t=5.08s
SLSQP	run 258/1000	MSE=2.125e-08	nit=12	nfev=41	t=6.39s
SLSQP	run 259/1000	MSE=1.760e-07	nit=12	nfev=40	t=5.95s
SLSQP	run 260/1000	MSE=1.107e-09	nit=8	nfev=27	t=4.14s
SLSQP	run 261/1000	MSE=1.330e-07	nit=12	nfev=40	t=6.02s
SLSQP	run 262/1000	MSE=4.414e-08	nit=9	nfev=29	t=4.15s
SLSQP	run 263/1000	MSE=8.070e-08	nit=12	nfev=38	t=5.73s
SLSQP	run 264/1000	MSE=2.163e-09	nit=14	nfev=47	t=7.07s
SLSQP	run 265/1000	MSE=9.757e-09	nit=12	nfev=40	t=6.26s
SLSQP	run 266/1000	MSE=3.001e-08	nit=9	nfev=31	t=4.55s
SLSQP	run 267/1000	MSE=1.872e-07	nit=10	nfev=32	t=4.77s
SLSQP	run 268/1000	MSE=1.653e-09	nit=11	nfev=35	t=5.18s
SLSQP	run 269/1000	MSE=4.807e-08	nit=13	nfev=41	t=6.01s
SLSQP	run 270/1000	MSE=2.309e-08	nit=16	nfev=54	t=8.24s
SLSQP	run 271/1000	MSE=1.725e-07	nit=8	nfev=26	t=3.89s
SLSQP	run 272/1000	MSE=1.902e-07	nit=10	nfev=33	t=4.68s
SLSQP	run 273/1000	MSE=4.894e-07	nit=13	nfev=43	t=6.65s
SLSQP	run 274/1000	MSE=3.782e-08	nit=8	nfev=27	t=4.42s
SLSQP	run 275/1000	MSE=5.213e-07	nit=16	nfev=53	t=7.84s

SLSQP	run 276/1000	MSE=4.512e-07	nit=11	nfev=37	t=5.68s
SLSQP	run 277/1000	MSE=6.192e-08	nit=12	nfev=38	t=5.44s
SLSQP	run 278/1000	MSE=1.121e-08	nit=10	nfev=35	t=5.35s
SLSQP	run 279/1000	MSE=1.079e-07	nit=11	nfev=35	t=5.27s
SLSQP	run 280/1000	MSE=4.181e-08	nit=7	nfev=23	t=3.31s
SLSQP	run 281/1000	MSE=1.246e-08	nit=10	nfev=32	t=4.86s
SLSQP	run 282/1000	MSE=3.694e-07	nit=12	nfev=38	t=5.65s
SLSQP	run 283/1000	MSE=7.645e-09	nit=13	nfev=45	t=6.66s
SLSQP	run 284/1000	MSE=7.976e-08	nit=11	nfev=36	t=5.59s
SLSQP	run 285/1000	MSE=4.456e-09	nit=8	nfev=27	t=4.09s
SLSQP	run 286/1000	MSE=1.921e-08	nit=15	nfev=51	t=7.46s
SLSQP	run 287/1000	MSE=7.226e-08	nit=13	nfev=43	t=6.50s
SLSQP	run 288/1000	MSE=1.868e-07	nit=10	nfev=33	t=4.95s
SLSQP	run 289/1000	MSE=2.578e-07	nit=10	nfev=32	t=4.75s
SLSQP	run 290/1000	MSE=1.232e-09	nit=11	nfev=37	t=5.69s
SLSQP	run 291/1000	MSE=3.634e-09	nit=10	nfev=32	t=4.82s
SLSQP	run 292/1000	MSE=3.340e-07	nit=10	nfev=33	t=5.04s
SLSQP	run 293/1000	MSE=1.798e-08	nit=11	nfev=36	t=5.42s
SLSQP	run 294/1000	MSE=5.241e-09	nit=7	nfev=26	t=3.70s
SLSQP	run 295/1000	MSE=4.483e-08	nit=13	nfev=43	t=6.51s
SLSQP	run 296/1000	MSE=4.015e-09	nit=11	nfev=38	t=5.73s
SLSQP	run 297/1000	MSE=1.915e-08	nit=8	nfev=26	t=3.75s
SLSQP	run 298/1000	MSE=2.449e-08	nit=12	nfev=39	t=5.71s
SLSQP	run 299/1000	MSE=5.043e-07	nit=11	nfev=36	t=5.44s
SLSQP	run 300/1000	MSE=6.864e-07	nit=10	nfev=34	t=4.91s
SLSQP	run 301/1000	MSE=1.162e-08	nit=11	nfev=37	t=5.78s
SLSQP	run 302/1000	MSE=2.512e-07	nit=10	nfev=33	t=4.97s
SLSQP	run 303/1000	MSE=1.377e-07	nit=17	nfev=58	t=8.67s
SLSQP	run 304/1000	MSE=5.941e-08	nit=11	nfev=35	t=5.20s
SLSQP	run 305/1000	MSE=1.169e-08	nit=14	nfev=45	t=6.70s
SLSQP	run 306/1000	MSE=8.999e-08	nit=10	nfev=33	t=5.01s
SLSQP	run 307/1000	MSE=3.347e-07	nit=12	nfev=40	t=5.91s
SLSQP	run 308/1000	MSE=4.320e-08	nit=6	nfev=23	t=3.38s
SLSQP	run 309/1000	MSE=1.385e-07	nit=10	nfev=33	t=5.09s
SLSQP	run 310/1000	MSE=5.722e-07	nit=11	nfev=35	t=5.18s
SLSQP	run 311/1000	MSE=1.650e-08	nit=12	nfev=41	t=5.93s
SLSQP	run 312/1000	MSE=5.288e-09	nit=8	nfev=27	t=4.16s
SLSQP	run 313/1000	MSE=3.076e-08	nit=11	nfev=37	t=5.58s
SLSQP	run 314/1000	MSE=6.571e-08	nit=15	nfev=51	t=7.73s
SLSQP	run 315/1000	MSE=1.048e-07	nit=6	nfev=22	t=3.25s
SLSQP	run 316/1000	MSE=4.372e-09	nit=16	nfev=56	t=8.22s
SLSQP	run 317/1000	MSE=1.169e-08	nit=16	nfev=53	t=7.85s
SLSQP	run 318/1000	MSE=1.427e-07	nit=9	nfev=29	t=4.43s
SLSQP	run 319/1000	MSE=6.620e-08	nit=9	nfev=31	t=4.62s
SLSQP	run 320/1000	MSE=1.715e-07	nit=12	nfev=39	t=5.74s
SLSQP	run 321/1000	MSE=2.634e-07	nit=10	nfev=36	t=5.57s
SLSQP	run 322/1000	MSE=1.859e-07	nit=11	nfev=38	t=5.52s
SLSQP	run 323/1000	MSE=1.003e-07	nit=10	nfev=33	t=4.95s

SLSQP	run 324/1000	MSE=3.492e-08	nit=14	nfev=45	t=7.05s
SLSQP	run 325/1000	MSE=1.836e-09	nit=5	nfev=20	t=2.93s
SLSQP	run 326/1000	MSE=1.083e-07	nit=7	nfev=23	t=3.50s
SLSQP	run 327/1000	MSE=3.012e-07	nit=8	nfev=26	t=3.81s
SLSQP	run 328/1000	MSE=5.311e-07	nit=10	nfev=33	t=5.03s
SLSQP	run 329/1000	MSE=4.434e-08	nit=16	nfev=53	t=7.87s
SLSQP	run 330/1000	MSE=4.971e-07	nit=17	nfev=58	t=8.85s
SLSQP	run 331/1000	MSE=4.905e-09	nit=8	nfev=27	t=3.99s
SLSQP	run 332/1000	MSE=4.538e-07	nit=16	nfev=55	t=8.09s
SLSQP	run 333/1000	MSE=6.044e-09	nit=9	nfev=31	t=4.84s
SLSQP	run 334/1000	MSE=3.684e-08	nit=11	nfev=37	t=5.54s
SLSQP	run 335/1000	MSE=8.099e-08	nit=15	nfev=50	t=7.47s
SLSQP	run 336/1000	MSE=3.054e-08	nit=14	nfev=48	t=7.04s
SLSQP	run 337/1000	MSE=6.049e-08	nit=7	nfev=24	t=3.65s
SLSQP	run 338/1000	MSE=2.458e-07	nit=9	nfev=30	t=4.49s
SLSQP	run 339/1000	MSE=1.490e-08	nit=14	nfev=47	t=7.14s
SLSQP	run 340/1000	MSE=3.763e-09	nit=15	nfev=52	t=8.00s
SLSQP	run 341/1000	MSE=9.395e-08	nit=14	nfev=47	t=6.95s
SLSQP	run 342/1000	MSE=4.361e-07	nit=12	nfev=41	t=6.02s
SLSQP	run 343/1000	MSE=3.544e-07	nit=11	nfev=36	t=5.45s
SLSQP	run 344/1000	MSE=2.578e-08	nit=9	nfev=30	t=4.55s
SLSQP	run 345/1000	MSE=3.715e-09	nit=13	nfev=45	t=6.79s
SLSQP	run 346/1000	MSE=2.837e-07	nit=10	nfev=34	t=5.27s
SLSQP	run 347/1000	MSE=4.519e-07	nit=9	nfev=33	t=4.91s
SLSQP	run 348/1000	MSE=4.401e-08	nit=7	nfev=26	t=3.77s
SLSQP	run 349/1000	MSE=4.356e-09	nit=12	nfev=40	t=5.95s
SLSQP	run 350/1000	MSE=1.530e-07	nit=13	nfev=44	t=6.65s
SLSQP	run 351/1000	MSE=6.174e-09	nit=11	nfev=34	t=4.82s
SLSQP	run 352/1000	MSE=1.498e-05	nit=8	nfev=29	t=4.30s
SLSQP	run 353/1000	MSE=1.302e-07	nit=16	nfev=56	t=8.45s
SLSQP	run 354/1000	MSE=3.293e-08	nit=18	nfev=63	t=9.47s
SLSQP	run 355/1000	MSE=1.629e-09	nit=8	nfev=28	t=4.44s
SLSQP	run 356/1000	MSE=1.825e-07	nit=8	nfev=27	t=3.82s
SLSQP	run 357/1000	MSE=8.705e-12	nit=8	nfev=27	t=4.09s
SLSQP	run 358/1000	MSE=2.900e-08	nit=8	nfev=26	t=3.78s
SLSQP	run 359/1000	MSE=5.602e-09	nit=12	nfev=38	t=5.72s
SLSQP	run 360/1000	MSE=2.752e-08	nit=7	nfev=27	t=3.90s
SLSQP	run 361/1000	MSE=2.643e-08	nit=8	nfev=28	t=4.39s
SLSQP	run 362/1000	MSE=3.948e-07	nit=11	nfev=37	t=5.79s
SLSQP	run 363/1000	MSE=6.483e-09	nit=18	nfev=60	t=8.78s
SLSQP	run 364/1000	MSE=3.864e-09	nit=7	nfev=24	t=3.45s
SLSQP	run 365/1000	MSE=3.582e-07	nit=13	nfev=42	t=6.43s
SLSQP	run 366/1000	MSE=1.427e-10	nit=11	nfev=38	t=5.93s
SLSQP	run 367/1000	MSE=1.424e-08	nit=11	nfev=36	t=5.27s
SLSQP	run 368/1000	MSE=7.524e-10	nit=10	nfev=32	t=4.76s
SLSQP	run 369/1000	MSE=1.405e-07	nit=12	nfev=40	t=6.18s
SLSQP	run 370/1000	MSE=7.410e-08	nit=11	nfev=36	t=5.21s
SLSQP	run 371/1000	MSE=1.596e-07	nit=12	nfev=38	t=5.97s

SLSQP	run 372/1000	MSE=2.020e-05	nit=11	nfev=40	t=6.57s
SLSQP	run 373/1000	MSE=5.964e-09	nit=15	nfev=50	t=7.38s
SLSQP	run 374/1000	MSE=8.889e-09	nit=6	nfev=21	t=3.11s
SLSQP	run 375/1000	MSE=3.516e-07	nit=11	nfev=36	t=5.54s
SLSQP	run 376/1000	MSE=5.595e-09	nit=10	nfev=35	t=5.26s
SLSQP	run 377/1000	MSE=4.721e-10	nit=10	nfev=36	t=5.47s
SLSQP	run 378/1000	MSE=1.789e-09	nit=10	nfev=34	t=5.32s
SLSQP	run 379/1000	MSE=1.188e-08	nit=12	nfev=38	t=5.66s
SLSQP	run 380/1000	MSE=1.234e-07	nit=10	nfev=33	t=5.02s
SLSQP	run 381/1000	MSE=6.547e-09	nit=10	nfev=32	t=5.07s
SLSQP	run 382/1000	MSE=1.842e-08	nit=12	nfev=38	t=5.70s
SLSQP	run 383/1000	MSE=7.372e-09	nit=11	nfev=36	t=5.39s
SLSQP	run 384/1000	MSE=5.138e-07	nit=12	nfev=40	t=5.93s
SLSQP	run 385/1000	MSE=2.983e-09	nit=13	nfev=43	t=6.62s
SLSQP	run 386/1000	MSE=3.340e-09	nit=7	nfev=24	t=3.45s
SLSQP	run 387/1000	MSE=4.771e-10	nit=9	nfev=31	t=4.74s
SLSQP	run 388/1000	MSE=1.990e-09	nit=8	nfev=26	t=3.89s
SLSQP	run 389/1000	MSE=8.211e-06	nit=5	nfev=18	t=2.64s
SLSQP	run 390/1000	MSE=7.691e-09	nit=17	nfev=57	t=8.60s
SLSQP	run 391/1000	MSE=1.061e-07	nit=10	nfev=33	t=4.78s
SLSQP	run 392/1000	MSE=4.467e-08	nit=11	nfev=35	t=5.51s
SLSQP	run 393/1000	MSE=5.505e-08	nit=11	nfev=35	t=5.05s
SLSQP	run 394/1000	MSE=1.767e-08	nit=14	nfev=48	t=7.33s
SLSQP	run 395/1000	MSE=3.841e-07	nit=12	nfev=39	t=5.96s
SLSQP	run 396/1000	MSE=1.324e-07	nit=10	nfev=35	t=5.17s
SLSQP	run 397/1000	MSE=6.906e-08	nit=11	nfev=36	t=5.33s
SLSQP	run 398/1000	MSE=4.542e-07	nit=13	nfev=44	t=6.54s
SLSQP	run 399/1000	MSE=1.230e-07	nit=10	nfev=32	t=4.93s
SLSQP	run 400/1000	MSE=3.065e-08	nit=11	nfev=36	t=5.14s
SLSQP	run 401/1000	MSE=3.620e-09	nit=15	nfev=52	t=7.84s
SLSQP	run 402/1000	MSE=2.559e-07	nit=9	nfev=31	t=4.69s
SLSQP	run 403/1000	MSE=6.837e-09	nit=15	nfev=51	t=7.62s
SLSQP	run 404/1000	MSE=8.485e-09	nit=12	nfev=40	t=5.96s
SLSQP	run 405/1000	MSE=6.222e-09	nit=12	nfev=38	t=5.64s
SLSQP	run 406/1000	MSE=8.186e-09	nit=9	nfev=29	t=4.42s
SLSQP	run 407/1000	MSE=1.204e-07	nit=17	nfev=60	t=8.84s
SLSQP	run 408/1000	MSE=3.548e-07	nit=13	nfev=44	t=6.70s
SLSQP	run 409/1000	MSE=1.713e-07	nit=13	nfev=43	t=6.61s
SLSQP	run 410/1000	MSE=7.086e-08	nit=14	nfev=45	t=6.59s
SLSQP	run 411/1000	MSE=2.466e-08	nit=9	nfev=32	t=4.62s
SLSQP	run 412/1000	MSE=5.416e-08	nit=11	nfev=35	t=5.28s
SLSQP	run 413/1000	MSE=2.626e-07	nit=13	nfev=43	t=6.55s
SLSQP	run 414/1000	MSE=6.537e-08	nit=12	nfev=38	t=5.46s
SLSQP	run 415/1000	MSE=7.704e-08	nit=13	nfev=41	t=6.25s
SLSQP	run 416/1000	MSE=1.665e-07	nit=10	nfev=33	t=5.32s
SLSQP	run 417/1000	MSE=1.760e-08	nit=7	nfev=23	t=3.37s
SLSQP	run 418/1000	MSE=2.915e-07	nit=15	nfev=50	t=7.50s
SLSQP	run 419/1000	MSE=3.106e-08	nit=12	nfev=40	t=6.08s

SLSQP	run 420/1000	MSE=2.524e-08	nit=13	nfev=43	t=6.40s
SLSQP	run 421/1000	MSE=2.430e-08	nit=14	nfev=49	t=7.26s
SLSQP	run 422/1000	MSE=7.341e-11	nit=11	nfev=35	t=5.04s
SLSQP	run 423/1000	MSE=3.109e-07	nit=18	nfev=60	t=9.02s
SLSQP	run 424/1000	MSE=1.906e-07	nit=14	nfev=47	t=7.04s
SLSQP	run 425/1000	MSE=1.394e-08	nit=7	nfev=25	t=3.78s
SLSQP	run 426/1000	MSE=1.492e-08	nit=16	nfev=54	t=8.20s
SLSQP	run 427/1000	MSE=1.053e-07	nit=11	nfev=34	t=4.87s
SLSQP	run 428/1000	MSE=1.385e-07	nit=12	nfev=39	t=5.98s
SLSQP	run 429/1000	MSE=7.233e-09	nit=15	nfev=50	t=7.38s
SLSQP	run 430/1000	MSE=1.133e-07	nit=17	nfev=58	t=8.70s
SLSQP	run 431/1000	MSE=3.230e-08	nit=10	nfev=33	t=4.94s
SLSQP	run 432/1000	MSE=9.070e-07	nit=11	nfev=37	t=5.51s
SLSQP	run 433/1000	MSE=1.285e-07	nit=9	nfev=30	t=4.65s
SLSQP	run 434/1000	MSE=1.678e-08	nit=9	nfev=30	t=4.30s
SLSQP	run 435/1000	MSE=5.426e-08	nit=10	nfev=34	t=5.09s
SLSQP	run 436/1000	MSE=1.034e-07	nit=10	nfev=33	t=5.00s
SLSQP	run 437/1000	MSE=9.606e-09	nit=7	nfev=26	t=6.13s
SLSQP	run 438/1000	MSE=4.619e-07	nit=12	nfev=40	t=8.33s
SLSQP	run 439/1000	MSE=5.474e-07	nit=9	nfev=29	t=4.49s
SLSQP	run 440/1000	MSE=2.180e-08	nit=10	nfev=33	t=4.92s
SLSQP	run 441/1000	MSE=4.067e-07	nit=8	nfev=27	t=4.16s
SLSQP	run 442/1000	MSE=3.153e-08	nit=15	nfev=49	t=7.57s
SLSQP	run 443/1000	MSE=3.501e-08	nit=16	nfev=53	t=7.99s
SLSQP	run 444/1000	MSE=6.610e-08	nit=6	nfev=20	t=2.81s
SLSQP	run 445/1000	MSE=3.601e-07	nit=9	nfev=29	t=4.41s
SLSQP	run 446/1000	MSE=5.499e-09	nit=7	nfev=26	t=3.83s
SLSQP	run 447/1000	MSE=6.395e-08	nit=12	nfev=39	t=5.96s
SLSQP	run 448/1000	MSE=2.691e-07	nit=9	nfev=31	t=4.55s
SLSQP	run 449/1000	MSE=3.527e-07	nit=12	nfev=42	t=6.39s
SLSQP	run 450/1000	MSE=2.612e-10	nit=18	nfev=62	t=9.13s
SLSQP	run 451/1000	MSE=1.236e-07	nit=14	nfev=46	t=6.84s
SLSQP	run 452/1000	MSE=1.539e-08	nit=11	nfev=38	t=5.66s
SLSQP	run 453/1000	MSE=1.956e-09	nit=10	nfev=32	t=4.83s
SLSQP	run 454/1000	MSE=5.883e-09	nit=12	nfev=39	t=5.89s
SLSQP	run 455/1000	MSE=1.341e-07	nit=19	nfev=65	t=9.55s
SLSQP	run 456/1000	MSE=3.057e-07	nit=7	nfev=24	t=3.67s
SLSQP	run 457/1000	MSE=1.419e-07	nit=11	nfev=37	t=5.38s
SLSQP	run 458/1000	MSE=4.892e-09	nit=15	nfev=51	t=7.73s
SLSQP	run 459/1000	MSE=4.926e-08	nit=9	nfev=30	t=4.54s
SLSQP	run 460/1000	MSE=5.039e-07	nit=14	nfev=46	t=6.89s
SLSQP	run 461/1000	MSE=3.101e-08	nit=13	nfev=41	t=6.23s
SLSQP	run 462/1000	MSE=2.942e-10	nit=6	nfev=23	t=3.37s
SLSQP	run 463/1000	MSE=4.772e-07	nit=14	nfev=48	t=7.09s
SLSQP	run 464/1000	MSE=6.600e-08	nit=9	nfev=29	t=4.27s
SLSQP	run 465/1000	MSE=1.329e-08	nit=8	nfev=28	t=4.04s
SLSQP	run 466/1000	MSE=4.559e-07	nit=14	nfev=48	t=7.07s
SLSQP	run 467/1000	MSE=2.648e-08	nit=10	nfev=32	t=4.75s

SLSQP	run 468/1000	MSE=1.622e-08	nit=9	nfev=31	t=4.67s
SLSQP	run 469/1000	MSE=2.621e-08	nit=6	nfev=23	t=3.45s
SLSQP	run 470/1000	MSE=2.465e-08	nit=14	nfev=48	t=7.54s
SLSQP	run 471/1000	MSE=2.331e-07	nit=14	nfev=44	t=6.47s
SLSQP	run 472/1000	MSE=4.990e-08	nit=12	nfev=39	t=5.88s
SLSQP	run 473/1000	MSE=1.505e-07	nit=8	nfev=26	t=3.75s
SLSQP	run 474/1000	MSE=3.136e-10	nit=7	nfev=26	t=4.00s
SLSQP	run 475/1000	MSE=1.905e-07	nit=11	nfev=37	t=5.47s
SLSQP	run 476/1000	MSE=1.273e-07	nit=8	nfev=28	t=4.20s
SLSQP	run 477/1000	MSE=1.260e-07	nit=9	nfev=29	t=4.34s
SLSQP	run 478/1000	MSE=6.430e-09	nit=17	nfev=58	t=8.73s
SLSQP	run 479/1000	MSE=1.623e-07	nit=10	nfev=33	t=4.80s
SLSQP	run 480/1000	MSE=1.517e-08	nit=10	nfev=35	t=5.40s
SLSQP	run 481/1000	MSE=2.955e-09	nit=14	nfev=47	t=7.04s
SLSQP	run 482/1000	MSE=9.706e-09	nit=13	nfev=43	t=6.33s
SLSQP	run 483/1000	MSE=1.229e-09	nit=13	nfev=43	t=6.87s
SLSQP	run 484/1000	MSE=1.526e-08	nit=12	nfev=39	t=5.63s
SLSQP	run 485/1000	MSE=2.787e-09	nit=12	nfev=39	t=5.86s
SLSQP	run 486/1000	MSE=2.001e-08	nit=9	nfev=29	t=4.73s
SLSQP	run 487/1000	MSE=5.177e-07	nit=12	nfev=40	t=5.77s
SLSQP	run 488/1000	MSE=1.349e-08	nit=8	nfev=27	t=4.09s
SLSQP	run 489/1000	MSE=1.884e-07	nit=8	nfev=28	t=4.02s
SLSQP	run 490/1000	MSE=5.043e-09	nit=10	nfev=34	t=5.16s
SLSQP	run 491/1000	MSE=9.325e-10	nit=8	nfev=29	t=4.28s
SLSQP	run 492/1000	MSE=1.625e-09	nit=6	nfev=21	t=3.28s
SLSQP	run 493/1000	MSE=1.892e-07	nit=9	nfev=30	t=4.36s
SLSQP	run 494/1000	MSE=4.693e-09	nit=14	nfev=45	t=6.86s
SLSQP	run 495/1000	MSE=8.913e-08	nit=11	nfev=36	t=5.79s
SLSQP	run 496/1000	MSE=2.849e-09	nit=7	nfev=24	t=3.46s
SLSQP	run 497/1000	MSE=2.152e-07	nit=9	nfev=29	t=4.44s
SLSQP	run 498/1000	MSE=2.526e-07	nit=13	nfev=42	t=6.33s
SLSQP	run 499/1000	MSE=4.151e-07	nit=15	nfev=50	t=7.37s
SLSQP	run 500/1000	MSE=4.247e-07	nit=13	nfev=42	t=6.32s
SLSQP	run 501/1000	MSE=5.808e-08	nit=8	nfev=26	t=3.87s
SLSQP	run 502/1000	MSE=1.941e-08	nit=13	nfev=44	t=6.52s
SLSQP	run 503/1000	MSE=3.443e-08	nit=11	nfev=35	t=5.21s
SLSQP	run 504/1000	MSE=1.978e-07	nit=13	nfev=43	t=6.46s
SLSQP	run 505/1000	MSE=1.534e-07	nit=6	nfev=20	t=3.06s
SLSQP	run 506/1000	MSE=5.015e-08	nit=15	nfev=52	t=7.79s
SLSQP	run 507/1000	MSE=4.106e-07	nit=12	nfev=40	t=5.85s
SLSQP	run 508/1000	MSE=5.139e-08	nit=9	nfev=31	t=4.68s
SLSQP	run 509/1000	MSE=1.009e-06	nit=8	nfev=26	t=3.71s
SLSQP	run 510/1000	MSE=1.217e-07	nit=10	nfev=32	t=4.77s
SLSQP	run 511/1000	MSE=4.631e-09	nit=5	nfev=18	t=2.68s
SLSQP	run 512/1000	MSE=7.489e-09	nit=15	nfev=52	t=7.63s
SLSQP	run 513/1000	MSE=2.145e-08	nit=14	nfev=45	t=6.90s
SLSQP	run 514/1000	MSE=9.098e-08	nit=8	nfev=27	t=4.32s
SLSQP	run 515/1000	MSE=8.419e-08	nit=15	nfev=49	t=7.16s

SLSQP	run 516/1000	MSE=3.020e-07	nit=11	nfev=36	t=5.19s
SLSQP	run 517/1000	MSE=2.413e-09	nit=14	nfev=48	t=7.15s
SLSQP	run 518/1000	MSE=3.320e-07	nit=10	nfev=32	t=4.84s
SLSQP	run 519/1000	MSE=8.115e-08	nit=13	nfev=44	t=6.59s
SLSQP	run 520/1000	MSE=1.570e-08	nit=15	nfev=50	t=7.38s
SLSQP	run 521/1000	MSE=1.541e-07	nit=7	nfev=24	t=3.45s
SLSQP	run 522/1000	MSE=3.665e-08	nit=15	nfev=49	t=7.34s
SLSQP	run 523/1000	MSE=1.001e-07	nit=16	nfev=55	t=8.13s
SLSQP	run 524/1000	MSE=1.910e-10	nit=7	nfev=26	t=3.73s
SLSQP	run 525/1000	MSE=2.819e-11	nit=15	nfev=49	t=7.21s
SLSQP	run 526/1000	MSE=1.467e-08	nit=13	nfev=41	t=6.16s
SLSQP	run 527/1000	MSE=9.944e-11	nit=6	nfev=22	t=3.15s
SLSQP	run 528/1000	MSE=8.762e-09	nit=8	nfev=27	t=3.97s
SLSQP	run 529/1000	MSE=5.724e-07	nit=10	nfev=32	t=4.88s
SLSQP	run 530/1000	MSE=6.077e-08	nit=17	nfev=59	t=8.68s
SLSQP	run 531/1000	MSE=3.594e-07	nit=16	nfev=53	t=7.88s
SLSQP	run 532/1000	MSE=1.522e-09	nit=8	nfev=25	t=3.55s
SLSQP	run 533/1000	MSE=2.172e-07	nit=12	nfev=39	t=5.87s
SLSQP	run 534/1000	MSE=9.337e-11	nit=17	nfev=57	t=8.67s
SLSQP	run 535/1000	MSE=9.329e-08	nit=9	nfev=30	t=4.57s
SLSQP	run 536/1000	MSE=1.378e-07	nit=10	nfev=33	t=4.73s
SLSQP	run 537/1000	MSE=1.808e-08	nit=8	nfev=27	t=4.09s
SLSQP	run 538/1000	MSE=1.329e-07	nit=9	nfev=29	t=4.18s
SLSQP	run 539/1000	MSE=1.073e-07	nit=9	nfev=31	t=4.75s
SLSQP	run 540/1000	MSE=6.428e-08	nit=10	nfev=32	t=4.60s
SLSQP	run 541/1000	MSE=1.687e-08	nit=7	nfev=25	t=3.54s
SLSQP	run 542/1000	MSE=1.571e-08	nit=8	nfev=26	t=3.79s
SLSQP	run 543/1000	MSE=1.648e-08	nit=12	nfev=41	t=6.11s
SLSQP	run 544/1000	MSE=2.622e-08	nit=8	nfev=27	t=4.23s
SLSQP	run 545/1000	MSE=4.203e-08	nit=12	nfev=42	t=6.09s
SLSQP	run 546/1000	MSE=2.107e-08	nit=12	nfev=40	t=6.05s
SLSQP	run 547/1000	MSE=8.104e-10	nit=7	nfev=25	t=3.78s
SLSQP	run 548/1000	MSE=3.431e-07	nit=16	nfev=54	t=7.93s
SLSQP	run 549/1000	MSE=4.038e-09	nit=15	nfev=52	t=7.86s
SLSQP	run 550/1000	MSE=4.041e-09	nit=11	nfev=38	t=5.71s
SLSQP	run 551/1000	MSE=1.646e-07	nit=11	nfev=34	t=5.04s
SLSQP	run 552/1000	MSE=1.651e-07	nit=13	nfev=43	t=6.57s
SLSQP	run 553/1000	MSE=1.654e-08	nit=13	nfev=41	t=6.10s
SLSQP	run 554/1000	MSE=2.602e-08	nit=11	nfev=35	t=5.46s
SLSQP	run 555/1000	MSE=3.443e-09	nit=7	nfev=24	t=3.51s
SLSQP	run 556/1000	MSE=1.192e-08	nit=12	nfev=38	t=5.83s
SLSQP	run 557/1000	MSE=2.711e-08	nit=14	nfev=46	t=6.75s
SLSQP	run 558/1000	MSE=3.930e-07	nit=13	nfev=42	t=6.00s
SLSQP	run 559/1000	MSE=4.502e-09	nit=12	nfev=39	t=5.81s
SLSQP	run 560/1000	MSE=9.512e-08	nit=8	nfev=27	t=4.11s
SLSQP	run 561/1000	MSE=3.434e-07	nit=11	nfev=38	t=5.50s
SLSQP	run 562/1000	MSE=4.700e-07	nit=10	nfev=34	t=5.05s
SLSQP	run 563/1000	MSE=1.676e-07	nit=12	nfev=38	t=5.79s

SLSQP	run 564/1000	MSE=4.310e-08	nit=15	nfev=50	t=7.45s
SLSQP	run 565/1000	MSE=2.247e-09	nit=13	nfev=43	t=6.46s
SLSQP	run 566/1000	MSE=3.576e-08	nit=12	nfev=39	t=5.80s
SLSQP	run 567/1000	MSE=2.160e-09	nit=8	nfev=28	t=4.29s
SLSQP	run 568/1000	MSE=1.256e-08	nit=12	nfev=39	t=5.90s
SLSQP	run 569/1000	MSE=3.092e-08	nit=7	nfev=24	t=3.44s
SLSQP	run 570/1000	MSE=3.529e-09	nit=13	nfev=44	t=6.64s
SLSQP	run 571/1000	MSE=2.344e-10	nit=15	nfev=50	t=7.45s
SLSQP	run 572/1000	MSE=9.249e-09	nit=12	nfev=39	t=5.80s
SLSQP	run 573/1000	MSE=3.796e-07	nit=15	nfev=52	t=7.69s
SLSQP	run 574/1000	MSE=4.871e-07	nit=7	nfev=25	t=3.73s
SLSQP	run 575/1000	MSE=1.464e-08	nit=14	nfev=46	t=7.03s
SLSQP	run 576/1000	MSE=4.986e-12	nit=12	nfev=40	t=6.26s
SLSQP	run 577/1000	MSE=1.768e-08	nit=13	nfev=42	t=6.30s
SLSQP	run 578/1000	MSE=3.043e-10	nit=13	nfev=42	t=6.05s
SLSQP	run 579/1000	MSE=2.050e-08	nit=12	nfev=41	t=6.11s
SLSQP	run 580/1000	MSE=1.181e-07	nit=7	nfev=24	t=3.68s
SLSQP	run 581/1000	MSE=1.550e-08	nit=12	nfev=38	t=5.50s
SLSQP	run 582/1000	MSE=4.991e-07	nit=16	nfev=53	t=7.98s
SLSQP	run 583/1000	MSE=1.383e-08	nit=8	nfev=27	t=4.15s
SLSQP	run 584/1000	MSE=2.759e-07	nit=10	nfev=34	t=5.02s
SLSQP	run 585/1000	MSE=5.008e-08	nit=6	nfev=20	t=2.94s
SLSQP	run 586/1000	MSE=7.491e-09	nit=9	nfev=30	t=4.33s
SLSQP	run 587/1000	MSE=6.220e-09	nit=16	nfev=54	t=8.04s
SLSQP	run 588/1000	MSE=2.769e-07	nit=7	nfev=23	t=3.51s
SLSQP	run 589/1000	MSE=3.755e-08	nit=6	nfev=21	t=3.01s
SLSQP	run 590/1000	MSE=2.147e-07	nit=9	nfev=30	t=4.60s
SLSQP	run 591/1000	MSE=3.060e-09	nit=14	nfev=44	t=6.86s
SLSQP	run 592/1000	MSE=8.339e-09	nit=6	nfev=21	t=3.02s
SLSQP	run 593/1000	MSE=1.785e-09	nit=11	nfev=36	t=5.59s
SLSQP	run 594/1000	MSE=4.894e-07	nit=7	nfev=24	t=3.46s
SLSQP	run 595/1000	MSE=3.580e-10	nit=7	nfev=25	t=3.83s
SLSQP	run 596/1000	MSE=1.665e-07	nit=11	nfev=37	t=5.43s
SLSQP	run 597/1000	MSE=6.980e-08	nit=10	nfev=33	t=5.00s
SLSQP	run 598/1000	MSE=1.187e-08	nit=12	nfev=40	t=6.04s
SLSQP	run 599/1000	MSE=1.753e-08	nit=9	nfev=28	t=4.03s
SLSQP	run 600/1000	MSE=8.748e-08	nit=12	nfev=41	t=6.30s
SLSQP	run 601/1000	MSE=7.958e-08	nit=10	nfev=33	t=5.04s
SLSQP	run 602/1000	MSE=1.718e-08	nit=13	nfev=42	t=6.22s
SLSQP	run 603/1000	MSE=1.008e-07	nit=15	nfev=52	t=7.72s
SLSQP	run 604/1000	MSE=4.309e-07	nit=8	nfev=26	t=4.00s
SLSQP	run 605/1000	MSE=4.943e-08	nit=13	nfev=43	t=6.48s
SLSQP	run 606/1000	MSE=5.118e-08	nit=11	nfev=36	t=5.14s
SLSQP	run 607/1000	MSE=1.880e-07	nit=14	nfev=47	t=6.95s
SLSQP	run 608/1000	MSE=2.392e-07	nit=17	nfev=58	t=8.63s
SLSQP	run 609/1000	MSE=9.106e-09	nit=16	nfev=54	t=8.10s
SLSQP	run 610/1000	MSE=1.263e-08	nit=5	nfev=19	t=2.90s
SLSQP	run 611/1000	MSE=2.208e-08	nit=12	nfev=42	t=6.22s

SLSQP	run 612/1000	MSE=9.487e-08	nit=8	nfev=27	t=4.12s
SLSQP	run 613/1000	MSE=5.054e-08	nit=9	nfev=29	t=4.40s
SLSQP	run 614/1000	MSE=6.861e-08	nit=14	nfev=46	t=6.66s
SLSQP	run 615/1000	MSE=2.680e-03	nit=8	nfev=27	t=4.18s
SLSQP	run 616/1000	MSE=4.817e-06	nit=6	nfev=20	t=2.90s
SLSQP	run 617/1000	MSE=4.714e-08	nit=7	nfev=23	t=3.67s
SLSQP	run 618/1000	MSE=8.087e-08	nit=11	nfev=37	t=5.35s
SLSQP	run 619/1000	MSE=1.082e-08	nit=7	nfev=28	t=4.37s
SLSQP	run 620/1000	MSE=2.934e-08	nit=6	nfev=22	t=3.19s
SLSQP	run 621/1000	MSE=1.928e-10	nit=8	nfev=27	t=4.14s
SLSQP	run 622/1000	MSE=1.448e-08	nit=8	nfev=26	t=3.82s
SLSQP	run 623/1000	MSE=1.536e-09	nit=11	nfev=35	t=5.25s
SLSQP	run 624/1000	MSE=7.731e-08	nit=9	nfev=29	t=4.42s
SLSQP	run 625/1000	MSE=5.095e-08	nit=11	nfev=36	t=5.26s
SLSQP	run 626/1000	MSE=5.129e-07	nit=9	nfev=30	t=4.52s
SLSQP	run 627/1000	MSE=4.749e-07	nit=17	nfev=58	t=8.55s
SLSQP	run 628/1000	MSE=7.014e-09	nit=11	nfev=36	t=5.50s
SLSQP	run 629/1000	MSE=7.696e-08	nit=8	nfev=28	t=4.02s
SLSQP	run 630/1000	MSE=1.684e-09	nit=15	nfev=48	t=7.07s
SLSQP	run 631/1000	MSE=2.687e-08	nit=10	nfev=33	t=5.03s
SLSQP	run 632/1000	MSE=3.170e-07	nit=18	nfev=62	t=9.31s
SLSQP	run 633/1000	MSE=6.951e-07	nit=15	nfev=52	t=7.74s
SLSQP	run 634/1000	MSE=8.839e-10	nit=9	nfev=29	t=4.27s
SLSQP	run 635/1000	MSE=2.165e-07	nit=8	nfev=28	t=4.20s
SLSQP	run 636/1000	MSE=4.485e-07	nit=7	nfev=25	t=3.60s
SLSQP	run 637/1000	MSE=7.164e-09	nit=8	nfev=28	t=4.25s
SLSQP	run 638/1000	MSE=3.917e-09	nit=14	nfev=48	t=7.13s
SLSQP	run 639/1000	MSE=1.857e-07	nit=10	nfev=32	t=4.91s
SLSQP	run 640/1000	MSE=2.919e-09	nit=13	nfev=45	t=6.40s
SLSQP	run 641/1000	MSE=4.676e-10	nit=9	nfev=31	t=4.71s
SLSQP	run 642/1000	MSE=8.620e-08	nit=12	nfev=41	t=6.11s
SLSQP	run 643/1000	MSE=9.727e-08	nit=13	nfev=44	t=6.60s
SLSQP	run 644/1000	MSE=4.124e-07	nit=10	nfev=33	t=4.87s
SLSQP	run 645/1000	MSE=6.055e-08	nit=17	nfev=56	t=8.25s
SLSQP	run 646/1000	MSE=1.518e-07	nit=12	nfev=39	t=5.82s
SLSQP	run 647/1000	MSE=8.891e-09	nit=10	nfev=35	t=5.26s
SLSQP	run 648/1000	MSE=7.920e-08	nit=14	nfev=46	t=6.91s
SLSQP	run 649/1000	MSE=5.268e-08	nit=13	nfev=43	t=6.22s
SLSQP	run 650/1000	MSE=9.938e-09	nit=13	nfev=44	t=6.60s
SLSQP	run 651/1000	MSE=3.062e-08	nit=9	nfev=30	t=4.50s
SLSQP	run 652/1000	MSE=4.094e-09	nit=13	nfev=44	t=6.48s
SLSQP	run 653/1000	MSE=6.422e-09	nit=15	nfev=48	t=7.33s
SLSQP	run 654/1000	MSE=4.571e-08	nit=14	nfev=46	t=6.76s
SLSQP	run 655/1000	MSE=1.442e-07	nit=11	nfev=38	t=5.83s
SLSQP	run 656/1000	MSE=2.672e-07	nit=6	nfev=20	t=2.91s
SLSQP	run 657/1000	MSE=9.416e-08	nit=10	nfev=33	t=4.97s
SLSQP	run 658/1000	MSE=8.052e-10	nit=9	nfev=29	t=4.39s
SLSQP	run 659/1000	MSE=3.044e-05	nit=6	nfev=20	t=3.18s

SLSQP	run 660/1000	MSE=4.748e-08	nit=13	nfev=41	t=6.20s
SLSQP	run 661/1000	MSE=5.960e-08	nit=8	nfev=27	t=3.89s
SLSQP	run 662/1000	MSE=2.683e-09	nit=16	nfev=53	t=8.04s
SLSQP	run 663/1000	MSE=1.895e-08	nit=10	nfev=34	t=5.70s
SLSQP	run 664/1000	MSE=4.340e-09	nit=16	nfev=53	t=8.26s
SLSQP	run 665/1000	MSE=1.237e-08	nit=14	nfev=47	t=7.18s
SLSQP	run 666/1000	MSE=3.021e-13	nit=18	nfev=63	t=9.18s
SLSQP	run 667/1000	MSE=6.829e-09	nit=8	nfev=26	t=3.75s
SLSQP	run 668/1000	MSE=5.506e-08	nit=11	nfev=35	t=5.04s
SLSQP	run 669/1000	MSE=5.030e-08	nit=11	nfev=35	t=5.42s
SLSQP	run 670/1000	MSE=5.738e-08	nit=8	nfev=28	t=4.03s
SLSQP	run 671/1000	MSE=1.243e-07	nit=8	nfev=26	t=3.93s
SLSQP	run 672/1000	MSE=2.425e-09	nit=15	nfev=52	t=7.71s
SLSQP	run 673/1000	MSE=1.729e-07	nit=11	nfev=38	t=5.71s
SLSQP	run 674/1000	MSE=9.240e-08	nit=12	nfev=42	t=6.25s
SLSQP	run 675/1000	MSE=2.974e-08	nit=10	nfev=33	t=4.91s
SLSQP	run 676/1000	MSE=6.094e-09	nit=14	nfev=46	t=6.86s
SLSQP	run 677/1000	MSE=1.705e-07	nit=7	nfev=24	t=3.46s
SLSQP	run 678/1000	MSE=3.864e-07	nit=7	nfev=23	t=3.47s
SLSQP	run 679/1000	MSE=1.172e-07	nit=16	nfev=54	t=8.10s
SLSQP	run 680/1000	MSE=2.368e-08	nit=15	nfev=50	t=7.50s
SLSQP	run 681/1000	MSE=5.061e-09	nit=8	nfev=28	t=4.13s
SLSQP	run 682/1000	MSE=1.972e-07	nit=9	nfev=29	t=4.32s
SLSQP	run 683/1000	MSE=2.457e-07	nit=13	nfev=43	t=6.52s
SLSQP	run 684/1000	MSE=2.741e-09	nit=14	nfev=46	t=6.87s
SLSQP	run 685/1000	MSE=2.620e-07	nit=8	nfev=27	t=4.13s
SLSQP	run 686/1000	MSE=2.683e-03	nit=9	nfev=30	t=4.41s
SLSQP	run 687/1000	MSE=2.138e-09	nit=7	nfev=25	t=3.58s
SLSQP	run 688/1000	MSE=5.028e-09	nit=14	nfev=43	t=6.39s
SLSQP	run 689/1000	MSE=2.320e-08	nit=12	nfev=38	t=5.71s
SLSQP	run 690/1000	MSE=3.527e-08	nit=18	nfev=61	t=9.04s
SLSQP	run 691/1000	MSE=1.140e-08	nit=10	nfev=35	t=5.01s
SLSQP	run 692/1000	MSE=2.164e-08	nit=17	nfev=57	t=8.61s
SLSQP	run 693/1000	MSE=1.199e-08	nit=10	nfev=32	t=4.58s
SLSQP	run 694/1000	MSE=3.790e-07	nit=12	nfev=41	t=6.10s
SLSQP	run 695/1000	MSE=1.814e-11	nit=10	nfev=34	t=5.38s
SLSQP	run 696/1000	MSE=1.146e-07	nit=13	nfev=42	t=6.15s
SLSQP	run 697/1000	MSE=4.179e-07	nit=9	nfev=30	t=4.55s
SLSQP	run 698/1000	MSE=2.612e-07	nit=13	nfev=41	t=6.51s
SLSQP	run 699/1000	MSE=1.220e-07	nit=11	nfev=38	t=5.65s
SLSQP	run 700/1000	MSE=1.459e-08	nit=11	nfev=39	t=5.78s
SLSQP	run 701/1000	MSE=8.169e-09	nit=10	nfev=35	t=5.21s
SLSQP	run 702/1000	MSE=1.362e-08	nit=10	nfev=31	t=4.62s
SLSQP	run 703/1000	MSE=2.338e-08	nit=16	nfev=52	t=7.86s
SLSQP	run 704/1000	MSE=4.667e-07	nit=12	nfev=37	t=5.42s
SLSQP	run 705/1000	MSE=3.027e-09	nit=12	nfev=41	t=6.27s
SLSQP	run 706/1000	MSE=3.399e-09	nit=7	nfev=25	t=3.62s
SLSQP	run 707/1000	MSE=8.507e-09	nit=14	nfev=48	t=7.10s

SLSQP	run 708/1000	MSE=4.616e-07	nit=10	nfev=34	t=5.37s
SLSQP	run 709/1000	MSE=3.094e-07	nit=17	nfev=55	t=8.15s
SLSQP	run 710/1000	MSE=3.609e-07	nit=17	nfev=56	t=8.53s
SLSQP	run 711/1000	MSE=1.438e-08	nit=13	nfev=42	t=6.41s
SLSQP	run 712/1000	MSE=8.993e-08	nit=12	nfev=37	t=5.38s
SLSQP	run 713/1000	MSE=1.339e-07	nit=17	nfev=58	t=8.59s
SLSQP	run 714/1000	MSE=1.301e-07	nit=12	nfev=40	t=6.09s
SLSQP	run 715/1000	MSE=3.177e-07	nit=9	nfev=30	t=4.49s
SLSQP	run 716/1000	MSE=1.139e-07	nit=13	nfev=41	t=5.89s
SLSQP	run 717/1000	MSE=3.505e-09	nit=14	nfev=49	t=7.25s
SLSQP	run 718/1000	MSE=7.262e-09	nit=14	nfev=45	t=6.68s
SLSQP	run 719/1000	MSE=3.346e-09	nit=10	nfev=35	t=5.44s
SLSQP	run 720/1000	MSE=1.004e-08	nit=14	nfev=46	t=6.77s
SLSQP	run 721/1000	MSE=2.171e-07	nit=15	nfev=50	t=7.35s
SLSQP	run 722/1000	MSE=2.695e-07	nit=12	nfev=40	t=5.86s
SLSQP	run 723/1000	MSE=2.068e-08	nit=8	nfev=28	t=4.24s
SLSQP	run 724/1000	MSE=8.856e-09	nit=13	nfev=43	t=6.50s
SLSQP	run 725/1000	MSE=2.333e-08	nit=12	nfev=40	t=5.81s
SLSQP	run 726/1000	MSE=2.545e-08	nit=12	nfev=38	t=5.73s
SLSQP	run 727/1000	MSE=9.178e-08	nit=11	nfev=35	t=5.18s
SLSQP	run 728/1000	MSE=5.376e-09	nit=15	nfev=51	t=7.61s
SLSQP	run 729/1000	MSE=2.896e-10	nit=10	nfev=33	t=4.89s
SLSQP	run 730/1000	MSE=8.226e-09	nit=14	nfev=46	t=6.97s
SLSQP	run 731/1000	MSE=1.552e-08	nit=8	nfev=27	t=4.17s
SLSQP	run 732/1000	MSE=1.673e-07	nit=7	nfev=23	t=3.28s
SLSQP	run 733/1000	MSE=6.672e-09	nit=17	nfev=56	t=8.43s
SLSQP	run 734/1000	MSE=2.205e-07	nit=8	nfev=27	t=4.38s
SLSQP	run 735/1000	MSE=4.850e-07	nit=10	nfev=33	t=4.81s
SLSQP	run 736/1000	MSE=3.595e-09	nit=11	nfev=36	t=5.28s
SLSQP	run 737/1000	MSE=2.064e-11	nit=16	nfev=53	t=7.83s
SLSQP	run 738/1000	MSE=5.822e-09	nit=14	nfev=45	t=6.70s
SLSQP	run 739/1000	MSE=6.639e-09	nit=12	nfev=39	t=5.66s
SLSQP	run 740/1000	MSE=8.789e-08	nit=10	nfev=35	t=5.31s
SLSQP	run 741/1000	MSE=1.279e-07	nit=16	nfev=56	t=8.37s
SLSQP	run 742/1000	MSE=4.527e-09	nit=11	nfev=36	t=5.41s
SLSQP	run 743/1000	MSE=1.789e-07	nit=8	nfev=27	t=3.93s
SLSQP	run 744/1000	MSE=2.209e-07	nit=9	nfev=29	t=4.40s
SLSQP	run 745/1000	MSE=2.972e-07	nit=6	nfev=22	t=3.30s
SLSQP	run 746/1000	MSE=1.024e-07	nit=13	nfev=43	t=6.29s
SLSQP	run 747/1000	MSE=1.277e-07	nit=7	nfev=25	t=3.59s
SLSQP	run 748/1000	MSE=4.034e-08	nit=10	nfev=33	t=4.91s
SLSQP	run 749/1000	MSE=5.055e-08	nit=8	nfev=26	t=4.02s
SLSQP	run 750/1000	MSE=7.764e-08	nit=12	nfev=40	t=5.75s
SLSQP	run 751/1000	MSE=2.608e-07	nit=11	nfev=36	t=5.53s
SLSQP	run 752/1000	MSE=3.481e-07	nit=10	nfev=32	t=4.90s
SLSQP	run 753/1000	MSE=3.728e-07	nit=17	nfev=57	t=8.36s
SLSQP	run 754/1000	MSE=2.394e-07	nit=9	nfev=31	t=4.59s
SLSQP	run 755/1000	MSE=5.491e-08	nit=13	nfev=42	t=6.36s

SLSQP	run 756/1000	MSE=9.431e-08	nit=14	nfev=45	t=6.91s
SLSQP	run 757/1000	MSE=4.148e-10	nit=9	nfev=31	t=4.46s
SLSQP	run 758/1000	MSE=4.601e-07	nit=8	nfev=27	t=4.08s
SLSQP	run 759/1000	MSE=6.963e-08	nit=10	nfev=34	t=5.12s
SLSQP	run 760/1000	MSE=3.742e-07	nit=9	nfev=33	t=4.83s
SLSQP	run 761/1000	MSE=1.483e-08	nit=9	nfev=31	t=4.59s
SLSQP	run 762/1000	MSE=8.581e-08	nit=6	nfev=21	t=3.06s
SLSQP	run 763/1000	MSE=1.232e-07	nit=9	nfev=29	t=4.55s
SLSQP	run 764/1000	MSE=1.126e-07	nit=12	nfev=38	t=5.40s
SLSQP	run 765/1000	MSE=5.812e-09	nit=13	nfev=44	t=6.53s
SLSQP	run 766/1000	MSE=3.930e-07	nit=10	nfev=33	t=4.91s
SLSQP	run 767/1000	MSE=7.719e-08	nit=11	nfev=34	t=4.95s
SLSQP	run 768/1000	MSE=8.914e-08	nit=17	nfev=58	t=8.85s
SLSQP	run 769/1000	MSE=8.745e-09	nit=12	nfev=42	t=6.07s
SLSQP	run 770/1000	MSE=3.561e-08	nit=9	nfev=29	t=4.32s
SLSQP	run 771/1000	MSE=1.524e-08	nit=16	nfev=56	t=8.24s
SLSQP	run 772/1000	MSE=1.286e-08	nit=9	nfev=29	t=4.44s
SLSQP	run 773/1000	MSE=1.865e-08	nit=10	nfev=34	t=5.06s
SLSQP	run 774/1000	MSE=3.211e-10	nit=9	nfev=32	t=4.82s
SLSQP	run 775/1000	MSE=2.691e-07	nit=11	nfev=36	t=5.91s
SLSQP	run 776/1000	MSE=2.107e-07	nit=8	nfev=27	t=4.56s
SLSQP	run 777/1000	MSE=8.178e-07	nit=6	nfev=20	t=3.52s
SLSQP	run 778/1000	MSE=7.935e-08	nit=16	nfev=56	t=9.89s
SLSQP	run 779/1000	MSE=2.393e-09	nit=18	nfev=60	t=9.25s
SLSQP	run 780/1000	MSE=2.547e-08	nit=11	nfev=37	t=5.50s
SLSQP	run 781/1000	MSE=3.983e-08	nit=10	nfev=32	t=4.86s
SLSQP	run 782/1000	MSE=2.002e-09	nit=8	nfev=30	t=4.63s
SLSQP	run 783/1000	MSE=4.348e-07	nit=14	nfev=45	t=6.67s
SLSQP	run 784/1000	MSE=3.353e-08	nit=11	nfev=36	t=5.30s
SLSQP	run 785/1000	MSE=8.867e-09	nit=11	nfev=39	t=5.73s
SLSQP	run 786/1000	MSE=4.316e-07	nit=13	nfev=44	t=6.59s
SLSQP	run 787/1000	MSE=1.812e-08	nit=12	nfev=40	t=5.77s
SLSQP	run 788/1000	MSE=4.451e-08	nit=9	nfev=29	t=4.73s
SLSQP	run 789/1000	MSE=4.878e-09	nit=16	nfev=55	t=8.24s
SLSQP	run 790/1000	MSE=4.431e-08	nit=13	nfev=43	t=6.44s
SLSQP	run 791/1000	MSE=1.876e-07	nit=17	nfev=58	t=8.54s
SLSQP	run 792/1000	MSE=4.735e-07	nit=8	nfev=26	t=3.75s
SLSQP	run 793/1000	MSE=2.766e-09	nit=12	nfev=40	t=5.94s
SLSQP	run 794/1000	MSE=2.959e-08	nit=12	nfev=38	t=5.66s
SLSQP	run 795/1000	MSE=1.517e-09	nit=15	nfev=50	t=7.49s
SLSQP	run 796/1000	MSE=4.056e-08	nit=16	nfev=54	t=7.98s
SLSQP	run 797/1000	MSE=7.750e-12	nit=13	nfev=41	t=6.09s
SLSQP	run 798/1000	MSE=6.908e-08	nit=13	nfev=41	t=6.07s
SLSQP	run 799/1000	MSE=1.895e-07	nit=11	nfev=36	t=5.73s
SLSQP	run 800/1000	MSE=1.825e-09	nit=10	nfev=34	t=5.21s
SLSQP	run 801/1000	MSE=3.414e-07	nit=14	nfev=49	t=7.23s
SLSQP	run 802/1000	MSE=1.090e-08	nit=15	nfev=51	t=7.64s
SLSQP	run 803/1000	MSE=2.875e-07	nit=17	nfev=59	t=8.68s

SLSQP	run 804/1000	MSE=1.655e-08	nit=16	nfev=56	t=8.58s
SLSQP	run 805/1000	MSE=3.683e-08	nit=9	nfev=29	t=4.49s
SLSQP	run 806/1000	MSE=5.069e-07	nit=11	nfev=36	t=5.71s
SLSQP	run 807/1000	MSE=6.972e-09	nit=8	nfev=31	t=4.58s
SLSQP	run 808/1000	MSE=8.150e-08	nit=10	nfev=33	t=5.21s
SLSQP	run 809/1000	MSE=1.003e-07	nit=13	nfev=42	t=6.33s
SLSQP	run 810/1000	MSE=1.628e-08	nit=16	nfev=54	t=8.05s
SLSQP	run 811/1000	MSE=2.232e-09	nit=18	nfev=59	t=8.87s
SLSQP	run 812/1000	MSE=4.923e-08	nit=13	nfev=42	t=6.26s
SLSQP	run 813/1000	MSE=7.989e-08	nit=10	nfev=31	t=4.53s
SLSQP	run 814/1000	MSE=3.063e-07	nit=9	nfev=30	t=4.55s
SLSQP	run 815/1000	MSE=1.562e-07	nit=9	nfev=29	t=4.15s
SLSQP	run 816/1000	MSE=2.175e-07	nit=13	nfev=43	t=6.60s
SLSQP	run 817/1000	MSE=1.135e-09	nit=10	nfev=32	t=4.82s
SLSQP	run 818/1000	MSE=3.509e-07	nit=11	nfev=37	t=5.43s
SLSQP	run 819/1000	MSE=6.690e-08	nit=11	nfev=37	t=5.55s
SLSQP	run 820/1000	MSE=9.062e-09	nit=12	nfev=37	t=5.63s
SLSQP	run 821/1000	MSE=2.777e-07	nit=8	nfev=26	t=3.77s
SLSQP	run 822/1000	MSE=1.679e-09	nit=9	nfev=30	t=4.64s
SLSQP	run 823/1000	MSE=9.053e-08	nit=6	nfev=21	t=3.05s
SLSQP	run 824/1000	MSE=2.133e-08	nit=11	nfev=36	t=5.36s
SLSQP	run 825/1000	MSE=2.114e-08	nit=13	nfev=42	t=6.22s
SLSQP	run 826/1000	MSE=3.222e-08	nit=10	nfev=33	t=4.98s
SLSQP	run 827/1000	MSE=2.508e-08	nit=12	nfev=39	t=5.87s
SLSQP	run 828/1000	MSE=4.183e-08	nit=11	nfev=36	t=5.49s
SLSQP	run 829/1000	MSE=2.863e-07	nit=11	nfev=35	t=5.06s
SLSQP	run 830/1000	MSE=4.320e-08	nit=11	nfev=36	t=5.39s
SLSQP	run 831/1000	MSE=1.811e-08	nit=12	nfev=38	t=5.66s
SLSQP	run 832/1000	MSE=4.047e-09	nit=11	nfev=36	t=5.18s
SLSQP	run 833/1000	MSE=1.944e-07	nit=13	nfev=42	t=6.67s
SLSQP	run 834/1000	MSE=1.343e-07	nit=15	nfev=50	t=7.40s
SLSQP	run 835/1000	MSE=1.558e-08	nit=9	nfev=29	t=4.31s
SLSQP	run 836/1000	MSE=1.957e-07	nit=12	nfev=40	t=5.95s
SLSQP	run 837/1000	MSE=4.904e-09	nit=14	nfev=47	t=7.08s
SLSQP	run 838/1000	MSE=2.877e-07	nit=9	nfev=29	t=4.50s
SLSQP	run 839/1000	MSE=2.922e-07	nit=9	nfev=32	t=4.57s
SLSQP	run 840/1000	MSE=5.622e-09	nit=14	nfev=47	t=7.01s
SLSQP	run 841/1000	MSE=1.167e-07	nit=11	nfev=35	t=5.26s
SLSQP	run 842/1000	MSE=3.430e-07	nit=10	nfev=33	t=4.99s
SLSQP	run 843/1000	MSE=1.998e-09	nit=11	nfev=36	t=5.38s
SLSQP	run 844/1000	MSE=2.220e-07	nit=13	nfev=43	t=6.46s
SLSQP	run 845/1000	MSE=3.443e-08	nit=13	nfev=42	t=6.22s
SLSQP	run 846/1000	MSE=4.261e-09	nit=6	nfev=20	t=2.93s
SLSQP	run 847/1000	MSE=2.561e-07	nit=8	nfev=27	t=4.14s
SLSQP	run 848/1000	MSE=3.445e-07	nit=12	nfev=41	t=6.13s
SLSQP	run 849/1000	MSE=3.213e-08	nit=11	nfev=36	t=5.36s
SLSQP	run 850/1000	MSE=5.513e-08	nit=15	nfev=50	t=7.41s
SLSQP	run 851/1000	MSE=5.303e-08	nit=11	nfev=38	t=5.81s

SLSQP	run 852/1000	MSE=6.708e-09	nit=14	nfev=46	t=6.89s
SLSQP	run 853/1000	MSE=3.522e-07	nit=17	nfev=58	t=8.70s
SLSQP	run 854/1000	MSE=3.924e-08	nit=18	nfev=60	t=8.95s
SLSQP	run 855/1000	MSE=1.604e-07	nit=12	nfev=40	t=5.92s
SLSQP	run 856/1000	MSE=4.905e-09	nit=10	nfev=35	t=5.16s
SLSQP	run 857/1000	MSE=4.835e-07	nit=15	nfev=51	t=7.70s
SLSQP	run 858/1000	MSE=8.929e-08	nit=13	nfev=44	t=6.73s
SLSQP	run 859/1000	MSE=3.661e-07	nit=10	nfev=33	t=5.05s
SLSQP	run 860/1000	MSE=3.465e-08	nit=15	nfev=52	t=7.65s
SLSQP	run 861/1000	MSE=8.788e-10	nit=12	nfev=41	t=6.10s
SLSQP	run 862/1000	MSE=5.157e-07	nit=8	nfev=28	t=4.09s
SLSQP	run 863/1000	MSE=3.047e-10	nit=10	nfev=34	t=5.28s
SLSQP	run 864/1000	MSE=2.530e-08	nit=10	nfev=32	t=4.59s
SLSQP	run 865/1000	MSE=7.819e-09	nit=15	nfev=51	t=7.60s
SLSQP	run 866/1000	MSE=1.733e-07	nit=11	nfev=36	t=5.43s
SLSQP	run 867/1000	MSE=1.287e-08	nit=14	nfev=47	t=7.04s
SLSQP	run 868/1000	MSE=3.853e-07	nit=7	nfev=23	t=3.57s
SLSQP	run 869/1000	MSE=2.052e-07	nit=13	nfev=41	t=6.04s
SLSQP	run 870/1000	MSE=2.335e-07	nit=12	nfev=38	t=5.71s
SLSQP	run 871/1000	MSE=2.950e-09	nit=12	nfev=41	t=6.13s
SLSQP	run 872/1000	MSE=2.041e-07	nit=12	nfev=40	t=5.79s
SLSQP	run 873/1000	MSE=3.769e-07	nit=10	nfev=33	t=5.20s
SLSQP	run 874/1000	MSE=2.341e-07	nit=14	nfev=48	t=7.16s
SLSQP	run 875/1000	MSE=1.505e-07	nit=14	nfev=47	t=6.91s
SLSQP	run 876/1000	MSE=2.613e-09	nit=11	nfev=37	t=5.49s
SLSQP	run 877/1000	MSE=1.166e-08	nit=9	nfev=30	t=4.38s
SLSQP	run 878/1000	MSE=7.293e-08	nit=8	nfev=28	t=4.33s
SLSQP	run 879/1000	MSE=3.164e-07	nit=10	nfev=33	t=4.99s
SLSQP	run 880/1000	MSE=3.345e-08	nit=10	nfev=34	t=5.24s
SLSQP	run 881/1000	MSE=8.520e-09	nit=7	nfev=26	t=3.79s
SLSQP	run 882/1000	MSE=4.001e-08	nit=18	nfev=61	t=9.27s
SLSQP	run 883/1000	MSE=5.100e-08	nit=13	nfev=42	t=6.10s
SLSQP	run 884/1000	MSE=8.103e-10	nit=8	nfev=28	t=4.44s
SLSQP	run 885/1000	MSE=8.433e-09	nit=7	nfev=27	t=3.85s
SLSQP	run 886/1000	MSE=2.566e-08	nit=16	nfev=54	t=8.04s
SLSQP	run 887/1000	MSE=6.844e-08	nit=10	nfev=32	t=4.80s
SLSQP	run 888/1000	MSE=1.520e-07	nit=8	nfev=27	t=4.07s
SLSQP	run 889/1000	MSE=4.633e-07	nit=10	nfev=34	t=5.57s
SLSQP	run 890/1000	MSE=2.901e-07	nit=12	nfev=42	t=6.25s
SLSQP	run 891/1000	MSE=4.417e-08	nit=8	nfev=27	t=3.92s
SLSQP	run 892/1000	MSE=5.585e-08	nit=11	nfev=36	t=5.31s
SLSQP	run 893/1000	MSE=2.119e-07	nit=12	nfev=40	t=5.98s
SLSQP	run 894/1000	MSE=2.399e-08	nit=16	nfev=53	t=8.00s
SLSQP	run 895/1000	MSE=3.327e-09	nit=13	nfev=44	t=6.64s
SLSQP	run 896/1000	MSE=5.050e-08	nit=11	nfev=36	t=5.39s
SLSQP	run 897/1000	MSE=1.475e-07	nit=15	nfev=49	t=7.01s
SLSQP	run 898/1000	MSE=3.409e-08	nit=15	nfev=49	t=7.41s
SLSQP	run 899/1000	MSE=2.806e-08	nit=14	nfev=49	t=7.46s

SLSQP	run 900/1000	MSE=5.024e-08	nit=8	nfev=28	t=4.19s
SLSQP	run 901/1000	MSE=5.268e-08	nit=6	nfev=21	t=3.23s
SLSQP	run 902/1000	MSE=2.677e-09	nit=11	nfev=36	t=5.50s
SLSQP	run 903/1000	MSE=4.425e-07	nit=5	nfev=18	t=2.55s
SLSQP	run 904/1000	MSE=1.522e-08	nit=9	nfev=30	t=4.58s
SLSQP	run 905/1000	MSE=8.566e-08	nit=9	nfev=30	t=4.44s
SLSQP	run 906/1000	MSE=3.447e-08	nit=8	nfev=28	t=4.38s
SLSQP	run 907/1000	MSE=4.107e-09	nit=15	nfev=52	t=7.64s
SLSQP	run 908/1000	MSE=4.230e-07	nit=10	nfev=32	t=4.63s
SLSQP	run 909/1000	MSE=1.595e-07	nit=9	nfev=29	t=4.30s
SLSQP	run 910/1000	MSE=1.275e-07	nit=10	nfev=35	t=5.26s
SLSQP	run 911/1000	MSE=1.313e-09	nit=14	nfev=44	t=6.82s
SLSQP	run 912/1000	MSE=4.145e-08	nit=8	nfev=27	t=3.81s
SLSQP	run 913/1000	MSE=2.772e-10	nit=13	nfev=43	t=6.67s
SLSQP	run 914/1000	MSE=7.124e-08	nit=9	nfev=33	t=4.79s
SLSQP	run 915/1000	MSE=2.029e-10	nit=17	nfev=56	t=8.25s
SLSQP	run 916/1000	MSE=2.342e-07	nit=6	nfev=22	t=3.48s
SLSQP	run 917/1000	MSE=3.069e-08	nit=10	nfev=32	t=4.62s
SLSQP	run 918/1000	MSE=4.277e-08	nit=10	nfev=34	t=4.82s
SLSQP	run 919/1000	MSE=7.772e-08	nit=12	nfev=37	t=5.50s
SLSQP	run 920/1000	MSE=3.388e-08	nit=11	nfev=37	t=5.44s
SLSQP	run 921/1000	MSE=1.448e-07	nit=10	nfev=33	t=4.95s
SLSQP	run 922/1000	MSE=4.358e-09	nit=13	nfev=44	t=6.60s
SLSQP	run 923/1000	MSE=6.075e-08	nit=13	nfev=41	t=6.02s
SLSQP	run 924/1000	MSE=1.635e-07	nit=11	nfev=36	t=5.16s
SLSQP	run 925/1000	MSE=2.190e-07	nit=16	nfev=53	t=7.79s
SLSQP	run 926/1000	MSE=2.247e-08	nit=9	nfev=31	t=4.63s
SLSQP	run 927/1000	MSE=3.922e-09	nit=13	nfev=43	t=6.56s
SLSQP	run 928/1000	MSE=1.035e-08	nit=12	nfev=39	t=5.58s
SLSQP	run 929/1000	MSE=8.846e-08	nit=8	nfev=26	t=3.98s
SLSQP	run 930/1000	MSE=3.136e-08	nit=13	nfev=43	t=6.52s
SLSQP	run 931/1000	MSE=2.353e-09	nit=11	nfev=37	t=5.53s
SLSQP	run 932/1000	MSE=1.055e-08	nit=11	nfev=38	t=5.63s
SLSQP	run 933/1000	MSE=5.292e-08	nit=10	nfev=33	t=5.01s
SLSQP	run 934/1000	MSE=3.867e-07	nit=11	nfev=36	t=5.51s
SLSQP	run 935/1000	MSE=1.874e-08	nit=8	nfev=28	t=4.06s
SLSQP	run 936/1000	MSE=1.623e-09	nit=12	nfev=38	t=5.68s
SLSQP	run 937/1000	MSE=7.512e-11	nit=9	nfev=33	t=5.00s
SLSQP	run 938/1000	MSE=5.949e-08	nit=14	nfev=48	t=7.06s
SLSQP	run 939/1000	MSE=6.918e-09	nit=11	nfev=37	t=5.54s
SLSQP	run 940/1000	MSE=3.715e-09	nit=9	nfev=34	t=5.07s
SLSQP	run 941/1000	MSE=6.760e-10	nit=10	nfev=33	t=4.71s
SLSQP	run 942/1000	MSE=7.691e-09	nit=11	nfev=36	t=5.38s
SLSQP	run 943/1000	MSE=5.604e-07	nit=10	nfev=33	t=5.00s
SLSQP	run 944/1000	MSE=2.922e-07	nit=9	nfev=30	t=4.48s
SLSQP	run 945/1000	MSE=3.450e-08	nit=10	nfev=32	t=4.70s
SLSQP	run 946/1000	MSE=3.109e-08	nit=13	nfev=44	t=6.61s
SLSQP	run 947/1000	MSE=1.843e-07	nit=15	nfev=49	t=7.11s

SLSQP	run 948/1000	MSE=3.719e-08	nit=17	nfev=56	t=8.46s
SLSQP	run 949/1000	MSE=1.122e-08	nit=11	nfev=35	t=5.01s
SLSQP	run 950/1000	MSE=2.858e-08	nit=15	nfev=50	t=7.45s
SLSQP	run 951/1000	MSE=4.146e-08	nit=7	nfev=23	t=3.57s
SLSQP	run 952/1000	MSE=9.481e-10	nit=11	nfev=39	t=5.68s
SLSQP	run 953/1000	MSE=2.235e-09	nit=5	nfev=19	t=3.24s
SLSQP	run 954/1000	MSE=3.269e-08	nit=13	nfev=44	t=6.41s
SLSQP	run 955/1000	MSE=7.676e-08	nit=13	nfev=45	t=6.66s
SLSQP	run 956/1000	MSE=1.926e-07	nit=5	nfev=17	t=2.61s
SLSQP	run 957/1000	MSE=3.084e-08	nit=20	nfev=70	
t=10.28s					
SLSQP	run 958/1000	MSE=9.160e-08	nit=13	nfev=43	t=6.37s
SLSQP	run 959/1000	MSE=5.350e-10	nit=10	nfev=34	t=5.05s
SLSQP	run 960/1000	MSE=4.463e-07	nit=10	nfev=33	t=4.88s
SLSQP	run 961/1000	MSE=4.222e-07	nit=8	nfev=26	t=3.72s
SLSQP	run 962/1000	MSE=3.261e-09	nit=14	nfev=49	t=7.33s
SLSQP	run 963/1000	MSE=1.556e-08	nit=8	nfev=30	t=4.69s
SLSQP	run 964/1000	MSE=4.977e-07	nit=8	nfev=26	t=3.84s
SLSQP	run 965/1000	MSE=4.557e-07	nit=8	nfev=26	t=4.04s
SLSQP	run 966/1000	MSE=3.206e-09	nit=10	nfev=35	t=5.02s
SLSQP	run 967/1000	MSE=6.677e-08	nit=9	nfev=33	t=4.92s
SLSQP	run 968/1000	MSE=9.967e-08	nit=14	nfev=45	t=6.66s
SLSQP	run 969/1000	MSE=1.373e-08	nit=12	nfev=38	t=6.74s
SLSQP	run 970/1000	MSE=3.575e-07	nit=14	nfev=48	t=7.78s
SLSQP	run 971/1000	MSE=9.960e-09	nit=14	nfev=44	t=6.32s
SLSQP	run 972/1000	MSE=2.082e-07	nit=16	nfev=56	t=8.54s
SLSQP	run 973/1000	MSE=3.309e-08	nit=17	nfev=57	t=8.63s
SLSQP	run 974/1000	MSE=1.697e-09	nit=19	nfev=63	t=9.56s
SLSQP	run 975/1000	MSE=7.654e-08	nit=10	nfev=33	t=4.70s
SLSQP	run 976/1000	MSE=2.091e-07	nit=7	nfev=26	t=4.00s
SLSQP	run 977/1000	MSE=5.947e-08	nit=15	nfev=50	t=7.35s
SLSQP	run 978/1000	MSE=2.057e-07	nit=10	nfev=33	t=4.79s
SLSQP	run 979/1000	MSE=5.064e-07	nit=12	nfev=40	t=6.14s
SLSQP	run 980/1000	MSE=5.921e-09	nit=14	nfev=45	t=6.62s
SLSQP	run 981/1000	MSE=8.435e-09	nit=12	nfev=42	t=6.13s
SLSQP	run 982/1000	MSE=6.377e-07	nit=6	nfev=20	t=2.91s
SLSQP	run 983/1000	MSE=4.763e-07	nit=14	nfev=47	t=7.05s
SLSQP	run 984/1000	MSE=4.617e-07	nit=9	nfev=29	t=4.48s
SLSQP	run 985/1000	MSE=2.305e-07	nit=9	nfev=29	t=4.15s
SLSQP	run 986/1000	MSE=1.539e-09	nit=10	nfev=32	t=4.94s
SLSQP	run 987/1000	MSE=6.910e-09	nit=15	nfev=50	t=7.48s
SLSQP	run 988/1000	MSE=2.828e-08	nit=10	nfev=33	t=4.78s
SLSQP	run 989/1000	MSE=2.800e-09	nit=12	nfev=39	t=5.98s
SLSQP	run 990/1000	MSE=2.551e-08	nit=7	nfev=24	t=3.44s
SLSQP	run 991/1000	MSE=7.027e-08	nit=14	nfev=46	t=6.76s
SLSQP	run 992/1000	MSE=9.003e-09	nit=5	nfev=18	t=2.75s
SLSQP	run 993/1000	MSE=1.159e-07	nit=12	nfev=39	t=5.77s
SLSQP	run 994/1000	MSE=2.511e-07	nit=8	nfev=26	t=3.96s

SLSQP	run 995/1000	MSE=6.105e-09	nit=10	nfev=33	t=4.96s
SLSQP	run 996/1000	MSE=3.649e-07	nit=13	nfev=44	t=6.54s
SLSQP	run 997/1000	MSE=2.946e-09	nit=13	nfev=45	t=6.63s
SLSQP	run 998/1000	MSE=7.743e-08	nit=12	nfev=41	t=6.13s
SLSQP	run 999/1000	MSE=2.741e-07	nit=12	nfev=38	t=5.72s
SLSQP	run 1000/1000	MSE=8.532e-08	nit=10	nfev=32	

t=4.65s

Scenario A - COBYLA (1000 runs)

COBYLA	run 1/1000	MSE=4.130e-05	nit=-1	nfev=131	t=19.81s
COBYLA	run 2/1000	MSE=4.659e-05	nit=-1	nfev=134	t=20.25s
COBYLA	run 3/1000	MSE=5.904e-05	nit=-1	nfev=395	t=60.50s
COBYLA	run 4/1000	MSE=6.246e-05	nit=-1	nfev=473	t=71.50s
COBYLA	run 5/1000	MSE=1.686e-05	nit=-1	nfev=95	t=14.45s
COBYLA	run 6/1000	MSE=1.024e-01	nit=-1	nfev=22	t=3.22s
COBYLA	run 7/1000	MSE=5.502e-05	nit=-1	nfev=132	t=19.63s
COBYLA	run 8/1000	MSE=1.878e-05	nit=-1	nfev=127	t=19.27s
COBYLA	run 9/1000	MSE=1.601e-04	nit=-1	nfev=529	t=80.18s
COBYLA	run 10/1000	MSE=4.745e-05	nit=-1	nfev=111	t=16.58s
COBYLA	run 11/1000	MSE=3.733e-05	nit=-1	nfev=575	t=91.58s
COBYLA	run 12/1000	MSE=5.006e-05	nit=-1	nfev=558	t=83.54s
COBYLA	run 13/1000	MSE=8.028e-05	nit=-1	nfev=96	t=14.65s
COBYLA	run 14/1000	MSE=1.110e-04	nit=-1	nfev=432	t=65.00s
COBYLA	run 15/1000	MSE=2.293e-05	nit=-1	nfev=181	t=27.19s
COBYLA	run 16/1000	MSE=2.612e-03	nit=-1	nfev=28	t=4.29s
COBYLA	run 17/1000	MSE=2.757e-05	nit=-1	nfev=570	t=86.92s
COBYLA	run 18/1000	MSE=2.336e-05	nit=-1	nfev=119	t=18.10s
COBYLA	run 19/1000	MSE=6.847e-05	nit=-1	nfev=146	t=21.91s
COBYLA	run 20/1000	MSE=3.757e-05	nit=-1	nfev=62	t=9.38s
COBYLA	run 21/1000	MSE=1.639e-05	nit=-1	nfev=38	t=5.94s
COBYLA	run 22/1000	MSE=3.193e-06	nit=-1	nfev=51	t=7.58s
COBYLA	run 23/1000	MSE=5.636e-05	nit=-1	nfev=521	t=79.15s
COBYLA	run 24/1000	MSE=4.422e-05	nit=-1	nfev=102	t=15.61s
COBYLA	run 25/1000	MSE=1.930e-05	nit=-1	nfev=105	t=16.27s
COBYLA	run 26/1000	MSE=9.052e-05	nit=-1	nfev=177	t=26.55s
COBYLA	run 27/1000	MSE=1.680e-05	nit=-1	nfev=125	t=18.87s
COBYLA	run 28/1000	MSE=4.347e-05	nit=-1	nfev=132	t=20.03s
COBYLA	run 29/1000	MSE=3.733e-05	nit=-1	nfev=575	t=86.69s
COBYLA	run 30/1000	MSE=2.667e-05	nit=-1	nfev=136	t=20.38s
COBYLA	run 31/1000	MSE=4.310e-06	nit=-1	nfev=46	t=7.22s
COBYLA	run 32/1000	MSE=1.063e-04	nit=-1	nfev=100	t=14.94s
COBYLA	run 33/1000	MSE=3.186e-05	nit=-1	nfev=39	t=5.90s
COBYLA	run 34/1000	MSE=4.523e-05	nit=-1	nfev=99	t=14.97s
COBYLA	run 35/1000	MSE=3.114e-05	nit=-1	nfev=159	t=23.98s
COBYLA	run 36/1000	MSE=1.423e-05	nit=-1	nfev=448	t=67.59s
COBYLA	run 37/1000	MSE=5.031e-05	nit=-1	nfev=505	t=76.25s
COBYLA	run 38/1000	MSE=1.494e-05	nit=-1	nfev=153	t=22.99s
COBYLA	run 39/1000	MSE=3.733e-05	nit=-1	nfev=575	t=86.60s

COBYLA	run 40/1000	MSE=3.715e-06	nit=-1	nfev=93	t=14.01s
COBYLA	run 41/1000	MSE=2.032e-05	nit=-1	nfev=199	t=29.77s
COBYLA	run 42/1000	MSE=6.735e-05	nit=-1	nfev=575	t=86.05s
COBYLA	run 43/1000	MSE=3.733e-05	nit=-1	nfev=575	t=86.39s
COBYLA	run 44/1000	MSE=4.461e-07	nit=-1	nfev=110	t=16.61s
COBYLA	run 45/1000	MSE=1.024e-01	nit=-1	nfev=23	t=3.43s
COBYLA	run 46/1000	MSE=4.204e-05	nit=-1	nfev=168	t=25.98s
COBYLA	run 47/1000	MSE=5.537e-05	nit=-1	nfev=49	t=7.07s
COBYLA	run 48/1000	MSE=1.402e-05	nit=-1	nfev=129	t=19.40s
COBYLA	run 49/1000	MSE=5.904e-05	nit=-1	nfev=394	t=60.02s
COBYLA	run 50/1000	MSE=1.073e-04	nit=-1	nfev=114	t=16.99s
COBYLA	run 51/1000	MSE=6.308e-05	nit=-1	nfev=454	t=68.14s
COBYLA	run 52/1000	MSE=4.463e-05	nit=-1	nfev=119	t=17.72s
COBYLA	run 53/1000	MSE=3.361e-05	nit=-1	nfev=56	t=8.41s
COBYLA	run 54/1000	MSE=1.115e-04	nit=-1	nfev=120	t=18.48s
COBYLA	run 55/1000	MSE=5.853e-05	nit=-1	nfev=100	t=15.06s
COBYLA	run 56/1000	MSE=3.989e-05	nit=-1	nfev=508	t=76.24s
COBYLA	run 57/1000	MSE=8.415e-06	nit=-1	nfev=91	t=13.75s
COBYLA	run 58/1000	MSE=1.134e-04	nit=-1	nfev=498	t=75.42s
COBYLA	run 59/1000	MSE=4.092e-05	nit=-1	nfev=115	t=17.50s
COBYLA	run 60/1000	MSE=2.684e-03	nit=-1	nfev=29	t=4.19s
COBYLA	run 61/1000	MSE=5.954e-05	nit=-1	nfev=557	t=83.88s
COBYLA	run 62/1000	MSE=6.172e-05	nit=-1	nfev=181	t=26.93s
COBYLA	run 63/1000	MSE=7.656e-05	nit=-1	nfev=252	t=37.69s
COBYLA	run 64/1000	MSE=9.983e-05	nit=-1	nfev=76	t=11.31s
COBYLA	run 65/1000	MSE=6.818e-06	nit=-1	nfev=129	t=19.53s
COBYLA	run 66/1000	MSE=1.963e-06	nit=-1	nfev=118	t=17.62s
COBYLA	run 67/1000	MSE=4.509e-05	nit=-1	nfev=509	t=76.22s
COBYLA	run 68/1000	MSE=2.557e-05	nit=-1	nfev=484	t=72.68s
COBYLA	run 69/1000	MSE=3.834e-07	nit=-1	nfev=97	t=14.59s
COBYLA	run 70/1000	MSE=8.087e-05	nit=-1	nfev=94	t=14.23s
COBYLA	run 71/1000	MSE=2.158e-05	nit=-1	nfev=164	t=24.89s
COBYLA	run 72/1000	MSE=4.893e-05	nit=-1	nfev=111	t=16.62s
COBYLA	run 73/1000	MSE=7.027e-05	nit=-1	nfev=503	t=75.71s
COBYLA	run 74/1000	MSE=6.957e-05	nit=-1	nfev=124	t=18.47s
COBYLA	run 75/1000	MSE=2.966e-05	nit=-1	nfev=81	t=12.05s
COBYLA	run 76/1000	MSE=2.555e-04	nit=-1	nfev=102	t=15.25s
COBYLA	run 77/1000	MSE=8.793e-05	nit=-1	nfev=142	t=21.46s
COBYLA	run 78/1000	MSE=1.985e-05	nit=-1	nfev=71	t=10.43s
COBYLA	run 79/1000	MSE=4.944e-05	nit=-1	nfev=459	t=69.15s
COBYLA	run 80/1000	MSE=1.178e-05	nit=-1	nfev=166	t=24.73s
COBYLA	run 81/1000	MSE=6.921e-05	nit=-1	nfev=288	t=43.19s
COBYLA	run 82/1000	MSE=8.343e-06	nit=-1	nfev=526	t=79.50s
COBYLA	run 83/1000	MSE=6.547e-05	nit=-1	nfev=113	t=17.32s
COBYLA	run 84/1000	MSE=4.841e-05	nit=-1	nfev=66	t=9.91s
COBYLA	run 85/1000	MSE=2.873e-05	nit=-1	nfev=135	t=20.48s
COBYLA	run 86/1000	MSE=2.757e-05	nit=-1	nfev=570	t=85.78s
COBYLA	run 87/1000	MSE=3.561e-06	nit=-1	nfev=65	t=9.78s

COBYLA	run 88/1000	MSE=1.275e-04	nit=-1	nfev=480	t=73.01s
COBYLA	run 89/1000	MSE=5.025e-05	nit=-1	nfev=203	t=30.60s
COBYLA	run 90/1000	MSE=4.921e-05	nit=-1	nfev=193	t=29.41s
COBYLA	run 91/1000	MSE=3.733e-05	nit=-1	nfev=575	t=86.10s
COBYLA	run 92/1000	MSE=1.063e-06	nit=-1	nfev=33	t=5.04s
COBYLA	run 93/1000	MSE=6.270e-05	nit=-1	nfev=198	t=29.85s
COBYLA	run 94/1000	MSE=5.762e-05	nit=-1	nfev=110	t=16.63s
COBYLA	run 95/1000	MSE=9.128e-07	nit=-1	nfev=34	t=4.94s
COBYLA	run 96/1000	MSE=2.732e-05	nit=-1	nfev=129	t=19.61s
COBYLA	run 97/1000	MSE=1.164e-05	nit=-1	nfev=128	t=19.05s
COBYLA	run 98/1000	MSE=4.272e-05	nit=-1	nfev=123	t=18.73s
COBYLA	run 99/1000	MSE=2.942e-05	nit=-1	nfev=133	t=20.70s
COBYLA	run 100/1000	MSE=1.600e-04	nit=-1	nfev=474	
t=71.89s					
COBYLA	run 101/1000	MSE=3.300e-05	nit=-1	nfev=134	
t=20.32s					
COBYLA	run 102/1000	MSE=2.635e-05	nit=-1	nfev=222	
t=33.46s					
COBYLA	run 103/1000	MSE=2.278e-05	nit=-1	nfev=528	
t=79.54s					
COBYLA	run 104/1000	MSE=7.593e-05	nit=-1	nfev=105	
t=15.86s					
COBYLA	run 105/1000	MSE=4.423e-05	nit=-1	nfev=128	
t=19.32s					
COBYLA	run 106/1000	MSE=5.904e-05	nit=-1	nfev=395	
t=59.66s					
COBYLA	run 107/1000	MSE=1.601e-04	nit=-1	nfev=529	
t=80.23s					
COBYLA	run 108/1000	MSE=2.364e-05	nit=-1	nfev=94	
t=14.05s					
COBYLA	run 109/1000	MSE=4.359e-05	nit=-1	nfev=473	
t=71.19s					
COBYLA	run 110/1000	MSE=1.025e-01	nit=-1	nfev=17	t=2.70s
COBYLA	run 111/1000	MSE=3.015e-05	nit=-1	nfev=122	
t=18.10s					
COBYLA	run 112/1000	MSE=3.558e-07	nit=-1	nfev=162	
t=24.34s					
COBYLA	run 113/1000	MSE=8.878e-05	nit=-1	nfev=193	
t=29.29s					
COBYLA	run 114/1000	MSE=5.713e-05	nit=-1	nfev=489	
t=73.87s					
COBYLA	run 115/1000	MSE=1.190e-05	nit=-1	nfev=152	
t=23.06s					
COBYLA	run 116/1000	MSE=1.368e-04	nit=-1	nfev=97	
t=14.36s					
COBYLA	run 117/1000	MSE=1.089e-04	nit=-1	nfev=119	
t=17.90s					
COBYLA	run 118/1000	MSE=8.867e-05	nit=-1	nfev=87	

t=12.96s						
COBYLA	run 119/1000	MSE=5.904e-05	nit=-1	nfev=395		
t=59.44s						
COBYLA	run 120/1000	MSE=1.852e-05	nit=-1	nfev=135		
t=20.11s						
COBYLA	run 121/1000	MSE=6.773e-05	nit=-1	nfev=122		
t=18.22s						
COBYLA	run 122/1000	MSE=1.601e-04	nit=-1	nfev=529		
t=79.53s						
COBYLA	run 123/1000	MSE=6.148e-06	nit=-1	nfev=118		
t=17.50s						
COBYLA	run 124/1000	MSE=3.565e-05	nit=-1	nfev=105		
t=16.14s						
COBYLA	run 125/1000	MSE=1.025e-01	nit=-1	nfev=17	t=2.84s	
COBYLA	run 126/1000	MSE=5.285e-05	nit=-1	nfev=431		
t=71.37s						
COBYLA	run 127/1000	MSE=9.816e-05	nit=-1	nfev=498		
t=75.54s						
COBYLA	run 128/1000	MSE=1.595e-05	nit=-1	nfev=112		
t=16.72s						
COBYLA	run 129/1000	MSE=3.385e-06	nit=-1	nfev=56	t=8.41s	
COBYLA	run 130/1000	MSE=2.311e-04	nit=-1	nfev=58	t=8.89s	
COBYLA	run 131/1000	MSE=6.754e-05	nit=-1	nfev=136		
t=20.46s						
COBYLA	run 132/1000	MSE=3.080e-05	nit=-1	nfev=126		
t=18.93s						
COBYLA	run 133/1000	MSE=5.079e-05	nit=-1	nfev=81		
t=12.26s						
COBYLA	run 134/1000	MSE=6.246e-05	nit=-1	nfev=473		
t=71.01s						
COBYLA	run 135/1000	MSE=4.025e-05	nit=-1	nfev=576		
t=87.50s						
COBYLA	run 136/1000	MSE=1.387e-06	nit=-1	nfev=107		
t=16.10s						
COBYLA	run 137/1000	MSE=1.618e-05	nit=-1	nfev=84		
t=13.06s						
COBYLA	run 138/1000	MSE=1.571e-04	nit=-1	nfev=553		
t=83.88s						
COBYLA	run 139/1000	MSE=5.959e-05	nit=-1	nfev=100		
t=15.06s						
COBYLA	run 140/1000	MSE=5.715e-05	nit=-1	nfev=121		
t=18.60s						
COBYLA	run 141/1000	MSE=5.619e-05	nit=-1	nfev=117		
t=17.54s						
COBYLA	run 142/1000	MSE=3.733e-05	nit=-1	nfev=575		
t=87.48s						
COBYLA	run 143/1000	MSE=9.478e-05	nit=-1	nfev=113		
t=17.35s						

COBYLA	run 144/1000	MSE=1.590e-04	nit=-1	nfev=74	
t=10.98s					
COBYLA	run 145/1000	MSE=9.429e-05	nit=-1	nfev=101	
t=15.13s					
COBYLA	run 146/1000	MSE=6.062e-05	nit=-1	nfev=156	
t=24.39s					
COBYLA	run 147/1000	MSE=4.508e-05	nit=-1	nfev=196	
t=29.45s					
COBYLA	run 148/1000	MSE=5.680e-05	nit=-1	nfev=117	
t=17.64s					
COBYLA	run 149/1000	MSE=1.099e-04	nit=-1	nfev=44	t=6.39s
COBYLA	run 150/1000	MSE=1.034e-04	nit=-1	nfev=161	
t=24.26s					
COBYLA	run 151/1000	MSE=6.668e-05	nit=-1	nfev=468	
t=70.62s					
COBYLA	run 152/1000	MSE=8.807e-09	nit=-1	nfev=69	
t=10.52s					
COBYLA	run 153/1000	MSE=1.044e-04	nit=-1	nfev=476	
t=71.68s					
COBYLA	run 154/1000	MSE=1.119e-04	nit=-1	nfev=505	
t=77.04s					
COBYLA	run 155/1000	MSE=5.343e-05	nit=-1	nfev=279	
t=41.62s					
COBYLA	run 156/1000	MSE=3.131e-05	nit=-1	nfev=164	
t=25.06s					
COBYLA	run 157/1000	MSE=1.565e-05	nit=-1	nfev=494	
t=74.38s					
COBYLA	run 158/1000	MSE=6.725e-05	nit=-1	nfev=416	
t=63.18s					
COBYLA	run 159/1000	MSE=5.904e-05	nit=-1	nfev=395	
t=61.19s					
COBYLA	run 160/1000	MSE=6.048e-05	nit=-1	nfev=553	
t=83.36s					
COBYLA	run 161/1000	MSE=9.625e-05	nit=-1	nfev=110	
t=16.44s					
COBYLA	run 162/1000	MSE=8.047e-05	nit=-1	nfev=111	
t=16.61s					
COBYLA	run 163/1000	MSE=2.533e-05	nit=-1	nfev=54	t=8.23s
COBYLA	run 164/1000	MSE=4.397e-05	nit=-1	nfev=112	
t=16.98s					
COBYLA	run 165/1000	MSE=6.533e-06	nit=-1	nfev=138	
t=20.68s					
COBYLA	run 166/1000	MSE=2.175e-05	nit=-1	nfev=241	
t=36.31s					
COBYLA	run 167/1000	MSE=4.991e-06	nit=-1	nfev=593	
t=88.54s					
COBYLA	run 168/1000	MSE=4.231e-05	nit=-1	nfev=445	
t=66.84s					

COBYLA	run 169/1000	MSE=8.456e-06	nit=-1	nfev=232	
t=34.96s					
COBYLA	run 170/1000	MSE=8.143e-05	nit=-1	nfev=433	
t=65.04s					
COBYLA	run 171/1000	MSE=3.635e-05	nit=-1	nfev=129	
t=19.56s					
COBYLA	run 172/1000	MSE=1.848e-05	nit=-1	nfev=180	
t=27.00s					
COBYLA	run 173/1000	MSE=3.377e-06	nit=-1	nfev=66	t=9.91s
COBYLA	run 174/1000	MSE=1.283e-05	nit=-1	nfev=116	
t=18.33s					
COBYLA	run 175/1000	MSE=1.692e-05	nit=-1	nfev=75	
t=11.21s					
COBYLA	run 176/1000	MSE=5.738e-05	nit=-1	nfev=502	
t=75.67s					
COBYLA	run 177/1000	MSE=5.114e-05	nit=-1	nfev=488	
t=73.29s					
COBYLA	run 178/1000	MSE=3.528e-05	nit=-1	nfev=134	
t=20.03s					
COBYLA	run 179/1000	MSE=6.246e-05	nit=-1	nfev=473	
t=70.98s					
COBYLA	run 180/1000	MSE=8.506e-05	nit=-1	nfev=179	
t=26.82s					
COBYLA	run 181/1000	MSE=1.884e-06	nit=-1	nfev=128	
t=19.20s					
COBYLA	run 182/1000	MSE=4.029e-05	nit=-1	nfev=138	
t=20.62s					
COBYLA	run 183/1000	MSE=5.639e-05	nit=-1	nfev=66	
t=10.19s					
COBYLA	run 184/1000	MSE=8.362e-06	nit=-1	nfev=50	t=7.36s
COBYLA	run 185/1000	MSE=2.895e-05	nit=-1	nfev=500	
t=74.89s					
COBYLA	run 186/1000	MSE=3.861e-05	nit=-1	nfev=124	
t=18.55s					
COBYLA	run 187/1000	MSE=9.942e-05	nit=-1	nfev=102	
t=15.32s					
COBYLA	run 188/1000	MSE=5.316e-06	nit=-1	nfev=58	t=8.69s
COBYLA	run 189/1000	MSE=5.080e-05	nit=-1	nfev=83	
t=12.44s					
COBYLA	run 190/1000	MSE=3.803e-05	nit=-1	nfev=194	
t=28.97s					
COBYLA	run 191/1000	MSE=2.241e-05	nit=-1	nfev=65	t=9.84s
COBYLA	run 192/1000	MSE=1.413e-04	nit=-1	nfev=501	
t=75.40s					
COBYLA	run 193/1000	MSE=5.597e-06	nit=-1	nfev=119	
t=18.04s					
COBYLA	run 194/1000	MSE=4.617e-05	nit=-1	nfev=113	
t=16.94s					

COBYLA	run 195/1000	MSE=2.235e-05	nit=-1	nfev=125	
t=18.82s					
COBYLA	run 196/1000	MSE=5.904e-05	nit=-1	nfev=395	
t=59.30s					
COBYLA	run 197/1000	MSE=2.800e-05	nit=-1	nfev=119	
t=17.84s					
COBYLA	run 198/1000	MSE=1.071e-04	nit=-1	nfev=447	
t=67.10s					
COBYLA	run 199/1000	MSE=6.562e-05	nit=-1	nfev=58	t=8.59s
COBYLA	run 200/1000	MSE=2.790e-05	nit=-1	nfev=134	
t=20.24s					
COBYLA	run 201/1000	MSE=6.308e-05	nit=-1	nfev=454	
t=68.18s					
COBYLA	run 202/1000	MSE=3.733e-05	nit=-1	nfev=575	
t=86.15s					
COBYLA	run 203/1000	MSE=5.789e-06	nit=-1	nfev=102	
t=15.31s					
COBYLA	run 204/1000	MSE=3.405e-05	nit=-1	nfev=95	
t=14.35s					
COBYLA	run 205/1000	MSE=2.308e-05	nit=-1	nfev=135	
t=20.23s					
COBYLA	run 206/1000	MSE=1.135e-05	nit=-1	nfev=170	
t=25.42s					
COBYLA	run 207/1000	MSE=5.904e-05	nit=-1	nfev=394	
t=58.93s					
COBYLA	run 208/1000	MSE=3.113e-05	nit=-1	nfev=417	
t=62.58s					
COBYLA	run 209/1000	MSE=5.157e-06	nit=-1	nfev=123	
t=18.28s					
COBYLA	run 210/1000	MSE=1.811e-05	nit=-1	nfev=97	
t=14.88s					
COBYLA	run 211/1000	MSE=5.658e-05	nit=-1	nfev=91	
t=13.52s					
COBYLA	run 212/1000	MSE=1.970e-05	nit=-1	nfev=497	
t=75.78s					
COBYLA	run 213/1000	MSE=6.668e-05	nit=-1	nfev=467	
t=70.27s					
COBYLA	run 214/1000	MSE=5.098e-05	nit=-1	nfev=438	
t=67.36s					
COBYLA	run 215/1000	MSE=4.531e-05	nit=-1	nfev=131	
t=19.49s					
COBYLA	run 216/1000	MSE=6.251e-05	nit=-1	nfev=525	
t=78.91s					
COBYLA	run 217/1000	MSE=5.992e-05	nit=-1	nfev=65	t=9.69s
COBYLA	run 218/1000	MSE=4.221e-05	nit=-1	nfev=118	
t=17.71s					
COBYLA	run 219/1000	MSE=1.351e-04	nit=-1	nfev=171	
t=25.46s					

COBYLA	run 220/1000	MSE=4.247e-05	nit=-1	nfev=543	
t=81.42s					
COBYLA	run 221/1000	MSE=1.460e-06	nit=-1	nfev=90	
t=13.32s					
COBYLA	run 222/1000	MSE=8.306e-05	nit=-1	nfev=110	
t=16.62s					
COBYLA	run 223/1000	MSE=3.636e-05	nit=-1	nfev=165	
t=25.03s					
COBYLA	run 224/1000	MSE=5.664e-05	nit=-1	nfev=88	
t=13.19s					
COBYLA	run 225/1000	MSE=3.733e-05	nit=-1	nfev=575	
t=86.52s					
COBYLA	run 226/1000	MSE=1.121e-04	nit=-1	nfev=584	
t=88.13s					
COBYLA	run 227/1000	MSE=3.968e-05	nit=-1	nfev=106	
t=15.91s					
COBYLA	run 228/1000	MSE=5.958e-05	nit=-1	nfev=566	
t=84.64s					
COBYLA	run 229/1000	MSE=8.909e-05	nit=-1	nfev=103	
t=16.26s					
COBYLA	run 230/1000	MSE=2.491e-05	nit=-1	nfev=132	
t=19.77s					
COBYLA	run 231/1000	MSE=4.634e-05	nit=-1	nfev=151	
t=22.69s					
COBYLA	run 232/1000	MSE=4.055e-06	nit=-1	nfev=507	
t=78.20s					
COBYLA	run 233/1000	MSE=4.650e-05	nit=-1	nfev=57	
t=10.81s					
COBYLA	run 234/1000	MSE=4.407e-05	nit=-1	nfev=554	
t=83.21s					
COBYLA	run 235/1000	MSE=2.445e-05	nit=-1	nfev=101	
t=15.09s					
COBYLA	run 236/1000	MSE=4.151e-05	nit=-1	nfev=116	
t=17.73s					
COBYLA	run 237/1000	MSE=4.938e-05	nit=-1	nfev=518	
t=77.28s					
COBYLA	run 238/1000	MSE=3.560e-05	nit=-1	nfev=107	
t=15.99s					
COBYLA	run 239/1000	MSE=2.913e-05	nit=-1	nfev=543	
t=82.28s					
COBYLA	run 240/1000	MSE=2.535e-05	nit=-1	nfev=211	
t=32.12s					
COBYLA	run 241/1000	MSE=3.733e-05	nit=-1	nfev=575	
t=86.20s					
COBYLA	run 242/1000	MSE=2.238e-05	nit=-1	nfev=576	
t=86.36s					
COBYLA	run 243/1000	MSE=4.742e-06	nit=-1	nfev=34	t=4.98s
COBYLA	run 244/1000	MSE=4.810e-05	nit=-1	nfev=77	

t=11.69s						
COBYLA	run 245/1000	MSE=6.668e-05	nit=-1	nfev=467		
t=69.67s						
COBYLA	run 246/1000	MSE=1.155e-04	nit=-1	nfev=379		
t=56.77s						
COBYLA	run 247/1000	MSE=6.496e-05	nit=-1	nfev=477		
t=71.15s						
COBYLA	run 248/1000	MSE=6.375e-05	nit=-1	nfev=134		
t=19.95s						
COBYLA	run 249/1000	MSE=1.264e-05	nit=-1	nfev=48	t=7.26s	
COBYLA	run 250/1000	MSE=2.859e-06	nit=-1	nfev=119		
t=17.65s						
COBYLA	run 251/1000	MSE=5.565e-05	nit=-1	nfev=435		
t=65.31s						
COBYLA	run 252/1000	MSE=5.412e-06	nit=-1	nfev=110		
t=17.67s						
COBYLA	run 253/1000	MSE=9.117e-05	nit=-1	nfev=197		
t=29.38s						
COBYLA	run 254/1000	MSE=4.097e-05	nit=-1	nfev=564		
t=84.93s						
COBYLA	run 255/1000	MSE=4.811e-05	nit=-1	nfev=469		
t=69.82s						
COBYLA	run 256/1000	MSE=3.352e-05	nit=-1	nfev=54	t=7.99s	
COBYLA	run 257/1000	MSE=4.021e-05	nit=-1	nfev=521		
t=78.61s						
COBYLA	run 258/1000	MSE=1.039e-04	nit=-1	nfev=175		
t=26.42s						
COBYLA	run 259/1000	MSE=4.937e-05	nit=-1	nfev=167		
t=24.95s						
COBYLA	run 260/1000	MSE=3.733e-05	nit=-1	nfev=575		
t=86.05s						
COBYLA	run 261/1000	MSE=4.749e-05	nit=-1	nfev=122		
t=18.09s						
COBYLA	run 262/1000	MSE=1.135e-04	nit=-1	nfev=251		
t=39.91s						
COBYLA	run 263/1000	MSE=3.137e-05	nit=-1	nfev=115		
t=17.50s						
COBYLA	run 264/1000	MSE=1.284e-05	nit=-1	nfev=130		
t=19.54s						
COBYLA	run 265/1000	MSE=2.243e-06	nit=-1	nfev=109		
t=16.41s						
COBYLA	run 266/1000	MSE=9.481e-06	nit=-1	nfev=36	t=5.35s	
COBYLA	run 267/1000	MSE=1.377e-04	nit=-1	nfev=109		
t=16.33s						
COBYLA	run 268/1000	MSE=1.246e-04	nit=-1	nfev=62	t=9.46s	
COBYLA	run 269/1000	MSE=1.386e-06	nit=-1	nfev=67		
t=10.07s						
COBYLA	run 270/1000	MSE=1.023e-04	nit=-1	nfev=603		

t=90.92s						
COBYLA	run 271/1000	MSE=1.263e-04	nit=-1	nfev=146		
t=21.94s						
COBYLA	run 272/1000	MSE=3.423e-05	nit=-1	nfev=100		
t=14.99s						
COBYLA	run 273/1000	MSE=7.656e-06	nit=-1	nfev=129		
t=20.04s						
COBYLA	run 274/1000	MSE=5.071e-05	nit=-1	nfev=111		
t=16.60s						
COBYLA	run 275/1000	MSE=1.318e-04	nit=-1	nfev=88		
t=13.23s						
COBYLA	run 276/1000	MSE=2.140e-04	nit=-1	nfev=482		
t=72.30s						
COBYLA	run 277/1000	MSE=1.163e-04	nit=-1	nfev=57	t=8.55s	
COBYLA	run 278/1000	MSE=1.601e-04	nit=-1	nfev=529		
t=79.15s						
COBYLA	run 279/1000	MSE=2.129e-05	nit=-1	nfev=97		
t=14.54s						
COBYLA	run 280/1000	MSE=1.636e-05	nit=-1	nfev=486		
t=72.75s						
COBYLA	run 281/1000	MSE=3.402e-05	nit=-1	nfev=118		
t=17.84s						
COBYLA	run 282/1000	MSE=8.104e-05	nit=-1	nfev=178		
t=27.16s						
COBYLA	run 283/1000	MSE=3.333e-05	nit=-1	nfev=135		
t=20.64s						
COBYLA	run 284/1000	MSE=1.216e-05	nit=-1	nfev=114		
t=17.16s						
COBYLA	run 285/1000	MSE=1.068e-06	nit=-1	nfev=77		
t=11.39s						
COBYLA	run 286/1000	MSE=5.268e-05	nit=-1	nfev=512		
t=76.66s						
COBYLA	run 287/1000	MSE=5.958e-05	nit=-1	nfev=566		
t=87.80s						
COBYLA	run 288/1000	MSE=3.073e-05	nit=-1	nfev=564		
t=85.07s						
COBYLA	run 289/1000	MSE=8.556e-05	nit=-1	nfev=562		
t=84.08s						
COBYLA	run 290/1000	MSE=5.446e-05	nit=-1	nfev=94		
t=14.14s						
COBYLA	run 291/1000	MSE=2.836e-05	nit=-1	nfev=584		
t=87.79s						
COBYLA	run 292/1000	MSE=4.128e-05	nit=-1	nfev=80		
t=12.13s						
COBYLA	run 293/1000	MSE=1.818e-05	nit=-1	nfev=565		
t=84.76s						
COBYLA	run 294/1000	MSE=3.989e-05	nit=-1	nfev=508		
t=75.90s						

COBYLA	run 295/1000	MSE=1.206e-05	nit=-1	nfev=99	
t=14.88s					
COBYLA	run 296/1000	MSE=1.155e-05	nit=-1	nfev=95	
t=14.45s					
COBYLA	run 297/1000	MSE=3.821e-05	nit=-1	nfev=131	
t=20.32s					
COBYLA	run 298/1000	MSE=1.428e-05	nit=-1	nfev=111	
t=16.55s					
COBYLA	run 299/1000	MSE=3.682e-05	nit=-1	nfev=130	
t=19.78s					
COBYLA	run 300/1000	MSE=5.122e-05	nit=-1	nfev=473	
t=71.61s					
COBYLA	run 301/1000	MSE=6.691e-05	nit=-1	nfev=545	
t=82.05s					
COBYLA	run 302/1000	MSE=3.873e-05	nit=-1	nfev=130	
t=19.69s					
COBYLA	run 303/1000	MSE=1.931e-05	nit=-1	nfev=117	
t=17.26s					
COBYLA	run 304/1000	MSE=5.630e-05	nit=-1	nfev=460	
t=69.98s					
COBYLA	run 305/1000	MSE=2.506e-05	nit=-1	nfev=128	
t=19.16s					
COBYLA	run 306/1000	MSE=2.658e-03	nit=-1	nfev=32	t=5.46s
COBYLA	run 307/1000	MSE=1.310e-04	nit=-1	nfev=98	
t=15.05s					
COBYLA	run 308/1000	MSE=2.785e-05	nit=-1	nfev=128	
t=19.36s					
COBYLA	run 309/1000	MSE=3.777e-05	nit=-1	nfev=216	
t=32.82s					
COBYLA	run 310/1000	MSE=1.209e-05	nit=-1	nfev=109	
t=16.43s					
COBYLA	run 311/1000	MSE=3.733e-05	nit=-1	nfev=575	
t=86.11s					
COBYLA	run 312/1000	MSE=4.973e-07	nit=-1	nfev=45	t=6.76s
COBYLA	run 313/1000	MSE=3.471e-05	nit=-1	nfev=150	
t=22.75s					
COBYLA	run 314/1000	MSE=3.114e-05	nit=-1	nfev=122	
t=18.25s					
COBYLA	run 315/1000	MSE=6.317e-05	nit=-1	nfev=139	
t=20.54s					
COBYLA	run 316/1000	MSE=3.082e-05	nit=-1	nfev=113	
t=17.15s					
COBYLA	run 317/1000	MSE=9.154e-05	nit=-1	nfev=137	
t=20.69s					
COBYLA	run 318/1000	MSE=6.269e-05	nit=-1	nfev=97	
t=14.62s					
COBYLA	run 319/1000	MSE=6.095e-05	nit=-1	nfev=130	
t=19.40s					

COBYLA	run 320/1000	MSE=5.310e-05	nit=-1	nfev=135	
t=20.74s					
COBYLA	run 321/1000	MSE=6.725e-05	nit=-1	nfev=416	
t=62.66s					
COBYLA	run 322/1000	MSE=8.114e-05	nit=-1	nfev=512	
t=76.96s					
COBYLA	run 323/1000	MSE=3.478e-05	nit=-1	nfev=86	
t=13.07s					
COBYLA	run 324/1000	MSE=4.614e-05	nit=-1	nfev=129	
t=19.31s					
COBYLA	run 325/1000	MSE=1.498e-04	nit=-1	nfev=414	
t=62.21s					
COBYLA	run 326/1000	MSE=1.155e-04	nit=-1	nfev=380	
t=56.80s					
COBYLA	run 327/1000	MSE=4.196e-05	nit=-1	nfev=190	
t=28.60s					
COBYLA	run 328/1000	MSE=2.664e-06	nit=-1	nfev=32	t=4.88s
COBYLA	run 329/1000	MSE=1.677e-05	nit=-1	nfev=529	
t=80.11s					
COBYLA	run 330/1000	MSE=4.733e-05	nit=-1	nfev=101	
t=15.44s					
COBYLA	run 331/1000	MSE=1.440e-06	nit=-1	nfev=30	t=4.42s
COBYLA	run 332/1000	MSE=4.509e-05	nit=-1	nfev=510	
t=76.51s					
COBYLA	run 333/1000	MSE=3.733e-05	nit=-1	nfev=575	
t=86.72s					
COBYLA	run 334/1000	MSE=1.925e-05	nit=-1	nfev=64	t=9.66s
COBYLA	run 335/1000	MSE=5.904e-05	nit=-1	nfev=395	
t=65.58s					
COBYLA	run 336/1000	MSE=4.299e-05	nit=-1	nfev=115	
t=17.64s					
COBYLA	run 337/1000	MSE=3.680e-05	nit=-1	nfev=508	
t=76.84s					
COBYLA	run 338/1000	MSE=1.024e-04	nit=-1	nfev=132	
t=20.79s					
COBYLA	run 339/1000	MSE=6.308e-05	nit=-1	nfev=454	
t=69.19s					
COBYLA	run 340/1000	MSE=4.237e-05	nit=-1	nfev=91	
t=13.68s					
COBYLA	run 341/1000	MSE=5.580e-05	nit=-1	nfev=46	t=6.84s
COBYLA	run 342/1000	MSE=2.937e-06	nit=-1	nfev=119	
t=18.07s					
COBYLA	run 343/1000	MSE=1.627e-05	nit=-1	nfev=126	
t=19.11s					
COBYLA	run 344/1000	MSE=2.074e-05	nit=-1	nfev=105	
t=15.63s					
COBYLA	run 345/1000	MSE=2.238e-05	nit=-1	nfev=575	
t=89.14s					

COBYLA	run 346/1000	MSE=2.663e-03	nit=-1	nfev=29	t=4.53s
COBYLA	run 347/1000	MSE=1.275e-04	nit=-1	nfev=480	
t=77.93s					
COBYLA	run 348/1000	MSE=3.989e-05	nit=-1	nfev=508	
t=77.18s					
COBYLA	run 349/1000	MSE=1.840e-04	nit=-1	nfev=97	
t=14.86s					
COBYLA	run 350/1000	MSE=4.064e-05	nit=-1	nfev=169	
t=25.91s					
COBYLA	run 351/1000	MSE=3.809e-05	nit=-1	nfev=140	
t=21.04s					
COBYLA	run 352/1000	MSE=4.848e-05	nit=-1	nfev=93	
t=14.03s					
COBYLA	run 353/1000	MSE=9.267e-05	nit=-1	nfev=102	
t=15.81s					
COBYLA	run 354/1000	MSE=5.904e-05	nit=-1	nfev=395	
t=59.47s					
COBYLA	run 355/1000	MSE=5.180e-06	nit=-1	nfev=66	
t=10.06s					
COBYLA	run 356/1000	MSE=1.074e-04	nit=-1	nfev=28	t=4.20s
COBYLA	run 357/1000	MSE=4.096e-06	nit=-1	nfev=107	
t=16.08s					
COBYLA	run 358/1000	MSE=7.088e-05	nit=-1	nfev=135	
t=20.42s					
COBYLA	run 359/1000	MSE=1.461e-05	nit=-1	nfev=111	
t=16.79s					
COBYLA	run 360/1000	MSE=7.543e-05	nit=-1	nfev=420	
t=63.59s					
COBYLA	run 361/1000	MSE=6.311e-05	nit=-1	nfev=510	
t=77.10s					
COBYLA	run 362/1000	MSE=8.617e-05	nit=-1	nfev=483	
t=72.73s					
COBYLA	run 363/1000	MSE=1.670e-05	nit=-1	nfev=53	t=7.97s
COBYLA	run 364/1000	MSE=4.323e-05	nit=-1	nfev=577	
t=86.73s					
COBYLA	run 365/1000	MSE=5.890e-05	nit=-1	nfev=136	
t=20.38s					
COBYLA	run 366/1000	MSE=2.879e-05	nit=-1	nfev=112	
t=16.74s					
COBYLA	run 367/1000	MSE=1.023e-01	nit=-1	nfev=26	t=4.14s
COBYLA	run 368/1000	MSE=1.104e-04	nit=-1	nfev=69	
t=10.61s					
COBYLA	run 369/1000	MSE=1.740e-05	nit=-1	nfev=118	
t=17.85s					
COBYLA	run 370/1000	MSE=5.031e-05	nit=-1	nfev=505	
t=76.49s					
COBYLA	run 371/1000	MSE=7.822e-05	nit=-1	nfev=143	
t=22.64s					

COBYLA	run 372/1000	MSE=9.630e-06	nit=-1	nfev=60	t=9.00s
COBYLA	run 373/1000	MSE=2.603e-05	nit=-1	nfev=524	
t=79.43s					
COBYLA	run 374/1000	MSE=7.010e-05	nit=-1	nfev=160	
t=23.99s					
COBYLA	run 375/1000	MSE=7.599e-05	nit=-1	nfev=187	
t=28.95s					
COBYLA	run 376/1000	MSE=1.134e-04	nit=-1	nfev=497	
t=75.80s					
COBYLA	run 377/1000	MSE=5.217e-05	nit=-1	nfev=516	
t=78.31s					
COBYLA	run 378/1000	MSE=4.103e-05	nit=-1	nfev=95	
t=14.22s					
COBYLA	run 379/1000	MSE=6.806e-05	nit=-1	nfev=117	
t=18.14s					
COBYLA	run 380/1000	MSE=6.248e-06	nit=-1	nfev=70	
t=10.35s					
COBYLA	run 381/1000	MSE=4.893e-05	nit=-1	nfev=122	
t=18.91s					
COBYLA	run 382/1000	MSE=2.882e-05	nit=-1	nfev=135	
t=20.13s					
COBYLA	run 383/1000	MSE=9.720e-05	nit=-1	nfev=123	
t=18.90s					
COBYLA	run 384/1000	MSE=9.590e-05	nit=-1	nfev=133	
t=20.24s					
COBYLA	run 385/1000	MSE=1.112e-04	nit=-1	nfev=108	
t=16.07s					
COBYLA	run 386/1000	MSE=1.329e-05	nit=-1	nfev=209	
t=31.39s					
COBYLA	run 387/1000	MSE=2.686e-03	nit=-1	nfev=30	t=4.41s
COBYLA	run 388/1000	MSE=6.154e-05	nit=-1	nfev=116	
t=17.81s					
COBYLA	run 389/1000	MSE=1.395e-01	nit=-1	nfev=33	t=5.01s
COBYLA	run 390/1000	MSE=1.065e-04	nit=-1	nfev=485	
t=74.23s					
COBYLA	run 391/1000	MSE=2.209e-05	nit=-1	nfev=111	
t=16.99s					
COBYLA	run 392/1000	MSE=4.593e-05	nit=-1	nfev=149	
t=22.47s					
COBYLA	run 393/1000	MSE=3.733e-05	nit=-1	nfev=575	
t=86.40s					
COBYLA	run 394/1000	MSE=1.600e-04	nit=-1	nfev=475	
t=71.76s					
COBYLA	run 395/1000	MSE=3.253e-05	nit=-1	nfev=98	
t=15.08s					
COBYLA	run 396/1000	MSE=6.395e-05	nit=-1	nfev=124	
t=18.50s					
COBYLA	run 397/1000	MSE=5.031e-05	nit=-1	nfev=505	

t=77.07s						
COBYLA	run 398/1000	MSE=3.074e-05	nit=-1	nfev=439		
t=66.35s						
COBYLA	run 399/1000	MSE=3.465e-07	nit=-1	nfev=56	t=8.68s	
COBYLA	run 400/1000	MSE=4.407e-05	nit=-1	nfev=554		
t=83.30s						
COBYLA	run 401/1000	MSE=2.115e-05	nit=-1	nfev=147		
t=22.16s						
COBYLA	run 402/1000	MSE=1.036e-04	nit=-1	nfev=90		
t=13.62s						
COBYLA	run 403/1000	MSE=3.345e-05	nit=-1	nfev=125		
t=18.99s						
COBYLA	run 404/1000	MSE=4.315e-06	nit=-1	nfev=140		
t=21.39s						
COBYLA	run 405/1000	MSE=5.318e-05	nit=-1	nfev=130		
t=19.40s						
COBYLA	run 406/1000	MSE=4.882e-05	nit=-1	nfev=152		
t=23.41s						
COBYLA	run 407/1000	MSE=7.436e-05	nit=-1	nfev=487		
t=73.56s						
COBYLA	run 408/1000	MSE=5.905e-07	nit=-1	nfev=94		
t=14.06s						
COBYLA	run 409/1000	MSE=2.537e-03	nit=-1	nfev=34	t=5.07s	
COBYLA	run 410/1000	MSE=6.108e-05	nit=-1	nfev=111		
t=16.91s						
COBYLA	run 411/1000	MSE=2.448e-05	nit=-1	nfev=163		
t=24.64s						
COBYLA	run 412/1000	MSE=6.150e-05	nit=-1	nfev=160		
t=23.96s						
COBYLA	run 413/1000	MSE=4.509e-05	nit=-1	nfev=510		
t=77.49s						
COBYLA	run 414/1000	MSE=7.536e-05	nit=-1	nfev=81		
t=12.03s						
COBYLA	run 415/1000	MSE=8.405e-05	nit=-1	nfev=98		
t=14.59s						
COBYLA	run 416/1000	MSE=4.600e-05	nit=-1	nfev=111		
t=16.69s						
COBYLA	run 417/1000	MSE=3.733e-05	nit=-1	nfev=575		
t=86.77s						
COBYLA	run 418/1000	MSE=3.113e-05	nit=-1	nfev=417		
t=62.92s						
COBYLA	run 419/1000	MSE=2.444e-05	nit=-1	nfev=113		
t=16.94s						
COBYLA	run 420/1000	MSE=5.822e-05	nit=-1	nfev=136		
t=20.47s						
COBYLA	run 421/1000	MSE=2.707e-05	nit=-1	nfev=245		
t=36.66s						
COBYLA	run 422/1000	MSE=3.733e-05	nit=-1	nfev=575		

t=86.39s	COBYLA	run 423/1000	MSE=1.038e-04	nit=-1	nfev=75	
t=11.33s	COBYLA	run 424/1000	MSE=1.147e-04	nit=-1	nfev=80	
t=12.55s	COBYLA	run 425/1000	MSE=7.378e-07	nit=-1	nfev=88	
t=13.21s	COBYLA	run 426/1000	MSE=1.468e-04	nit=-1	nfev=89	
t=13.66s	COBYLA	run 427/1000	MSE=1.563e-05	nit=-1	nfev=110	
t=16.50s	COBYLA	run 428/1000	MSE=9.343e-05	nit=-1	nfev=94	
t=14.22s	COBYLA	run 429/1000	MSE=2.087e-04	nit=-1	nfev=178	
t=26.74s	COBYLA	run 430/1000	MSE=4.696e-05	nit=-1	nfev=96	
t=14.57s	COBYLA	run 431/1000	MSE=2.742e-05	nit=-1	nfev=58	t=8.62s
	COBYLA	run 432/1000	MSE=4.169e-05	nit=-1	nfev=164	
t=24.59s	COBYLA	run 433/1000	MSE=2.757e-05	nit=-1	nfev=570	
t=85.68s	COBYLA	run 434/1000	MSE=7.495e-05	nit=-1	nfev=187	
t=28.25s	COBYLA	run 435/1000	MSE=3.887e-05	nit=-1	nfev=157	
t=23.75s	COBYLA	run 436/1000	MSE=2.681e-03	nit=-1	nfev=36	t=5.42s
	COBYLA	run 437/1000	MSE=3.989e-05	nit=-1	nfev=508	
t=75.92s	COBYLA	run 438/1000	MSE=7.962e-05	nit=-1	nfev=100	
t=15.18s	COBYLA	run 439/1000	MSE=1.896e-04	nit=-1	nfev=97	
t=14.53s	COBYLA	run 440/1000	MSE=3.989e-05	nit=-1	nfev=508	
t=76.98s	COBYLA	run 441/1000	MSE=8.281e-05	nit=-1	nfev=132	
t=20.11s	COBYLA	run 442/1000	MSE=5.153e-05	nit=-1	nfev=408	
t=61.68s	COBYLA	run 443/1000	MSE=4.097e-05	nit=-1	nfev=564	
t=85.90s	COBYLA	run 444/1000	MSE=5.502e-05	nit=-1	nfev=573	
t=86.29s	COBYLA	run 445/1000	MSE=1.202e-06	nit=-1	nfev=78	
t=11.66s	COBYLA	run 446/1000	MSE=4.939e-05	nit=-1	nfev=532	
t=80.32s	COBYLA	run 447/1000	MSE=2.085e-05	nit=-1	nfev=112	

t=16.83s						
COBYLA	run 448/1000	MSE=5.607e-05	nit=-1	nfev=228		
t=34.21s						
COBYLA	run 449/1000	MSE=1.001e-04	nit=-1	nfev=524		
t=83.86s						
COBYLA	run 450/1000	MSE=3.131e-05	nit=-1	nfev=126		
t=20.50s						
COBYLA	run 451/1000	MSE=5.904e-05	nit=-1	nfev=395		
t=59.89s						
COBYLA	run 452/1000	MSE=6.245e-05	nit=-1	nfev=132		
t=19.57s						
COBYLA	run 453/1000	MSE=7.655e-06	nit=-1	nfev=34	t=5.23s	
COBYLA	run 454/1000	MSE=6.004e-06	nit=-1	nfev=168		
t=25.39s						
COBYLA	run 455/1000	MSE=9.673e-05	nit=-1	nfev=73		
t=11.15s						
COBYLA	run 456/1000	MSE=5.300e-05	nit=-1	nfev=29	t=4.35s	
COBYLA	run 457/1000	MSE=4.967e-05	nit=-1	nfev=96		
t=14.58s						
COBYLA	run 458/1000	MSE=5.904e-05	nit=-1	nfev=394		
t=59.44s						
COBYLA	run 459/1000	MSE=2.057e-05	nit=-1	nfev=105		
t=15.89s						
COBYLA	run 460/1000	MSE=2.540e-03	nit=-1	nfev=41	t=6.14s	
COBYLA	run 461/1000	MSE=7.503e-05	nit=-1	nfev=129		
t=19.69s						
COBYLA	run 462/1000	MSE=1.826e-04	nit=-1	nfev=137		
t=21.03s						
COBYLA	run 463/1000	MSE=6.246e-05	nit=-1	nfev=473		
t=70.81s						
COBYLA	run 464/1000	MSE=8.173e-05	nit=-1	nfev=140		
t=21.11s						
COBYLA	run 465/1000	MSE=5.708e-05	nit=-1	nfev=442		
t=65.99s						
COBYLA	run 466/1000	MSE=2.171e-05	nit=-1	nfev=106		
t=15.86s						
COBYLA	run 467/1000	MSE=4.091e-05	nit=-1	nfev=153		
t=23.17s						
COBYLA	run 468/1000	MSE=1.677e-05	nit=-1	nfev=557		
t=83.79s						
COBYLA	run 469/1000	MSE=1.301e-06	nit=-1	nfev=78		
t=11.57s						
COBYLA	run 470/1000	MSE=5.588e-05	nit=-1	nfev=478		
t=71.53s						
COBYLA	run 471/1000	MSE=5.904e-05	nit=-1	nfev=395		
t=58.81s						
COBYLA	run 472/1000	MSE=3.720e-05	nit=-1	nfev=110		
t=16.59s						

COBYLA	run 473/1000	MSE=1.996e-05	nit=-1	nfev=102	
t=15.43s					
COBYLA	run 474/1000	MSE=3.989e-05	nit=-1	nfev=508	
t=76.13s					
COBYLA	run 475/1000	MSE=9.922e-05	nit=-1	nfev=131	
t=19.65s					
COBYLA	run 476/1000	MSE=6.267e-05	nit=-1	nfev=147	
t=22.11s					
COBYLA	run 477/1000	MSE=7.713e-05	nit=-1	nfev=217	
t=34.01s					
COBYLA	run 478/1000	MSE=2.689e-03	nit=-1	nfev=31	t=4.65s
COBYLA	run 479/1000	MSE=5.174e-05	nit=-1	nfev=155	
t=24.59s					
COBYLA	run 480/1000	MSE=8.114e-05	nit=-1	nfev=512	
t=76.49s					
COBYLA	run 481/1000	MSE=1.502e-04	nit=-1	nfev=107	
t=16.06s					
COBYLA	run 482/1000	MSE=4.929e-05	nit=-1	nfev=91	
t=13.45s					
COBYLA	run 483/1000	MSE=2.717e-05	nit=-1	nfev=227	
t=33.97s					
COBYLA	run 484/1000	MSE=2.981e-05	nit=-1	nfev=148	
t=22.40s					
COBYLA	run 485/1000	MSE=5.754e-05	nit=-1	nfev=76	
t=11.21s					
COBYLA	run 486/1000	MSE=1.485e-04	nit=-1	nfev=65	t=9.89s
COBYLA	run 487/1000	MSE=7.261e-05	nit=-1	nfev=163	
t=24.35s					
COBYLA	run 488/1000	MSE=8.994e-05	nit=-1	nfev=213	
t=31.69s					
COBYLA	run 489/1000	MSE=4.640e-05	nit=-1	nfev=131	
t=19.75s					
COBYLA	run 490/1000	MSE=1.423e-05	nit=-1	nfev=447	
t=67.15s					
COBYLA	run 491/1000	MSE=7.626e-05	nit=-1	nfev=132	
t=20.19s					
COBYLA	run 492/1000	MSE=3.400e-05	nit=-1	nfev=182	
t=27.69s					
COBYLA	run 493/1000	MSE=4.030e-05	nit=-1	nfev=209	
t=31.16s					
COBYLA	run 494/1000	MSE=2.904e-05	nit=-1	nfev=106	
t=15.98s					
COBYLA	run 495/1000	MSE=5.033e-06	nit=-1	nfev=74	
t=11.06s					
COBYLA	run 496/1000	MSE=6.607e-05	nit=-1	nfev=506	
t=76.76s					
COBYLA	run 497/1000	MSE=1.183e-04	nit=-1	nfev=102	
t=15.17s					

COBYLA	run 498/1000	MSE=4.625e-06	nit=-1	nfev=55	t=8.18s
COBYLA	run 499/1000	MSE=9.415e-06	nit=-1	nfev=140	
t=22.22s					
COBYLA	run 500/1000	MSE=3.733e-05	nit=-1	nfev=575	
t=86.19s					
COBYLA	run 501/1000	MSE=2.551e-03	nit=-1	nfev=32	t=4.61s
COBYLA	run 502/1000	MSE=6.486e-05	nit=-1	nfev=111	
t=16.88s					
COBYLA	run 503/1000	MSE=4.109e-05	nit=-1	nfev=44	t=6.98s
COBYLA	run 504/1000	MSE=3.299e-05	nit=-1	nfev=141	
t=21.28s					
COBYLA	run 505/1000	MSE=3.568e-05	nit=-1	nfev=139	
t=20.60s					
COBYLA	run 506/1000	MSE=6.342e-05	nit=-1	nfev=106	
t=15.88s					
COBYLA	run 507/1000	MSE=4.661e-06	nit=-1	nfev=108	
t=16.16s					
COBYLA	run 508/1000	MSE=1.899e-05	nit=-1	nfev=508	
t=76.75s					
COBYLA	run 509/1000	MSE=2.045e-07	nit=-1	nfev=97	
t=14.51s					
COBYLA	run 510/1000	MSE=1.131e-04	nit=-1	nfev=551	
t=82.65s					
COBYLA	run 511/1000	MSE=5.958e-05	nit=-1	nfev=566	
t=84.46s					
COBYLA	run 512/1000	MSE=4.927e-05	nit=-1	nfev=134	
t=19.98s					
COBYLA	run 513/1000	MSE=5.472e-05	nit=-1	nfev=132	
t=19.73s					
COBYLA	run 514/1000	MSE=4.509e-05	nit=-1	nfev=509	
t=76.18s					
COBYLA	run 515/1000	MSE=6.147e-05	nit=-1	nfev=82	
t=12.30s					
COBYLA	run 516/1000	MSE=6.048e-05	nit=-1	nfev=553	
t=84.62s					
COBYLA	run 517/1000	MSE=5.023e-05	nit=-1	nfev=83	
t=12.40s					
COBYLA	run 518/1000	MSE=3.144e-05	nit=-1	nfev=218	
t=32.51s					
COBYLA	run 519/1000	MSE=9.951e-05	nit=-1	nfev=110	
t=16.36s					
COBYLA	run 520/1000	MSE=4.949e-05	nit=-1	nfev=85	
t=12.62s					
COBYLA	run 521/1000	MSE=1.061e-04	nit=-1	nfev=116	
t=17.19s					
COBYLA	run 522/1000	MSE=7.433e-05	nit=-1	nfev=111	
t=16.51s					
COBYLA	run 523/1000	MSE=7.027e-05	nit=-1	nfev=503	

t=75.25s						
COBYLA	run 524/1000	MSE=6.748e-05	nit=-1	nfev=501		
t=75.13s						
COBYLA	run 525/1000	MSE=5.312e-05	nit=-1	nfev=135		
t=20.07s						
COBYLA	run 526/1000	MSE=6.283e-05	nit=-1	nfev=154		
t=23.04s						
COBYLA	run 527/1000	MSE=1.065e-04	nit=-1	nfev=514		
t=77.19s						
COBYLA	run 528/1000	MSE=6.116e-05	nit=-1	nfev=425		
t=63.41s						
COBYLA	run 529/1000	MSE=5.264e-05	nit=-1	nfev=167		
t=24.94s						
COBYLA	run 530/1000	MSE=4.414e-05	nit=-1	nfev=133		
t=19.88s						
COBYLA	run 531/1000	MSE=4.168e-06	nit=-1	nfev=28	t=4.53s	
COBYLA	run 532/1000	MSE=5.958e-05	nit=-1	nfev=566		
t=85.00s						
COBYLA	run 533/1000	MSE=5.904e-05	nit=-1	nfev=395		
t=59.39s						
COBYLA	run 534/1000	MSE=2.651e-03	nit=-1	nfev=31	t=4.64s	
COBYLA	run 535/1000	MSE=6.306e-05	nit=-1	nfev=144		
t=21.52s						
COBYLA	run 536/1000	MSE=4.450e-05	nit=-1	nfev=474		
t=85.79s						
COBYLA	run 537/1000	MSE=6.422e-05	nit=-1	nfev=469		
t=83.95s						
COBYLA	run 538/1000	MSE=1.141e-04	nit=-1	nfev=478		
t=91.75s						
COBYLA	run 539/1000	MSE=1.114e-04	nit=-1	nfev=99		
t=23.00s						
COBYLA	run 540/1000	MSE=1.034e-06	nit=-1	nfev=40	t=7.68s	
COBYLA	run 541/1000	MSE=8.371e-05	nit=-1	nfev=108		
t=21.11s						
COBYLA	run 542/1000	MSE=2.647e-05	nit=-1	nfev=69		
t=11.91s						
COBYLA	run 543/1000	MSE=8.087e-05	nit=-1	nfev=82		
t=14.86s						
COBYLA	run 544/1000	MSE=1.019e-04	nit=-1	nfev=29	t=4.89s	
COBYLA	run 545/1000	MSE=5.904e-05	nit=-1	nfev=394		
t=70.80s						
COBYLA	run 546/1000	MSE=1.369e-05	nit=-1	nfev=110		
t=18.17s						
COBYLA	run 547/1000	MSE=2.050e-05	nit=-1	nfev=551		
t=90.63s						
COBYLA	run 548/1000	MSE=7.447e-05	nit=-1	nfev=103		
t=16.72s						
COBYLA	run 549/1000	MSE=3.259e-05	nit=-1	nfev=148		

t=23.31s						
COBYLA	run 550/1000	MSE=6.151e-05	nit=-1	nfev=186		
t=29.63s						
COBYLA	run 551/1000	MSE=7.023e-05	nit=-1	nfev=159		
t=25.27s						
COBYLA	run 552/1000	MSE=4.343e-07	nit=-1	nfev=103		
t=16.65s						
COBYLA	run 553/1000	MSE=4.659e-06	nit=-1	nfev=31	t=4.80s	
COBYLA	run 554/1000	MSE=2.580e-05	nit=-1	nfev=530		
t=84.97s						
COBYLA	run 555/1000	MSE=1.180e-04	nit=-1	nfev=497		
t=82.09s						
COBYLA	run 556/1000	MSE=5.904e-05	nit=-1	nfev=394		
t=65.34s						
COBYLA	run 557/1000	MSE=3.518e-05	nit=-1	nfev=169		
t=26.95s						
COBYLA	run 558/1000	MSE=7.553e-07	nit=-1	nfev=90		
t=14.37s						
COBYLA	run 559/1000	MSE=3.620e-05	nit=-1	nfev=115		
t=18.16s						
COBYLA	run 560/1000	MSE=9.328e-05	nit=-1	nfev=68		
t=10.79s						
COBYLA	run 561/1000	MSE=4.495e-05	nit=-1	nfev=142		
t=22.96s						
COBYLA	run 562/1000	MSE=6.504e-05	nit=-1	nfev=436		
t=68.76s						
COBYLA	run 563/1000	MSE=4.457e-05	nit=-1	nfev=151		
t=23.96s						
COBYLA	run 564/1000	MSE=7.360e-05	nit=-1	nfev=592		
t=100.54s						
COBYLA	run 565/1000	MSE=5.807e-05	nit=-1	nfev=107		
t=17.16s						
COBYLA	run 566/1000	MSE=4.242e-05	nit=-1	nfev=104		
t=16.44s						
COBYLA	run 567/1000	MSE=2.003e-04	nit=-1	nfev=196		
t=30.45s						
COBYLA	run 568/1000	MSE=2.663e-03	nit=-1	nfev=47	t=7.39s	
COBYLA	run 569/1000	MSE=1.023e-01	nit=-1	nfev=31	t=5.09s	
COBYLA	run 570/1000	MSE=4.824e-05	nit=-1	nfev=538		
t=85.02s						
COBYLA	run 571/1000	MSE=4.737e-05	nit=-1	nfev=130		
t=21.21s						
COBYLA	run 572/1000	MSE=9.054e-05	nit=-1	nfev=141		
t=22.55s						
COBYLA	run 573/1000	MSE=3.220e-05	nit=-1	nfev=116		
t=18.55s						
COBYLA	run 574/1000	MSE=5.961e-05	nit=-1	nfev=511		
t=81.66s						

COBYLA	run 575/1000	MSE=2.641e-05	nit=-1	nfev=152	
t=24.51s					
COBYLA	run 576/1000	MSE=7.507e-05	nit=-1	nfev=513	
t=81.37s					
COBYLA	run 577/1000	MSE=5.904e-05	nit=-1	nfev=395	
t=63.08s					
COBYLA	run 578/1000	MSE=1.017e-05	nit=-1	nfev=126	
t=20.53s					
COBYLA	run 579/1000	MSE=2.624e-04	nit=-1	nfev=87	
t=14.19s					
COBYLA	run 580/1000	MSE=3.733e-05	nit=-1	nfev=575	
t=91.51s					
COBYLA	run 581/1000	MSE=2.967e-05	nit=-1	nfev=86	
t=13.59s					
COBYLA	run 582/1000	MSE=2.910e-05	nit=-1	nfev=134	
t=21.03s					
COBYLA	run 583/1000	MSE=4.114e-05	nit=-1	nfev=73	
t=12.00s					
COBYLA	run 584/1000	MSE=4.925e-05	nit=-1	nfev=121	
t=19.09s					
COBYLA	run 585/1000	MSE=4.192e-05	nit=-1	nfev=484	
t=76.70s					
COBYLA	run 586/1000	MSE=3.805e-05	nit=-1	nfev=112	
t=17.74s					
COBYLA	run 587/1000	MSE=2.837e-05	nit=-1	nfev=125	
t=20.10s					
COBYLA	run 588/1000	MSE=3.733e-05	nit=-1	nfev=575	
t=95.51s					
COBYLA	run 589/1000	MSE=2.986e-05	nit=-1	nfev=202	
t=33.60s					
COBYLA	run 590/1000	MSE=9.474e-05	nit=-1	nfev=430	
t=71.03s					
COBYLA	run 591/1000	MSE=4.892e-06	nit=-1	nfev=131	
t=23.15s					
COBYLA	run 592/1000	MSE=1.023e-01	nit=-1	nfev=31	t=5.25s
COBYLA	run 593/1000	MSE=1.003e-04	nit=-1	nfev=111	
t=20.32s					
COBYLA	run 594/1000	MSE=1.685e-05	nit=-1	nfev=98	
t=16.84s					
COBYLA	run 595/1000	MSE=1.565e-05	nit=-1	nfev=494	
t=86.91s					
COBYLA	run 596/1000	MSE=6.332e-05	nit=-1	nfev=130	
t=25.84s					
COBYLA	run 597/1000	MSE=7.643e-05	nit=-1	nfev=174	
t=31.42s					
COBYLA	run 598/1000	MSE=6.702e-05	nit=-1	nfev=73	
t=13.37s					
COBYLA	run 599/1000	MSE=6.422e-05	nit=-1	nfev=469	

t=103.17s						
COBYLA	run 600/1000	MSE=5.805e-05	nit=-1	nfev=513		
t=88.94s						
COBYLA	run 601/1000	MSE=9.874e-05	nit=-1	nfev=510		
t=83.98s						
COBYLA	run 602/1000	MSE=5.904e-05	nit=-1	nfev=395		
t=65.10s						
COBYLA	run 603/1000	MSE=3.482e-05	nit=-1	nfev=499		
t=84.36s						
COBYLA	run 604/1000	MSE=1.985e-05	nit=-1	nfev=502		
t=86.93s						
COBYLA	run 605/1000	MSE=1.008e-05	nit=-1	nfev=435		
t=77.39s						
COBYLA	run 606/1000	MSE=8.733e-05	nit=-1	nfev=120		
t=20.09s						
COBYLA	run 607/1000	MSE=3.651e-05	nit=-1	nfev=234		
t=41.50s						
COBYLA	run 608/1000	MSE=3.134e-07	nit=-1	nfev=34	t=5.77s	
COBYLA	run 609/1000	MSE=7.027e-05	nit=-1	nfev=503		
t=84.28s						
COBYLA	run 610/1000	MSE=5.516e-06	nit=-1	nfev=52	t=9.66s	
COBYLA	run 611/1000	MSE=1.423e-05	nit=-1	nfev=447		
t=71.98s						
COBYLA	run 612/1000	MSE=2.923e-05	nit=-1	nfev=107		
t=16.10s						
COBYLA	run 613/1000	MSE=3.998e-05	nit=-1	nfev=440		
t=66.00s						
COBYLA	run 614/1000	MSE=3.472e-05	nit=-1	nfev=106		
t=15.97s						
COBYLA	run 615/1000	MSE=6.385e-05	nit=-1	nfev=164		
t=25.93s						
COBYLA	run 616/1000	MSE=2.858e-05	nit=-1	nfev=65		
t=10.32s						
COBYLA	run 617/1000	MSE=3.624e-05	nit=-1	nfev=150		
t=23.23s						
COBYLA	run 618/1000	MSE=5.226e-05	nit=-1	nfev=238		
t=37.11s						
COBYLA	run 619/1000	MSE=6.812e-05	nit=-1	nfev=131		
t=20.04s						
COBYLA	run 620/1000	MSE=1.403e-05	nit=-1	nfev=30	t=4.85s	
COBYLA	run 621/1000	MSE=6.867e-05	nit=-1	nfev=483		
t=82.50s						
COBYLA	run 622/1000	MSE=4.055e-06	nit=-1	nfev=507		
t=92.49s						
COBYLA	run 623/1000	MSE=3.733e-05	nit=-1	nfev=575		
t=104.20s						
COBYLA	run 624/1000	MSE=2.040e-05	nit=-1	nfev=147		
t=23.79s						

COBYLA	run 625/1000	MSE=1.525e-05	nit=-1	nfev=35	t=5.67s
COBYLA	run 626/1000	MSE=2.592e-03	nit=-1	nfev=31	t=5.52s
COBYLA	run 627/1000	MSE=3.562e-05	nit=-1	nfev=80	
t=13.85s					
COBYLA	run 628/1000	MSE=1.125e-04	nit=-1	nfev=182	
t=32.87s					
COBYLA	run 629/1000	MSE=3.733e-05	nit=-1	nfev=575	
t=102.60s					
COBYLA	run 630/1000	MSE=8.474e-05	nit=-1	nfev=110	
t=19.61s					
COBYLA	run 631/1000	MSE=2.499e-03	nit=-1	nfev=29	t=4.83s
COBYLA	run 632/1000	MSE=1.106e-04	nit=-1	nfev=103	
t=17.44s					
COBYLA	run 633/1000	MSE=4.865e-05	nit=-1	nfev=87	
t=15.56s					
COBYLA	run 634/1000	MSE=5.088e-05	nit=-1	nfev=598	
t=98.49s					
COBYLA	run 635/1000	MSE=8.299e-05	nit=-1	nfev=105	
t=19.78s					
COBYLA	run 636/1000	MSE=2.746e-05	nit=-1	nfev=313	
t=53.50s					
COBYLA	run 637/1000	MSE=1.110e-04	nit=-1	nfev=500	
t=86.95s					
COBYLA	run 638/1000	MSE=1.631e-04	nit=-1	nfev=150	
t=25.61s					
COBYLA	run 639/1000	MSE=3.831e-05	nit=-1	nfev=263	
t=44.87s					
COBYLA	run 640/1000	MSE=5.685e-05	nit=-1	nfev=518	
t=87.59s					
COBYLA	run 641/1000	MSE=3.733e-05	nit=-1	nfev=575	
t=98.13s					
COBYLA	run 642/1000	MSE=2.477e-06	nit=-1	nfev=74	
t=12.35s					
COBYLA	run 643/1000	MSE=3.722e-05	nit=-1	nfev=140	
t=25.67s					
COBYLA	run 644/1000	MSE=5.426e-05	nit=-1	nfev=114	
t=19.53s					
COBYLA	run 645/1000	MSE=9.379e-05	nit=-1	nfev=492	
t=79.60s					
COBYLA	run 646/1000	MSE=9.972e-06	nit=-1	nfev=142	
t=23.17s					
COBYLA	run 647/1000	MSE=5.161e-05	nit=-1	nfev=122	
t=21.29s					
COBYLA	run 648/1000	MSE=3.806e-05	nit=-1	nfev=126	
t=21.85s					
COBYLA	run 649/1000	MSE=3.663e-05	nit=-1	nfev=661	
t=110.17s					
COBYLA	run 650/1000	MSE=4.596e-05	nit=-1	nfev=73	

t=11.24s	COBYLA	run 651/1000	MSE=1.469e-05	nit=-1	nfev=119	
t=17.95s	COBYLA	run 652/1000	MSE=3.618e-05	nit=-1	nfev=126	
t=19.44s	COBYLA	run 653/1000	MSE=5.904e-05	nit=-1	nfev=395	
t=59.98s	COBYLA	run 654/1000	MSE=2.997e-05	nit=-1	nfev=135	
t=20.15s	COBYLA	run 655/1000	MSE=4.033e-05	nit=-1	nfev=109	
t=16.63s	COBYLA	run 656/1000	MSE=5.020e-05	nit=-1	nfev=499	
t=82.07s	COBYLA	run 657/1000	MSE=2.855e-05	nit=-1	nfev=133	
t=20.74s	COBYLA	run 658/1000	MSE=2.910e-05	nit=-1	nfev=130	
t=19.83s	COBYLA	run 659/1000	MSE=1.604e-05	nit=-1	nfev=112	
t=17.07s	COBYLA	run 660/1000	MSE=1.833e-05	nit=-1	nfev=165	
t=25.10s	COBYLA	run 661/1000	MSE=5.961e-05	nit=-1	nfev=511	
t=77.68s	COBYLA	run 662/1000	MSE=6.145e-05	nit=-1	nfev=112	
t=17.40s	COBYLA	run 663/1000	MSE=4.557e-05	nit=-1	nfev=39	t=5.90s
	COBYLA	run 664/1000	MSE=2.683e-05	nit=-1	nfev=499	
t=75.90s	COBYLA	run 665/1000	MSE=6.308e-05	nit=-1	nfev=454	
t=69.63s	COBYLA	run 666/1000	MSE=3.113e-05	nit=-1	nfev=417	
t=62.25s	COBYLA	run 667/1000	MSE=1.142e-04	nit=-1	nfev=538	
t=80.99s	COBYLA	run 668/1000	MSE=6.562e-05	nit=-1	nfev=67	
t=10.36s	COBYLA	run 669/1000	MSE=9.432e-05	nit=-1	nfev=102	
t=15.15s	COBYLA	run 670/1000	MSE=7.494e-05	nit=-1	nfev=519	
t=81.63s	COBYLA	run 671/1000	MSE=1.403e-04	nit=-1	nfev=145	
t=27.28s	COBYLA	run 672/1000	MSE=5.904e-05	nit=-1	nfev=394	
t=66.71s	COBYLA	run 673/1000	MSE=5.098e-05	nit=-1	nfev=437	
t=74.79s	COBYLA	run 674/1000	MSE=2.454e-04	nit=-1	nfev=28	t=4.73s
	COBYLA	run 675/1000	MSE=6.897e-05	nit=-1	nfev=77	

t=12.66s						
COBYLA	run 676/1000	MSE=3.020e-05	nit=-1	nfev=197		
t=33.53s						
COBYLA	run 677/1000	MSE=6.107e-05	nit=-1	nfev=507		
t=93.90s						
COBYLA	run 678/1000	MSE=1.134e-04	nit=-1	nfev=497		
t=86.28s						
COBYLA	run 679/1000	MSE=9.213e-06	nit=-1	nfev=146		
t=25.28s						
COBYLA	run 680/1000	MSE=5.904e-05	nit=-1	nfev=394		
t=65.51s						
COBYLA	run 681/1000	MSE=1.134e-04	nit=-1	nfev=497		
t=83.21s						
COBYLA	run 682/1000	MSE=6.723e-05	nit=-1	nfev=114		
t=20.00s						
COBYLA	run 683/1000	MSE=3.465e-05	nit=-1	nfev=122		
t=22.57s						
COBYLA	run 684/1000	MSE=2.870e-05	nit=-1	nfev=107		
t=17.73s						
COBYLA	run 685/1000	MSE=8.554e-06	nit=-1	nfev=77		
t=13.11s						
COBYLA	run 686/1000	MSE=6.882e-05	nit=-1	nfev=149		
t=26.11s						
COBYLA	run 687/1000	MSE=5.387e-05	nit=-1	nfev=523		
t=85.43s						
COBYLA	run 688/1000	MSE=3.253e-05	nit=-1	nfev=86		
t=16.41s						
COBYLA	run 689/1000	MSE=1.061e-04	nit=-1	nfev=103		
t=17.22s						
COBYLA	run 690/1000	MSE=2.710e-05	nit=-1	nfev=234		
t=42.55s						
COBYLA	run 691/1000	MSE=5.391e-05	nit=-1	nfev=117		
t=20.13s						
COBYLA	run 692/1000	MSE=1.972e-05	nit=-1	nfev=153		
t=25.39s						
COBYLA	run 693/1000	MSE=6.192e-05	nit=-1	nfev=123		
t=20.90s						
COBYLA	run 694/1000	MSE=6.379e-05	nit=-1	nfev=98		
t=17.73s						
COBYLA	run 695/1000	MSE=7.923e-05	nit=-1	nfev=554		
t=94.74s						
COBYLA	run 696/1000	MSE=1.806e-05	nit=-1	nfev=98		
t=26.35s						
COBYLA	run 697/1000	MSE=5.807e-05	nit=-1	nfev=139		
t=30.00s						
COBYLA	run 698/1000	MSE=5.903e-05	nit=-1	nfev=65		
t=11.02s						
COBYLA	run 699/1000	MSE=4.627e-07	nit=-1	nfev=60	t=9.72s	

COBYLA	run 700/1000	MSE=8.760e-05	nit=-1	nfev=174	
t=33.39s					
COBYLA	run 701/1000	MSE=4.010e-06	nit=-1	nfev=104	
t=19.01s					
COBYLA	run 702/1000	MSE=2.679e-03	nit=-1	nfev=31	t=5.40s
COBYLA	run 703/1000	MSE=5.585e-05	nit=-1	nfev=200	
t=39.10s					
COBYLA	run 704/1000	MSE=1.076e-04	nit=-1	nfev=131	
t=23.74s					
COBYLA	run 705/1000	MSE=2.682e-05	nit=-1	nfev=38	t=6.61s
COBYLA	run 706/1000	MSE=3.698e-05	nit=-1	nfev=545	
t=93.85s					
COBYLA	run 707/1000	MSE=1.989e-05	nit=-1	nfev=141	
t=24.04s					
COBYLA	run 708/1000	MSE=1.635e-05	nit=-1	nfev=88	
t=15.63s					
COBYLA	run 709/1000	MSE=1.980e-05	nit=-1	nfev=167	
t=30.27s					
COBYLA	run 710/1000	MSE=4.869e-05	nit=-1	nfev=283	
t=49.72s					
COBYLA	run 711/1000	MSE=5.120e-05	nit=-1	nfev=76	
t=14.50s					
COBYLA	run 712/1000	MSE=2.629e-05	nit=-1	nfev=146	
t=25.30s					
COBYLA	run 713/1000	MSE=2.942e-05	nit=-1	nfev=126	
t=23.48s					
COBYLA	run 714/1000	MSE=9.968e-05	nit=-1	nfev=135	
t=22.78s					
COBYLA	run 715/1000	MSE=1.573e-05	nit=-1	nfev=524	
t=94.52s					
COBYLA	run 716/1000	MSE=3.805e-05	nit=-1	nfev=153	
t=29.42s					
COBYLA	run 717/1000	MSE=6.246e-05	nit=-1	nfev=473	
t=87.08s					
COBYLA	run 718/1000	MSE=4.926e-05	nit=-1	nfev=508	
t=98.25s					
COBYLA	run 719/1000	MSE=7.448e-05	nit=-1	nfev=82	
t=13.40s					
COBYLA	run 720/1000	MSE=2.035e-05	nit=-1	nfev=152	
t=27.24s					
COBYLA	run 721/1000	MSE=2.571e-05	nit=-1	nfev=143	
t=27.09s					
COBYLA	run 722/1000	MSE=4.143e-05	nit=-1	nfev=106	
t=19.02s					
COBYLA	run 723/1000	MSE=4.533e-05	nit=-1	nfev=129	
t=27.16s					
COBYLA	run 724/1000	MSE=2.996e-05	nit=-1	nfev=95	
t=14.95s					

COBYLA t=35.79s	run 725/1000	MSE=6.387e-05	nit=-1	nfev=229
COBYLA t=22.08s	run 726/1000	MSE=1.755e-06	nit=-1	nfev=127
COBYLA t=21.13s	run 727/1000	MSE=1.404e-04	nit=-1	nfev=133
COBYLA t=14.44s	run 728/1000	MSE=4.864e-05	nit=-1	nfev=93
COBYLA t=80.71s	run 729/1000	MSE=6.607e-05	nit=-1	nfev=506
COBYLA t=23.78s	run 730/1000	MSE=1.873e-05	nit=-1	nfev=149
COBYLA t=89.65s	run 731/1000	MSE=3.733e-05	nit=-1	nfev=575
COBYLA t=64.94s	run 732/1000	MSE=1.248e-04	nit=-1	nfev=417
COBYLA t=60.95s	run 733/1000	MSE=5.904e-05	nit=-1	nfev=395
COBYLA t=66.20s	run 734/1000	MSE=9.474e-05	nit=-1	nfev=430
COBYLA t=61.14s	run 735/1000	MSE=1.752e-04	nit=-1	nfev=393
COBYLA t=17.34s	run 736/1000	MSE=8.521e-05	nit=-1	nfev=111
COBYLA t=14.86s	run 737/1000	MSE=1.815e-04	nit=-1	nfev=94
COBYLA t=22.60s	run 738/1000	MSE=2.041e-05	nit=-1	nfev=145
COBYLA t=22.60s	run 739/1000	MSE=3.943e-05	nit=-1	nfev=141
COBYLA t=21.19s	run 740/1000	MSE=5.384e-05	nit=-1	nfev=138
COBYLA t=76.35s	run 741/1000	MSE=5.523e-05	nit=-1	nfev=491
COBYLA t=90.66s	run 742/1000	MSE=2.757e-05	nit=-1	nfev=570
COBYLA t=17.62s	run 743/1000	MSE=8.173e-05	nit=-1	nfev=113
COBYLA t=78.79s	run 744/1000	MSE=4.055e-06	nit=-1	nfev=507
COBYLA t=91.69s	run 745/1000	MSE=1.749e-05	nit=-1	nfev=592
COBYLA t=13.01s	run 746/1000	MSE=8.231e-06	nit=-1	nfev=86
COBYLA t=66.79s	run 747/1000	MSE=3.687e-05	nit=-1	nfev=365
COBYLA t=25.92s	run 748/1000	MSE=4.655e-05	nit=-1	nfev=166

COBYLA	run 749/1000	MSE=7.894e-05	nit=-1	nfev=130	
t=20.40s					
COBYLA	run 750/1000	MSE=9.644e-05	nit=-1	nfev=46	t=7.11s
COBYLA	run 751/1000	MSE=9.506e-05	nit=-1	nfev=498	
t=89.81s					
COBYLA	run 752/1000	MSE=4.748e-06	nit=-1	nfev=27	t=4.23s
COBYLA	run 753/1000	MSE=4.537e-05	nit=-1	nfev=462	
t=82.43s					
COBYLA	run 754/1000	MSE=1.565e-05	nit=-1	nfev=494	
t=80.00s					
COBYLA	run 755/1000	MSE=2.671e-03	nit=-1	nfev=34	t=5.30s
COBYLA	run 756/1000	MSE=5.904e-05	nit=-1	nfev=395	
t=59.68s					
COBYLA	run 757/1000	MSE=2.762e-04	nit=-1	nfev=467	
t=70.73s					
COBYLA	run 758/1000	MSE=5.961e-05	nit=-1	nfev=511	
t=78.71s					
COBYLA	run 759/1000	MSE=2.624e-03	nit=-1	nfev=45	t=6.66s
COBYLA	run 760/1000	MSE=7.517e-05	nit=-1	nfev=41	t=6.14s
COBYLA	run 761/1000	MSE=6.767e-05	nit=-1	nfev=93	
t=14.11s					
COBYLA	run 762/1000	MSE=5.961e-05	nit=-1	nfev=511	
t=78.65s					
COBYLA	run 763/1000	MSE=2.346e-05	nit=-1	nfev=92	
t=13.85s					
COBYLA	run 764/1000	MSE=4.948e-05	nit=-1	nfev=99	
t=14.83s					
COBYLA	run 765/1000	MSE=3.143e-05	nit=-1	nfev=95	
t=14.50s					
COBYLA	run 766/1000	MSE=3.357e-05	nit=-1	nfev=200	
t=30.20s					
COBYLA	run 767/1000	MSE=2.117e-05	nit=-1	nfev=90	
t=13.49s					
COBYLA	run 768/1000	MSE=1.484e-05	nit=-1	nfev=91	
t=13.45s					
COBYLA	run 769/1000	MSE=5.309e-05	nit=-1	nfev=97	
t=14.60s					
COBYLA	run 770/1000	MSE=5.031e-05	nit=-1	nfev=176	
t=26.71s					
COBYLA	run 771/1000	MSE=6.308e-05	nit=-1	nfev=454	
t=68.22s					
COBYLA	run 772/1000	MSE=1.400e-05	nit=-1	nfev=30	t=4.97s
COBYLA	run 773/1000	MSE=1.032e-04	nit=-1	nfev=124	
t=18.65s					
COBYLA	run 774/1000	MSE=1.612e-04	nit=-1	nfev=98	
t=14.80s					
COBYLA	run 775/1000	MSE=3.928e-09	nit=-1	nfev=106	
t=15.83s					

COBYLA	run 776/1000	MSE=2.759e-05	nit=-1	nfev=563	
t=85.26s					
COBYLA	run 777/1000	MSE=6.701e-05	nit=-1	nfev=137	
t=20.76s					
COBYLA	run 778/1000	MSE=7.507e-05	nit=-1	nfev=514	
t=77.35s					
COBYLA	run 779/1000	MSE=2.549e-05	nit=-1	nfev=442	
t=78.68s					
COBYLA	run 780/1000	MSE=1.811e-04	nit=-1	nfev=104	
t=23.23s					
COBYLA	run 781/1000	MSE=3.393e-05	nit=-1	nfev=451	
t=80.28s					
COBYLA	run 782/1000	MSE=8.079e-06	nit=-1	nfev=112	
t=18.67s					
COBYLA	run 783/1000	MSE=1.848e-05	nit=-1	nfev=131	
t=22.00s					
COBYLA	run 784/1000	MSE=5.489e-06	nit=-1	nfev=156	
t=26.24s					
COBYLA	run 785/1000	MSE=1.565e-05	nit=-1	nfev=494	
t=83.75s					
COBYLA	run 786/1000	MSE=2.548e-03	nit=-1	nfev=29	t=4.69s
COBYLA	run 787/1000	MSE=4.667e-05	nit=-1	nfev=123	
t=20.44s					
COBYLA	run 788/1000	MSE=4.938e-05	nit=-1	nfev=518	
t=87.65s					
COBYLA	run 789/1000	MSE=1.659e-05	nit=-1	nfev=118	
t=20.45s					
COBYLA	run 790/1000	MSE=5.904e-05	nit=-1	nfev=395	
t=65.94s					
COBYLA	run 791/1000	MSE=3.582e-05	nit=-1	nfev=145	
t=24.10s					
COBYLA	run 792/1000	MSE=2.683e-03	nit=-1	nfev=28	t=4.78s
COBYLA	run 793/1000	MSE=2.851e-05	nit=-1	nfev=76	
t=12.67s					
COBYLA	run 794/1000	MSE=7.314e-06	nit=-1	nfev=58	t=9.75s
COBYLA	run 795/1000	MSE=3.368e-05	nit=-1	nfev=96	
t=16.36s					
COBYLA	run 796/1000	MSE=4.295e-07	nit=-1	nfev=73	
t=12.39s					
COBYLA	run 797/1000	MSE=1.636e-05	nit=-1	nfev=486	
t=80.90s					
COBYLA	run 798/1000	MSE=4.301e-05	nit=-1	nfev=123	
t=20.59s					
COBYLA	run 799/1000	MSE=5.904e-05	nit=-1	nfev=395	
t=66.33s					
COBYLA	run 800/1000	MSE=4.407e-05	nit=-1	nfev=554	
t=92.82s					
COBYLA	run 801/1000	MSE=6.601e-06	nit=-1	nfev=80	

t=13.37s						
COBYLA	run 802/1000	MSE=5.904e-05	nit=-1	nfev=394		
t=66.21s						
COBYLA	run 803/1000	MSE=1.565e-04	nit=-1	nfev=198		
t=33.86s						
COBYLA	run 804/1000	MSE=2.238e-05	nit=-1	nfev=576		
t=96.63s						
COBYLA	run 805/1000	MSE=1.661e-05	nit=-1	nfev=84		
t=14.20s						
COBYLA	run 806/1000	MSE=5.904e-05	nit=-1	nfev=395		
t=66.48s						
COBYLA	run 807/1000	MSE=6.246e-05	nit=-1	nfev=473		
t=79.71s						
COBYLA	run 808/1000	MSE=1.152e-04	nit=-1	nfev=104		
t=17.49s						
COBYLA	run 809/1000	MSE=2.145e-05	nit=-1	nfev=550		
t=89.22s						
COBYLA	run 810/1000	MSE=2.620e-03	nit=-1	nfev=37	t=5.79s	
COBYLA	run 811/1000	MSE=1.589e-04	nit=-1	nfev=110		
t=17.15s						
COBYLA	run 812/1000	MSE=1.652e-05	nit=-1	nfev=485		
t=75.20s						
COBYLA	run 813/1000	MSE=3.568e-05	nit=-1	nfev=146		
t=22.27s						
COBYLA	run 814/1000	MSE=6.757e-05	nit=-1	nfev=495		
t=76.99s						
COBYLA	run 815/1000	MSE=2.836e-05	nit=-1	nfev=584		
t=90.13s						
COBYLA	run 816/1000	MSE=3.945e-05	nit=-1	nfev=109		
t=17.16s						
COBYLA	run 817/1000	MSE=6.370e-05	nit=-1	nfev=557		
t=85.91s						
COBYLA	run 818/1000	MSE=5.373e-05	nit=-1	nfev=128		
t=19.79s						
COBYLA	run 819/1000	MSE=6.731e-05	nit=-1	nfev=84		
t=13.40s						
COBYLA	run 820/1000	MSE=7.886e-05	nit=-1	nfev=156		
t=30.46s						
COBYLA	run 821/1000	MSE=8.605e-05	nit=-1	nfev=124		
t=25.86s						
COBYLA	run 822/1000	MSE=6.376e-05	nit=-1	nfev=217		
t=39.11s						
COBYLA	run 823/1000	MSE=6.409e-05	nit=-1	nfev=101		
t=17.36s						
COBYLA	run 824/1000	MSE=2.552e-03	nit=-1	nfev=43	t=8.32s	
COBYLA	run 825/1000	MSE=4.622e-06	nit=-1	nfev=77		
t=13.58s						
COBYLA	run 826/1000	MSE=4.398e-05	nit=-1	nfev=93		

t=15.28s					
COBYLA	run 827/1000	MSE=1.117e-04	nit=-1	nfev=108	
t=19.06s					
COBYLA	run 828/1000	MSE=8.266e-05	nit=-1	nfev=165	
t=29.67s					
COBYLA	run 829/1000	MSE=1.166e-04	nit=-1	nfev=568	
t=100.56s					
COBYLA	run 830/1000	MSE=3.310e-07	nit=-1	nfev=117	
t=23.23s					
COBYLA	run 831/1000	MSE=3.580e-05	nit=-1	nfev=77	
t=30.15s					
COBYLA	run 832/1000	MSE=5.182e-05	nit=-1	nfev=96	
t=19.00s					
COBYLA	run 833/1000	MSE=2.694e-05	nit=-1	nfev=131	
t=26.94s					
COBYLA	run 834/1000	MSE=5.904e-05	nit=-1	nfev=395	
t=73.93s					
COBYLA	run 835/1000	MSE=4.509e-05	nit=-1	nfev=509	
t=95.01s					
COBYLA	run 836/1000	MSE=1.251e-04	nit=-1	nfev=123	
t=22.39s					
COBYLA	run 837/1000	MSE=7.507e-05	nit=-1	nfev=513	
t=87.61s					
COBYLA	run 838/1000	MSE=2.019e-05	nit=-1	nfev=560	
t=100.02s					
COBYLA	run 839/1000	MSE=1.201e-04	nit=-1	nfev=94	
t=15.27s					
COBYLA	run 840/1000	MSE=8.354e-05	nit=-1	nfev=131	
t=21.82s					
COBYLA	run 841/1000	MSE=3.733e-05	nit=-1	nfev=575	
t=96.47s					
COBYLA	run 842/1000	MSE=4.725e-05	nit=-1	nfev=140	
t=25.66s					
COBYLA	run 843/1000	MSE=5.601e-06	nit=-1	nfev=563	
t=127.49s					
COBYLA	run 844/1000	MSE=5.904e-05	nit=-1	nfev=395	
t=103.72s					
COBYLA	run 845/1000	MSE=5.731e-05	nit=-1	nfev=120	
t=28.80s					
COBYLA	run 846/1000	MSE=4.025e-05	nit=-1	nfev=577	
t=170.17s					
COBYLA	run 847/1000	MSE=2.688e-03	nit=-1	nfev=29	
t=10.22s					
COBYLA	run 848/1000	MSE=2.505e-05	nit=-1	nfev=108	
t=39.43s					
COBYLA	run 849/1000	MSE=1.880e-05	nit=-1	nfev=141	
t=35.33s					
COBYLA	run 850/1000	MSE=4.464e-05	nit=-1	nfev=59	

t=14.44s						
COBYLA	run 851/1000	MSE=2.620e-05	nit=-1	nfev=157		
t=36.55s						
COBYLA	run 852/1000	MSE=1.493e-05	nit=-1	nfev=102		
t=23.72s						
COBYLA	run 853/1000	MSE=1.331e-04	nit=-1	nfev=174		
t=41.73s						
COBYLA	run 854/1000	MSE=4.423e-05	nit=-1	nfev=136		
t=37.14s						
COBYLA	run 855/1000	MSE=3.129e-08	nit=-1	nfev=87		
t=20.12s						
COBYLA	run 856/1000	MSE=8.114e-05	nit=-1	nfev=512		
t=122.59s						
COBYLA	run 857/1000	MSE=8.323e-06	nit=-1	nfev=106		
t=24.67s						
COBYLA	run 858/1000	MSE=5.800e-06	nit=-1	nfev=66		
t=22.27s						
COBYLA	run 859/1000	MSE=4.903e-06	nit=-1	nfev=55		
t=13.50s						
COBYLA	run 860/1000	MSE=3.495e-05	nit=-1	nfev=128		
t=35.68s						
COBYLA	run 861/1000	MSE=1.067e-05	nit=-1	nfev=105		
t=23.13s						
COBYLA	run 862/1000	MSE=2.788e-05	nit=-1	nfev=145		
t=31.61s						
COBYLA	run 863/1000	MSE=2.776e-05	nit=-1	nfev=166		
t=47.84s						
COBYLA	run 864/1000	MSE=6.363e-05	nit=-1	nfev=134		
t=27.50s						
COBYLA	run 865/1000	MSE=7.413e-05	nit=-1	nfev=462		
t=78.92s						
COBYLA	run 866/1000	MSE=3.316e-05	nit=-1	nfev=488		
t=91.15s						
COBYLA	run 867/1000	MSE=4.021e-05	nit=-1	nfev=521		
t=122.96s						
COBYLA	run 868/1000	MSE=3.698e-05	nit=-1	nfev=545		
t=150.09s						
COBYLA	run 869/1000	MSE=5.136e-05	nit=-1	nfev=60		
t=21.17s						
COBYLA	run 870/1000	MSE=1.431e-06	nit=-1	nfev=99		
t=27.44s						
COBYLA	run 871/1000	MSE=7.609e-05	nit=-1	nfev=172		
t=44.70s						
COBYLA	run 872/1000	MSE=2.270e-04	nit=-1	nfev=32	t=8.44s	
COBYLA	run 873/1000	MSE=1.091e-04	nit=-1	nfev=176		
t=51.52s						
COBYLA	run 874/1000	MSE=6.246e-05	nit=-1	nfev=473		
t=132.86s						

COBYLA	run 875/1000	MSE=9.833e-05	nit=-1	nfev=485	
t=110.72s					
COBYLA	run 876/1000	MSE=1.666e-05	nit=-1	nfev=102	
t=20.52s					
COBYLA	run 877/1000	MSE=1.607e-05	nit=-1	nfev=131	
t=26.25s					
COBYLA	run 878/1000	MSE=2.455e-05	nit=-1	nfev=590	
t=118.64s					
COBYLA	run 879/1000	MSE=2.556e-05	nit=-1	nfev=28	t=5.16s
COBYLA	run 880/1000	MSE=3.695e-05	nit=-1	nfev=104	
t=21.06s					
COBYLA	run 881/1000	MSE=2.680e-03	nit=-1	nfev=29	t=6.80s
COBYLA	run 882/1000	MSE=1.794e-06	nit=-1	nfev=30	t=7.42s
COBYLA	run 883/1000	MSE=5.935e-06	nit=-1	nfev=150	
t=36.06s					
COBYLA	run 884/1000	MSE=3.733e-05	nit=-1	nfev=575	
t=111.15s					
COBYLA	run 885/1000	MSE=7.494e-05	nit=-1	nfev=519	
t=103.26s					
COBYLA	run 886/1000	MSE=6.274e-05	nit=-1	nfev=201	
t=40.21s					
COBYLA	run 887/1000	MSE=7.027e-05	nit=-1	nfev=503	
t=99.55s					
COBYLA	run 888/1000	MSE=5.285e-05	nit=-1	nfev=431	
t=79.62s					
COBYLA	run 889/1000	MSE=2.274e-06	nit=-1	nfev=92	
t=17.27s					
COBYLA	run 890/1000	MSE=1.970e-05	nit=-1	nfev=497	
t=93.06s					
COBYLA	run 891/1000	MSE=5.207e-05	nit=-1	nfev=85	
t=16.64s					
COBYLA	run 892/1000	MSE=6.607e-05	nit=-1	nfev=506	
t=96.73s					
COBYLA	run 893/1000	MSE=8.204e-07	nit=-1	nfev=94	
t=16.79s					
COBYLA	run 894/1000	MSE=3.663e-05	nit=-1	nfev=661	
t=136.64s					
COBYLA	run 895/1000	MSE=4.422e-05	nit=-1	nfev=475	
t=88.30s					
COBYLA	run 896/1000	MSE=3.100e-05	nit=-1	nfev=158	
t=30.64s					
COBYLA	run 897/1000	MSE=6.931e-06	nit=-1	nfev=134	
t=24.20s					
COBYLA	run 898/1000	MSE=6.817e-06	nit=-1	nfev=97	
t=17.23s					
COBYLA	run 899/1000	MSE=3.969e-05	nit=-1	nfev=62	
t=11.23s					
COBYLA	run 900/1000	MSE=5.053e-05	nit=-1	nfev=526	

t=92.70s						
COBYLA	run 901/1000	MSE=2.665e-06	nit=-1	nfev=35	t=5.76s	
COBYLA	run 902/1000	MSE=6.794e-05	nit=-1	nfev=114		
t=18.78s						
COBYLA	run 903/1000	MSE=2.328e-05	nit=-1	nfev=45	t=7.50s	
COBYLA	run 904/1000	MSE=2.469e-05	nit=-1	nfev=130		
t=21.99s						
COBYLA	run 905/1000	MSE=2.692e-05	nit=-1	nfev=82		
t=14.24s						
COBYLA	run 906/1000	MSE=3.186e-06	nit=-1	nfev=80		
t=13.53s						
COBYLA	run 907/1000	MSE=1.037e-05	nit=-1	nfev=137		
t=22.77s						
COBYLA	run 908/1000	MSE=9.874e-06	nit=-1	nfev=134		
t=22.44s						
COBYLA	run 909/1000	MSE=3.132e-05	nit=-1	nfev=220		
t=36.91s						
COBYLA	run 910/1000	MSE=1.036e-05	nit=-1	nfev=189		
t=33.03s						
COBYLA	run 911/1000	MSE=3.488e-05	nit=-1	nfev=363		
t=70.09s						
COBYLA	run 912/1000	MSE=3.733e-05	nit=-1	nfev=575		
t=113.03s						
COBYLA	run 913/1000	MSE=6.882e-05	nit=-1	nfev=196		
t=41.75s						
COBYLA	run 914/1000	MSE=2.981e-05	nit=-1	nfev=457		
t=94.95s						
COBYLA	run 915/1000	MSE=1.762e-05	nit=-1	nfev=465		
t=98.15s						
COBYLA	run 916/1000	MSE=3.989e-05	nit=-1	nfev=508		
t=105.70s						
COBYLA	run 917/1000	MSE=6.724e-05	nit=-1	nfev=481		
t=115.90s						
COBYLA	run 918/1000	MSE=4.292e-05	nit=-1	nfev=131		
t=26.76s						
COBYLA	run 919/1000	MSE=4.717e-05	nit=-1	nfev=148		
t=28.83s						
COBYLA	run 920/1000	MSE=6.501e-05	nit=-1	nfev=103		
t=19.56s						
COBYLA	run 921/1000	MSE=4.815e-05	nit=-1	nfev=130		
t=31.20s						
COBYLA	run 922/1000	MSE=1.432e-04	nit=-1	nfev=81		
t=15.10s						
COBYLA	run 923/1000	MSE=6.860e-05	nit=-1	nfev=118		
t=23.73s						
COBYLA	run 924/1000	MSE=1.880e-06	nit=-1	nfev=75		
t=12.59s						
COBYLA	run 925/1000	MSE=2.035e-05	nit=-1	nfev=147		

t=25.12s	COBYLA	run 926/1000	MSE=9.806e-05	nit=-1	nfev=82	
t=16.17s	COBYLA	run 927/1000	MSE=6.989e-06	nit=-1	nfev=79	
t=15.85s	COBYLA	run 928/1000	MSE=2.602e-05	nit=-1	nfev=156	
t=26.50s	COBYLA	run 929/1000	MSE=9.474e-05	nit=-1	nfev=430	
t=75.17s	COBYLA	run 930/1000	MSE=9.179e-06	nit=-1	nfev=151	
t=34.17s	COBYLA	run 931/1000	MSE=4.835e-05	nit=-1	nfev=493	
t=88.02s	COBYLA	run 932/1000	MSE=2.296e-05	nit=-1	nfev=116	
t=21.95s	COBYLA	run 933/1000	MSE=7.544e-05	nit=-1	nfev=95	
t=16.29s	COBYLA	run 934/1000	MSE=1.202e-04	nit=-1	nfev=101	
t=16.71s	COBYLA	run 935/1000	MSE=1.167e-04	nit=-1	nfev=478	
t=82.34s	COBYLA	run 936/1000	MSE=9.657e-05	nit=-1	nfev=555	
t=95.85s	COBYLA	run 937/1000	MSE=7.645e-05	nit=-1	nfev=527	
t=93.74s	COBYLA	run 938/1000	MSE=4.019e-05	nit=-1	nfev=158	
t=28.91s	COBYLA	run 939/1000	MSE=1.492e-04	nit=-1	nfev=50	t=8.85s
	COBYLA	run 940/1000	MSE=2.762e-04	nit=-1	nfev=467	
t=79.48s	COBYLA	run 941/1000	MSE=4.509e-05	nit=-1	nfev=509	
t=91.56s	COBYLA	run 942/1000	MSE=3.574e-05	nit=-1	nfev=74	
t=12.65s	COBYLA	run 943/1000	MSE=2.757e-05	nit=-1	nfev=570	
t=102.07s	COBYLA	run 944/1000	MSE=4.707e-05	nit=-1	nfev=104	
t=17.68s	COBYLA	run 945/1000	MSE=4.407e-05	nit=-1	nfev=554	
t=100.20s	COBYLA	run 946/1000	MSE=1.842e-05	nit=-1	nfev=176	
t=30.32s	COBYLA	run 947/1000	MSE=7.118e-06	nit=-1	nfev=104	
t=17.53s	COBYLA	run 948/1000	MSE=9.355e-05	nit=-1	nfev=54	t=8.93s
	COBYLA	run 949/1000	MSE=5.463e-06	nit=-1	nfev=125	
t=21.01s	COBYLA	run 950/1000	MSE=5.904e-05	nit=-1	nfev=394	

t=66.15s						
COBYLA	run 951/1000	MSE=5.685e-05	nit=-1	nfev=466		
t=78.28s						
COBYLA	run 952/1000	MSE=3.618e-06	nit=-1	nfev=107		
t=18.25s						
COBYLA	run 953/1000	MSE=9.682e-05	nit=-1	nfev=142		
t=24.69s						
COBYLA	run 954/1000	MSE=2.436e-05	nit=-1	nfev=107		
t=17.85s						
COBYLA	run 955/1000	MSE=2.101e-05	nit=-1	nfev=174		
t=28.62s						
COBYLA	run 956/1000	MSE=5.031e-05	nit=-1	nfev=505		
t=84.21s						
COBYLA	run 957/1000	MSE=1.044e-04	nit=-1	nfev=119		
t=19.73s						
COBYLA	run 958/1000	MSE=5.904e-05	nit=-1	nfev=395		
t=65.15s						
COBYLA	run 959/1000	MSE=2.676e-03	nit=-1	nfev=33	t=5.19s	
COBYLA	run 960/1000	MSE=4.407e-05	nit=-1	nfev=554		
t=93.05s						
COBYLA	run 961/1000	MSE=1.275e-04	nit=-1	nfev=480		
t=87.31s						
COBYLA	run 962/1000	MSE=8.046e-05	nit=-1	nfev=534		
t=92.84s						
COBYLA	run 963/1000	MSE=6.411e-05	nit=-1	nfev=122		
t=21.18s						
COBYLA	run 964/1000	MSE=5.904e-05	nit=-1	nfev=394		
t=68.89s						
COBYLA	run 965/1000	MSE=4.097e-05	nit=-1	nfev=562		
t=97.75s						
COBYLA	run 966/1000	MSE=4.917e-05	nit=-1	nfev=153		
t=25.75s						
COBYLA	run 967/1000	MSE=3.802e-05	nit=-1	nfev=154		
t=25.90s						
COBYLA	run 968/1000	MSE=1.696e-05	nit=-1	nfev=522		
t=87.57s						
COBYLA	run 969/1000	MSE=2.773e-05	nit=-1	nfev=148		
t=25.12s						
COBYLA	run 970/1000	MSE=7.826e-05	nit=-1	nfev=475		
t=80.11s						
COBYLA	run 971/1000	MSE=8.061e-05	nit=-1	nfev=137		
t=23.45s						
COBYLA	run 972/1000	MSE=5.904e-05	nit=-1	nfev=395		
t=69.56s						
COBYLA	run 973/1000	MSE=1.427e-04	nit=-1	nfev=124		
t=25.96s						
COBYLA	run 974/1000	MSE=2.795e-05	nit=-1	nfev=175		
t=30.06s						

COBYLA	run 975/1000	MSE=6.757e-05	nit=-1	nfev=518	
t=89.00s					
COBYLA	run 976/1000	MSE=4.939e-05	nit=-1	nfev=532	
t=92.46s					
COBYLA	run 977/1000	MSE=5.904e-05	nit=-1	nfev=394	
t=65.95s					
COBYLA	run 978/1000	MSE=4.724e-05	nit=-1	nfev=99	
t=17.05s					
COBYLA	run 979/1000	MSE=2.645e-05	nit=-1	nfev=176	
t=30.11s					
COBYLA	run 980/1000	MSE=2.499e-05	nit=-1	nfev=122	
t=22.50s					
COBYLA	run 981/1000	MSE=1.639e-04	nit=-1	nfev=490	
t=99.54s					
COBYLA	run 982/1000	MSE=1.421e-05	nit=-1	nfev=32	t=5.51s
COBYLA	run 983/1000	MSE=6.242e-05	nit=-1	nfev=116	
t=21.11s					
COBYLA	run 984/1000	MSE=7.110e-07	nit=-1	nfev=132	
t=24.61s					
COBYLA	run 985/1000	MSE=1.946e-05	nit=-1	nfev=114	
t=18.51s					
COBYLA	run 986/1000	MSE=8.346e-05	nit=-1	nfev=124	
t=20.97s					
COBYLA	run 987/1000	MSE=2.311e-03	nit=-1	nfev=66	
t=13.40s					
COBYLA	run 988/1000	MSE=1.013e-04	nit=-1	nfev=120	
t=21.49s					
COBYLA	run 989/1000	MSE=5.904e-05	nit=-1	nfev=395	
t=66.61s					
COBYLA	run 990/1000	MSE=1.104e-04	nit=-1	nfev=500	
t=82.30s					
COBYLA	run 991/1000	MSE=3.978e-05	nit=-1	nfev=124	
t=20.24s					
COBYLA	run 992/1000	MSE=2.252e-05	nit=-1	nfev=71	
t=12.00s					
COBYLA	run 993/1000	MSE=1.352e-06	nit=-1	nfev=116	
t=20.97s					
COBYLA	run 994/1000	MSE=1.636e-05	nit=-1	nfev=486	
t=86.64s					
COBYLA	run 995/1000	MSE=3.678e-06	nit=-1	nfev=119	
t=21.29s					
COBYLA	run 996/1000	MSE=5.275e-05	nit=-1	nfev=511	
t=86.17s					
COBYLA	run 997/1000	MSE=7.507e-05	nit=-1	nfev=513	
t=91.54s					
COBYLA	run 998/1000	MSE=1.210e-05	nit=-1	nfev=87	
t=16.34s					
COBYLA	run 999/1000	MSE=1.374e-04	nit=-1	nfev=113	

t=18.56s

COBYLA

run 1000/1000 MSE=2.176e-05 nit=-1 nfev=503

t=83.88s

Scenario A - trust-constr (1000 runs)

trust-constr	run 1/1000	MSE=4.994e-15	nit=23	nfev=54	t=9.35s
trust-constr	run 2/1000	MSE=5.049e-15	nit=19	nfev=39	t=8.59s
trust-constr	run 3/1000	MSE=7.545e-14	nit=69	nfev=183	t=32.21s
trust-constr	run 4/1000	MSE=1.379e-12	nit=35	nfev=84	t=13.53s
trust-constr	run 5/1000	MSE=1.787e-12	nit=44	nfev=123	t=19.87s
trust-constr	run 6/1000	MSE=7.590e-14	nit=27	nfev=63	t=10.46s
trust-constr	run 7/1000	MSE=7.667e-14	nit=77	nfev=210	t=39.21s
trust-constr	run 8/1000	MSE=9.960e-12	nit=28	nfev=66	t=15.15s
trust-constr	run 9/1000	MSE=1.266e-12	nit=27	nfev=60	t=10.80s
trust-constr	run 10/1000	MSE=7.545e-14	nit=41	nfev=102	t=15.37s
trust-constr	run 11/1000	MSE=1.266e-12	nit=24	nfev=51	t=7.58s
trust-constr	run 12/1000	MSE=1.266e-12	nit=25	nfev=54	t=8.41s
trust-constr	run 13/1000	MSE=1.379e-12	nit=30	nfev=69	t=10.26s
trust-constr	run 14/1000	MSE=9.975e-12	nit=26	nfev=60	t=8.92s
trust-constr	run 15/1000	MSE=9.975e-12	nit=26	nfev=60	t=9.01s
trust-constr	run 16/1000	MSE=1.379e-12	nit=39	nfev=102	t=15.30s
trust-constr	run 17/1000	MSE=7.599e-14	nit=31	nfev=75	t=11.68s
trust-constr	run 18/1000	MSE=4.869e-16	nit=80	nfev=219	t=33.08s
trust-constr	run 19/1000	MSE=3.864e-15	nit=84	nfev=228	t=33.89s
trust-constr	run 20/1000	MSE=8.684e-16	nit=80	nfev=216	t=32.35s
trust-constr	run 21/1000	MSE=1.266e-12	nit=48	nfev=129	t=19.47s
trust-constr	run 22/1000	MSE=1.379e-12	nit=43	nfev=108	t=16.10s
trust-constr	run 23/1000	MSE=5.000e-15	nit=19	nfev=39	t=6.64s
trust-constr	run 24/1000	MSE=7.607e-14	nit=27	nfev=60	t=8.98s
trust-constr	run 25/1000	MSE=1.266e-12	nit=31	nfev=75	t=11.20s
trust-constr	run 26/1000	MSE=4.974e-15	nit=20	nfev=42	t=6.35s
trust-constr	run 27/1000	MSE=9.974e-12	nit=97	nfev=276	t=41.27s
trust-constr	run 28/1000	MSE=7.597e-14	nit=67	nfev=174	t=26.56s
trust-constr	run 29/1000	MSE=3.297e-15	nit=84	nfev=231	t=34.31s
trust-constr	run 30/1000	MSE=9.995e-12	nit=46	nfev=123	t=18.16s
trust-constr	run 31/1000	MSE=7.614e-14	nit=22	nfev=45	t=7.41s
trust-constr	run 32/1000	MSE=6.825e-16	nit=77	nfev=210	t=31.86s
trust-constr	run 33/1000	MSE=9.995e-12	nit=31	nfev=78	t=11.87s
trust-constr	run 34/1000	MSE=9.975e-12	nit=28	nfev=66	t=10.29s
trust-constr	run 35/1000	MSE=7.618e-14	nit=30	nfev=75	t=11.47s
trust-constr	run 36/1000	MSE=1.379e-12	nit=28	nfev=66	t=9.78s
trust-constr	run 37/1000	MSE=1.266e-12	nit=31	nfev=75	t=10.97s
trust-constr	run 38/1000	MSE=5.010e-15	nit=86	nfev=249	t=36.74s
trust-constr	run 39/1000	MSE=1.066e-15	nit=99	nfev=273	t=40.77s
trust-constr	run 40/1000	MSE=1.379e-12	nit=25	nfev=54	t=8.10s
trust-constr	run 41/1000	MSE=6.068e-13	nit=37	nfev=87	t=13.01s
trust-constr	run 42/1000	MSE=1.001e-11	nit=27	nfev=66	t=9.83s
trust-constr	run 43/1000	MSE=1.266e-12	nit=25	nfev=54	t=7.96s

trust-constr	run 44/1000	MSE=1.266e-12	nit=34	nfev=84	t=12.33s
trust-constr	run 45/1000	MSE=6.066e-13	nit=34	nfev=81	t=12.02s
trust-constr	run 46/1000	MSE=3.477e-15	nit=85	nfev=231	t=34.18s
trust-constr	run 47/1000	MSE=1.379e-12	nit=33	nfev=78	t=11.69s
trust-constr	run 48/1000	MSE=1.001e-11	nit=25	nfev=57	t=8.39s
trust-constr	run 49/1000	MSE=7.608e-14	nit=25	nfev=54	t=8.06s
trust-constr	run 50/1000	MSE=1.379e-12	nit=39	nfev=99	t=14.39s
trust-constr	run 51/1000	MSE=6.067e-13	nit=32	nfev=75	t=11.28s
trust-constr	run 52/1000	MSE=9.591e-16	nit=80	nfev=219	t=40.09s
trust-constr	run 53/1000	MSE=7.617e-14	nit=68	nfev=186	t=32.30s
trust-constr	run 54/1000	MSE=7.641e-16	nit=108	nfev=300	t=48.96s
trust-constr	run 55/1000	MSE=7.621e-14	nit=52	nfev=138	t=22.09s
trust-constr	run 56/1000	MSE=7.585e-14	nit=27	nfev=60	t=9.61s
trust-constr	run 57/1000	MSE=9.999e-12	nit=24	nfev=54	t=8.93s
trust-constr	run 58/1000	MSE=2.553e-16	nit=74	nfev=198	t=35.37s
trust-constr	run 59/1000	MSE=1.509e+00	nit=175	nfev=174	t=29.25s
trust-constr	run 60/1000	MSE=5.073e-15	nit=32	nfev=81	t=13.04s
trust-constr	run 61/1000	MSE=1.266e-12	nit=27	nfev=60	t=9.47s
trust-constr	run 62/1000	MSE=1.379e-12	nit=34	nfev=84	t=13.36s
trust-constr	run 63/1000	MSE=1.379e-12	nit=26	nfev=57	t=10.15s
trust-constr	run 64/1000	MSE=7.601e-14	nit=27	nfev=60	t=10.36s
trust-constr	run 65/1000	MSE=7.606e-14	nit=42	nfev=108	t=19.24s
trust-constr	run 66/1000	MSE=6.489e-13	nit=29	nfev=63	t=11.10s
trust-constr	run 67/1000	MSE=1.379e-12	nit=34	nfev=81	t=13.19s
trust-constr	run 68/1000	MSE=2.978e-15	nit=91	nfev=255	t=43.93s
trust-constr	run 69/1000	MSE=9.977e-12	nit=53	nfev=144	t=25.76s
trust-constr	run 70/1000	MSE=9.975e-12	nit=27	nfev=63	t=11.44s
trust-constr	run 71/1000	MSE=4.230e-15	nit=92	nfev=252	t=44.66s
trust-constr	run 72/1000	MSE=1.379e-12	nit=54	nfev=147	t=27.56s
trust-constr	run 73/1000	MSE=1.266e-12	nit=29	nfev=66	t=10.93s
trust-constr	run 74/1000	MSE=7.320e-16	nit=105	nfev=291	t=51.89s
trust-constr	run 75/1000	MSE=9.981e-12	nit=27	nfev=66	t=11.07s
trust-constr	run 76/1000	MSE=7.606e-14	nit=25	nfev=57	t=12.73s
trust-constr	run 77/1000	MSE=1.379e-12	nit=25	nfev=54	t=9.91s
trust-constr	run 78/1000	MSE=9.996e-12	nit=33	nfev=90	t=15.43s
trust-constr	run 79/1000	MSE=5.008e-15	nit=19	nfev=39	t=6.39s
trust-constr	run 80/1000	MSE=8.033e-16	nit=79	nfev=219	t=36.83s
trust-constr	run 81/1000	MSE=1.787e-12	nit=26	nfev=57	t=9.74s
trust-constr	run 82/1000	MSE=8.128e-16	nit=95	nfev=264	t=40.85s
trust-constr	run 83/1000	MSE=1.787e-12	nit=29	nfev=69	t=10.50s
trust-constr	run 84/1000	MSE=3.037e-15	nit=82	nfev=222	t=33.53s
trust-constr	run 85/1000	MSE=1.379e-12	nit=29	nfev=66	t=9.97s
trust-constr	run 86/1000	MSE=7.769e-14	nit=41	nfev=114	t=17.39s
trust-constr	run 87/1000	MSE=9.975e-12	nit=31	nfev=78	t=12.00s
trust-constr	run 88/1000	MSE=1.266e-12	nit=28	nfev=63	t=9.56s
trust-constr	run 89/1000	MSE=7.641e-14	nit=81	nfev=225	t=34.44s
trust-constr	run 90/1000	MSE=1.787e-12	nit=30	nfev=72	t=10.77s
trust-constr	run 91/1000	MSE=1.266e-12	nit=30	nfev=69	t=12.13s

trust-constr	run 92/1000	MSE=1.379e-12	nit=24	nfev=54	t=8.98s
trust-constr	run 93/1000	MSE=1.589e-11	nit=26	nfev=60	t=9.05s
trust-constr	run 94/1000	MSE=7.599e-14	nit=42	nfev=108	t=16.37s
trust-constr	run 95/1000	MSE=9.983e-12	nit=24	nfev=54	t=8.39s
trust-constr	run 96/1000	MSE=7.600e-14	nit=23	nfev=48	t=7.39s
trust-constr	run 97/1000	MSE=1.379e-12	nit=34	nfev=81	t=12.23s
trust-constr	run 98/1000	MSE=6.067e-13	nit=33	nfev=75	t=12.06s
trust-constr	run 99/1000	MSE=1.379e-12	nit=31	nfev=72	t=11.84s
trust-constr	run 100/1000	MSE=9.978e-12	nit=22	nfev=54	t=9.66s
trust-constr	run 101/1000	MSE=9.973e-12	nit=97	nfev=276	
t=48.61s					
trust-constr	run 102/1000	MSE=6.067e-13	nit=46	nfev=120	
t=21.14s					
trust-constr	run 103/1000	MSE=2.041e-15	nit=110	nfev=315	
t=48.45s					
trust-constr	run 104/1000	MSE=9.277e-16	nit=112	nfev=312	
t=47.90s					
trust-constr	run 105/1000	MSE=1.379e-12	nit=27	nfev=60	t=9.09s
trust-constr	run 106/1000	MSE=7.607e-14	nit=25	nfev=54	t=8.32s
trust-constr	run 107/1000	MSE=1.266e-12	nit=29	nfev=66	
t=10.01s					
trust-constr	run 108/1000	MSE=7.599e-14	nit=70	nfev=192	
t=30.09s					
trust-constr	run 109/1000	MSE=1.001e-11	nit=27	nfev=66	
t=10.20s					
trust-constr	run 110/1000	MSE=1.379e-12	nit=26	nfev=57	t=8.87s
trust-constr	run 111/1000	MSE=7.613e-14	nit=109	nfev=306	
t=46.71s					
trust-constr	run 112/1000	MSE=7.620e-14	nit=60	nfev=153	
t=23.47s					
trust-constr	run 113/1000	MSE=6.067e-13	nit=35	nfev=81	
t=12.52s					
trust-constr	run 114/1000	MSE=2.344e-16	nit=89	nfev=246	
t=38.36s					
trust-constr	run 115/1000	MSE=1.379e-12	nit=26	nfev=57	t=8.62s
trust-constr	run 116/1000	MSE=1.379e-12	nit=29	nfev=69	
t=10.35s					
trust-constr	run 117/1000	MSE=4.952e-15	nit=30	nfev=72	
t=11.16s					
trust-constr	run 118/1000	MSE=2.679e-16	nit=75	nfev=201	
t=30.89s					
trust-constr	run 119/1000	MSE=3.877e-15	nit=87	nfev=240	
t=39.19s					
trust-constr	run 120/1000	MSE=1.787e-12	nit=32	nfev=75	
t=12.24s					
trust-constr	run 121/1000	MSE=1.379e-12	nit=24	nfev=51	t=8.37s
trust-constr	run 122/1000	MSE=1.266e-12	nit=26	nfev=57	
t=10.64s					

trust-constr	run 123/1000	MSE=7.606e-14	nit=21	nfev=42	t=6.98s
trust-constr	run 124/1000	MSE=4.771e-17	nit=88	nfev=240	
t=39.59s					
trust-constr	run 125/1000	MSE=1.379e-12	nit=26	nfev=57	t=9.04s
trust-constr	run 126/1000	MSE=9.965e-12	nit=29	nfev=69	
t=11.87s					
trust-constr	run 127/1000	MSE=7.603e-14	nit=25	nfev=57	t=9.51s
trust-constr	run 128/1000	MSE=4.110e-16	nit=82	nfev=225	
t=34.84s					
trust-constr	run 129/1000	MSE=1.266e-12	nit=23	nfev=48	t=7.54s
trust-constr	run 130/1000	MSE=1.841e-15	nit=80	nfev=216	
t=35.27s					
trust-constr	run 131/1000	MSE=4.817e-15	nit=28	nfev=69	
t=11.45s					
trust-constr	run 132/1000	MSE=1.108e-15	nit=82	nfev=222	
t=36.42s					
trust-constr	run 133/1000	MSE=6.280e-16	nit=127	nfev=357	
t=57.31s					
trust-constr	run 134/1000	MSE=1.379e-12	nit=31	nfev=72	
t=13.71s					
trust-constr	run 135/1000	MSE=9.885e-14	nit=31	nfev=69	
t=12.41s					
trust-constr	run 136/1000	MSE=4.970e-15	nit=19	nfev=39	t=6.24s
trust-constr	run 137/1000	MSE=6.067e-13	nit=27	nfev=57	t=9.60s
trust-constr	run 138/1000	MSE=5.233e-16	nit=90	nfev=249	
t=41.55s					
trust-constr	run 139/1000	MSE=6.067e-13	nit=37	nfev=90	
t=15.22s					
trust-constr	run 140/1000	MSE=6.066e-13	nit=27	nfev=57	t=9.63s
trust-constr	run 141/1000	MSE=8.214e-14	nit=64	nfev=165	
t=27.98s					
trust-constr	run 142/1000	MSE=1.266e-12	nit=28	nfev=63	
t=10.55s					
trust-constr	run 143/1000	MSE=1.379e-12	nit=23	nfev=48	t=8.05s
trust-constr	run 144/1000	MSE=7.649e-14	nit=58	nfev=147	
t=24.60s					
trust-constr	run 145/1000	MSE=6.069e-13	nit=30	nfev=66	
t=11.13s					
trust-constr	run 146/1000	MSE=7.606e-14	nit=21	nfev=42	t=7.22s
trust-constr	run 147/1000	MSE=1.379e-12	nit=25	nfev=54	t=9.28s
trust-constr	run 148/1000	MSE=4.823e-15	nit=28	nfev=69	
t=11.57s					
trust-constr	run 149/1000	MSE=1.379e-12	nit=31	nfev=72	
t=11.95s					
trust-constr	run 150/1000	MSE=1.379e-12	nit=32	nfev=78	
t=13.09s					
trust-constr	run 151/1000	MSE=6.067e-13	nit=29	nfev=63	
t=10.39s					

trust-constr t=37.11s	run 152/1000	MSE=6.809e-16	nit=78	nfev=213	
trust-constr t=11.28s	run 153/1000	MSE=1.266e-12	nit=29	nfev=66	
trust-constr t=36.08s	run 154/1000	MSE=5.598e-16	nit=76	nfev=204	
trust-constr t=37.01s	run 155/1000	MSE=4.893e-16	nit=75	nfev=201	
trust-constr t=10.74s	run 156/1000	MSE=7.606e-14	nit=27	nfev=60	
trust-constr t=11.38s	run 157/1000	MSE=9.883e-14	nit=29	nfev=63	
trust-constr t=11.33s	run 158/1000	MSE=1.266e-12	nit=28	nfev=63	
trust-constr t=16.48s	run 159/1000	MSE=7.625e-14	nit=35	nfev=93	
trust-constr t=15.77s	run 160/1000	MSE=7.608e-14	nit=34	nfev=84	
trust-constr t=14.01s	run 161/1000	MSE=1.379e-12	nit=31	nfev=75	
trust-constr t=46.31s	run 162/1000	MSE=1.001e-11	nit=90	nfev=255	
trust-constr t=10.71s	run 163/1000	MSE=1.379e-12	nit=25	nfev=57	
trust-constr t=10.06s	run 164/1000	MSE=7.632e-14	nit=25	nfev=54	
trust-constr	run 165/1000	MSE=1.379e-12	nit=23	nfev=48	t=9.83s
trust-constr t=12.49s	run 166/1000	MSE=1.787e-12	nit=27	nfev=60	
trust-constr	run 167/1000	MSE=1.266e-12	nit=27	nfev=60	t=9.83s
trust-constr	run 168/1000	MSE=1.266e-12	nit=27	nfev=60	t=9.95s
trust-constr t=34.40s	run 169/1000	MSE=4.784e-16	nit=75	nfev=201	
trust-constr t=11.91s	run 170/1000	MSE=1.379e-12	nit=31	nfev=75	
trust-constr t=27.48s	run 171/1000	MSE=5.986e-14	nit=64	nfev=165	
trust-constr t=29.59s	run 172/1000	MSE=1.379e-12	nit=66	nfev=177	
trust-constr	run 173/1000	MSE=1.379e-12	nit=25	nfev=54	t=8.29s
trust-constr t=21.50s	run 174/1000	MSE=1.379e-12	nit=53	nfev=138	
trust-constr t=12.11s	run 175/1000	MSE=1.379e-12	nit=31	nfev=75	
trust-constr t=11.79s	run 176/1000	MSE=9.885e-14	nit=34	nfev=78	
trust-constr t=20.41s	run 177/1000	MSE=1.266e-12	nit=48	nfev=123	

trust-constr t=66.04s	run 178/1000	MSE=1.266e-12	nit=137	nfev=393	
trust-constr t=55.04s	run 179/1000	MSE=1.379e-12	nit=114	nfev=324	
trust-constr t=11.11s	run 180/1000	MSE=4.905e-13	nit=33	nfev=72	
trust-constr	run 181/1000	MSE=1.379e-12	nit=26	nfev=57	t=9.25s
trust-constr t=17.07s	run 182/1000	MSE=4.897e-13	nit=43	nfev=102	
trust-constr t=32.66s	run 183/1000	MSE=7.456e-14	nit=74	nfev=201	
trust-constr	run 184/1000	MSE=1.379e-12	nit=26	nfev=57	t=9.13s
trust-constr	run 185/1000	MSE=1.266e-12	nit=23	nfev=48	t=8.17s
trust-constr t=40.60s	run 186/1000	MSE=2.830e-15	nit=86	nfev=237	
trust-constr t=17.42s	run 187/1000	MSE=4.910e-15	nit=42	nfev=111	
trust-constr	run 188/1000	MSE=1.003e-11	nit=24	nfev=54	t=8.68s
trust-constr	run 189/1000	MSE=1.379e-12	nit=26	nfev=57	t=9.25s
trust-constr t=10.73s	run 190/1000	MSE=1.379e-12	nit=29	nfev=66	
trust-constr	run 191/1000	MSE=1.379e-12	nit=27	nfev=60	t=9.45s
trust-constr t=13.22s	run 192/1000	MSE=1.379e-12	nit=29	nfev=69	
trust-constr t=43.56s	run 193/1000	MSE=8.699e-16	nit=98	nfev=273	
trust-constr t=14.05s	run 194/1000	MSE=5.223e-15	nit=35	nfev=90	
trust-constr	run 195/1000	MSE=9.986e-12	nit=22	nfev=48	t=7.41s
trust-constr t=10.75s	run 196/1000	MSE=6.067e-13	nit=30	nfev=66	
trust-constr t=11.20s	run 197/1000	MSE=9.987e-12	nit=29	nfev=72	
trust-constr t=15.53s	run 198/1000	MSE=9.999e-12	nit=36	nfev=96	
trust-constr	run 199/1000	MSE=4.811e-15	nit=26	nfev=60	t=9.72s
trust-constr t=23.80s	run 200/1000	MSE=1.379e-12	nit=49	nfev=135	
trust-constr t=12.00s	run 201/1000	MSE=9.971e-12	nit=31	nfev=75	
trust-constr t=15.68s	run 202/1000	MSE=1.266e-12	nit=38	nfev=99	
trust-constr t=16.69s	run 203/1000	MSE=7.604e-14	nit=38	nfev=99	
trust-constr t=18.10s	run 204/1000	MSE=7.616e-14	nit=37	nfev=90	
trust-constr t=23.61s	run 205/1000	MSE=7.630e-14	nit=52	nfev=138	

trust-constr t=26.33s	run 206/1000	MSE=7.613e-14	nit=62	nfev=162	
trust-constr t=15.57s	run 207/1000	MSE=1.379e-12	nit=37	nfev=96	
trust-constr t=10.84s	run 208/1000	MSE=6.066e-13	nit=30	nfev=66	
trust-constr t=14.71s	run 209/1000	MSE=4.992e-15	nit=34	nfev=87	
trust-constr t=12.97s	run 210/1000	MSE=7.615e-14	nit=32	nfev=78	
trust-constr t=10.65s	run 211/1000	MSE=4.801e-15	nit=27	nfev=63	
trust-constr t=10.30s	run 212/1000	MSE=1.266e-12	nit=25	nfev=54	
trust-constr	run 213/1000	MSE=7.597e-14	nit=21	nfev=42	t=7.37s
trust-constr t=13.76s	run 214/1000	MSE=9.885e-14	nit=35	nfev=84	
trust-constr t=32.80s	run 215/1000	MSE=2.060e-15	nit=76	nfev=204	
trust-constr t=22.94s	run 216/1000	MSE=6.944e-14	nit=57	nfev=144	
trust-constr t=24.14s	run 217/1000	MSE=7.602e-14	nit=40	nfev=108	
trust-constr	run 218/1000	MSE=7.613e-14	nit=19	nfev=36	t=7.62s
trust-constr t=61.86s	run 219/1000	MSE=8.431e-16	nit=140	nfev=399	
trust-constr	run 220/1000	MSE=7.599e-14	nit=21	nfev=42	t=6.29s
trust-constr t=11.88s	run 221/1000	MSE=6.066e-13	nit=34	nfev=78	
trust-constr	run 222/1000	MSE=7.606e-14	nit=26	nfev=57	t=8.33s
trust-constr t=13.25s	run 223/1000	MSE=1.379e-12	nit=35	nfev=90	
trust-constr	run 224/1000	MSE=7.606e-14	nit=24	nfev=54	t=8.01s
trust-constr	run 225/1000	MSE=1.787e-12	nit=28	nfev=63	t=9.22s
trust-constr	run 226/1000	MSE=1.266e-12	nit=29	nfev=66	t=9.66s
trust-constr t=11.77s	run 227/1000	MSE=1.379e-12	nit=32	nfev=78	
trust-constr t=10.49s	run 228/1000	MSE=7.628e-14	nit=31	nfev=72	
trust-constr t=10.68s	run 229/1000	MSE=1.002e-11	nit=29	nfev=72	
trust-constr t=18.40s	run 230/1000	MSE=7.611e-14	nit=47	nfev=123	
trust-constr	run 231/1000	MSE=1.788e-12	nit=28	nfev=63	t=9.31s
trust-constr t=14.13s	run 232/1000	MSE=6.067e-13	nit=39	nfev=96	
trust-constr t=32.29s	run 233/1000	MSE=8.149e-16	nit=79	nfev=219	

trust-constr t=30.00s	run 234/1000	MSE=5.319e-15	nit=76	nfev=204	
trust-constr t=44.40s	run 235/1000	MSE=7.610e-14	nit=106	nfev=300	
trust-constr	run 236/1000	MSE=6.067e-13	nit=29	nfev=63	t=9.12s
trust-constr	run 237/1000	MSE=1.266e-12	nit=27	nfev=60	t=8.81s
trust-constr t=38.80s	run 238/1000	MSE=9.750e-16	nit=87	nfev=240	
trust-constr	run 239/1000	MSE=1.266e-12	nit=27	nfev=60	t=9.38s
trust-constr	run 240/1000	MSE=7.606e-14	nit=21	nfev=42	t=6.83s
trust-constr t=41.74s	run 241/1000	MSE=3.836e-15	nit=92	nfev=252	
trust-constr t=10.09s	run 242/1000	MSE=9.984e-12	nit=26	nfev=60	
trust-constr t=17.28s	run 243/1000	MSE=4.715e-15	nit=40	nfev=105	
trust-constr t=12.98s	run 244/1000	MSE=1.379e-12	nit=32	nfev=75	
trust-constr	run 245/1000	MSE=5.155e-15	nit=20	nfev=42	t=7.52s
trust-constr t=15.11s	run 246/1000	MSE=7.608e-14	nit=37	nfev=93	
trust-constr t=33.82s	run 247/1000	MSE=2.549e-16	nit=75	nfev=201	
trust-constr t=13.82s	run 248/1000	MSE=7.610e-14	nit=34	nfev=84	
trust-constr	run 249/1000	MSE=7.605e-14	nit=21	nfev=42	t=6.52s
trust-constr	run 250/1000	MSE=9.996e-12	nit=24	nfev=54	t=8.60s
trust-constr t=10.77s	run 251/1000	MSE=9.887e-14	nit=30	nfev=66	
trust-constr t=65.66s	run 252/1000	MSE=4.988e-16	nit=137	nfev=390	
trust-constr t=10.64s	run 253/1000	MSE=1.787e-12	nit=26	nfev=57	
trust-constr	run 254/1000	MSE=7.604e-14	nit=24	nfev=51	t=9.46s
trust-constr t=10.76s	run 255/1000	MSE=7.625e-14	nit=26	nfev=57	
trust-constr t=14.18s	run 256/1000	MSE=7.610e-14	nit=34	nfev=84	
trust-constr t=11.93s	run 257/1000	MSE=1.266e-12	nit=29	nfev=66	
trust-constr t=25.25s	run 258/1000	MSE=6.892e-14	nit=57	nfev=144	
trust-constr t=32.31s	run 259/1000	MSE=7.605e-14	nit=72	nfev=195	
trust-constr t=12.09s	run 260/1000	MSE=9.883e-14	nit=32	nfev=72	
trust-constr	run 261/1000	MSE=1.379e-12	nit=23	nfev=48	t=7.60s
trust-constr	run 262/1000	MSE=5.082e-15	nit=23	nfev=51	t=8.45s

trust-constr t=53.34s	run 263/1000	MSE=2.764e-16	nit=106	nfev=294	
trust-constr t=37.06s	run 264/1000	MSE=1.379e-12	nit=76	nfev=210	
trust-constr t=33.53s	run 265/1000	MSE=2.447e-15	nit=74	nfev=198	
trust-constr t=11.18s	run 266/1000	MSE=1.001e-11	nit=27	nfev=66	
trust-constr	run 267/1000	MSE=4.936e-15	nit=20	nfev=42	t=7.37s
trust-constr t=20.18s	run 268/1000	MSE=7.601e-14	nit=45	nfev=117	
trust-constr	run 269/1000	MSE=1.379e-12	nit=25	nfev=54	t=9.18s
trust-constr t=10.44s	run 270/1000	MSE=5.049e-15	nit=25	nfev=57	
trust-constr t=49.40s	run 271/1000	MSE=1.379e-12	nit=102	nfev=288	
trust-constr t=14.14s	run 272/1000	MSE=1.379e-12	nit=34	nfev=87	
trust-constr	run 273/1000	MSE=1.379e-12	nit=26	nfev=57	t=9.75s
trust-constr t=31.46s	run 274/1000	MSE=7.659e-14	nit=65	nfev=180	
trust-constr t=49.77s	run 275/1000	MSE=3.928e-16	nit=108	nfev=300	
trust-constr	run 276/1000	MSE=1.266e-12	nit=25	nfev=54	t=8.67s
trust-constr t=14.53s	run 277/1000	MSE=4.814e-15	nit=35	nfev=90	
trust-constr	run 278/1000	MSE=9.883e-14	nit=27	nfev=57	t=9.10s
trust-constr t=33.11s	run 279/1000	MSE=2.871e-16	nit=77	nfev=207	
trust-constr	run 280/1000	MSE=9.883e-14	nit=29	nfev=63	t=9.97s
trust-constr t=43.76s	run 281/1000	MSE=8.430e-16	nit=99	nfev=276	
trust-constr t=32.66s	run 282/1000	MSE=2.594e-16	nit=75	nfev=201	
trust-constr	run 283/1000	MSE=1.379e-12	nit=27	nfev=60	t=9.90s
trust-constr	run 284/1000	MSE=9.980e-12	nit=22	nfev=48	t=7.45s
trust-constr t=31.59s	run 285/1000	MSE=6.623e-16	nit=75	nfev=201	
trust-constr t=35.18s	run 286/1000	MSE=8.028e-14	nit=81	nfev=222	
trust-constr t=18.56s	run 287/1000	MSE=1.266e-12	nit=43	nfev=117	
trust-constr t=14.30s	run 288/1000	MSE=1.379e-12	nit=36	nfev=90	
trust-constr t=12.20s	run 289/1000	MSE=6.067e-13	nit=33	nfev=78	
trust-constr t=36.54s	run 290/1000	MSE=7.382e-16	nit=83	nfev=228	

trust-constr t=12.04s	run 291/1000	MSE=9.884e-14	nit=32	nfev=72	
trust-constr t=10.81s	run 292/1000	MSE=1.001e-11	nit=28	nfev=69	
trust-constr t=20.08s	run 293/1000	MSE=7.632e-14	nit=46	nfev=126	
trust-constr	run 294/1000	MSE=5.182e-15	nit=20	nfev=42	t=7.16s
trust-constr	run 295/1000	MSE=1.379e-12	nit=26	nfev=57	t=9.09s
trust-constr	run 296/1000	MSE=1.788e-12	nit=22	nfev=45	t=7.16s
trust-constr t=16.96s	run 297/1000	MSE=1.787e-12	nit=41	nfev=108	
trust-constr t=10.55s	run 298/1000	MSE=7.615e-14	nit=28	nfev=66	
trust-constr t=16.56s	run 299/1000	MSE=7.612e-14	nit=40	nfev=105	
trust-constr t=12.33s	run 300/1000	MSE=9.882e-14	nit=32	nfev=78	
trust-constr t=11.44s	run 301/1000	MSE=6.066e-13	nit=31	nfev=72	
trust-constr t=17.07s	run 302/1000	MSE=1.379e-12	nit=40	nfev=105	
trust-constr t=15.20s	run 303/1000	MSE=7.575e-14	nit=38	nfev=96	
trust-constr t=17.31s	run 304/1000	MSE=9.993e-12	nit=40	nfev=108	
trust-constr t=31.26s	run 305/1000	MSE=1.379e-12	nit=73	nfev=198	
trust-constr t=12.95s	run 306/1000	MSE=1.379e-12	nit=32	nfev=78	
trust-constr t=82.84s	run 307/1000	MSE=5.053e-15	nit=171	nfev=498	
trust-constr	run 308/1000	MSE=1.379e-12	nit=25	nfev=54	t=9.38s
trust-constr t=10.42s	run 309/1000	MSE=1.787e-12	nit=26	nfev=57	
trust-constr t=18.75s	run 310/1000	MSE=7.627e-14	nit=45	nfev=114	
trust-constr t=13.74s	run 311/1000	MSE=1.266e-12	nit=34	nfev=84	
trust-constr t=13.22s	run 312/1000	MSE=1.266e-12	nit=30	nfev=72	
trust-constr t=45.36s	run 313/1000	MSE=5.656e-16	nit=96	nfev=264	
trust-constr t=30.19s	run 314/1000	MSE=7.674e-14	nit=60	nfev=153	
trust-constr	run 315/1000	MSE=1.379e-12	nit=25	nfev=54	t=8.89s
trust-constr t=24.94s	run 316/1000	MSE=7.955e-14	nit=59	nfev=150	
trust-constr	run 317/1000	MSE=7.611e-14	nit=20	nfev=39	t=6.51s

trust-constr t=25.90s	run 318/1000	MSE=7.608e-14	nit=58	nfev=156	
trust-constr t=13.39s	run 319/1000	MSE=7.609e-14	nit=33	nfev=81	
trust-constr t=13.00s	run 320/1000	MSE=1.379e-12	nit=32	nfev=75	
trust-constr t=10.83s	run 321/1000	MSE=7.605e-14	nit=28	nfev=66	
trust-constr t=14.80s	run 322/1000	MSE=1.266e-12	nit=35	nfev=87	
trust-constr t=16.58s	run 323/1000	MSE=1.379e-12	nit=36	nfev=90	
trust-constr t=67.52s	run 324/1000	MSE=3.893e-15	nit=131	nfev=369	
trust-constr t=10.84s	run 325/1000	MSE=1.266e-12	nit=27	nfev=60	
trust-constr t=11.06s	run 326/1000	MSE=9.886e-14	nit=31	nfev=69	
trust-constr t=36.52s	run 327/1000	MSE=9.994e-15	nit=79	nfev=213	
trust-constr t=10.68s	run 328/1000	MSE=7.606e-14	nit=25	nfev=54	
trust-constr	run 329/1000	MSE=7.612e-14	nit=22	nfev=45	t=8.68s
trust-constr	run 330/1000	MSE=7.600e-14	nit=24	nfev=51	t=9.06s
trust-constr t=12.51s	run 331/1000	MSE=1.266e-12	nit=30	nfev=75	
trust-constr t=36.24s	run 332/1000	MSE=2.373e-15	nit=79	nfev=216	
trust-constr t=14.77s	run 333/1000	MSE=1.379e-12	nit=37	nfev=93	
trust-constr t=10.54s	run 334/1000	MSE=1.379e-12	nit=27	nfev=60	
trust-constr t=14.42s	run 335/1000	MSE=9.974e-12	nit=31	nfev=78	
trust-constr t=10.47s	run 336/1000	MSE=1.379e-12	nit=25	nfev=54	
trust-constr t=11.61s	run 337/1000	MSE=1.266e-12	nit=29	nfev=66	
trust-constr t=10.04s	run 338/1000	MSE=1.589e-11	nit=26	nfev=60	
trust-constr t=11.25s	run 339/1000	MSE=6.067e-13	nit=31	nfev=72	
trust-constr	run 340/1000	MSE=7.606e-14	nit=25	nfev=54	t=9.00s
trust-constr t=12.40s	run 341/1000	MSE=1.000e-11	nit=27	nfev=66	
trust-constr	run 342/1000	MSE=1.379e-12	nit=26	nfev=57	t=9.98s
trust-constr t=13.48s	run 343/1000	MSE=7.627e-14	nit=29	nfev=66	

trust-constr t=15.73s	run 344/1000	MSE=4.897e-13	nit=33	nfev=72	
trust-constr t=14.54s	run 345/1000	MSE=5.004e-15	nit=28	nfev=69	
trust-constr t=10.59s	run 346/1000	MSE=1.266e-12	nit=25	nfev=57	
trust-constr t=14.12s	run 347/1000	MSE=1.266e-12	nit=31	nfev=72	
trust-constr t=36.19s	run 348/1000	MSE=4.140e-16	nit=72	nfev=192	
trust-constr t=40.46s	run 349/1000	MSE=6.309e-16	nit=82	nfev=225	
trust-constr t=74.93s	run 350/1000	MSE=1.266e-12	nit=145	nfev=414	
trust-constr t=38.27s	run 351/1000	MSE=1.379e-12	nit=80	nfev=225	
trust-constr t=21.94s	run 352/1000	MSE=1.379e-12	nit=41	nfev=105	
trust-constr t=14.39s	run 353/1000	MSE=4.898e-13	nit=31	nfev=66	
trust-constr t=19.59s	run 354/1000	MSE=6.068e-13	nit=36	nfev=90	
trust-constr t=39.88s	run 355/1000	MSE=5.066e-16	nit=77	nfev=210	
trust-constr t=16.56s	run 356/1000	MSE=9.976e-12	nit=33	nfev=84	
trust-constr t=10.84s	run 357/1000	MSE=9.951e-12	nit=27	nfev=63	
trust-constr t=10.16s	run 358/1000	MSE=1.379e-12	nit=26	nfev=57	
trust-constr t=44.13s	run 359/1000	MSE=7.548e-14	nit=80	nfev=219	
trust-constr t=34.86s	run 360/1000	MSE=7.557e-14	nit=75	nfev=201	
trust-constr t=12.14s	run 361/1000	MSE=1.266e-12	nit=31	nfev=72	
trust-constr t=30.16s	run 362/1000	MSE=7.595e-14	nit=65	nfev=171	
trust-constr t=45.56s	run 363/1000	MSE=3.034e-15	nit=92	nfev=252	
trust-constr t=12.16s	run 364/1000	MSE=1.266e-12	nit=29	nfev=66	
trust-constr t=52.30s	run 365/1000	MSE=4.170e-16	nit=106	nfev=297	
trust-constr t=41.81s	run 366/1000	MSE=2.949e-16	nit=86	nfev=234	
trust-constr	run 367/1000	MSE=4.971e-15	nit=22	nfev=48	t=8.04s
trust-constr	run 368/1000	MSE=7.602e-14	nit=62	nfev=171	

t=30.23s						
trust-constr	run 369/1000	MSE=1.379e-12	nit=45	nfev=117		
t=19.45s						
trust-constr	run 370/1000	MSE=9.986e-12	nit=33	nfev=84		
t=15.29s						
trust-constr	run 371/1000	MSE=6.066e-13	nit=64	nfev=171		
t=29.21s						
trust-constr	run 372/1000	MSE=1.379e-12	nit=64	nfev=174		
t=29.80s						
trust-constr	run 373/1000	MSE=7.608e-14	nit=24	nfev=54	t=9.18s	
trust-constr	run 374/1000	MSE=1.266e-12	nit=27	nfev=60		
t=10.11s						
trust-constr	run 375/1000	MSE=1.379e-12	nit=25	nfev=54	t=9.48s	
trust-constr	run 376/1000	MSE=1.266e-12	nit=52	nfev=138		
t=23.91s						
trust-constr	run 377/1000	MSE=1.266e-12	nit=34	nfev=81		
t=13.26s						
trust-constr	run 378/1000	MSE=7.600e-14	nit=67	nfev=177		
t=29.97s						
trust-constr	run 379/1000	MSE=4.586e-16	nit=75	nfev=201		
t=35.78s						
trust-constr	run 380/1000	MSE=6.068e-13	nit=30	nfev=69		
t=11.79s						
trust-constr	run 381/1000	MSE=7.613e-14	nit=69	nfev=198		
t=33.91s						
trust-constr	run 382/1000	MSE=7.605e-14	nit=35	nfev=84		
t=14.63s						
trust-constr	run 383/1000	MSE=7.459e-14	nit=79	nfev=213		
t=35.62s						
trust-constr	run 384/1000	MSE=9.974e-12	nit=32	nfev=81		
t=14.06s						
trust-constr	run 385/1000	MSE=7.617e-14	nit=26	nfev=57		
t=10.80s						
trust-constr	run 386/1000	MSE=9.974e-12	nit=30	nfev=72		
t=12.20s						
trust-constr	run 387/1000	MSE=1.266e-12	nit=27	nfev=60		
t=12.17s						
trust-constr	run 388/1000	MSE=7.611e-14	nit=24	nfev=51		
t=10.45s						
trust-constr	run 389/1000	MSE=4.508e-15	nit=74	nfev=198		
t=38.44s						
trust-constr	run 390/1000	MSE=7.607e-14	nit=23	nfev=48	t=9.43s	
trust-constr	run 391/1000	MSE=1.589e-11	nit=27	nfev=63		
t=11.52s						
trust-constr	run 392/1000	MSE=9.974e-12	nit=56	nfev=153		
t=30.69s						
trust-constr	run 393/1000	MSE=7.602e-14	nit=33	nfev=81		
t=16.08s						

trust-constr	run 394/1000	MSE=1.379e-12	nit=32	nfev=78	
t=13.39s					
trust-constr	run 395/1000	MSE=7.610e-14	nit=78	nfev=216	
t=36.90s					
trust-constr	run 396/1000	MSE=4.557e-16	nit=88	nfev=240	
t=40.38s					
trust-constr	run 397/1000	MSE=1.379e-12	nit=32	nfev=78	
t=13.32s					
trust-constr	run 398/1000	MSE=3.292e-15	nit=86	nfev=237	
t=43.83s					
trust-constr	run 399/1000	MSE=7.612e-14	nit=59	nfev=150	
t=31.18s					
trust-constr	run 400/1000	MSE=6.067e-13	nit=31	nfev=69	
t=15.26s					
trust-constr	run 401/1000	MSE=4.981e-15	nit=19	nfev=39	t=8.08s
trust-constr	run 402/1000	MSE=7.606e-14	nit=31	nfev=75	
t=15.72s					
trust-constr	run 403/1000	MSE=4.779e-15	nit=22	nfev=48	
t=11.04s					
trust-constr	run 404/1000	MSE=1.379e-12	nit=38	nfev=93	
t=17.91s					
trust-constr	run 405/1000	MSE=4.089e-16	nit=87	nfev=237	
t=45.81s					
trust-constr	run 406/1000	MSE=7.639e-14	nit=54	nfev=144	
t=27.13s					
trust-constr	run 407/1000	MSE=1.379e-12	nit=26	nfev=57	t=9.85s
trust-constr	run 408/1000	MSE=1.379e-12	nit=26	nfev=60	
t=11.16s					
trust-constr	run 409/1000	MSE=2.346e-16	nit=81	nfev=222	
t=43.73s					
trust-constr	run 410/1000	MSE=5.119e-16	nit=160	nfev=456	
t=87.06s					
trust-constr	run 411/1000	MSE=6.458e-14	nit=75	nfev=150	
t=26.45s					
trust-constr	run 412/1000	MSE=1.379e-12	nit=32	nfev=75	
t=13.64s					
trust-constr	run 413/1000	MSE=3.087e-16	nit=76	nfev=204	
t=36.80s					
trust-constr	run 414/1000	MSE=7.605e-14	nit=38	nfev=93	
t=17.73s					
trust-constr	run 415/1000	MSE=2.749e-16	nit=75	nfev=201	
t=33.75s					
trust-constr	run 416/1000	MSE=1.379e-12	nit=28	nfev=66	
t=10.96s					
trust-constr	run 417/1000	MSE=9.883e-14	nit=29	nfev=63	
t=10.31s					
trust-constr	run 418/1000	MSE=7.601e-14	nit=21	nfev=42	t=7.23s
trust-constr	run 419/1000	MSE=1.379e-12	nit=39	nfev=99	

t=16.35s						
trust-constr	run 420/1000	MSE=6.067e-13	nit=48	nfev=126		
t=23.25s						
trust-constr	run 421/1000	MSE=6.068e-13	nit=25	nfev=51	t=8.94s	
trust-constr	run 422/1000	MSE=9.885e-14	nit=42	nfev=105		
t=18.19s						
trust-constr	run 423/1000	MSE=3.186e-16	nit=76	nfev=204		
t=33.67s						
trust-constr	run 424/1000	MSE=1.409e-16	nit=74	nfev=198		
t=33.27s						
trust-constr	run 425/1000	MSE=7.598e-14	nit=22	nfev=45	t=7.40s	
trust-constr	run 426/1000	MSE=1.982e-12	nit=21	nfev=48	t=9.23s	
trust-constr	run 427/1000	MSE=7.610e-14	nit=45	nfev=117		
t=22.38s						
trust-constr	run 428/1000	MSE=7.604e-14	nit=81	nfev=234		
t=41.77s						
trust-constr	run 429/1000	MSE=5.238e-15	nit=27	nfev=66		
t=11.79s						
trust-constr	run 430/1000	MSE=3.507e-16	nit=87	nfev=240		
t=43.42s						
trust-constr	run 431/1000	MSE=4.703e-14	nit=90	nfev=300		
t=51.97s						
trust-constr	run 432/1000	MSE=1.266e-12	nit=50	nfev=138		
t=25.26s						
trust-constr	run 433/1000	MSE=1.391e-01	nit=127	nfev=120		
t=19.96s						
trust-constr	run 434/1000	MSE=1.378e-12	nit=41	nfev=108		
t=19.67s						
trust-constr	run 435/1000	MSE=1.379e-12	nit=27	nfev=60		
t=11.43s						
trust-constr	run 436/1000	MSE=9.994e-12	nit=29	nfev=72		
t=13.73s						
trust-constr	run 437/1000	MSE=7.622e-14	nit=58	nfev=147		
t=27.97s						
trust-constr	run 438/1000	MSE=6.066e-13	nit=25	nfev=51		
t=10.06s						
trust-constr	run 439/1000	MSE=6.067e-13	nit=37	nfev=90		
t=17.75s						
trust-constr	run 440/1000	MSE=6.065e-13	nit=32	nfev=78		
t=14.39s						
trust-constr	run 441/1000	MSE=6.066e-13	nit=40	nfev=99		
t=18.16s						
trust-constr	run 442/1000	MSE=3.840e-16	nit=76	nfev=204		
t=36.91s						
trust-constr	run 443/1000	MSE=4.935e-15	nit=23	nfev=51	t=8.73s	
trust-constr	run 444/1000	MSE=1.266e-12	nit=26	nfev=57	t=9.72s	
trust-constr	run 445/1000	MSE=7.006e-14	nit=58	nfev=147		
t=26.32s						

trust-constr t=23.34s	run 446/1000	MSE=1.266e-12	nit=49	nfev=129	
trust-constr t=22.01s	run 447/1000	MSE=1.379e-12	nit=43	nfev=111	
trust-constr t=17.13s	run 448/1000	MSE=1.379e-12	nit=37	nfev=99	
trust-constr	run 449/1000	MSE=1.266e-12	nit=26	nfev=57	t=9.86s
trust-constr t=24.69s	run 450/1000	MSE=7.674e-14	nit=58	nfev=147	
trust-constr t=36.48s	run 451/1000	MSE=1.068e-15	nit=80	nfev=216	
trust-constr t=44.66s	run 452/1000	MSE=9.970e-16	nit=87	nfev=240	
trust-constr t=13.11s	run 453/1000	MSE=1.379e-12	nit=25	nfev=57	
trust-constr t=48.06s	run 454/1000	MSE=6.821e-14	nit=67	nfev=177	
trust-constr t=36.75s	run 455/1000	MSE=7.626e-14	nit=57	nfev=144	
trust-constr	run 456/1000	MSE=9.973e-12	nit=21	nfev=45	t=8.35s
trust-constr t=23.02s	run 457/1000	MSE=1.379e-12	nit=45	nfev=117	
trust-constr t=28.46s	run 458/1000	MSE=6.067e-13	nit=56	nfev=147	
trust-constr t=14.85s	run 459/1000	MSE=1.787e-12	nit=33	nfev=78	
trust-constr t=79.72s	run 460/1000	MSE=9.956e-12	nit=133	nfev=387	
trust-constr t=27.26s	run 461/1000	MSE=7.607e-14	nit=46	nfev=120	
trust-constr t=22.25s	run 462/1000	MSE=1.787e-12	nit=45	nfev=114	
trust-constr t=16.99s	run 463/1000	MSE=9.974e-12	nit=33	nfev=90	
trust-constr t=12.08s	run 464/1000	MSE=1.379e-12	nit=24	nfev=51	
trust-constr t=12.51s	run 465/1000	MSE=1.266e-12	nit=30	nfev=69	
trust-constr	run 466/1000	MSE=7.618e-14	nit=21	nfev=42	t=7.96s
trust-constr t=30.60s	run 467/1000	MSE=7.607e-14	nit=58	nfev=153	
trust-constr t=10.97s	run 468/1000	MSE=1.379e-12	nit=25	nfev=54	
trust-constr t=45.54s	run 469/1000	MSE=7.700e-16	nit=86	nfev=234	
trust-constr t=14.58s	run 470/1000	MSE=1.379e-12	nit=32	nfev=78	
trust-constr	run 471/1000	MSE=1.978e-12	nit=26	nfev=66	

t=13.63s						
trust-constr	run 472/1000	MSE=4.789e-15	nit=31	nfev=75		
t=14.60s						
trust-constr	run 473/1000	MSE=4.956e-15	nit=94	nfev=264		
t=51.92s						
trust-constr	run 474/1000	MSE=6.311e-16	nit=80	nfev=216		
t=41.37s						
trust-constr	run 475/1000	MSE=7.625e-14	nit=34	nfev=84		
t=15.24s						
trust-constr	run 476/1000	MSE=1.379e-12	nit=39	nfev=96		
t=16.97s						
trust-constr	run 477/1000	MSE=4.897e-13	nit=36	nfev=81		
t=14.51s						
trust-constr	run 478/1000	MSE=4.972e-15	nit=75	nfev=210		
t=39.99s						
trust-constr	run 479/1000	MSE=7.603e-14	nit=38	nfev=102		
t=23.53s						
trust-constr	run 480/1000	MSE=5.354e-14	nit=78	nfev=210		
t=41.98s						
trust-constr	run 481/1000	MSE=4.729e-16	nit=146	nfev=417		
t=92.39s						
trust-constr	run 482/1000	MSE=5.787e-16	nit=95	nfev=267		
t=55.05s						
trust-constr	run 483/1000	MSE=1.379e-12	nit=37	nfev=99		
t=21.08s						
trust-constr	run 484/1000	MSE=1.379e-12	nit=23	nfev=48	t=8.34s	
trust-constr	run 485/1000	MSE=5.581e-16	nit=84	nfev=231		
t=40.91s						
trust-constr	run 486/1000	MSE=7.593e-14	nit=93	nfev=258		
t=45.00s						
trust-constr	run 487/1000	MSE=6.066e-13	nit=25	nfev=51	t=8.74s	
trust-constr	run 488/1000	MSE=9.978e-12	nit=31	nfev=78		
t=13.42s						
trust-constr	run 489/1000	MSE=1.787e-12	nit=23	nfev=48	t=8.13s	
trust-constr	run 490/1000	MSE=9.851e-16	nit=79	nfev=216		
t=39.93s						
trust-constr	run 491/1000	MSE=9.693e-17	nit=86	nfev=237		
t=40.32s						
trust-constr	run 492/1000	MSE=1.266e-12	nit=27	nfev=60	t=9.96s	
trust-constr	run 493/1000	MSE=1.787e-12	nit=33	nfev=81		
t=14.81s						
trust-constr	run 494/1000	MSE=1.379e-12	nit=63	nfev=171		
t=37.50s						
trust-constr	run 495/1000	MSE=4.993e-15	nit=20	nfev=42	t=8.07s	
trust-constr	run 496/1000	MSE=1.265e-12	nit=28	nfev=63		
t=11.22s						
trust-constr	run 497/1000	MSE=7.591e-14	nit=47	nfev=120		
t=21.33s						

trust-constr	run 498/1000	MSE=6.067e-13	nit=26	nfev=54	t=9.02s
trust-constr	run 499/1000	MSE=4.928e-15	nit=22	nfev=48	t=8.08s
trust-constr	run 500/1000	MSE=4.798e-15	nit=22	nfev=48	t=8.53s
trust-constr	run 501/1000	MSE=6.066e-13	nit=62	nfev=171	
t=31.06s					
trust-constr	run 502/1000	MSE=6.067e-13	nit=46	nfev=117	
t=25.29s					
trust-constr	run 503/1000	MSE=2.965e-16	nit=135	nfev=384	
t=90.18s					
trust-constr	run 504/1000	MSE=4.922e-15	nit=26	nfev=60	
t=11.72s					
trust-constr	run 505/1000	MSE=1.379e-12	nit=41	nfev=111	
t=25.30s					
trust-constr	run 506/1000	MSE=1.379e-12	nit=32	nfev=75	
t=15.22s					
trust-constr	run 507/1000	MSE=1.379e-12	nit=28	nfev=63	
t=12.55s					
trust-constr	run 508/1000	MSE=4.897e-13	nit=33	nfev=75	
t=15.45s					
trust-constr	run 509/1000	MSE=1.787e-12	nit=25	nfev=54	
t=10.45s					
trust-constr	run 510/1000	MSE=6.065e-13	nit=45	nfev=117	
t=22.01s					
trust-constr	run 511/1000	MSE=1.266e-12	nit=27	nfev=60	
t=11.99s					
trust-constr	run 512/1000	MSE=6.144e-16	nit=76	nfev=204	
t=42.04s					
trust-constr	run 513/1000	MSE=1.058e-15	nit=82	nfev=222	
t=44.94s					
trust-constr	run 514/1000	MSE=6.065e-13	nit=33	nfev=78	
t=14.43s					
trust-constr	run 515/1000	MSE=1.379e-12	nit=70	nfev=189	
t=36.96s					
trust-constr	run 516/1000	MSE=1.266e-12	nit=29	nfev=69	
t=13.66s					
trust-constr	run 517/1000	MSE=3.773e-16	nit=76	nfev=207	
t=45.37s					
trust-constr	run 518/1000	MSE=7.602e-14	nit=20	nfev=39	t=7.71s
trust-constr	run 519/1000	MSE=4.899e-13	nit=31	nfev=69	
t=15.88s					
trust-constr	run 520/1000	MSE=1.615e-15	nit=84	nfev=228	
t=45.65s					
trust-constr	run 521/1000	MSE=9.975e-12	nit=22	nfev=48	t=8.40s
trust-constr	run 522/1000	MSE=6.066e-13	nit=91	nfev=255	
t=45.77s					
trust-constr	run 523/1000	MSE=7.604e-14	nit=21	nfev=42	t=7.76s
trust-constr	run 524/1000	MSE=4.892e-15	nit=37	nfev=96	
t=18.25s					

trust-constr t=42.61s	run 525/1000	MSE=1.432e-15	nit=81	nfev=219	
trust-constr t=16.86s	run 526/1000	MSE=7.606e-14	nit=34	nfev=84	
trust-constr t=14.11s	run 527/1000	MSE=1.266e-12	nit=27	nfev=60	
trust-constr t=51.26s	run 528/1000	MSE=3.908e-16	nit=97	nfev=270	
trust-constr t=17.55s	run 529/1000	MSE=1.001e-11	nit=38	nfev=99	
trust-constr t=12.44s	run 530/1000	MSE=1.379e-12	nit=31	nfev=72	
trust-constr t=28.74s	run 531/1000	MSE=7.602e-14	nit=56	nfev=150	
trust-constr t=11.55s	run 532/1000	MSE=1.266e-12	nit=26	nfev=57	
trust-constr t=30.68s	run 533/1000	MSE=9.970e-12	nit=58	nfev=162	
trust-constr t=12.80s	run 534/1000	MSE=6.067e-13	nit=29	nfev=66	
trust-constr	run 535/1000	MSE=9.977e-12	nit=25	nfev=57	t=9.82s
trust-constr t=10.59s	run 536/1000	MSE=1.266e-12	nit=28	nfev=63	
trust-constr t=12.56s	run 537/1000	MSE=1.266e-12	nit=29	nfev=66	
trust-constr	run 538/1000	MSE=1.379e-12	nit=25	nfev=54	t=9.74s
trust-constr	run 539/1000	MSE=1.788e-12	nit=23	nfev=48	t=8.70s
trust-constr t=12.38s	run 540/1000	MSE=9.971e-12	nit=29	nfev=69	
trust-constr t=22.32s	run 541/1000	MSE=6.067e-13	nit=38	nfev=99	
trust-constr t=14.18s	run 542/1000	MSE=6.488e-13	nit=33	nfev=75	
trust-constr t=80.94s	run 543/1000	MSE=2.778e-15	nit=144	nfev=411	
trust-constr t=27.56s	run 544/1000	MSE=6.066e-13	nit=53	nfev=138	
trust-constr t=34.21s	run 545/1000	MSE=1.379e-12	nit=59	nfev=159	
trust-constr	run 546/1000	MSE=1.787e-12	nit=23	nfev=48	t=8.99s
trust-constr t=12.66s	run 547/1000	MSE=1.266e-12	nit=27	nfev=60	
trust-constr t=24.97s	run 548/1000	MSE=9.979e-12	nit=49	nfev=132	
trust-constr t=14.14s	run 549/1000	MSE=7.616e-14	nit=26	nfev=57	
trust-constr t=10.72s	run 550/1000	MSE=7.609e-14	nit=25	nfev=57	

trust-constr t=21.38s	run 551/1000	MSE=7.606e-14	nit=46	nfev=117	
trust-constr t=10.07s	run 552/1000	MSE=1.379e-12	nit=26	nfev=57	
trust-constr t=41.58s	run 553/1000	MSE=7.614e-14	nit=75	nfev=201	
trust-constr t=16.62s	run 554/1000	MSE=7.599e-14	nit=34	nfev=87	
trust-constr t=14.90s	run 555/1000	MSE=1.266e-12	nit=27	nfev=60	
trust-constr t=37.27s	run 556/1000	MSE=2.782e-16	nit=81	nfev=222	
trust-constr	run 557/1000	MSE=7.602e-14	nit=24	nfev=51	t=8.31s
trust-constr	run 558/1000	MSE=9.996e-12	nit=24	nfev=54	t=9.18s
trust-constr t=12.44s	run 559/1000	MSE=7.612e-14	nit=31	nfev=75	
trust-constr t=17.63s	run 560/1000	MSE=4.894e-13	nit=43	nfev=105	
trust-constr t=16.25s	run 561/1000	MSE=1.787e-12	nit=37	nfev=99	
trust-constr	run 562/1000	MSE=7.613e-14	nit=22	nfev=45	t=7.71s
trust-constr t=25.78s	run 563/1000	MSE=1.379e-12	nit=56	nfev=153	
trust-constr t=32.37s	run 564/1000	MSE=1.712e-15	nit=74	nfev=198	
trust-constr t=33.84s	run 565/1000	MSE=5.494e-16	nit=74	nfev=198	
trust-constr t=13.31s	run 566/1000	MSE=7.607e-14	nit=34	nfev=81	
trust-constr t=14.75s	run 567/1000	MSE=1.379e-12	nit=35	nfev=90	
trust-constr	run 568/1000	MSE=7.619e-14	nit=25	nfev=57	t=9.53s
trust-constr t=10.06s	run 569/1000	MSE=1.266e-12	nit=28	nfev=63	
trust-constr t=34.85s	run 570/1000	MSE=5.468e-16	nit=77	nfev=207	
trust-constr t=26.41s	run 571/1000	MSE=6.067e-13	nit=60	nfev=162	
trust-constr t=15.18s	run 572/1000	MSE=6.064e-13	nit=37	nfev=90	
trust-constr t=48.10s	run 573/1000	MSE=9.971e-12	nit=92	nfev=258	
trust-constr	run 574/1000	MSE=7.609e-14	nit=24	nfev=51	t=8.34s
trust-constr t=30.62s	run 575/1000	MSE=5.328e-14	nit=70	nfev=186	
trust-constr t=42.30s	run 576/1000	MSE=1.405e-15	nit=77	nfev=207	
trust-constr	run 577/1000	MSE=4.342e-16	nit=81	nfev=219	

t=38.95s						
trust-constr	run 578/1000	MSE=7.617e-14	nit=35	nfev=84		
t=14.98s						
trust-constr	run 579/1000	MSE=7.606e-14	nit=25	nfev=57	t=9.95s	
trust-constr	run 580/1000	MSE=1.266e-12	nit=32	nfev=78		
t=14.04s						
trust-constr	run 581/1000	MSE=4.812e-15	nit=33	nfev=84		
t=17.67s						
trust-constr	run 582/1000	MSE=2.212e-15	nit=84	nfev=228		
t=51.01s						
trust-constr	run 583/1000	MSE=6.067e-13	nit=27	nfev=57		
t=10.13s						
trust-constr	run 584/1000	MSE=6.066e-13	nit=32	nfev=75		
t=15.61s						
trust-constr	run 585/1000	MSE=6.941e-11	nit=26	nfev=60		
t=13.55s						
trust-constr	run 586/1000	MSE=1.618e-15	nit=101	nfev=282		
t=51.02s						
trust-constr	run 587/1000	MSE=1.199e-15	nit=81	nfev=219		
t=36.87s						
trust-constr	run 588/1000	MSE=1.266e-12	nit=31	nfev=72		
t=11.55s						
trust-constr	run 589/1000	MSE=1.266e-12	nit=25	nfev=54	t=9.37s	
trust-constr	run 590/1000	MSE=9.971e-12	nit=29	nfev=69		
t=11.64s						
trust-constr	run 591/1000	MSE=7.606e-14	nit=21	nfev=42	t=7.10s	
trust-constr	run 592/1000	MSE=1.265e-12	nit=28	nfev=63		
t=10.83s						
trust-constr	run 593/1000	MSE=3.877e-16	nit=83	nfev=228		
t=38.25s						
trust-constr	run 594/1000	MSE=5.001e-15	nit=21	nfev=45	t=7.82s	
trust-constr	run 595/1000	MSE=9.899e-14	nit=33	nfev=78		
t=13.72s						
trust-constr	run 596/1000	MSE=9.973e-12	nit=31	nfev=75		
t=12.72s						
trust-constr	run 597/1000	MSE=6.067e-13	nit=65	nfev=171		
t=30.06s						
trust-constr	run 598/1000	MSE=7.602e-14	nit=38	nfev=93		
t=16.25s						
trust-constr	run 599/1000	MSE=1.266e-12	nit=29	nfev=66		
t=11.47s						
trust-constr	run 600/1000	MSE=3.888e-16	nit=76	nfev=204		
t=33.56s						
trust-constr	run 601/1000	MSE=6.066e-13	nit=43	nfev=111		
t=18.14s						
trust-constr	run 602/1000	MSE=4.775e-15	nit=26	nfev=60	t=9.63s	
trust-constr	run 603/1000	MSE=9.980e-12	nit=29	nfev=69		
t=11.54s						

trust-constr t=10.88s	run 604/1000	MSE=1.266e-12	nit=29	nfev=66	
trust-constr t=36.44s	run 605/1000	MSE=1.000e-11	nit=68	nfev=195	
trust-constr t=12.59s	run 606/1000	MSE=9.984e-12	nit=28	nfev=66	
trust-constr t=15.68s	run 607/1000	MSE=1.266e-12	nit=36	nfev=93	
trust-constr t=12.15s	run 608/1000	MSE=6.066e-13	nit=32	nfev=72	
trust-constr	run 609/1000	MSE=4.750e-15	nit=22	nfev=48	t=7.89s
trust-constr t=33.92s	run 610/1000	MSE=1.452e-15	nit=73	nfev=195	
trust-constr t=49.03s	run 611/1000	MSE=2.870e-15	nit=107	nfev=303	
trust-constr	run 612/1000	MSE=4.900e-13	nit=29	nfev=60	t=9.05s
trust-constr	run 613/1000	MSE=1.266e-12	nit=27	nfev=60	t=9.34s
trust-constr t=21.58s	run 614/1000	MSE=4.983e-15	nit=50	nfev=135	
trust-constr t=22.81s	run 615/1000	MSE=7.204e-14	nit=57	nfev=144	
trust-constr t=14.50s	run 616/1000	MSE=4.897e-13	nit=39	nfev=93	
trust-constr t=20.28s	run 617/1000	MSE=9.999e-12	nit=47	nfev=126	
trust-constr t=11.92s	run 618/1000	MSE=1.787e-12	nit=29	nfev=66	
trust-constr t=18.50s	run 619/1000	MSE=4.947e-15	nit=35	nfev=90	
trust-constr t=15.11s	run 620/1000	MSE=1.379e-12	nit=38	nfev=96	
trust-constr t=13.05s	run 621/1000	MSE=7.604e-14	nit=33	nfev=78	
trust-constr t=12.83s	run 622/1000	MSE=1.266e-12	nit=31	nfev=75	
trust-constr	run 623/1000	MSE=4.962e-15	nit=19	nfev=39	t=6.84s
trust-constr t=24.44s	run 624/1000	MSE=6.066e-13	nit=58	nfev=153	
trust-constr t=10.79s	run 625/1000	MSE=1.379e-12	nit=30	nfev=72	
trust-constr	run 626/1000	MSE=1.003e-11	nit=27	nfev=63	t=9.18s
trust-constr t=12.32s	run 627/1000	MSE=4.939e-15	nit=33	nfev=81	
trust-constr t=15.63s	run 628/1000	MSE=6.067e-13	nit=39	nfev=99	
trust-constr t=15.16s	run 629/1000	MSE=1.266e-12	nit=40	nfev=105	
trust-constr	run 630/1000	MSE=7.608e-14	nit=21	nfev=42	t=6.59s

trust-constr	run 631/1000	MSE=1.004e-11	nit=27	nfev=63	t=9.08s
trust-constr	run 632/1000	MSE=7.620e-14	nit=23	nfev=48	t=6.94s
trust-constr	run 633/1000	MSE=1.379e-12	nit=122	nfev=348	
t=51.20s					
trust-constr	run 634/1000	MSE=6.943e-11	nit=26	nfev=60	t=8.78s
trust-constr	run 635/1000	MSE=1.266e-12	nit=22	nfev=45	t=6.31s
trust-constr	run 636/1000	MSE=1.266e-12	nit=28	nfev=63	t=9.34s
trust-constr	run 637/1000	MSE=1.266e-12	nit=31	nfev=75	
t=11.00s					
trust-constr	run 638/1000	MSE=4.801e-15	nit=29	nfev=72	
t=10.52s					
trust-constr	run 639/1000	MSE=1.786e-12	nit=26	nfev=57	t=8.26s
trust-constr	run 640/1000	MSE=5.693e-14	nit=70	nfev=189	
t=27.60s					
trust-constr	run 641/1000	MSE=1.266e-12	nit=27	nfev=60	t=8.88s
trust-constr	run 642/1000	MSE=9.974e-12	nit=29	nfev=69	
t=10.09s					
trust-constr	run 643/1000	MSE=6.067e-13	nit=28	nfev=60	t=8.86s
trust-constr	run 644/1000	MSE=1.379e-12	nit=31	nfev=75	
t=11.30s					
trust-constr	run 645/1000	MSE=1.379e-12	nit=28	nfev=63	t=9.36s
trust-constr	run 646/1000	MSE=9.984e-12	nit=55	nfev=150	
t=21.81s					
trust-constr	run 647/1000	MSE=1.102e-15	nit=87	nfev=240	
t=35.22s					
trust-constr	run 648/1000	MSE=7.613e-14	nit=23	nfev=48	t=6.96s
trust-constr	run 649/1000	MSE=6.069e-13	nit=31	nfev=69	
t=10.50s					
trust-constr	run 650/1000	MSE=4.848e-15	nit=18	nfev=36	t=5.27s
trust-constr	run 651/1000	MSE=4.899e-13	nit=28	nfev=57	t=8.61s
trust-constr	run 652/1000	MSE=7.606e-14	nit=23	nfev=48	t=7.03s
trust-constr	run 653/1000	MSE=7.607e-14	nit=25	nfev=54	t=8.09s
trust-constr	run 654/1000	MSE=1.379e-12	nit=84	nfev=237	
t=37.48s					
trust-constr	run 655/1000	MSE=1.266e-12	nit=38	nfev=93	
t=15.43s					
trust-constr	run 656/1000	MSE=6.941e-11	nit=26	nfev=60	t=9.89s
trust-constr	run 657/1000	MSE=1.787e-12	nit=43	nfev=114	
t=18.60s					
trust-constr	run 658/1000	MSE=1.379e-12	nit=29	nfev=66	
t=10.63s					
trust-constr	run 659/1000	MSE=1.379e-12	nit=33	nfev=78	
t=12.81s					
trust-constr	run 660/1000	MSE=7.610e-14	nit=34	nfev=81	
t=13.90s					
trust-constr	run 661/1000	MSE=2.334e-15	nit=91	nfev=255	
t=44.90s					
trust-constr	run 662/1000	MSE=1.379e-12	nit=34	nfev=87	

t=16.58s						
trust-constr	run 663/1000	MSE=1.379e-12	nit=36	nfev=93		
t=16.91s						
trust-constr	run 664/1000	MSE=4.981e-15	nit=20	nfev=42	t=7.37s	
trust-constr	run 665/1000	MSE=1.379e-12	nit=31	nfev=75		
t=14.02s						
trust-constr	run 666/1000	MSE=7.544e-14	nit=65	nfev=171		
t=28.81s						
trust-constr	run 667/1000	MSE=1.266e-12	nit=26	nfev=57	t=9.05s	
trust-constr	run 668/1000	MSE=7.510e-14	nit=21	nfev=42	t=7.58s	
trust-constr	run 669/1000	MSE=1.000e-11	nit=41	nfev=105		
t=18.86s						
trust-constr	run 670/1000	MSE=4.930e-15	nit=25	nfev=57		
t=10.90s						
trust-constr	run 671/1000	MSE=6.067e-13	nit=34	nfev=78		
t=14.13s						
trust-constr	run 672/1000	MSE=7.609e-14	nit=30	nfev=69		
t=12.71s						
trust-constr	run 673/1000	MSE=8.770e-16	nit=93	nfev=255		
t=43.51s						
trust-constr	run 674/1000	MSE=7.611e-14	nit=27	nfev=60		
t=10.23s						
trust-constr	run 675/1000	MSE=1.002e-11	nit=29	nfev=72		
t=12.58s						
trust-constr	run 676/1000	MSE=9.994e-12	nit=21	nfev=45	t=8.04s	
trust-constr	run 677/1000	MSE=9.883e-14	nit=35	nfev=81		
t=15.04s						
trust-constr	run 678/1000	MSE=9.971e-12	nit=28	nfev=66		
t=11.45s						
trust-constr	run 679/1000	MSE=1.379e-12	nit=26	nfev=57	t=9.20s	
trust-constr	run 680/1000	MSE=4.048e-16	nit=85	nfev=231		
t=38.06s						
trust-constr	run 681/1000	MSE=6.487e-13	nit=45	nfev=117		
t=19.83s						
trust-constr	run 682/1000	MSE=4.639e-16	nit=77	nfev=207		
t=35.89s						
trust-constr	run 683/1000	MSE=1.379e-12	nit=56	nfev=156		
t=24.95s						
trust-constr	run 684/1000	MSE=6.062e-13	nit=133	nfev=378		
t=59.55s						
trust-constr	run 685/1000	MSE=1.379e-12	nit=26	nfev=57	t=9.26s	
trust-constr	run 686/1000	MSE=7.601e-14	nit=21	nfev=42	t=7.32s	
trust-constr	run 687/1000	MSE=1.266e-12	nit=27	nfev=60		
t=11.34s						
trust-constr	run 688/1000	MSE=2.786e-15	nit=114	nfev=318		
t=52.30s						
trust-constr	run 689/1000	MSE=7.611e-14	nit=38	nfev=96		
t=14.12s						

trust-constr	run 690/1000	MSE=6.066e-13	nit=27	nfev=57	t=9.29s
trust-constr	run 691/1000	MSE=6.064e-13	nit=29	nfev=66	
t=10.21s					
trust-constr	run 692/1000	MSE=6.067e-13	nit=47	nfev=120	
t=17.89s					
trust-constr	run 693/1000	MSE=6.067e-13	nit=27	nfev=57	t=8.76s
trust-constr	run 694/1000	MSE=1.266e-12	nit=38	nfev=96	
t=16.25s					
trust-constr	run 695/1000	MSE=1.266e-12	nit=29	nfev=66	
t=10.13s					
trust-constr	run 696/1000	MSE=2.383e-16	nit=75	nfev=201	
t=32.10s					
trust-constr	run 697/1000	MSE=1.588e-11	nit=28	nfev=66	
t=10.88s					
trust-constr	run 698/1000	MSE=9.991e-12	nit=72	nfev=204	
t=31.88s					
trust-constr	run 699/1000	MSE=6.066e-13	nit=28	nfev=60	t=9.73s
trust-constr	run 700/1000	MSE=9.342e-16	nit=74	nfev=198	
t=28.89s					
trust-constr	run 701/1000	MSE=9.990e-12	nit=25	nfev=57	t=8.51s
trust-constr	run 702/1000	MSE=9.972e-12	nit=92	nfev=261	
t=40.79s					
trust-constr	run 703/1000	MSE=9.056e-16	nit=82	nfev=222	
t=34.14s					
trust-constr	run 704/1000	MSE=7.271e-16	nit=79	nfev=213	
t=36.72s					
trust-constr	run 705/1000	MSE=4.446e-16	nit=81	nfev=222	
t=37.16s					
trust-constr	run 706/1000	MSE=9.886e-14	nit=27	nfev=57	t=8.51s
trust-constr	run 707/1000	MSE=9.979e-12	nit=33	nfev=84	
t=13.37s					
trust-constr	run 708/1000	MSE=5.925e-17	nit=84	nfev=228	
t=34.46s					
trust-constr	run 709/1000	MSE=7.616e-14	nit=58	nfev=147	
t=23.24s					
trust-constr	run 710/1000	MSE=1.379e-12	nit=24	nfev=51	t=8.37s
trust-constr	run 711/1000	MSE=7.603e-14	nit=34	nfev=81	
t=12.86s					
trust-constr	run 712/1000	MSE=7.115e-14	nit=75	nfev=201	
t=33.66s					
trust-constr	run 713/1000	MSE=7.603e-14	nit=41	nfev=105	
t=18.03s					
trust-constr	run 714/1000	MSE=6.068e-13	nit=27	nfev=57	
t=11.17s					
trust-constr	run 715/1000	MSE=1.266e-12	nit=29	nfev=66	
t=10.40s					
trust-constr	run 716/1000	MSE=7.605e-14	nit=28	nfev=63	
t=11.02s					

trust-constr t=16.32s	run 717/1000	MSE=1.379e-12	nit=36	nfev=87	
trust-constr	run 718/1000	MSE=7.606e-14	nit=20	nfev=39	t=7.17s
trust-constr	run 719/1000	MSE=1.379e-12	nit=25	nfev=54	t=8.31s
trust-constr t=23.90s	run 720/1000	MSE=7.624e-14	nit=59	nfev=153	
trust-constr t=33.82s	run 721/1000	MSE=4.591e-16	nit=79	nfev=213	
trust-constr	run 722/1000	MSE=1.379e-12	nit=25	nfev=54	t=9.16s
trust-constr t=39.57s	run 723/1000	MSE=4.665e-16	nit=84	nfev=228	
trust-constr t=42.32s	run 724/1000	MSE=6.613e-16	nit=92	nfev=252	
trust-constr	run 725/1000	MSE=1.379e-12	nit=23	nfev=48	t=8.21s
trust-constr t=40.44s	run 726/1000	MSE=2.052e-16	nit=89	nfev=246	
trust-constr t=22.82s	run 727/1000	MSE=6.563e-14	nit=58	nfev=147	
trust-constr t=39.26s	run 728/1000	MSE=1.805e-16	nit=76	nfev=204	
trust-constr t=39.48s	run 729/1000	MSE=9.826e-16	nit=82	nfev=222	
trust-constr t=42.50s	run 730/1000	MSE=1.154e-15	nit=82	nfev=225	
trust-constr t=49.01s	run 731/1000	MSE=5.587e-16	nit=100	nfev=279	
trust-constr t=12.84s	run 732/1000	MSE=1.266e-12	nit=29	nfev=66	
trust-constr	run 733/1000	MSE=7.616e-14	nit=24	nfev=51	t=8.38s
trust-constr t=18.46s	run 734/1000	MSE=1.787e-12	nit=41	nfev=114	
trust-constr t=12.91s	run 735/1000	MSE=1.379e-12	nit=30	nfev=69	
trust-constr t=13.21s	run 736/1000	MSE=1.379e-12	nit=33	nfev=81	
trust-constr t=36.17s	run 737/1000	MSE=2.259e-15	nit=82	nfev=222	
trust-constr t=29.52s	run 738/1000	MSE=6.953e-14	nit=69	nfev=183	
trust-constr t=22.91s	run 739/1000	MSE=7.129e-14	nit=57	nfev=144	
trust-constr t=215.30s	run 740/1000	MSE=7.602e-14	nit=450	nfev=1332	
trust-constr	run 741/1000	MSE=1.379e-12	nit=27	nfev=60	t=9.87s
trust-constr t=35.11s	run 742/1000	MSE=1.281e-15	nit=81	nfev=219	
trust-constr t=16.77s	run 743/1000	MSE=1.379e-12	nit=41	nfev=105	

trust-constr	run 744/1000	MSE=7.585e-14	nit=24	nfev=51	t=7.93s
trust-constr	run 745/1000	MSE=7.603e-14	nit=34	nfev=87	
t=13.94s					
trust-constr	run 746/1000	MSE=6.066e-13	nit=27	nfev=57	t=9.26s
trust-constr	run 747/1000	MSE=6.942e-11	nit=23	nfev=51	t=8.10s
trust-constr	run 748/1000	MSE=1.379e-12	nit=66	nfev=180	
t=28.69s					
trust-constr	run 749/1000	MSE=1.379e-12	nit=29	nfev=72	
t=11.37s					
trust-constr	run 750/1000	MSE=8.107e-16	nit=79	nfev=213	
t=33.84s					
trust-constr	run 751/1000	MSE=9.886e-14	nit=31	nfev=69	
t=10.93s					
trust-constr	run 752/1000	MSE=1.379e-12	nit=68	nfev=186	
t=29.74s					
trust-constr	run 753/1000	MSE=7.381e-15	nit=76	nfev=204	
t=32.20s					
trust-constr	run 754/1000	MSE=9.889e-14	nit=33	nfev=75	
t=11.93s					
trust-constr	run 755/1000	MSE=7.622e-14	nit=49	nfev=132	
t=20.90s					
trust-constr	run 756/1000	MSE=7.603e-14	nit=28	nfev=63	t=9.84s
trust-constr	run 757/1000	MSE=6.942e-11	nit=26	nfev=60	t=9.78s
trust-constr	run 758/1000	MSE=1.266e-12	nit=34	nfev=84	
t=20.69s					
trust-constr	run 759/1000	MSE=1.269e-15	nit=81	nfev=219	
t=38.99s					
trust-constr	run 760/1000	MSE=7.574e-14	nit=27	nfev=60	
t=10.46s					
trust-constr	run 761/1000	MSE=7.600e-14	nit=32	nfev=78	
t=13.64s					
trust-constr	run 762/1000	MSE=1.266e-12	nit=27	nfev=60	
t=10.48s					
trust-constr	run 763/1000	MSE=7.609e-14	nit=52	nfev=135	
t=22.58s					
trust-constr	run 764/1000	MSE=1.379e-12	nit=66	nfev=180	
t=32.01s					
trust-constr	run 765/1000	MSE=6.068e-13	nit=28	nfev=60	t=9.79s
trust-constr	run 766/1000	MSE=1.000e-11	nit=21	nfev=45	
t=10.65s					
trust-constr	run 767/1000	MSE=7.602e-14	nit=65	nfev=177	
t=32.09s					
trust-constr	run 768/1000	MSE=4.745e-15	nit=23	nfev=51	
t=11.14s					
trust-constr	run 769/1000	MSE=1.586e-11	nit=23	nfev=51	
t=10.88s					
trust-constr	run 770/1000	MSE=1.379e-12	nit=31	nfev=72	
t=13.26s					

trust-constr t=11.74s	run 771/1000	MSE=6.066e-13	nit=33	nfev=75	
trust-constr t=44.53s	run 772/1000	MSE=5.210e-16	nit=107	nfev=297	
trust-constr t=10.73s	run 773/1000	MSE=9.975e-12	nit=27	nfev=66	
trust-constr t=37.11s	run 774/1000	MSE=1.590e-15	nit=81	nfev=225	
trust-constr	run 775/1000	MSE=7.611e-14	nit=21	nfev=42	t=6.39s
trust-constr	run 776/1000	MSE=1.378e-12	nit=25	nfev=54	t=7.83s
trust-constr t=12.25s	run 777/1000	MSE=1.379e-12	nit=30	nfev=69	
trust-constr t=10.99s	run 778/1000	MSE=6.067e-13	nit=30	nfev=69	
trust-constr	run 779/1000	MSE=7.603e-14	nit=22	nfev=45	t=7.05s
trust-constr t=38.79s	run 780/1000	MSE=5.813e-16	nit=80	nfev=216	
trust-constr t=16.66s	run 781/1000	MSE=9.977e-12	nit=37	nfev=96	
trust-constr t=39.36s	run 782/1000	MSE=4.237e-15	nit=82	nfev=225	
trust-constr	run 783/1000	MSE=7.605e-14	nit=23	nfev=48	t=8.13s
trust-constr t=21.65s	run 784/1000	MSE=9.980e-12	nit=48	nfev=126	
trust-constr t=13.22s	run 785/1000	MSE=1.266e-12	nit=32	nfev=75	
trust-constr t=48.85s	run 786/1000	MSE=1.266e-12	nit=97	nfev=273	
trust-constr t=16.51s	run 787/1000	MSE=7.617e-14	nit=39	nfev=99	
trust-constr t=12.63s	run 788/1000	MSE=1.266e-12	nit=27	nfev=60	
trust-constr t=12.21s	run 789/1000	MSE=7.602e-14	nit=23	nfev=51	
trust-constr t=11.58s	run 790/1000	MSE=7.606e-14	nit=22	nfev=45	
trust-constr t=12.95s	run 791/1000	MSE=1.379e-12	nit=26	nfev=57	
trust-constr t=12.01s	run 792/1000	MSE=7.611e-14	nit=25	nfev=57	
trust-constr t=11.76s	run 793/1000	MSE=1.379e-12	nit=26	nfev=60	
trust-constr	run 794/1000	MSE=7.606e-14	nit=21	nfev=42	t=8.11s
trust-constr	run 795/1000	MSE=7.612e-14	nit=23	nfev=48	t=8.47s
trust-constr t=10.29s	run 796/1000	MSE=1.379e-12	nit=26	nfev=57	
trust-constr t=38.41s	run 797/1000	MSE=3.503e-15	nit=77	nfev=207	

trust-constr t=13.28s	run 798/1000	MSE=9.975e-12	nit=31	nfev=75	
trust-constr t=13.30s	run 799/1000	MSE=9.975e-12	nit=29	nfev=72	
trust-constr t=10.39s	run 800/1000	MSE=5.019e-15	nit=25	nfev=57	
trust-constr t=10.29s	run 801/1000	MSE=1.379e-12	nit=25	nfev=57	
trust-constr t=15.71s	run 802/1000	MSE=4.921e-15	nit=33	nfev=87	
trust-constr t=34.57s	run 803/1000	MSE=7.776e-14	nit=68	nfev=180	
trust-constr t=14.49s	run 804/1000	MSE=6.067e-13	nit=32	nfev=75	
trust-constr t=37.55s	run 805/1000	MSE=4.232e-16	nit=76	nfev=204	
trust-constr t=15.52s	run 806/1000	MSE=1.379e-12	nit=35	nfev=87	
trust-constr t=12.49s	run 807/1000	MSE=1.787e-12	nit=29	nfev=69	
trust-constr t=34.17s	run 808/1000	MSE=2.608e-16	nit=75	nfev=201	
trust-constr	run 809/1000	MSE=7.601e-14	nit=20	nfev=39	t=6.65s
trust-constr t=27.31s	run 810/1000	MSE=7.671e-14	nit=62	nfev=159	
trust-constr t=35.33s	run 811/1000	MSE=1.078e-15	nit=79	nfev=213	
trust-constr t=24.08s	run 812/1000	MSE=7.068e-14	nit=57	nfev=144	
trust-constr t=31.70s	run 813/1000	MSE=1.965e-12	nit=64	nfev=189	
trust-constr t=17.01s	run 814/1000	MSE=4.924e-15	nit=37	nfev=102	
trust-constr	run 815/1000	MSE=1.266e-12	nit=24	nfev=51	t=8.54s
trust-constr t=40.62s	run 816/1000	MSE=1.266e-12	nit=86	nfev=246	
trust-constr t=11.01s	run 817/1000	MSE=4.875e-15	nit=28	nfev=66	
trust-constr t=29.48s	run 818/1000	MSE=7.196e-14	nit=66	nfev=174	
trust-constr t=14.85s	run 819/1000	MSE=7.674e-14	nit=33	nfev=81	
trust-constr t=28.61s	run 820/1000	MSE=7.611e-14	nit=49	nfev=129	
trust-constr t=26.69s	run 821/1000	MSE=7.607e-14	nit=53	nfev=141	
trust-constr t=15.36s	run 822/1000	MSE=4.904e-13	nit=37	nfev=87	

trust-constr	run 823/1000	MSE=1.379e-12	nit=25	nfev=54	t=9.86s
trust-constr	run 824/1000	MSE=7.664e-14	nit=70	nfev=183	
t=31.75s					
trust-constr	run 825/1000	MSE=7.615e-14	nit=51	nfev=135	
t=25.17s					
trust-constr	run 826/1000	MSE=1.379e-12	nit=27	nfev=63	
t=11.31s					
trust-constr	run 827/1000	MSE=6.066e-13	nit=64	nfev=171	
t=29.12s					
trust-constr	run 828/1000	MSE=7.606e-14	nit=21	nfev=42	t=6.93s
trust-constr	run 829/1000	MSE=1.266e-12	nit=26	nfev=57	t=9.73s
trust-constr	run 830/1000	MSE=1.588e-11	nit=21	nfev=45	t=7.74s
trust-constr	run 831/1000	MSE=4.880e-16	nit=87	nfev=243	
t=43.65s					
trust-constr	run 832/1000	MSE=7.607e-14	nit=30	nfev=72	
t=12.57s					
trust-constr	run 833/1000	MSE=1.379e-12	nit=83	nfev=228	
t=39.33s					
trust-constr	run 834/1000	MSE=5.002e-15	nit=23	nfev=51	
t=10.87s					
trust-constr	run 835/1000	MSE=7.607e-14	nit=23	nfev=48	t=8.23s
trust-constr	run 836/1000	MSE=7.604e-14	nit=42	nfev=114	
t=19.03s					
trust-constr	run 837/1000	MSE=7.634e-14	nit=20	nfev=39	t=7.52s
trust-constr	run 838/1000	MSE=6.067e-13	nit=31	nfev=72	
t=12.21s					
trust-constr	run 839/1000	MSE=4.866e-15	nit=32	nfev=84	
t=14.10s					
trust-constr	run 840/1000	MSE=3.456e-15	nit=78	nfev=210	
t=35.65s					
trust-constr	run 841/1000	MSE=7.604e-14	nit=29	nfev=66	
t=11.74s					
trust-constr	run 842/1000	MSE=7.610e-14	nit=30	nfev=72	
t=15.18s					
trust-constr	run 843/1000	MSE=1.266e-12	nit=25	nfev=54	
t=11.66s					
trust-constr	run 844/1000	MSE=7.605e-14	nit=28	nfev=63	
t=12.15s					
trust-constr	run 845/1000	MSE=4.871e-16	nit=94	nfev=258	
t=48.19s					
trust-constr	run 846/1000	MSE=1.266e-12	nit=28	nfev=63	
t=10.04s					
trust-constr	run 847/1000	MSE=4.678e-13	nit=97	nfev=264	
t=44.20s					
trust-constr	run 848/1000	MSE=1.266e-12	nit=38	nfev=96	
t=16.88s					
trust-constr	run 849/1000	MSE=4.850e-15	nit=28	nfev=69	
t=13.16s					

trust-constr	run 850/1000	MSE=7.609e-14	nit=26	nfev=57	t=9.66s
trust-constr	run 851/1000	MSE=4.439e-16	nit=121	nfev=342	
t=66.32s					
trust-constr	run 852/1000	MSE=6.171e-16	nit=81	nfev=219	
t=41.83s					
trust-constr	run 853/1000	MSE=4.892e-15	nit=64	nfev=174	
t=46.00s					
trust-constr	run 854/1000	MSE=7.606e-14	nit=21	nfev=42	t=7.09s
trust-constr	run 855/1000	MSE=1.000e-11	nit=22	nfev=48	t=8.35s
trust-constr	run 856/1000	MSE=5.449e-16	nit=87	nfev=237	
t=42.70s					
trust-constr	run 857/1000	MSE=6.067e-13	nit=25	nfev=51	t=9.15s
trust-constr	run 858/1000	MSE=7.606e-14	nit=21	nfev=42	
t=11.32s					
trust-constr	run 859/1000	MSE=3.154e-16	nit=75	nfev=201	
t=52.69s					
trust-constr	run 860/1000	MSE=1.379e-12	nit=40	nfev=105	
t=25.00s					
trust-constr	run 861/1000	MSE=9.975e-12	nit=27	nfev=66	
t=12.15s					
trust-constr	run 862/1000	MSE=1.787e-12	nit=27	nfev=60	
t=11.52s					
trust-constr	run 863/1000	MSE=4.782e-15	nit=32	nfev=81	
t=15.75s					
trust-constr	run 864/1000	MSE=1.466e+00	nit=1000	nfev=3003	
t=506.65s					
trust-constr	run 865/1000	MSE=5.163e-15	nit=20	nfev=42	t=6.80s
trust-constr	run 866/1000	MSE=1.001e-11	nit=26	nfev=60	
t=11.08s					
trust-constr	run 867/1000	MSE=1.266e-12	nit=26	nfev=57	
t=10.44s					
trust-constr	run 868/1000	MSE=9.884e-14	nit=30	nfev=66	
t=10.95s					
trust-constr	run 869/1000	MSE=1.912e-16	nit=75	nfev=201	
t=33.16s					
trust-constr	run 870/1000	MSE=7.605e-14	nit=21	nfev=42	t=6.99s
trust-constr	run 871/1000	MSE=1.379e-12	nit=33	nfev=84	
t=14.03s					
trust-constr	run 872/1000	MSE=7.596e-14	nit=62	nfev=159	
t=26.37s					
trust-constr	run 873/1000	MSE=6.521e-14	nit=66	nfev=171	
t=29.24s					
trust-constr	run 874/1000	MSE=1.379e-12	nit=37	nfev=93	
t=15.18s					
trust-constr	run 875/1000	MSE=3.361e-15	nit=84	nfev=228	
t=39.73s					
trust-constr	run 876/1000	MSE=7.614e-16	nit=90	nfev=246	
t=42.68s					

trust-constr	run 877/1000	MSE=1.787e-12	nit=25	nfev=54	t=9.32s
trust-constr	run 878/1000	MSE=1.266e-12	nit=30	nfev=69	
t=11.76s					
trust-constr	run 879/1000	MSE=6.449e-16	nit=80	nfev=219	
t=34.98s					
trust-constr	run 880/1000	MSE=1.379e-12	nit=33	nfev=81	
t=15.51s					
trust-constr	run 881/1000	MSE=9.969e-12	nit=26	nfev=63	
t=12.22s					
trust-constr	run 882/1000	MSE=7.612e-14	nit=53	nfev=141	
t=34.55s					
trust-constr	run 883/1000	MSE=7.584e-14	nit=91	nfev=249	
t=51.90s					
trust-constr	run 884/1000	MSE=1.266e-12	nit=28	nfev=63	
t=14.37s					
trust-constr	run 885/1000	MSE=7.606e-14	nit=36	nfev=87	
t=24.99s					
trust-constr	run 886/1000	MSE=1.379e-12	nit=24	nfev=51	
t=10.55s					
trust-constr	run 887/1000	MSE=7.609e-14	nit=54	nfev=141	
t=31.71s					
trust-constr	run 888/1000	MSE=1.379e-12	nit=28	nfev=69	
t=14.08s					
trust-constr	run 889/1000	MSE=1.379e-12	nit=25	nfev=54	
t=11.50s					
trust-constr	run 890/1000	MSE=6.942e-11	nit=26	nfev=60	
t=12.67s					
trust-constr	run 891/1000	MSE=1.787e-12	nit=29	nfev=66	
t=11.95s					
trust-constr	run 892/1000	MSE=1.266e-12	nit=26	nfev=57	
t=10.67s					
trust-constr	run 893/1000	MSE=1.379e-12	nit=25	nfev=54	
t=10.26s					
trust-constr	run 894/1000	MSE=6.068e-13	nit=28	nfev=60	
t=11.22s					
trust-constr	run 895/1000	MSE=7.609e-14	nit=21	nfev=42	t=7.78s
trust-constr	run 896/1000	MSE=9.974e-12	nit=54	nfev=150	
t=33.00s					
trust-constr	run 897/1000	MSE=7.599e-14	nit=25	nfev=54	
t=13.94s					
trust-constr	run 898/1000	MSE=1.379e-12	nit=23	nfev=48	
t=10.37s					
trust-constr	run 899/1000	MSE=6.064e-13	nit=32	nfev=75	
t=16.97s					
trust-constr	run 900/1000	MSE=1.266e-12	nit=30	nfev=72	
t=17.21s					
trust-constr	run 901/1000	MSE=9.283e-15	nit=77	nfev=207	
t=44.03s					

trust-constr t=32.86s	run 902/1000	MSE=1.379e-12	nit=56	nfev=147	
trust-constr t=14.37s	run 903/1000	MSE=4.897e-13	nit=32	nfev=72	
trust-constr t=19.83s	run 904/1000	MSE=4.897e-13	nit=33	nfev=72	
trust-constr t=18.50s	run 905/1000	MSE=1.379e-12	nit=34	nfev=84	
trust-constr t=11.96s	run 906/1000	MSE=1.266e-12	nit=27	nfev=63	
trust-constr t=39.50s	run 907/1000	MSE=3.923e-15	nit=78	nfev=210	
trust-constr t=19.84s	run 908/1000	MSE=1.379e-12	nit=41	nfev=102	
trust-constr	run 909/1000	MSE=7.611e-14	nit=21	nfev=42	t=7.84s
trust-constr t=11.72s	run 910/1000	MSE=1.379e-12	nit=26	nfev=57	
trust-constr t=49.08s	run 911/1000	MSE=5.560e-15	nit=84	nfev=231	
trust-constr t=15.81s	run 912/1000	MSE=1.266e-12	nit=31	nfev=75	
trust-constr t=11.34s	run 913/1000	MSE=5.099e-15	nit=25	nfev=57	
trust-constr t=15.73s	run 914/1000	MSE=7.870e-14	nit=34	nfev=81	
trust-constr t=11.92s	run 915/1000	MSE=9.987e-12	nit=26	nfev=63	
trust-constr	run 916/1000	MSE=4.858e-15	nit=19	nfev=39	t=7.71s
trust-constr t=19.83s	run 917/1000	MSE=1.379e-12	nit=41	nfev=105	
trust-constr t=15.23s	run 918/1000	MSE=9.970e-12	nit=32	nfev=78	
trust-constr t=65.62s	run 919/1000	MSE=6.068e-16	nit=117	nfev=330	
trust-constr t=41.00s	run 920/1000	MSE=4.825e-16	nit=87	nfev=240	
trust-constr t=34.48s	run 921/1000	MSE=2.644e-16	nit=75	nfev=201	
trust-constr t=38.59s	run 922/1000	MSE=4.026e-16	nit=81	nfev=225	
trust-constr t=12.35s	run 923/1000	MSE=9.972e-12	nit=30	nfev=72	
trust-constr	run 924/1000	MSE=9.996e-12	nit=24	nfev=54	t=9.35s
trust-constr	run 925/1000	MSE=7.603e-14	nit=20	nfev=39	t=7.08s
trust-constr t=14.49s	run 926/1000	MSE=1.379e-12	nit=34	nfev=81	
trust-constr	run 927/1000	MSE=1.379e-12	nit=25	nfev=54	t=9.26s
trust-constr	run 928/1000	MSE=4.940e-15	nit=48	nfev=129	

t=23.02s						
trust-constr	run 929/1000	MSE=1.379e-12	nit=26	nfev=57		
t=10.98s						
trust-constr	run 930/1000	MSE=7.995e-15	nit=102	nfev=285		
t=50.87s						
trust-constr	run 931/1000	MSE=9.883e-14	nit=28	nfev=60		
t=10.91s						
trust-constr	run 932/1000	MSE=6.477e-15	nit=73	nfev=198		
t=34.58s						
trust-constr	run 933/1000	MSE=1.379e-12	nit=28	nfev=66		
t=16.56s						
trust-constr	run 934/1000	MSE=1.379e-12	nit=52	nfev=138		
t=25.26s						
trust-constr	run 935/1000	MSE=1.265e-12	nit=26	nfev=57	t=9.31s	
trust-constr	run 936/1000	MSE=1.266e-12	nit=30	nfev=69		
t=11.23s						
trust-constr	run 937/1000	MSE=4.416e-16	nit=90	nfev=249		
t=48.18s						
trust-constr	run 938/1000	MSE=1.379e-12	nit=46	nfev=126		
t=22.28s						
trust-constr	run 939/1000	MSE=9.971e-12	nit=33	nfev=84		
t=15.91s						
trust-constr	run 940/1000	MSE=1.252e-16	nit=82	nfev=231		
t=45.77s						
trust-constr	run 941/1000	MSE=9.974e-12	nit=33	nfev=84		
t=17.99s						
trust-constr	run 942/1000	MSE=4.899e-13	nit=35	nfev=81		
t=17.67s						
trust-constr	run 943/1000	MSE=7.604e-14	nit=29	nfev=69		
t=12.11s						
trust-constr	run 944/1000	MSE=1.379e-12	nit=33	nfev=81		
t=13.47s						
trust-constr	run 945/1000	MSE=5.101e-14	nit=75	nfev=204		
t=36.89s						
trust-constr	run 946/1000	MSE=9.992e-12	nit=27	nfev=66		
t=11.88s						
trust-constr	run 947/1000	MSE=7.628e-14	nit=60	nfev=153		
t=28.08s						
trust-constr	run 948/1000	MSE=7.621e-14	nit=26	nfev=57		
t=11.60s						
trust-constr	run 949/1000	MSE=4.875e-15	nit=20	nfev=42	t=9.58s	
trust-constr	run 950/1000	MSE=4.439e-15	nit=88	nfev=243		
t=43.42s						
trust-constr	run 951/1000	MSE=1.266e-12	nit=27	nfev=60	t=9.82s	
trust-constr	run 952/1000	MSE=6.487e-13	nit=30	nfev=66		
t=10.67s						
trust-constr	run 953/1000	MSE=1.266e-12	nit=27	nfev=60	t=9.90s	
trust-constr	run 954/1000	MSE=1.379e-12	nit=31	nfev=75		

t=15.19s						
trust-constr	run 955/1000	MSE=9.889e-14	nit=33	nfev=75		
t=13.79s						
trust-constr	run 956/1000	MSE=1.266e-12	nit=23	nfev=48	t=8.85s	
trust-constr	run 957/1000	MSE=9.973e-12	nit=42	nfev=108		
t=21.05s						
trust-constr	run 958/1000	MSE=7.605e-14	nit=29	nfev=66		
t=12.39s						
trust-constr	run 959/1000	MSE=1.000e-11	nit=27	nfev=66		
t=12.26s						
trust-constr	run 960/1000	MSE=7.931e-14	nit=66	nfev=171		
t=32.37s						
trust-constr	run 961/1000	MSE=1.266e-12	nit=29	nfev=66		
t=13.18s						
trust-constr	run 962/1000	MSE=8.054e-16	nit=84	nfev=228		
t=44.43s						
trust-constr	run 963/1000	MSE=9.978e-12	nit=23	nfev=51	t=9.27s	
trust-constr	run 964/1000	MSE=2.880e-16	nit=77	nfev=207		
t=37.98s						
trust-constr	run 965/1000	MSE=1.266e-12	nit=30	nfev=69		
t=12.45s						
trust-constr	run 966/1000	MSE=1.563e-15	nit=92	nfev=255		
t=49.22s						
trust-constr	run 967/1000	MSE=1.379e-12	nit=34	nfev=81		
t=16.70s						
trust-constr	run 968/1000	MSE=3.187e-16	nit=85	nfev=234		
t=44.43s						
trust-constr	run 969/1000	MSE=1.466e-15	nit=94	nfev=264		
t=43.96s						
trust-constr	run 970/1000	MSE=6.069e-13	nit=36	nfev=87		
t=13.87s						
trust-constr	run 971/1000	MSE=6.067e-13	nit=28	nfev=60	t=9.86s	
trust-constr	run 972/1000	MSE=1.379e-12	nit=28	nfev=66		
t=10.88s						
trust-constr	run 973/1000	MSE=7.605e-14	nit=58	nfev=147		
t=29.17s						
trust-constr	run 974/1000	MSE=4.842e-15	nit=25	nfev=57		
t=11.50s						
trust-constr	run 975/1000	MSE=1.266e-12	nit=169	nfev=489		
t=116.51s						
trust-constr	run 976/1000	MSE=7.698e-14	nit=74	nfev=201		
t=40.84s						
trust-constr	run 977/1000	MSE=2.719e-16	nit=74	nfev=198		
t=40.82s						
trust-constr	run 978/1000	MSE=1.379e-12	nit=58	nfev=159		
t=31.39s						
trust-constr	run 979/1000	MSE=9.986e-12	nit=22	nfev=48	t=9.73s	
trust-constr	run 980/1000	MSE=7.609e-14	nit=68	nfev=186		

t=36.64s	trust-constr	run 981/1000	MSE=6.068e-13	nit=34	nfev=78	
t=16.88s	trust-constr	run 982/1000	MSE=1.786e-12	nit=28	nfev=66	
t=15.25s	trust-constr	run 983/1000	MSE=7.593e-14	nit=156	nfev=444	
t=85.50s	trust-constr	run 984/1000	MSE=6.894e-14	nit=58	nfev=147	
t=28.03s	trust-constr	run 985/1000	MSE=2.755e-15	nit=98	nfev=270	
t=53.36s	trust-constr	run 986/1000	MSE=4.063e-15	nit=116	nfev=324	
t=64.32s	trust-constr	run 987/1000	MSE=7.599e-14	nit=28	nfev=63	
t=13.18s	trust-constr	run 988/1000	MSE=1.379e-12	nit=28	nfev=63	
t=12.45s	trust-constr	run 989/1000	MSE=1.379e-12	nit=41	nfev=108	
t=23.01s	trust-constr	run 990/1000	MSE=1.266e-12	nit=28	nfev=63	
t=14.03s	trust-constr	run 991/1000	MSE=9.987e-16	nit=87	nfev=237	
t=48.39s	trust-constr	run 992/1000	MSE=1.266e-12	nit=24	nfev=51	
t=10.58s	trust-constr	run 993/1000	MSE=9.995e-12	nit=24	nfev=54	
t=11.57s	trust-constr	run 994/1000	MSE=1.266e-12	nit=35	nfev=93	
t=20.37s	trust-constr	run 995/1000	MSE=4.979e-15	nit=20	nfev=42	t=7.74s
	trust-constr	run 996/1000	MSE=5.321e-15	nit=19	nfev=39	t=8.59s
	trust-constr	run 997/1000	MSE=5.943e-16	nit=75	nfev=201	
t=42.95s	trust-constr	run 998/1000	MSE=9.986e-12	nit=57	nfev=153	
t=34.76s	trust-constr	run 999/1000	MSE=4.514e-16	nit=104	nfev=294	
t=61.75s	trust-constr	run 1000/1000	MSE=6.067e-13	nit=45	nfev=111	
t=25.41s						

```

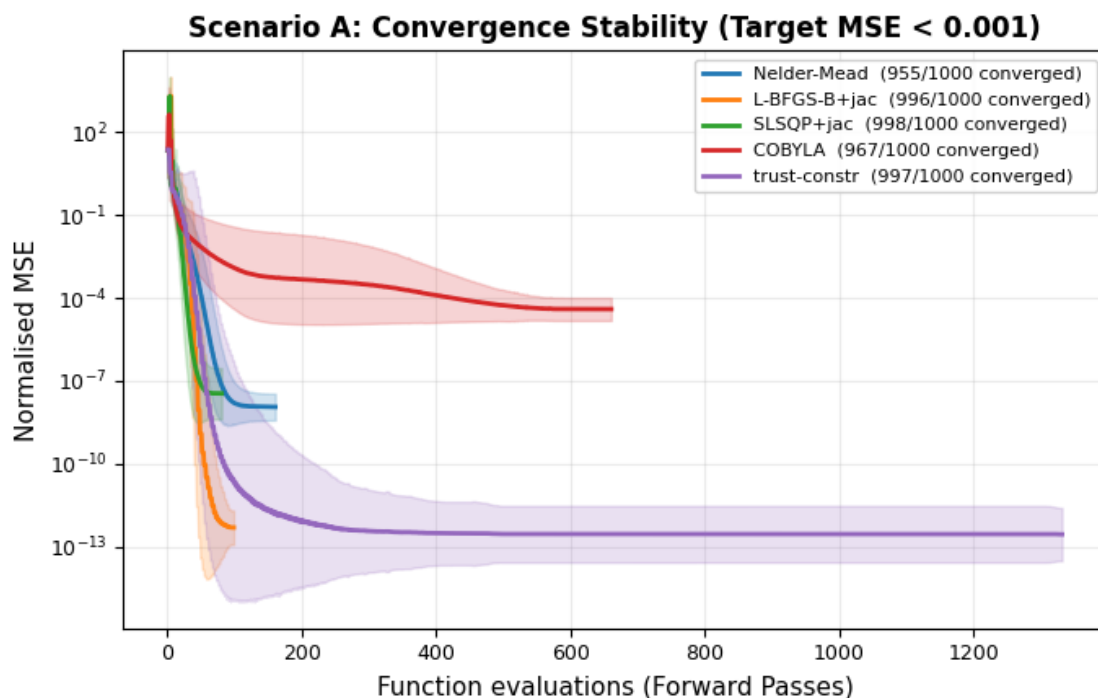
=====
=====
SCENARIO A: Robustness (Tol < 0.001) & Discovered Physics State vs Reference
=====
=====
method_tag  conv_rate nit_mean nfev_mean wall_time_mean final_mse_mean I_1_ref
I_1      I_3_ref   I_3      alpha_x   alpha_y
      COBYLA      97%    237.8    237.8      38.879 s      3.46e-05    0.8218 A

```

```

0.7581 A 1.0430 A 0.5891 A -7.10e-04 -2.08e-03
L-BFGS-B+jac 100% 12.0 54.3 8.110 s 4.95e-13 0.8218 A
0.7581 A 1.0430 A 0.5891 A -1.23e-07 -1.33e-07
Nelder-Mead 96% 39.7 75.8 11.625 s 8.41e-09 0.8218 A
0.2913 A 1.0430 A 0.2791 A 8.36e-06 1.66e-05
SLSQP+jac 100% 11.3 37.9 5.683 s 3.47e-08 0.8218 A
0.2913 A 1.0430 A 0.2791 A -6.89e-07 -3.59e-07
trust-constr 100% 47.1 120.9 21.316 s 1.52e-13 0.8218 A
0.7581 A 1.0430 A 0.5891 A -1.04e-07 -2.12e-07
=====
=====

```



Scenario A: Best Optimised Quadrupole Currents per Method

Method	Best MSE	I (A)	I2 (A)
Nelder-Mead	1.7298e-10	0.2913	0.2791
L-BFGS-B+jac	1.7676e-16	0.7581	0.5891
SLSQP+jac	3.0209e-13	0.2913	0.2791
COBYLA	3.9280e-09	0.7581	0.5891
trust-constr	4.7706e-17	0.7581	0.5891

Reference FELsim S1 currents: I=0.8218 A, I2=1.0430 A

Overall best across all methods/runs:

Method: trust-constr, MSE=4.7706e-17

I=0.75807 A, I2=0.58906 A

```
[5]: # save the results
experiment_save_path = os.path.join(CURRENT_DIR, "../results/
↳scenario_A_results_v4.pkl")
save_results(results_A, df_A, final_summary, experiment_save_path)
```

```
[9]: # load the results
results_A, df_A, final_summary = load_results(experiment_save_path)
```

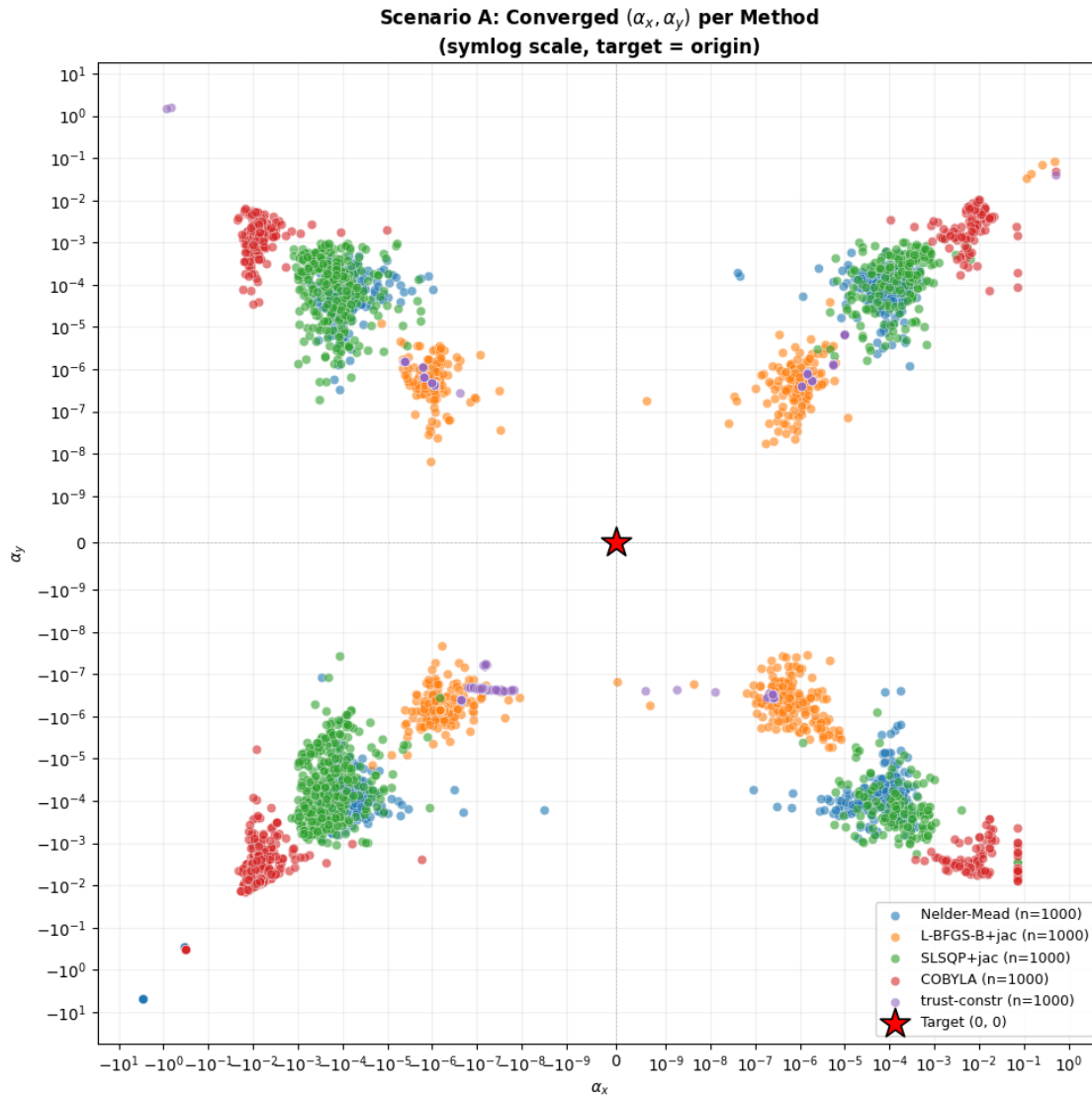
experiment results loaded from c:\Users\yilu2\my_files\research\FELsim\experiment\../results/scenario_A_results_v4.pkl

give the (alpha_x,alpha_y) scatter plot and MSE heatmap with iteration trajectories

```
[12]: import importlib, experiments_utils
importlib.reload(experiments_utils)
from experiments_utils import *
```

c:\Users\yilu2\my_files\research\FELsim\backend

```
[13]: # scatter plot
# parameter setting
N_GRID = 100
plot_MSE(df_A)
```

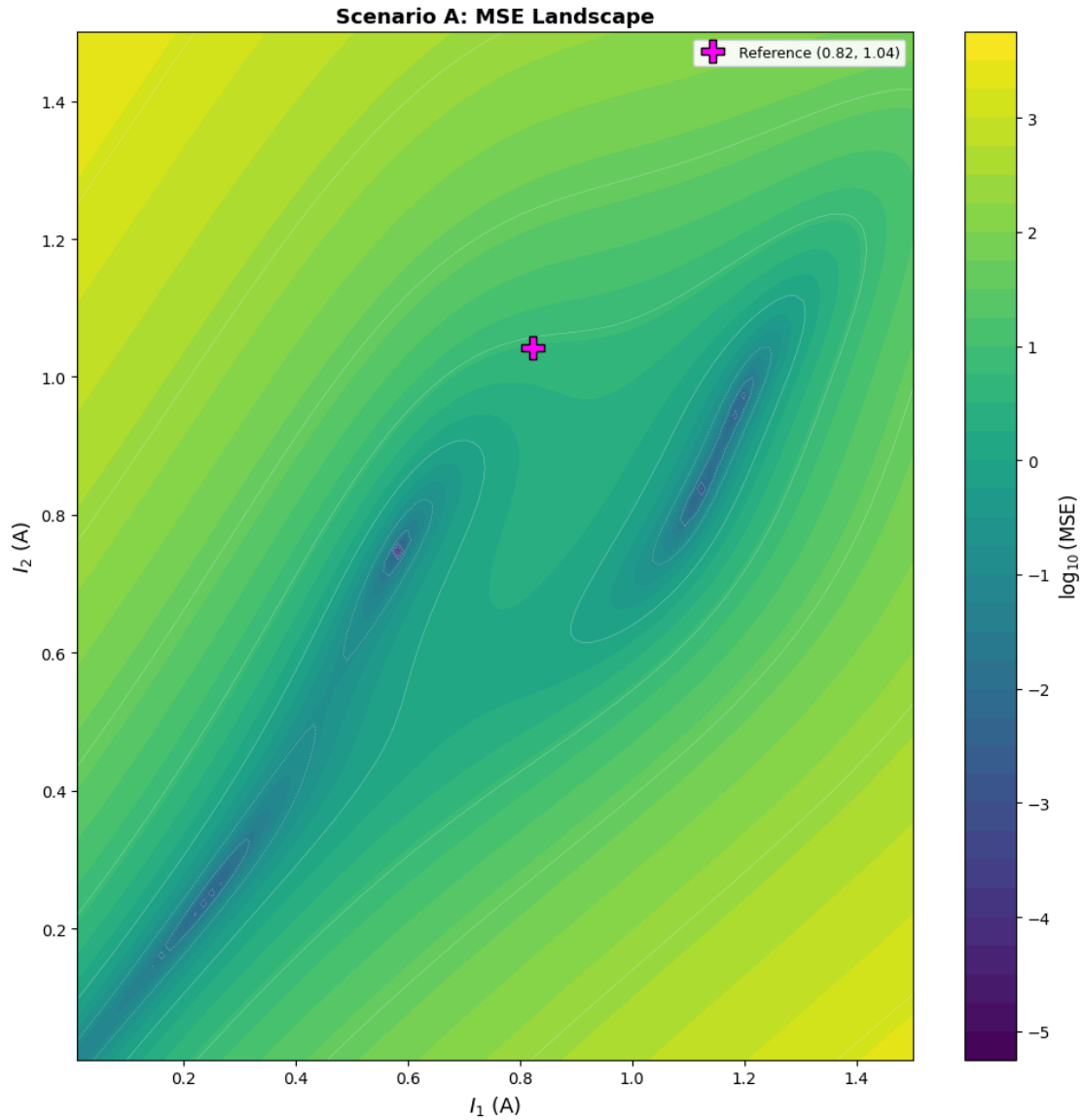


```
[14]: # save the landscape information
landscape_save_path = os.path.join(CURRENT_DIR, "../results/
↳scenario_A_landscape_v4.pkl")
save_landscape(CURRENT_BOUNDS, A_BEAMLINE_LEN, A_VARS, A_OBJ, N_GRID,
↳landscape_save_path)
plot_contour(landscape_save_path)
```

```
row 5/100    104s
row 10/100   199s
row 15/100   290s
row 20/100   382s
row 25/100   473s
row 30/100   567s
```

row 35/100	658s
row 40/100	750s
row 45/100	840s
row 50/100	929s
row 55/100	1020s
row 60/100	1110s
row 65/100	1201s
row 70/100	1291s
row 75/100	1381s
row 80/100	1470s
row 85/100	1560s
row 90/100	1650s
row 95/100	1740s
row 100/100	1829s

Grid 100 100, MSE range [6.38e-06, 3.21e+03]



```
[15]: N_STARTS = 10
trajectories_save_path = os.path.join(CURRENT_DIR, "../results/
↪scenario_A_trajectories_v4.pkl")
save_trajectories(A_BEAMLINE_LEN, N_STARTS, A_VARS, A_OBJ, METHODS_A, ↪
↪METHOD_OPTIONS, trajectories_save_path)
plot_trajectories(landscape_save_path, trajectories_save_path)
```

10 start points:

```
[[0.277 0.963]
 [0.706 0.562]
 [0.539 1.188]
 [1.359 0.274]]
```

```

[0.983 0.454]
[1.451 1.381]
[0.957 1.132]
[0.778 1.241]
[0.678 0.515]
[0.424 0.347]]
Nelder-Mead      start1 (0.28,0.96)  steps= 44  MSE=2.21e-09  x=[0.2457 0.2501]
Nelder-Mead      start2 (0.71,0.56)  steps= 26  MSE=1.97e-08  x=[0.5815 0.747 ]
Nelder-Mead      start3 (0.54,1.19)  steps= 39  MSE=2.47e-08  x=[1.1177 0.8281]
Nelder-Mead      start4 (1.36,0.27)  steps= 59  MSE=1.96e-09  x=[0.2456 0.2501]
Nelder-Mead      start5 (0.98,0.45)  steps= 33  MSE=2.98e-08  x=[0.5815 0.747 ]
Nelder-Mead      start6 (1.45,1.38)  steps= 42  MSE=3.90e-08  x=[1.1177 0.8281]
Nelder-Mead      start7 (0.96,1.13)  steps= 25  MSE=1.46e-08  x=[1.1902 0.9582]
Nelder-Mead      start8 (0.78,1.24)  steps= 30  MSE=1.65e-09  x=[1.1177 0.8281]
Nelder-Mead      start9 (0.68,0.51)  steps= 29  MSE=5.30e-08  x=[0.5815 0.747 ]
Nelder-Mead      start10 (0.42,0.35)  steps= 37  MSE=6.12e-09  x=[0.2456 0.2501]
L-BFGS-B         start1 (0.28,0.96)  steps= 11  MSE=1.40e-12  x=[0.5815 0.7469]
L-BFGS-B         start2 (0.71,0.56)  steps= 10  MSE=1.02e-13  x=[0.5815 0.7469]
L-BFGS-B         start3 (0.54,1.19)  steps= 10  MSE=1.55e-12  x=[0.5815 0.7469]
L-BFGS-B         start4 (1.36,0.27)  steps= 15  MSE=3.07e-12  x=[1.1177 0.8281]
L-BFGS-B         start5 (0.98,0.45)  steps= 17  MSE=6.84e-13  x=[1.1902 0.9583]
L-BFGS-B         start6 (1.45,1.38)  steps= 17  MSE=1.93e-12  x=[1.1902 0.9583]
L-BFGS-B         start7 (0.96,1.13)  steps= 19  MSE=5.87e-12  x=[1.1902 0.9583]
L-BFGS-B         start8 (0.78,1.24)  steps= 16  MSE=1.34e-12  x=[1.1177 0.8281]
L-BFGS-B         start9 (0.68,0.51)  steps= 9   MSE=3.18e-14  x=[0.5815 0.7469]
L-BFGS-B         start10 (0.42,0.35)  steps= 13  MSE=3.60e-13  x=[0.2457 0.2501]
SLSQP            start1 (0.28,0.96)  steps= 15  MSE=7.25e-08  x=[1.1177 0.828 ]
SLSQP            start2 (0.71,0.56)  steps= 7   MSE=3.09e-08  x=[0.5815 0.7469]
SLSQP            start3 (0.54,1.19)  steps= 14  MSE=1.28e-09  x=[1.1177 0.8281]
SLSQP            start4 (1.36,0.27)  steps= 15  MSE=1.01e-07  x=[0.2457 0.2501]
SLSQP            start5 (0.98,0.45)  steps= 10  MSE=6.32e-08  x=[0.5815 0.7469]
SLSQP            start6 (1.45,1.38)  steps= 14  MSE=1.90e-07  x=[1.1902 0.9583]
SLSQP            start7 (0.96,1.13)  steps= 11  MSE=1.44e-08  x=[1.1177 0.8281]
SLSQP            start8 (0.78,1.24)  steps= 12  MSE=3.36e-08  x=[1.1902 0.9583]
SLSQP            start9 (0.68,0.51)  steps= 9   MSE=7.34e-09  x=[0.5815 0.7469]
SLSQP            start10 (0.42,0.35)  steps= 10  MSE=4.00e-09  x=[0.2457 0.2501]
COBYLA           start1 (0.28,0.96)  steps= 99  MSE=3.02e-05  x=[1.1192 0.8303]
COBYLA           start2 (0.71,0.56)  steps= 44  MSE=9.76e-06  x=[0.5819 0.7477]
COBYLA           start3 (0.54,1.19)  steps=604  MSE=6.22e-05  x=[0.2415 0.2456]
COBYLA           start4 (1.36,0.27)  steps= 77  MSE=1.45e-05  x=[1.191  0.9596]
COBYLA           start5 (0.98,0.45)  steps= 44  MSE=9.28e-06  x=[1.1896 0.9571]
COBYLA           start6 (1.45,1.38)  steps= 89  MSE=9.69e-06  x=[1.1169 0.8269]
COBYLA           start7 (0.96,1.13)  steps= 57  MSE=5.63e-05  x=[1.1918 0.961 ]
COBYLA           start8 (0.78,1.24)  steps=109  MSE=4.73e-05  x=[0.242  0.2462]
COBYLA           start9 (0.68,0.51)  steps= 18  MSE=6.17e-02  x=[0.0112 0.0101]
COBYLA           start10 (0.42,0.35)  steps=529  MSE=4.97e-05  x=[0.2419 0.2461]
trust-constr      start1 (0.28,0.96)  steps= 23  MSE=5.77e-14  x=[0.5815 0.7469]
trust-constr      start2 (0.71,0.56)  steps= 80  MSE=1.11e-15  x=[0.5815 0.7469]

```

trust-constr	start3	(0.54,1.19)	steps= 22	MSE=5.76e-14	x=[0.5815 0.7469]
trust-constr	start4	(1.36,0.27)	steps= 31	MSE=5.77e-14	x=[0.5815 0.7469]
trust-constr	start5	(0.98,0.45)	steps= 34	MSE=2.79e-13	x=[1.1177 0.8281]
trust-constr	start6	(1.45,1.38)	steps= 27	MSE=7.54e-13	x=[1.1177 0.8281]
trust-constr	start7	(0.96,1.13)	steps= 29	MSE=1.04e-12	x=[1.1902 0.9583]
trust-constr	start8	(0.78,1.24)	steps= 26	MSE=5.76e-14	x=[0.5815 0.7469]
trust-constr	start9	(0.68,0.51)	steps= 24	MSE=9.55e-15	x=[0.5815 0.7469]
trust-constr	start10	(0.42,0.35)	steps= 30	MSE=2.81e-12	x=[0.2457 0.2501]

trajectories saved (5 methods * 10 starts)

